



Artificial Lift Field Operations Reference Manual - Downhole (ALFORM-D)

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Revision History

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1.0	31-Jan-2008	AL SS FSM was split into ALFORM-D and ALFORM-S.	Author: Alan Brown, Olesya Pearce
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2.2	19-Aug-2009	Revisions are made per InTouch tickets 4861729, 4844505, 4835962, 4815444, 4837612	Author: Olesya Pearce
2.3	11-Nov-2009	Revisions are made per InTouch tickets 4914602, 4903576.	Author: Olesya Pearce
2.4	06-Jan-2010	Updated the winding code for the 540-Series Motors. Additional revisions made per InTouch tickets 4903576, 4914602, 4779243, 4944615, 4948757, 4988310, 4998249	Author: Charleton Walker
2.5	01-Apr-2010	Added the Maximus appendix section, updated shimming procedures and updated HOTLINE 550 pothead installation procedure. Change title from "Shimming (Thrust Setting) Procedures for Submersible Pumps" to "Pump Shimming and Installation."	Author: Charleton Walker
2.6	24-Jun-2010	Corrected working voltage in the Personal Protective Equipment section and the "V" hole background color in the Modular Protector LSBPBSL section. Corrected protector clamp part number and Maximus gauge installation torque. Revisions made per 5075921, 5075869, 5075846, 5080490, 5080455	Author: Charleton Walker
2.7	09-Jul-2010	Updated the Equipment Serial Numbering System, Hand Tools, Installation Tools and Maximus Equipment. Added Mechanical Lifting Equipment section and removed references to the connection tables in the Technical and Engineering Information section. Revisions made per 5088789, 5101717.	Author: Charleton Walker

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Version	Date	Description	Prepared by
2.8	26-Aug-2010	Updated the Additional installation tools table in the Maximus section.	Author: Tyas Adi Pramana, Sergey Choudnovets
2.9	30-Nov-2010	Updated the Protector Installation section in the Maximus appendix. Added the oil filling for the tandem protector (PFSB on top 66L on bottom) to the Wellsite Procedures section.	Author: Tyas Adi Pramana, Sergey Choudnovets
3.0	20-Jul-2011	Updated gauge information in Wellsite Procedures section to cover standard ESP gauges. Removed Phoenix remote sensor installation procedures. Added the Hotline Gauge information to the Non-Standard Applications appendix section. Removed AL FSM references, changed load ratings from tons to lb. InTouch tickets 5396610, 5511441, 5416774.	Author: Tyas Adi Pramana, Sergey Choudnovets
3.1	05-Mar-2012	Updated Cables section, removed obsolete cable specifications. Added Trident MLE installation instructions to the Maximus section. Added the PL plugs and redress kits to the Logging Bypass section. Added notes to verify vent plug installation to the protector oil filling procedures in the Wellsite Procedures section. Removed oil filling procedures and added HotlineSA3 specifications in the Non-Standard Applications section. Updated Parts, Tools and Materials section and Equipment Serial Numbering System section.	Author: Tyas Adi Pramana, Sergey Choudnovets
3.2	27-Feb-2013	Reiterated vent plug torque and drain-and-fill valve torque in the Wellsite Procedures section. Removed the "-" from the part number references, updated the installation tools table, updated the lifting clamp torque settings in the Parts, Tools and Materials section. Corrected vent plug torque value in oil filling procedures for protectors section of the Wellsite Procedures section. Updated the cable nomenclature, specifications and testing procedures in the Cables section. Added reference to the HotlineSA3 MLE procedure (6058300) to the Non-Standard Applications section.	Author: Tyas Adi Pramana, Sergey Choudnovets

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QHSE

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QHSE

Schlumberger QHSE information is accessible via the Schlumberger QHSE Policy and Standards and OFS QHSE Standards Reference Page, InTouchSupport.com content ID 3312250.

This section is a collection of all QHSE Danger, Warning, and Caution notes found throughout this manual. The user must read and understand this entire section before installing or servicing any Artificial Lift equipment. Failure to read and follow QHSE directives is non-compliant with Schlumberger QHSE policy and consequences may include sanctions up to and including termination.

1.1

Danger to Personnel

**Warning**

Potential Severity:	Serious
Potential Loss:	Personnel
Hazard Category:	Machinery equipment hand tools

If a substitute extension bar is used, the front-end diameter must not be greater than 0.75 in. The extension bar and opener head must maintain a loose engagement to avoid personal injury during use.

**Danger**

Potential Severity:	Catastrophic
Potential Loss:	Personnel
Hazard Category:	Electrical

Ensure the power is off before working at the junction box.

**Danger**

Potential Severity:	Major
Potential Loss:	Personnel
Hazard Category:	Fire flammable

Never use a megger on the rig floor. A spark generated by a megger can ignite gasses emitted from the wellhead. Complete this test while the motor is in the shipping box. Additional megger tests can be completed safely at the cable reel once the cable is connected to the motor.

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**Danger**

Potential Severity: Major
 Potential Loss: Personnel
 Hazard Category: Electrical

Always discharge each winding after the megger test. The residual high-voltage charge can shock personnel and/or damage other meters.

**Danger**

Potential Severity: Major
 Potential Loss: Personnel
 Hazard Category: Temperature

Always use caution when handling heated equipment.

**Warning**

Potential Severity: Serious
 Potential Loss: Assets, Personnel
 Hazard Category: Electrical

Notify all personnel in the immediate area that electrical equipment will be energized up to the junction box. Place a flashing beacon on the junction box to further alert personnel in the area of danger.

**Warning**

Potential Severity: Serious
 Potential Loss: Personnel
 Hazard Category: Toxic corrosive hazardous substances

Lead is a recognized hazard to occupational health. The WCP-QHSE-S003 Personnel Lead Protection Standard is available from InTouchSupport.com (content ID 3846334).

**Warning**

Potential Severity: Serious
 Potential Loss: Assets, Personnel
 Hazard Category: Explosives

Gas accumulation in the switchboard cabinet can cause an explosion. A high-voltage junction box is required for all submersible pump applications. A junction box will preventing gas migration through the cable into the switchboard.

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**Danger**

Potential Severity: Catastrophic
 Potential Loss: Personnel
 Hazard Category: Electrical

High voltage is present in transformers. Always wear your hot gloves and other personal protective equipment when working with energized equipment.

**Danger**

Potential Severity: Catastrophic
 Potential Loss: Personnel
 Hazard Category: Electrical

High voltage is present in transformers. Megger tests must be preformed by qualified personnel. All power must be disconnected and locked out. Both primary and secondary bushings must be disconnected.

**Danger**

Potential Severity: Catastrophic
 Potential Loss: Personnel
 Hazard Category: Electrical

The voltage must be properly applied to the transformer. Failure to do so can result in serious personal injury, death or property damage. All tests must be preformed by qualified personnel.

**Danger**

Potential Severity: Major
 Potential Loss: Personnel
 Hazard Category: Fire flammable

Be aware of the temperature and pressure of the wellhead and piping before loosening any connection. Liquids can easily flash to vapors when the pressure is relieved and cause severe burns.

**Danger**

Potential Severity: Major
 Potential Loss: Personnel
 Hazard Category: Fire flammable

Before loosening any connection on the pumping unit, make certain the temperature is cool to the touch. If there are pockets of contained liquids under pressure, these could flash to vapors and cause a severe injury.

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1.2 Danger to Equipment



Danger

Potential Severity: Major
Potential Loss: Assets
Hazard Category: Machinery equipment hand tools

Certain concentrations of H₂S will cause embrittlement and subsequent failure of stainless steel bands.



Danger

Potential Severity: Major
Potential Loss: Assets
Hazard Category: Electrical

If non-armored cable is used, cable clamp saddles must be used to prevent cable damage by the metal bands.



Warning

Potential Severity: Serious
Potential Loss: Assets
Hazard Category: Machinery equipment hand tools

Ensure the cable sheave is attached securely with a chain and has a cable or chain safety backup.



Danger

Potential Severity: Major
Potential Loss: Assets
Hazard Category: Electrical

The switchboard case should be grounded to the well casing.



Danger

Potential Severity: Major
Potential Loss: Assets
Hazard Category: Electrical

Extreme care should be used in handling the switchboard. When uncrating, pry bars should be used instead of a hammer. The interior of the switchboard must be kept free of moisture. Always examine internal components for possible shipping damage.

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**Danger**

Potential Severity: Major
 Potential Loss: Assets
 Hazard Category: Electrical

Always perform a final visual inspection of the motor terminals for proper spacing in the slot and check for phase resistance after assembling the UMB. Refer to [Figure 4-6](#). For further details refer to [InTouch Content ID 3769116](#).

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Any cable damage should be located and repaired prior to installation. Never take a chance that unrepaired damage will have no effect.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Do not use test equipment with output voltages greater than 1500 volts DC to test PSI units. This can damage the unit.

**Caution**

Potential Severity: Light
 Potential Loss: Assets, Personnel
 Hazard Category: Electrical

Be careful do not to pull the three-phase terminal ring. Connected is a signal wire leading down to the instrument. Pulling the terminal ring may result in damage to the wire.

**Caution**

Potential Severity: Light
 Potential Loss: Assets, Personnel
 Hazard Category: Electrical

Use extreme caution when installing or removing adapters from the instrument. Pull the adapter straight away from its mating piece (along the tool longitudinal axis) if the adapter moves sideways and impacts a connector, it could damage the connector or wire.

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**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Use care not to damage the terminal insulators or O-rings.

**Warning**

Potential Severity: Serious
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

The lower part of string should be covered while shimming.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

If a protector is placed in a horizontal position without first being drained, water will flow to the other section. If stored, this can cause a seal to malfunction on the next installation, resulting in a burned motor.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Never reuse emery cloth when abrading different materials such as copper, insulation or jackets. This will embed copper into the insulation and degrade the integrity of the splice.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Cuts in the conductor insulation are not permitted past the bare conductor area. If the insulation is damaged when removing the jacket, remove the damaged area and start a new splice.

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**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Be careful not to nick or ring cut the copper when removing the insulation. This can create a hot spot which can result in a failure.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Do not remove the alloy coating covering the copper conductor.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Improper installation of splicing sleeves can result in splice failure.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Avoid heat damage to the cable under the soldered area. Do not use a torch. Cool the soldered joint immediately with a water-wetted cloth. Avoid contact with hot surfaces.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

When cutting with the hacksaw be careful not to damage the insulation on the adjacent conductors.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Remove the Kapton® insulating tape from the conductor on motor lead extensions using a scraping tool and 150-grit emery cloth.

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**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

No voids or pin holes should exist in the solder. This is critical.

**Warning**

Potential Severity: Serious
 Potential Loss: Assets
 Hazard Category: Electrical

Do not shut down a unit with the disconnect switch. Doing so shuts the unit down against a load and could cause arcing across the disconnects.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Do not attempt to service the switchboard with power connected. Open the main disconnect switch before opening the switchboard and use a voltmeter to verify no voltage is present before servicing.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Always make a visual inspection, making certain all three switch contacts are open.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Before pulling, make insulation and continuity test to make sure problem is not on the surface.

**Warning**

Potential Severity: Serious
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

Do not perform a MIST test on a motor filled with Reda No. 7 oil as serious personal injury or equipment damage may occur.

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**Warning**

Potential Severity: Serious
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

Use caution not to damage the shaft spline or flange connections during lifting.

1.2.1

Transformer QHSE

The following precautions must be followed. Failure to do so can result in death, serious injury or property damage.

**Danger**

Potential Severity: Catastrophic
 Potential Loss: Personnel
 Hazard Category: Electrical

Only trained and qualified personnel should service transformers. Dangerous voltages, pressures and other hazards are present that can result in death or serious injury.

Improperly performed work can result in catastrophic failure or explosion of the transformer.

All sources of power to the transformer must be disconnected.

All sources of power must be locked out and disconnected.

1. Before installing a transformer, be sure the transformer kVA, voltage, winding configuration and other characteristics are suitable for the application. This information is listed on the transformer nameplate.
2. Always lift the transformer by all four tank lifting lugs. Ensure the weight of the transformer does not exceed the capacity of the equipment used to lift it.
3. Do not lay a transformer on its side. Doing so will cause air to be trapped in the windings resulting in the failure of the transformer.
4. Solidly ground the transformer tank before it is energized.
5. Never operate a tap switch or a delta-wye switch while the transformer is energized. The transformer can violently fail.
6. Before connecting any equipment to the secondary of the transformer, measure the secondary voltage to ensure that it is suitable for the equipment to be connected. Since deadly voltages will be present, use suitable measuring equipment, hot gloves and observe all safety practices.
7. When a transformer will be serviced, all sources of power must be disconnected and locked out. The HV (high-voltage) and LV (low-voltage) terminals should be grounded when you are positive there is no voltage

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present at these terminals. The HV and LV leads should then be disconnected from the transformer bushings. Remove the grounds placed on the terminals prior to reapplying power.

8. The transformer tank cover or inspection access cover must not be removed when conditions exist that may allow moisture, dirt, foreign objects or other contamination to enter the transformer tank. You should remove loose items from your pockets prior to opening the cover. The entry of these materials in the transformer tank may result in the premature failure of the transformer.



Danger

Potential Severity: Major
 Potential Loss: Assets
 Hazard Category: Electrical

All pressure must be relieved from the transformer tank prior to removing the cover. This can be accomplished by pulling the ring on the pressure relief valve.

The transformer must be resealed so that it is pressure tight. The transformer should be pressure tested to 7 lbf/in² (48.3 kPa) with dry nitrogen after it is resealed. Thoroughly inspect the transformer for leaks.

9. If the level of the insulating liquid must be lowered to perform service, under no circumstances should the level of oil be any lower than 3 in (7.6 cm) above the top of the transformer coils. This helps prevent the coils from being exposed to air or moisture.

The liquid removed from the transformer must be stored in clean, tightly sealed barrels. The hoses and pump used to transfer oil must be clean. If there is a doubt, obtain enough extra insulating fluid of the same type as in the transformer to thoroughly flush the hoses and pump. Do not reuse this fluid.

When the transformer is refilled, take care to ensure that air is not forced into the windings of the transformer. Air that enters the coils will cause the transformer to fail.

After the transformer is refilled, wait at least three hours before energizing the transformer. This allows time for air in the insulating fluid to dissipate. Prior to energizing the transformer, all switches should be operated to dislodge any trapped air.

A transformer with a winding voltage that exceeds 5,000 V should be vacuum filled. Contact the transformer manufacture for specific instructions.

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1.2.2 QHSE for Extended Runlife Protector

This section reviews the manufacturing and field handling procedures and the potential hazards shop and field service personnel may experience while assembling, testing, installing, pulling and dismantling an Extended Runlife Protector.

1.2.2.1 HSE summary

The Extended Runlife Protector has undergone independent HSE reviews by SRC and BPC engineering, manufacturing and QHSE personnel. The findings of the review are

- No hazardous materials such as radioactive or explosive materials are used in any of the Extended Runlife Protectors.
- No electrical components are used in any of the Extended Runlife Protectors.
- No high-pressure components are used in any of the Extended Runlife Protectors.
- Only normal wellbore temperatures are experienced in the Extended Runlife Protectors.
- All of the Extended Runlife Protectors contain hydraulic oil and require normal hygiene precautions.
- The Extended Runlife Protectors can be heavy. Manual lifting is not recommended. Standard lifting methods are provided in the *Wellsite Procedures*.

1.2.2.2 Applicable standards

Standards applicable to the assembly, testing, installing, pulling and dismantling of the Extended Runlife Protector are shown in [Table 1-1](#).

Table 1-1: Applicable Extended Runlife Protector assembly, testing, installing, pulling and dismantling standards.

Standard	Description
ED-105	O-ring installation in oil well equipment
ED-164	Chain torque wrench tighten of threaded joints
ED-174	Joint weld specification
GED-001	Equipment serial numbering system
MPI-53112	Protector assembly, modular

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Standard	Description
MPI-59102	Install shipping cap
MPI-59105	Paint unit, install nameplate, patent sticker and install screen
MPI-59110	Spot weld joints
MPI-2015017	MPI, ASSEMBLY, 3 CHAMBER ADVANCED PROTECTOR ASSEMBLY, 540
MPI-2015020	MPI, OIL-FILLING, 3 CHAMBER ADVANCED PROTECTOR, 540
MPI-2015022	MPI, DISMANTLE, 3 CHAMBER ADVANCED PROTECTOR, 540
QAD-3054	Material handling
QAI and TI-2027	SPS vibration analysis for vertically and horizontally tested equipment
QAI and TI-2028	Helium leak test
QAI and TI-2013	Modular protector testing, 400 and 540 series
QAI and TI-3019	Assembly quality plan
100010378	MPI, Assembly, Advanced Protectors, Four-Chambers, 540
100012094	MPI, Oil Filling, Four-Chamber Advanced Protector, 540
100012248	MPI, Dismantle, Four-Chamber Advanced Protector, 540

1.2.2.3 Assembly

Quality

All material handling shall be performed in accordance with QAD 3054. General assembly shall follow QAI and TI 3019 Assembly Quality Plans.

- **Oil** — All current motor fluid used in protectors is considered nonhazardous.
- **Thread lube** — Current lubrication used on housings and head, base and body threads is food grade and is considered nonhazardous.
- **Loctite®** — Thread locking chemicals are considered nonhazardous.
- **Teflon® tape** — Nonhazardous.
- **Bellows** — The bellows are constructed of Inconel® 625. This material is already common within the system and not considered hazardous.

Safety

- **Handling and lifting** — During the course of assembly, several awkward and/or heavy components will require lifting. The most notable of these is the housing and the assembled unit. In each case, lifting should be done with a lifting sling and a hoist in accordance with MPI 57101.

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Other heavy items such as the head, base and shaft should be lifted and transported using proper stepping, handling and lifting procedures.

When possible all shop transport of long heavy objects should be done using a housing dolly.

- **Making up joints** — Protector joints are routinely tightened to only 200 ft-lbf. Appropriate documentation and tooling is available and in place.

1.2.2.4 Final test

Quality

Standard protector test parameters and procedures will apply except for the following:

- **Valve pressure setting** — An increase from 10 psi to 20 psi requires a change in the testing procedure for both the individual valve and the top-level assembly after the valve is installed. (See QAI 2013.)
- **Metal bellows** — A pressure and vacuum test needs to be performed on the bellows.

Safety

- **Lifting for test** — Lifting should be done with a lifting sling and a hoist in accordance with MPI 57101.
- **Pinch points** — When making flange to flange connections, keep fingers clear while mating the units.

1.2.2.5 Shipping

Quality

All material handling should be performed in accordance with QAD 3054. Boxing and/or loading of equipment should be done in accordance with MAT-B019.

Safety

The protector should not be lifted by hand. Lifting of the unit either for loading in a shipping box or loading onto a truck shall be done with a hoist and sling in accordance with MPI 57101 or with a forklift in accordance with QAD 3054.

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1.2.2.6 Equipment installation

Quality

The unit installation and wellsite procedures will be in accordance with current practices. Unit flange bolts will be tightened in accordance with the torque requirements in ED-119.

Safety

- **Lifting** — This unit should not be lifted by hand. All lifting clamps shall be used in accordance with FSB 119/rd and lifting should be performed in accordance with procedures.



Warning

Potential Severity:	Serious
Potential Loss:	Assets
Hazard Category:	Machinery equipment hand tools

Use caution not to damage the shaft spline or flange connections during lifting.

- **Pinch points** — When making flange-to-flange connections, keep fingers clear while lowering unit.

1.2.2.7 Equipment pull

Quality

Equipment pulls should be performed in accordance with procedures provided in the *Wellsite Procedures* .

1.2.2.8 Dismantle

Quality

The Extended Runlife Protector shall be dismantled per appropriate manufacturing process instruction (MPI).

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Parts, Tools and Materials

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2 Parts, Tools and Materials

The procedures covering the shelf life and storage for parts, components, raw materials and assembled equipment are described in [InTouch Content ID 5088035](#).

2.1 Personal Protective Equipment

Personal protective equipment descriptions and part numbers are shown in [Table 2-1](#).

Table 2-1: Personal Protective Equipment Descriptions and Part Numbers

Description	Part Number
Glove: (Hot) Lineman, 17 kV rating, inside Insulator (size 9 Woman/Small Man)	2004986
Glove: Leather Protective for Lineman (Hot) Glove (size 9)	2004987
Glove: Leather Protective for Rubber Gloves, Stringer 520–6 (size 10)	518068
Glove: Leather Protective for Lineman (Hot) Glove (size 10)	514760
Glove: (Hot) Lineman, 17 kV rating, inside Insulator (size 11 Large Man)	2004991
Glove: Leather Protective for Lineman (Hot) Glove (size 11)	2004992
Glove: (Hot) Lineman, 17 kV rating, inside Insulator (size 12 Extra Large)	2004993
Glove: Leather Protective for Lineman (Hot) Glove (size 12)	2004994
Glove Bag: Canvas, for Hot Gloves	509588
Glove: Talc Power	2004997
Glove: Tester (portable)	2004995
Glove: Tester (bench top)	2004996
Safety Glasses: (sideshields with clear lens)	2005001
Face Shield Headgear: (with ratchet take-up/adjustment)	2005002
Face Shield: (polycarbonate lens for Crew 103 headgear)	2005003
Chemical Splash Goggles: (clear lens, perforated sides)	2005005
Ear Muff: (High attenuating, dielectric, folding)	2005006
Hard Hat: MSA V-Gard with suspension SLB logo (cap style)	2005012
Hard Hat: MSA V-Gard with suspension SLB logo (full brim style)	2005013
Personal H2S detector: MSA "cricket" (Lo alarm 5 ppm Hi alarm 15 ppm)	2005020
Escape Pack: (High exertion fresh air pack, 5 Minute capacity)	2005021
EEBA Carrying Case: (North Safety Manufacturing P/N 885)	2005022
Survey Meter: (model 3 with microRoentgen/hour face)	2005014

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Description	Part Number
Nal Gamma Scintillator: Model 44-2 1 in X 1 in (Use with Model 3 Meter)	2005015
Radiation Detector Target: (1 microCurie Cesium 137 yellow disk)	2005016
Target Holder: (use with Ludlum target P/N 47-1532)	2005017
Radiation Detector carrying Case: (Model 3 Ludlum meter)	2005018
Scintillation Pocket Survey Meter: (Model 2401-S cesium)	2005019
Tyvek Suit: (disposable w/feet XL)	2005009
Tyvek Suit: (disposable w/feet XXL)	2005010
Tyvek Suit: (disposable w/feet XXXL)	2005011
Lockout Tag: Danger, Do Not Operate (3/8-in grommet for padlock)	2004999
Lockout Safety Hasp: (1½ in jaw accommodates up to 6 padlocks)	2004998
Lockout Padlock: (One key per lock, Key retained by lock when unlocked)	2005000

2.2

Electric Test Equipment

Electric test equipment descriptions and part numbers are shown in [Table 2-2](#).

Table 2-2: Electric Test Equipment

Description	Part Number
Ampmeter: Current Clamp-On, AC Fluke 801-600 (Fluke Model 83 & 87)	856856
Battery: Nicad Pack for Scopemeter Fluke Model 99B, PM 9086/001	1129360
Battery: Triplet 1.5 Volt	568782
Battery: Triplet 30 Volt	568790
Case: Hard case for Fluke Scopemeter Model 99B	1129410
Case: Leather Carrying for Triplet Model 639	511048
Case: Pelican #1500S, 6-7/8 X 15-1/4" X 18-1/2"	856872
Case: Soft Case for Fluke Scopemeter Model 97	1129378
Filter: Voltage, for use with Speedstar VSD	1229053
Fuse: for Fluke Model 83 & 87, 250V 2 Amp, AGX-2 (Buss), GGX-2 (Gould)	905224
Fuse: Triplet 1 Amp, AGC-1	682716
Fuse: Triplet 2 Amp, AGC-2	743229
Indicator: Motor Rotation, with Case	845768
Indicator: Phase Sequence, 600-4160 Volt, W/ Case; TSW-45	845776
Kit: Scopemeter, Includes Scopemeter Fluke Model 99, Hard Carrying Case & Amprobe Adapter	1129386
Kit: Test Meters, Includes 845784, 1043918, 856856, 856864, 856872	492264
Lead: Replacement, Fluke Y8134	856849

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Description	Part Number
Lead: Yokogawa Megger, (Ground)	857367
Lead: Yokogawa Megger, (Probe)	857375
Leads: Fluke Model #83	1043926
Leads: Triplett (Old Style) Male Conn, Type 3	515932
Leads: Triplett (New Style) Female Conn, Type 4 W/Alligator Clips	818583
Line Voltage Adapt/Battery Charger: for Fluke Scopemeter, 115/230V, N. American Style Plug	1157254
Megger: 5KV 120V, 50/60Hz, 250VA, Biddle Catalog #210400	976332
Megger: Yokogawa, 1000V & Leads, Complete W/Case	845784
Multimeter: Fluke Model #83, C/W Holster, Test Leads & C25 Case	1043918
Multimeter: Fluke Model #87, True Rms, C/W Holster, Test Leads & C25 Case	1048685
Multimeter: Triplett Model 630, 600 Volt Max. Leads Included	522797
Power Divider: 10 To 1 Voltage Ratio, For Use With Fluke Multimeters, Weston TS-W-22	856864
Probe: Fluke Scopemeter, Alligator Clip, 600V, PM 9084/001	1157221
Probe: Fluke Scopemeter, Hook-Style Clip, 600V, PM 9085/001	1157239
RS 232-C Optically Isolated Interface Adapter: Fluke PM 9080/001	1157247
Scopemeter: Fluke Model 99B, 50 MHZ Dual Trace, C/W Probes, Test Leads, Battery & Charger	1129394
Tester: Insulation, 230V 50HZ, Model HD B12	940502
Tester: Insulation, AC/DC, Hipot Hipotronic #HD-125, 10KV, 120V 60HZ	817411
Tester: Oil Dielectric, Hipotronics OC60D W/TCDE, Test Cell	564112
Variac: 0-140 Volt	830620

2.3

Installation Tools

Torque wrench operating instructions are described in [InTouch content ID 5244288](#).

Installation tools descriptions and part numbers are shown in [Table 2-3](#).

Table 2-3: Installation Tools Descriptions and Part Numbers

Description	Part Number
Tool Set, Field Service: Pump, 338-387 Series; Motor, 375 Series	0912048
Tool Set, Field Service: Pump, 387, 400, 450 Series; Motor, 456 Series	0912071
Tool Set, Field Service: Pump, 500-562 Series; Motor 540 Series	0912063
Tool Set, Field Service: Pump, 675 Series; Motor 738 Series	0912055
Adapter, Alemite: Oil Filling, Protector and Motor All Series	0001479

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Description	Part Number
Alemite: Pin Type Fitting, 1/8 in NPT, 31/32 in Long, A336	0551481
Alemite, Bushing: Reducing, Hex, 1/4 in MNPT X 1/8 in FNPT, Zinc-Plated, McMaster 4638K651 (use for Alemite Connection)	0910471
Alemite, Hose: Assembly, A-908-A, 48	0554170
Bit, Double-Ended: Special Bondhus 1/4 in X 3 in Long	0832675
Bit, Double-Ended: Special Bondhus 5/16 in X 3 in Long	0832683
Bit, Hex: .38 in Drive X .312 in Short Bit Socket	0747105
Bit, Hex: 1/4 in Socket X 3/8 in Drive, BA-8	0747055
Blade, Hacksaw: HHS 12 in, 18 Tooth	0569236
Blade: Surform, Stanley 21-293	0569293
Brush, Scratch: Lutz #30 #1777	0570614
Brush, Tube: Osborn, .75 in 842	0570770
Cleaner: Hand Citrus, w/ Pumice, 4-1/2 Lb Can Vin-Dotco 145	0573162
Drill/Driver: 5.2 Amp 3/8 in VSR with Storage Bag	0574327
File, Bastard: Square 10 in Nicholson	0579037
File: Stanley Surform 295	0579292
Flashlight: Eveready 2 Cells Torch-Light 1259	0582825
Frame, Hacksaw: 12 in 90 Douglas	0579458
Freon, Electronic Liquid: FLURINERT FC40 or FC43 11Lb Container (1 Gallon) 3M S	0734616
Link: .75 in Master Assembly, Grade 8, Crosby 1014306	0902262
Pick, Ice: Shank Diameter Not Greater than 0.2 in, Overall Length 7-10 in	0581355
Pliers, Long Nose: 6-1/2 in Length, 1-5/16 in Jaw Length	0586131
Pliers, Ring P105 IRR	0586222
Pliers: Crescent R220C	0586172
Pliers: Retaining Ring, 28 in Long, Internal, c/w Tip	0586248
Pliers: Stake On, T&B #WT-111M or Klein Tool #1005 or Equal	0517987
Pump: Tire Hand	0557496
Screwdriver: Blade, 5/16 in X 6 in, Zinc Plated	0588566
Screwdriver: Phillips, #1 X 6 in or Equal	1035484
Screwdriver: Phillips, #2 X 8 in, Non Insulated, Alloy Steel, Chrome Plate	1035476
Sling: 2 in W X 4 ft L, Type 5, Liftex EN1-62, incl. Manufacturing Load Certification, Vertical SWL 4400 Lbs	0589085
Snip, Band Cutter: Cut Straight to Left, WISS® M1R	0022178
Socket: 1/2 in 12 Point, Universal Joint, 3/8 in Drive, Williams BU-1216	0908046
Socket: 3/4 in Wrench 375-450-540-738 Series	0326041
Stickopaint, White: B Markal	0585828
Stove Pipe Assembly: Standard 9-5/8 in, 7 in, Suitable up to 562 Series ESP	A030031-231

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Description	Part Number
Stove Pipe Assembly: Standard 9-5/8 in, 9 in	A030059-231
Stripper, Wire: Ideal, Category No. 45-091	0563015
Tap: 1/2 in - 20 NF Hand Bottom Semi 2-1/2 Chamfer SH-47 Thread H3 4FL Greenfield 5303	0591016
Tap: 3/8 in - 24 NF Hand Bottom Semi 2-1/2 Chamfer SH-47 Thread H3 4FL Greenfield 5303	0590992
Tap: 5/16 in - 24 NF Hand Bottom Semi 2-1/2 Chamfer SH-47 Thread H3 4FL Greenfield 5303	0590984
Tap: 7/16 in - 20 NF Hand Bottom Semi 2-1/2 Chamfer Sh-47 Thread H3 4FL Greenfield 5303	0591008
Tap: 9/16 in - 18 NF Hand Bottom Semi H3 4FL EDP 25542T Hanson Whitney or Equal	0591024
Tool Box: 20 in L X 10.5 in H X 8.75 in W, w/ Tray, Contico 8200GY or Equivalent	1035369
Tool Box: Aluminum, Welded 1/8 in, 22 in Long x 10 in High x 10 in Wide, Special No Tray	0908731
Tool Box: Plastic, 22 in Long c/w Lift-Out Tray, Craftsman Mechanics 09-06508	1192632
Tool: Bit, Ballpoint, 3/8 X 1/4 Drive, 3 In L, Bondhus 10814	0747071
Tool: Cap Pressure Test, Flat Cable Assembly, 1 Direct Connect Pothead, 738 Series Dominator	2007371
Tool: Cap Pressure Test, Type B, Flat Cable Assembly, 375 Series	0305128
Tool: Cap Pressure Test, Type B, Flat Cable Assembly, 456 Series	0305136
Tool: Cap Pressure Test, Type B, Flat Cable Assembly, 540 Series	0305144
Tool: Cap Pressure Test, Type B, Flat Cable Assembly, 738 Series	0305151
Tool: Cap Pressure Test, Type X, Flat Cable Assembly, 456 Series	0888024
Tool: Cap Pressure Test, Type X, Flat Cable Assembly, 540 Series	0888032
Tool: Test Cap, Bolt-on, Maximus, 562 Series	100133248
Tool: Test Cap, Bolt-on, Maximus, 456 Series	100127330
Tool: Chisel, Cold, 1 in	0572867
Tool: Chisel, Cold, 1/2 in	0572826
Tool: Field, 5 Gallon Fluid Can For Use With Pump	0020974
Tool: Fold Up Hex Key Set (Allen Hex Set Small)	0593483
Tool: Gage Assembly, Field Pressure	0912923
Tool: Hammer, Ball Pein, 16 Oz	0580258
Tool: Hammer, Ball Pein, 24 Oz	0580266
Tool: Hammer, Ball Pein, 8 Oz	0580233
Tool: Handle, Ratchet 3/8 in Drive	0580415
Tool: Handle, Speed 3/8 in Drive	0580431
Tool: Hex Key, 1/4 in	0593624
Tool: Hex Key, 5/8 in	0593723
Tool: Hex Key, 7/16 in	0593665

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Description	Part Number
Tool: Installation Pothead Bolts (Torque Wrench Adapter)	0881466
Tool: Jack, Bottle, 6 Ton, In-Line Pmp, 8-3/4 in Minimum Lift Height, 5-1/4 in Hydraulic Stroke, 5-3/4 in X 3-1/2 in Base, Serrated Saddle, Grainger# 4GU76, or Equivalent	100392921
Tool: Jack & Plate, Hydraulic Type, Field Service Only, 375/450/540/562	100541677
Tool: Jack & Plate, Hydraulic Type, Field Service Only, 738 Series	100541675
Tool: Kit, Pressure Test, Flat Cable Without Pressure Test Caps, All Series Flat	0007955
Tool: Knife, Pocket	0582742
Tool: Lead, Scoring, .75 in Diameter X 6.75 in Long, Electrical Area	2005155
Tool: Pliers, Diagonal Cutting, High Leverage Head, 8 in Length, 13/16 in Jaw Length	0586115
Tool: Pliers, Diagonal, 6 in	0586099
Tool: Pliers, Motor Lead, 375 Series	0021923
Tool: Pliers, Motor Lead, 450, 540 Series	0008847
Tool: Pliers, Motor Lead, 738 Series	0032268
Tool: Pliers, Needle Nose, 2-11/16 in X 11/16 in Jaw, 7-15/32 in Overall Length	0586081
Tool: Proto 9853 Screw Starter or Equivalent (Pencil Magnet)	0592055
Tool: Screw Driver, 1/4 in X 4 in Flat	0588533
Tool: Shaft Setting, CT Pump to LT Pump D4300N/D5800N, 400 Series	2005330
Tool: Shaft Setting, Field Service, Protector to Intake, 338, 387, & 400	1088996
Tool: Shaft Setting, Field Service, Pump to Protector 738 Series	0046672
Tool: Shaft Setting, Field Service, Pump to Pump, 338	1089044
Tool: Shaft Setting, Field Service, Pump to Pump, 387	1089036
Tool: Shaft Setting, Field Service, Pump to Pump, 400	1089028
Tool: Shaft Setting, Field Service, Pump to Pump, 562	1083948
Tool: Shaft Setting, Field Service, Pump to Pump, 675/675, 675/562 AGH	1050400
Tool: Shaft Setting, Field Service, Pump to Pump, 725	1320045
Tool: Shaft Setting, Field Service, Pump to Pump, 862	1089432
Tool: Shaft Setting, Field Service, Pump to Pump, 862 HPH-H	100444696
Tool: Shaft Setting, Field Service, Pump, Protector, Intake Assembly 540 Series	1083930
Tool: Shaft Setting, Pump To Protector, Field Service, Pump, 950	0354597
Tool: Shaft Setting, Pump to Pump, Field Service, Pump, 66 C, 650	0048504
Tool: Socket Wrench, 9/16 in 6 point, 3/8 in Drive, William B-618	0961367
Tool: Socket, 1/2 in, 3/8 in Drive, 12 Point	1157759
Tool: Socket, 3/4 In, 3/8 in Drive, 12 Point	0908038
Tool: Socket, 5/8 in, 3/8 in Drive	0961359
Tool: Socket, Deep, 1-1/8 in, 1/2 in Drive, 3-1/4 in Long	101168703
Tool: Socket, Deep, 1-1/2 in, 1/2 in Drive, 3-1/2 in Long	101168675
Tool: Socket, Deep, 1-7/8 in, 1/2 in Drive, 3-1/2 in Long	101229115

Private

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Description	Part Number
Tool: Spline Alignment, 1.12 PD-27 Inv. Spline, UT - CT/LT, Motor 456 Dominator	2007272
Tool: Spline Alignment, 1.125-27 Inv, Promotor, 456	100188779
Tool: Spline Alignment, 1.12-6B Spline, Tandem Motors 456 Series	0455360
Tool: Spline Alignment, 1.188 SAE-6B Motors, 540	0044263
Tool: Spline Alignment, 1.37-21 Inv, Promotor, 562	100189508
Tool: Spline Alignment, 1.375-21 Tooth Involute, Motor Tandem, 562	1079920
Tool: Spline Alignment, 2.00-20 Spline Motors, LT-UT 738 Dominator	2006419
Tool: Spline Line Up, 1-3/8 in SAE-6B, 540 Series	0320937
Tool: Spline Line-Up, .69-6A Spline, Motor, 375	0044115
Tool: Spline Line-Up, 1.125-6B Shaft, 540	0044297
Tool: Spline Line-Up, 456 Series	0044271
Tool: Spline Line-Up, 738 Series	0044289
Tool: Subassembly, Fluid Filling Pump	0001586
Tool: Test Cap, Bolt-On, Maximus, 562	100133248
Tool: Test Cap, Bolt-On, Promotor, 456	100127330
Tool: Torque Wrench, Adjustable, 3/8 in Drive, 30-250 In-Lbs, Rubber Handle	100323750
Tool: Torque Wrench, .50 in Drive, 10-80 Ft/Lbs w/ .62 in X .75 in & .56 in X .50 in, S/A	0035634
Tool: Torque Wrench Micrometer TP 1/2 in TP, 50-250 ft-lb	2003754
Tool: Wrench Drain and Fill Valve 3/4 in 12 Point - 1/2 in Drive Motors, Protectors	1300870
Tool: Wrench, .875-6B Spline, Shop & Field, 387, 400, 540, 562 Series	0882316
Tool: Wrench, .87-6A Spline, Male And Female, 456 Maximus LT/S Protector and 456 Maximus UT/S Motor	100642812
Tool: Wrench, 1.00-6B Spline Wrench Shop & Field, 387, 400, 540, 562	0882324
Tool: Wrench, 1.19-6B Spline, Male And Female, 540 Maximus LT/S Protector and 562 Maximus UT/S Motor	100644839
Tool: Wrench, 1.19-6B Spline, Pump, Assembly	0886085
Tool: Wrench, 1.375-21 Spline, Male & Female Test, Well & Field Service 562 Series	1214840
Tool: Wrench, 1.375-6B Spline, Pump, Assembly, 862 Series	1087592
Tool: Wrench, 1-1/8 in Spline, Motor, 540 Series	0042663
Tool: Wrench, 1-3/16 in Spline, Pump, 738 Series	0013243-T
Tool: Wrench, 2.00-16C Spline, Pump	0886101
Tool: Wrench, 7/8 in X 1 in Spline	0007591
Tool: Wrench, Adjustable, 10 in	0593780
Tool: Wrench, Adjustable, 12 in	0593798
Tool: Wrench, Adjustable, 8 in	0593772
Tool: Wrench, Adjustable, Chrome, 4 in	0593855
Tool: Wrench, Adjustable, Length 6 in	0593764

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Description	Part Number
Tool: Wrench, Air Impact Cooper Tool MP2265, 10-55 Ft/Lb Range	1086354
Tool: Wrench, Box 0625 - .563 in Double Head, Double Offset, 12 Point	0001164
Tool: Wrench, Box End, 5/8 in X 3/4 in, Proto 1131	0047233
Tool: Wrench, Box End, 9/16 in X 1/2 in, Proto 1125	0047225
Tool: Wrench, Box Open End 15/16 in, William #1168 or Equal	0593996
Tool: Wrench, Cable Flange Flat Cable Elect Dept 375	0020735
Tool: Wrench, Combination, 1 in, 14 in Long	7003544
Tool: Wrench, Combination, 1/2 in, Length 8 in	0593426
Tool: Wrench, Combination, 11/16 in, 10-1/4 in Long	0593467
Tool: Wrench, Combination, 1-1/8 in, 15-7/8 in Long	7069446
Tool: Wrench, Combination, 13/16 in, Length 11-3/4 in	0593475
Tool: Wrench, Combination, 3/4 in Length 11 in	0593459
Tool: Wrench, Combination, 7/16 in, 7 in Long	0593418
Tool: Wrench, Combination, 9/16 in, 8-7/8 in Long	0593434
Tool: Wrench, Combination, 5/8 in, 9-1/2 in Long	0594044
Tool: Wrench, Flat Cable Flange, 738 Series	0031500
Tool: Wrench, Open End, 9/16 in X 5/8 in	0593442
Tool: Wrench, Packing Gland	0001396
Tool: Wrench, Pipe 18 in, RIDGID or Equal	0593947
Tool: Wrench, Speed, .312 Hex	0012963
Tool: Wrench, Speed, .375 Hex	0317149
Tool: Wrench, Speed, Socket 5/16 in Hex Extension	0033084
Tool: Wrench, Spline 1.12 PD-27 Inv Spline Motor, 456 Dominator	2007273
Tool: Wrench, Spline, 1.19-6B X 1.50-6B	0039594
Tool: Wrench, Torque, 3/8 in Drive, 20-100 Ft/Lbs, with Adapter	0593962
Wrench, Allen: 1/2 in, Short Arm	0593681
Wrench, Allen: 3/4 in	0594028
Wrench, Allen: 3/8 in, Short Arm	0593657
Wrench, Allen: 5/16 in, Long Arm	0593640
Wrench, Allen: 5/32 in, Short Arm	0593582
Wrench, Allen: 7/16 in, Long Arm	0593673
Wrench, Chain: 18 in X 2 1/2 in, RIDGID	0594069
Wrench, Chain: Heavy Duty 14 in X 2 in, RIDGID C-14 or Equal	0594051
Wrench, Hex Set: Ekland 51	0593509
Wrench, Torque: 1/2 in Beam Type, 0-150 Ft/Lbs, Armstrong 64-740 Adapter	0976274
Wrench, Torque: 1/2 in Drive, Proto 6008C, 10-80 Ft/Lbs Setting	1156926
Wrench, Vise Grip: 10R	0593756

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Description	Part Number
Wrench, Vise Grip: 7R	0593749
Wrench: 3/8 in Open End X 3/8 in Box End ,Snap-On OEX12B	0839985
Wrench: Box End, 13/16 in X 7/8 in, Alloy Steel, Black Oxide, Length 13-7/8 in	0593814

2.4

Mechanical Lifting Equipment

Mechanical lifting equipment for ESP installations is shown in the following tables.

- [Table 2-4: Lifting Clamps](#)
- [Table 2-5: Motor Tail Clamps](#)
- [Table 2-6: Lifting Chains](#)
- [Table 2-7: Centering Hooks](#)
- [Table 2-8: Cable Sheaves](#)



Note

See [OneCAT](#) or [InTouch Content ID 5095846](#) for ESP equipment weight information.

Table 2-4: Lifting Clamps

Part Number	Description	SWL Rating (lbf)	Bolt Torque Setting (lbf-ft)
Motor Clamps			
0609016	Clamp: Motor, 375 Series	9,000	100
0607010	Clamp: Motor, 456 Series	9,000	100
0004218	Clamp: Motor, 540 or 562 Series UT and 540 LT	12,300	100
1072842	Clamp: Motor, 562 Series, CT and LT	12,500	100
1220060	Clamp: Motor, 562 Series, Friction (to use with Direct Connect Pothead)	12,500	200
2005841	Clamp: Motor, 738 Series Dominator, Friction UT, CT & LT (to use with Direct Connect Pothead)	12,000	200
0607572	Clamp: Motor, 738 Series Dominator (to use with Standard Pothead)	12,300	100
Protector Clamps			
101094477	Clamp: Protector, 325 Series, Friction	12500	100

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Part Number	Description	SWL Rating (lbf)	Bolt Torque Setting (lbf-ft)
0041236	Clamp: Protector, 325/375 Series, Shoulder (same clamp as Pump 338 Series)	9,000	100
0609016	Clamp: Protector, 375 Series, Shoulder (same clamp as Motor 375 Series)	9,000	100
0046748	Clamp: Protector, 387 Series, Friction (same clamp as Pump 387 Series)	9,000	200
0028803	Clamp: Protector, 400 Series, Friction (same clamp as Pump 400 Series)	9,000	200
0027441	Clamp: Protector, 540/562 Series, Friction (same clamp as Pump 540 Series)	12,500	200
2006630	Clamp: Protector, 738 Series, Friction (same clamps as Pump 675 Series)	12,300	200
Pump Clamps			
0041236	Clamp: Pump, 338 Series, Shoulder	9,000	100
0046748	Clamp: Pump, 387 Series, Friction	9,000	200
1142298	Clamp: Pump, 387 Series, Shoulder	9,000	100
0028803	Clamp: Pump, 400 Series, Friction	9,000	200
0035808	Clamp: Pump, 400 Series, Shoulder	9,000	100
1148899	Clamp: Pump, 538 Series, Friction	12,500	200
101067259	Clamp: Pump, 538 Series, Shoulder	12,500	100
101067283	Clamp: Pump, 540 Series, Shoulder	12,500	100
0611327	Clamp: Pump, 562 Series, CT Shoulder	12,500	100
1220060	Clamp: Pump, 562 Series, Friction	12,500	200
0031906	Clamp: Pump, 675 Series, Friction	12,500	200
0042051	Clamp: Pump, 675 Series, Shoulder	12,000	100
1320003	Clamp: Pump, 725 Series, Shoulder	12,300	100
0041590	Clamp: Pump, 862 Series, Friction	17,000	200
Pump Clamps			
0044495	Clamp: Pump, 950 Series, Friction	16,200	200
1301290	Clamp: Pump, 1000 Series, Friction	16,200	200
PCP Pump Clamps			
1301274	Clamp: Pump, PCP 400 Series, Friction	9,000	200
1230754	Clamp: Pump, PCP 540 Series, Friction	12,500	200
1282235	Clamp: Pump, PCP 6.88 in ID, Friction	12,500	200

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Table 2-5: Motor Tail Clamps

Part Number	Description	SWL Rating (lbf)	Bolt Torque Setting (lbf-ft)
2003224	Clamp: Motor Tail, 456 Series / 456 Dominator	3,000	100
0001636	Clamp: Motor Tail, 540 Series	3,000	100
1055797	Clamp: Motor Tail, 562 Series	3,000	100
0320911	Clamp: Motor Tail, 738 Series	4,000	100

Table 2-6: Lifting Chains

Part Number	Description	WLL Rating at 60° Angle (lbf)
0902437	Chain: Lifting Equipment Clamp, Chain Length 22 in	12,300
2010793	Chain: Lifting Equipment Clamp, Chain Length 37 in	12,300
1301316	Chain: Lifting Equipment Clamp, Chain Length 23 in	18,000
100367280	Chain: Lifting Equipment Clamp, Spreader, Chain Length 33 in (to use on equipments with compensating shipping cap)	13,900

Table 2-7: Centering Hooks

Part Number	Description	SWL Rating (lbf)
0334482	Centering Hooks Assembly: to use with 3.5 in or smaller tubing elevator (applicable to all series Lifting Clamps)	11,250
100433456	Centering Hooks Assembly: to use with 3.5 in or smaller tubing elevator (applicable to all series Lifting Clamps)	18,000
100461915	Centering Hooks Assembly: to use with 5.5 in tubing elevator (applicable to all series Lifting Clamps)	25,000

Table 2-8: Cable Sheaves

Part Number	Description	SWL Rating (lbf)
0043950	Cable Sheave: Absorber, Wheel and Fork, 39 in Wheel (to use for round cable installation)	650
0044602	Cable Sheave: Absorber, Wheel and Fork, 54 in Wheel (to use for flat cable installation)	650

2.5

Hand Tools for Motors

Tool sets, description and part number for pumps to motors are shown in [Table 2-9](#).

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Table 2-9: Hand Tool Sets

Description	Part Number
Tool Set, Field Service: Pump, 338-387 Series; Motor, 375 Series	0912048
Tool Set, Field Service: Pump, 387, 400, 450 Series; Motor, 456 Series	0912071
Tool Set, Field Service: Pump, 540-562 Series; Motor, 540 Series	0912063
Tool Set, Field Service: Pump, 650- and 675 Series; Motor, 738 Series	0912055
Refer to OneCAT for the complete list of tools, descriptions and part numbers.	

2.6 Installation Materials

Installation materials descriptions and part numbers are shown in [Table 2-10](#).

Table 2-10: Installation Materials Descriptions and Part Numbers

Description	Part Number
Antigalling Compound: Food Grade, 8 oz. c/w Brush Top	849737
Capscrew: Monel, Hex Head, 5/16 in-24UNF X 1-1/4 in	846162
Capscrew: Monel, Socket Head, 3/8 in-24UNF X 1-1/2 in	118182
Capscrew: Monel, Socket Head, 7/16 in-20UNF X 1-1/2 in	118190
Capscrew: Monel, Socket Head, 1/2 in-20UNF X 1-7/8 in	267427
Capscrew: Monel, Socket Head, 9/16 in-18UNF X 1-7/8 in	210484
Capscrew: Steel, Hex Head, 5/16 in-24UNF X 1-1/4 in	89268
Capscrew: Steel, Socket Head, 3/8 in- 24UNF X 1-1/2 in	68965
Capscrew: Steel, Socket Head, 7/16 in-20UNF X 1-1/2 in	68973
Capscrew: Steel, Socket Head, 1/2 in-20UNF X 1-7/8 in	266932
Capscrew: Steel, Socket Head, 9/16 in-18UNF X 1-7/8 in	256503
Capscrew: Steel, Socket Head, 5/8 in-18UNF X 2-1/4 in	105916
Cleaner: CRC	2005332
Drain And Fill Valve: Monel	170811 ¹
Drain And Fill Valve: Steel	23853 ¹
Drain And Fill Valve: Monel	2003058 ¹
Drain And Fill Valve: Steel	1312974 ¹
Flakeline Kit (used to repair monel coating shelf life 6 months)	601971
Flat Cable Guard: 8 Ft. 3kv, Galv.	251488
Gasket: Break-Off Plug	46144
Gasket: Lead, Drain And Fill Valve	23903
Gasket: Lead, Vent Plug	15008
Grease, Lubricant: Dow Corning #4, 5.3 oz Tube	553420

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Description	Part Number
Grease, Lubricant Dow Corning #111, 5.3 oz Tube	100386898
Lockwasher: Monel, 5/16 in	310078
Lockwasher: Monel, 3/8 in	309740
Lockwasher: Monel, 7/16 in	310086
Lockwasher: Monel, 1/2 in	310151
Lockwasher: Monel, 9/16 in	310094
Lockwasher: Monel, 5/8 in	824300
Lockwasher: Steel, 5/16 in	306985
Lockwasher: Steel, 3/8 in	306993
Lockwasher: Steel, 7/16 in	307223
Lockwasher: Steel, 1/2 in	307009
Lockwasher: Steel, 9/16 in	307017
Lockwasher: Steel, 5/8 in	307025
Lockwasher: 375-series motor	0310086
Lockwasher: 456-, 540-, and 562-series motor	0309740
Lockwasher: 738-series motor	0310086
Nut: 375-series motor	0310185
Nut: 456-, 540-, and 562-series motor	0861062
Nut: 738-series motor	0310177
Oil, Low Temperature Motor Jack, Univis J13, 5 Gal Pail	044780
Oil: REDA No. 2 (5 Gal. Can)	556548
Oil: REDA No. 2 (55 Gal. Drum)	556498
Oil: REDA No. 3 (5 Gal. Can)	979062
Oil: REDA No. 3 (55 Gal. Drum)	976373
Oil: REDA No. 5 (5 Gal Can)	7004856
Oil: REDA No. 5 (55 Gal Drum)	7004872
Oil: REDA No. 6 (5 Gal Can)	7007248
Oil: REDA No. 6 (55 Gal Drum)	7005556
Shim: 1/200 in To Fit .62 in, .68 in, .87 in Shafts	7005218
Shim: 1/200 in To Fit 1.00 in, 1.187 in Shafts	7005192
Shim: 1/200 in To Fit 1.37 in, 1.50 in Shafts	7005200
Shim: 1/64 in To Fit .62 in, .68 in, .87 in Shafts	1146497
Shim: 1/64 in To Fit 1.00 in, 1.187 in Shafts	614461
Shim: 1/64 in To Fit 1.37 in, 1.50 in Shafts	1146521
Shim: 1/32 in To Fit 0.62 in, 0 .68 in, 0 .87 in Shafts	1146505
Shim: 1/32 in To Fit 1.00 in, 1.187 in Shafts	614479
Shim: 1/32 in To Fit 1.37 in, 1.50 in Shafts	1146539

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Description	Part Number
Shim: 1/16 in To Fit 0.62 in, 0.68 in, 0.87 in Shafts	1146513
Shim: 1/16 in To Fit 1.00 in, 1.187 " Shafts	614487
Shim: 1/16 in To Fit 1.37 in, 1.50 in Shafts	1146547
Shipping Gasket: 338 Series Motor Head	245233
Shipping Gasket: 375 Series Motor Head	50203
Shipping Gasket: 375 Series Pothole Flange, Type B	217216
Shipping Gasket: 375 Series Tandem Motor	83089
Shipping Gasket: 400 Series Motor Head	233718
Shipping Gasket: 450 Series Pump	104661
Shipping Gasket: 456 Series Motor Head	269910
Shipping Gasket: 456 Series Pothole Flange, Type B	217224
Shipping Gasket: 540 Series Motor Head	72462
Shipping Gasket: 540 Series Pump	68833
Shipping Gasket: 540/562 Series Pothole Flange, Type B	217232
Shipping Gasket: 562 Series Motor Head	274878
Shipping Gasket: 650 Series Motor Head	195776
Shipping Gasket: 738 Series Motor Head	339077
Shipping Gasket: 738 Series Pothole Flange, Type B	217240
Shipping Gasket: 862 Series Motor Head	329003
Shipping Gasket: 950 Series Pump	247437
Stud: 375-series motor	0861039
Stud: 456-, 540-, and 562-series motor	0861047
Stud: 738-series motor	0861054
Tape: 1 in Teflon	521427
Vent Plug: Steel, Standard, 0.375 in-24UNF X .44 in	23879 ¹
Vent Plug: Monel™, Standard, 0.375 in-24UNF X .44 in	170803 ¹
Vent Plug: Monel, Long, .375-24UNF-3 X .475 in	1020221 ¹
Vent Plug: Monel, Short, .375-24UNF-3 X .25 in	876821 ¹
Vent Plug: Steel, Short, 0.375 in-24UNF X .215 in	7019508 ¹
Vent Plug: Monel Short, 0.375 in-24UNF X .215 in	2004318 ¹
Vent Plug: Monel, Extra Long, .375-24UNF-3 X .63 in	100398227 ¹
Refer to OneCAT for the complete list of tools, descriptions and part numbers.	

¹ For the correct valve and plug part number, refer to the bill of materials in OneCat for your specific equipment part number.

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2.7 Splicing Tools

Splicing tools descriptions and part numbers are shown in [Table 2-11](#).

Table 2-11: Splicing Tools Descriptions and Part Numbers

Description	Part Number
Crimping Gage: Nicopress	831420
Crimping Tool: Nicopress NTSC #53-XPJ (for #10 solid through #1/7 STR cable sizes) c/w Crimping Gage and Adjusting Allen Tool	521708
Crimping Tool: Nicopress #3-F6 (for 2/0 and 1/0 cable sizes) c/w Crimping Gage and Adjusting Allen Tool	521732
Cutter: Cable Shears, Klein	854034
Drum Adapter: 1/4 in	569319
Drum: (For Drill)	569301
Iron: Soldering, 300 Watt, 110V 60/50HZ	582486
Iron: Soldering, 300 Watt, 220V 60/50HZ	743757
Table: Cable Splicing, Field Type	33381

2.8 Splicing Materials

Splicing materials descriptions and part numbers are shown in [Table 2-12](#).

Splicing sleeves descriptions and part numbers are shown in [Table 2-13](#).

Table 2-12: Splicing Materials Descriptions and Part Numbers

Description	Part Number
Aarmor: Galv, Size 2, 1.20 in ID X .025 in Thickness X .500 in Width	548339
Aarmor: Galv, Size 4, 1.00 in ID X .025 in Thickness X .500 in Width	548321
Aarmor: Monel, 1.115 in ID X .020 in Thickness X .500 in Width	548354
Emery Cloth: 50 Grit, 50 yd Roll	566117
Emery Cloth: 100 Grit	566125
Emery Cloth: 150 Grit	566133
Impervoiil Filler: 0.30 in Dia X 15 lb Roll	513713
Lead Sheet: 1/32 in X 10 in X 5 ft.	689307
Solder: Acid Core, 5 Lb. Roll	562678
Solder: Swifsolder Liquid Solder, 1 lb Jar	895243
Tape: Black Electrical, .75 in X 66 ft Roll, Scotch #88	521377
Tape: Glass, 1 in X 60 Yd	684811

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Description	Part Number
Tape: High Modulus, 1 in X 10 yd Roll, REDA (shelf life of 2 year at 77 degF)	748137
Tape: High Temp TFE, 1 in X .0035 in X 10 yd Roll, REDA (shelf life of 1.5 year at 77 degF)	823435
Tape: Splicing .75 in X 30 Ft Roll, Scotch #23	521435

2.8.1

Splicing Sleeves

Table 2-13: Splicing Sleeves Descriptions and Part Numbers

Conductor Size ¹	2/0 AWG/7	1/0 AWG/7 or 2/0 AWG/7 C/S	1 AWG/7 or 1/0 AWG/7 C/S	2 AWG/7, 1 AWG/1, 1 AWG/7 C/S	2 AWG/1 or 2 AWG/7 C/S	4 AWG/1	5 AWG/1 (16 mm ²)	6 AWG/1
2/0 AWG/7	253005	367144	267310	367151		7010457		
1/0 AWG/7 or 2/0 AWG/7 C/S	367144	247460	288233	246470	100574376	283853		
1 AWG/7 or 1/0 AWG/7 C/S	267310	288233	519405	519363	100444357	519454	519397	993055
2 AWG/7, 1 AWG/1, 1 AWG/7 C/S	367151	246470	519363	519355	977546	519488	519330	519413
2 AWG/1 or 2 AWG/7 C/S				977546	977538	977553		977561
4 AWG/1	7010457	283853	519454	519488	977553	519496	519512	519546
5 AWG/1 (16 mm ²)			519397	519330		519512	100112632	519645
6 AWG/1			993055	519413	977561	519546	519645	519314

¹ Refer to Table 5-8 for conductor description.

2.9

Banding Tools and Materials

Banding tools and materials descriptions and part numbers are shown in Table 2-14.

Table 2-14: Banding Tools and Materials Descriptions and Part Numbers

Description	Part Number
Banding Material Dispenser: Signode, DF10B	836754

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Description	Part Number
Banding Material: 302 Stainless Steel, 0.025 in X 3/4 in Continuous, 85 lbm Roll	849158
Banding Material: 400 Monel, 0.025 in X 3/4 in Continuous, 100 lbm Roll	871426
Banding Material: Carbon Steel, 0.025 in X 3/4 in Continuous, 100 lbm Roll	849133
Banding Material: Stainless Steel, 0.015 X 3/8 in Continuous, 40 lbm Roll	679480
Banding Tool: Pneumatic, Signode PNSC34	852624
Bands: Carbon Steel, 22 in	67587
Bands: Carbon Steel, 32 in	67595
Bands: Monel, 22 in	185421
Bands: Monel, 32 in	185439
Bands: Monel, 42 in	257725
Bands: Stainless Steel, 22 in	208496
Bands: Stainless Steel, 32 in	208504
Bands: Stainless Steel, 42 in	256966
Cable Clamp Saddles: For #1 To #1/0 Cable	163352
Cable Clamp Saddles: For #2 To #1 Cable	163345
Cable Clamp Saddles: For #4 To #2 Cable	163337
Cable Clamp Saddles: For #6 To #4 Cable	163329
Cable Clamp Saddles: For #8 To #6 Cable	160242
Capscrew: Button Head Socket, #10-24 X 3/8 in, Signode P-23099	852665
Capscrew: Socket Head, 1/4-20 X 1 3/4 in, Signode P-86181	852681
Hose Assembly: Signode P-23085	852657
Lasalle Cable Protector: Installation Kit #A08903	939488
Nut: Thin Flexloc, 1/4 in - 20, Signode P-5319	852640
Retainer Strap: Signode P-23008	852731
Sealer: 3/4 in	560763
Sealer: 3/8 in	560748
Sealer: Repair Kit for P/N 560763	1160217
Seals: 302 Stainless Steel, for 3/4 in Banding Material, 1000 Per Box	849166
Seals: 302 Stainless Steel, for 3/8 in Banding Material	560813
Seals: 400 Monel, for 3/4 in Banding Material	871418
Seals: Carbon Steel for 3/4 in Banding Material, 5000 Per Box	849141
Stretchers: 3/4 in	562983
Stretchers: 3/8 in	562975
Stretchers: Repair Kit for P/N 562983	1160274
Superband: Stainless, 1-1/4 in X 36 in	1036292

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2.10

Check Valves

Check Valves descriptions and part numbers are shown in [Table 2-15](#).

Table 2-15: Check Valves Descriptions and Part Numbers

Description	Carbon Steel Complete Assembly	Redalloy Complete Assembly	Component Part Number for use with Carbon Steel or Redalloy		
			Poppet	Retainer	Snap Ring
2-3/8 in 8 rd. EUE tubing	216440	464016	189662	215228	865659
2-3/8 in 10 rd. EUE tubing	215244		189662	215228	865659
2-3/8 in 10 rd. reg tubing	209585		189662		26161
2-7/8 in 8 rd. EUE tubing	215517	462739	189209	215178	558197
2-7/8 in 10 rd. EUE tubing	50237		189209		68676
2-7/8 in 10 rd reg tubing	110627	1044601	189209	264382	557900
3-1/2 in 8 rd. EUE tubing	77172	497750	115865	197855	497768
3-1/2 in 10 rd. reg. tubing	77404		115865	197855	497768
4-1/2 in 8 rd. EUE tubing	203547	1000769	189621	198051	1142082
4-1/2 in 8 rd. reg. tubing	180414		189621	198051	1142082
5-1/2 in 8rd long-thread casing	262121				
7 in 8 rd. long-thread casing	298968				

2.11

Bleeder Valves

Bleeder valves descriptions and part numbers are shown in [Table 2-16](#).

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Table 2-16: Bleeder Valves Descriptions and Part Numbers

Description	Carbon Steel Complete	Redalloy Complete	Break-Off Plug (Monel)
2-3/8 in 8 rd. EUE tubing	73551	463976	189589
2-3/8 in 10 thd. EUE tubing	35725		189589
2-3/8 in 10 rd. reg. tubing	209601		189589
2-7/8 in 8 rd. EUE tubing	69328	462754	189183
2-7/8 in 10 thd. EUE tubing	57828		189183
2-7/8 in 10 rd. reg. tubing	110601	1044627	189183
3-1/2 in 8 rd. EUE tubing	77198	497735	189562
3-1/2 in 10 rd. reg. tubing	77388		189562
4-1/2 in 8 rd. EUE tubing	203562	1000785	189571
4-1/2 in 8 rd. reg. tubing	180364		189571
7 in bleeder only	298943		189605

2.12

Combination Check and Bleeder Valves

Combination check and bleeder valves descriptions and part numbers are shown in [Table 2-17: Combination Check and Bleeder Valves Descriptions and Part Numbers](#).

Table 2-17: Combination Check and Bleeder Valves Descriptions and Part Numbers

Description	Part Number
2 $\frac{3}{8}$ in 8 rd. EUE Combination, Complete	999433
Body, 2 $\frac{3}{8}$ in 8 rd EUE	999425
Poppet, Ni-Resist	189662
O-ring, HSN, Size 2-224	997593
Snap Ring, 2-3/16 in	558171
Retaining Valve Seat	215228
Seat, Check Valve	189662
Break-off Plug	999482
2-7/8 in 8 rd. EUE Combination, Complete	999458
Body, 2-7/8 in 8 rd EUE	999441
Poppet, Ni-Resist	189209
O-ring, HSN, Size 2-226	1044841
Snap Ring, 2 $\frac{3}{4}$ in	558197

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Description	Part Number
Retaining Valve Seat	215178
Seat, Check Valve	215186
Break-off Plug	999482
3½ in 8 rd. EUE Combination, Complete	999474
Body, 3½ in 8 rd EUE	999466
Poppet, Ni-Resist	115865
Sealing Ring	197475
Snap Ring, 3-1/4 in	558288
Retaining Valve Seat	197855
Seat, Check Valve	197962
Break-off Plug	999490
4½ in 8 rd. EUE Combination, Complete	257857
Body, 4½ in 8 rd EUE	257840
Poppet, Ni-Resist	189621
Sealing Ring	198069
Snap Ring, 4-1/4 in	1142082
Retaining Valve Seat	198051
4½ in 8 rd. Reg. Tubing, Combination, Complete	229229
Body, 4½ in 8 rd Regular Tubing	229211
Poppet, Ni-Resist	189621
Sealing Ring	198069
Snap Ring, 4-1/4 in	1142082
Retaining Valve Seat	198051
Seat, Check Valve	198044
Break-off Plug, Monel	189597
5½ in 8 rd. csg. Combination, Complete	262071
Body, 5½ in 8 rd. Csg.	262063
Poppet, Ni-Resist	227033
Sealing Ring	197863
Snap Ring, 5-1/4 in	1142090
Break-off Plug, Monel	189597
6⅝ in 8 rd. csg. Combination, Complete	251462
Body, 6⅝ in 8 rd. Csg.	251454
Lock Pin	356246
Poppet, Ni-Resist	253096
Snap Ring, 6 in	1142108
Sealing Ring	204255

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Description	Part Number
Seat	172403
Neoprene Ring-Seat (2 Reg'd)	236224
Snap Ring-Seat	172395
Break-off Plug, Monel	189605
7 in 8 rd. csg. Combination, Complete	298976
Body, 7 in 8 rd. csg	298950
Lock Pin	356238
Poppet, Ni-Resist	253096
Snap Ring, for Poppet	1142108
Sealing Ring	204255
Seat	172403
Neoprene Ring-Seat (2 Reg'd)	236224
Snap Ring-Seat	172395
Break-off Plug, Monel	189605
 Note 6 $\frac{3}{8}$ in and 7 in use the same internal parts	

2.13

EFT Wellhead (Old Seaboard) Parts

Component descriptions and part numbers for 6-in 2000/3000 GT-6 adapter assembly and FC tubing head assembly are shown in [Table 2-18](#).

Table 2-18: EFT Wellhead (Old Seaboard) Part Descriptions and Part Numbers

Description	Part Number
Adapter 6D 2M/3M Type GT-6 Double Drilled	830794
Tubing Head, 7-in 8 rd 3M Type FC W/Two, 2-in LP Outlets	TBA
EFT Tubing Hanger 6 in x 2 $\frac{3}{8}$ in	830810
EFT Tubing Hanger 6 in x 2 $\frac{7}{8}$ in	830836
EFT Mini Mandrel 3000-V 120 A	830828
EFT Upper Pigtail Connector 3000 V, 70 A	729722
EFT Lower Pigtail Connector 3000 V, 70 A	739730
Boot	830844
O-ring Lower EFT (One Required)	830869
O-ring Upper EFT (One Required)	830851

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Description	Part Number
Cap, THD, 8 $\frac{5}{8}$ -in IRD Type FC (One Required)	830802
"T" Seal, For EFT Mandrel (Two Required)	830877

2.14

Hercules Companion Flanges and Nipples

Hercules companion flanges and nipples part descriptions and part numbers are shown in [Table 2-19](#).

Table 2-19: Hercules Companion Flanges and Nipples Part Descriptions and Part Numbers

6-in 600 Series Flange (2000 PSI) with 5 $\frac{1}{2}$ -in 8rd casing thread	552067
6-in 600 Series Flange (2000 PSI) with 7-in 8rd casing thread	552083
6 in - 900 Series Flange (3000 PSI) with 5 $\frac{1}{2}$ -in 8 rd casing thread	552091
6-in 900 Series Flange (3000 PSI) with 7-in 8 rd casing thread	552109
5 $\frac{1}{2}$ -in 8rd short casing nipple	555441
7-in 8rd short casing nipple	555458

2.15

Hercules HSM Wellheads

The HSM design permits positioning the cable and pack-off assembly prior to being lowered through the BOP into the body of the tubing head. The pack-off is affected by tightening the six bolts of the assembly around a high strength alloy steel mandrel nipple. Pack-off procedures are provided in the cables section.

2.15.1

Application Guide and Specifications

Refer to [Figure 2-1](#) and [Table 2-20](#) for HSM wellhead specifications.

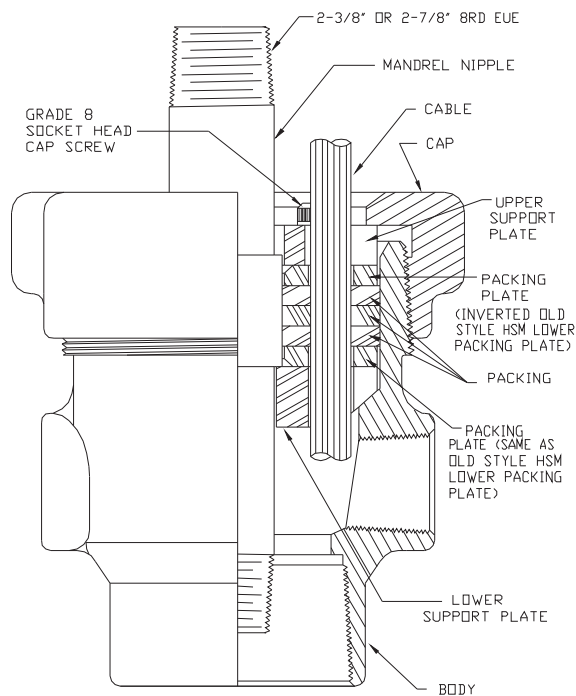


Figure 2-1: HSM Wellhead Drawing

Table 2-20: HSM Wellheads Specifications

Specifications	4.5 in	5.5 in	7 in
Working Pressure	N/A	1500 psi	N/A
Test Pressure (Body)	N/A	3000 psi	N/A
Recommended Slip Load	N/A	80,000 lbm	N/A
Maximum Slip Load	N/A	160,000 lbm	N/A
Packing Rating	N/A	1500 psi	N/A
Pack-Off Type	N/A	Bolted Mandrel Nipple	N/A
Inner String	N/A	2 in - 2.5 in EUE Only	N/A
Cap Thread	N/A	8-5/8 in - 8rd	N/A
Bottom Casing Thread	N/A	8rd	N/A
Casing Connection	N/A	Female	N/A
Minimum Bore	4 in	4 15/16 in	6 7/16 in
Side Outlet Size	N/A	2 in	N/A
Material	N/A	Ductile Iron	N/A
Suspension	N/A	Mandrel	N/A
Height	N/A	12-1/2 in	N/A
Weight	N/A	105 lbm	N/A
Full Opening	N/A	All Sizes	N/A

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2.15.2 HSM Wellhead Part Numbers

Refer to [Table 2-21](#) and [Table 2-22](#) for HSM wellhead part numbers.

Table 2-21: Hercules HSM Part Numbers

Description		Part No
Tubing Head Assembly		Parent
1	4-1/2 X 2 X 13/16 RCH	837005
2	4-1/2 X 2 X 15/16 RCH	837013
3	4-1/2 X 2 X 1-1/8 RCH	837021
4	4-1/2 X 2 X 7/16 RCH	837039
5	4-1/2 X 2 X 1/2 RCH	837047
6	4-1/2 X 2 X 5/8 RCH	837054
7	4-1/2 X 2-1/2 X 13/16 RCH	837062
8	4-1/2 X 2-1/2 X 15/16 RCH	837070
9	4-1/2 X 2-1/2 X 1-1/8 RCH	837088
10	4-1/2 X 2-1/2 X 7/16 RCH	837096
11	4-1/2 X 2-1/2 X 1/2 RCH	837104
12	4-1/2 X 2-1/2 X 5/8 RCH	837112
13	5-1/2 X 2 X 13/16 RCH	837120
14	5-1/2 X 2 X 15/16 RCH	837138
15	5-1/2 X 2 X 1-1/8 RCH	837146
16	5-1/2 X 2 X 1-1/4 RCH	837153
17	5-1/2 X 2 X 7/16 RCH	837161
18	5-1/2 X 2 X 1/2 RCH	837179
19	5-1/2 X 2 X 5/8 RCH	837187
20	5-1/2 X 2-1/2 X 13/16 RCH	837195
21	5-1/2 X 2-1/2 X 15/16 RCH	837203
22	5-1/2 X 2-1/2 X 1-1/8 RCH	837211
23	5-1/2 X 2-1/2 X 7/16 RCH	837229
24	5-1/2 X 2-1/2 X 1/2 RCH	837237
25	5-1/2 X 2-1/2 X 5/8 RCH	837245
26	7 X 2 X 13/16 RCH	837252
27	7 X 2 X 15/16 RCH	837260
28	7 X 2 X 1-1/8 RCH	837278
29	7 X 2 X 1-1/4 RCH	837286

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Description		Part No
Tubing Head Assembly		Parent
30	7 X 2 X 7/16 RCH	837294
31	7 X 2 X 1/2 RCH	837302
32	7 X 2 X 5/8 RCH	837310
33	7 X 2-1/2 X 13/16 RCH	837328
34	7 X 2-1/2 X 15/16 RCH	837336
35	7 X 2-1/2 X 1-1/8 RCH	837344
36	7 X 2-1/2 X 1-1/4 RCH	837351
37	7 X 2-1/2 X 7/16 RCH	537369
38	7 X 2-1/2 X 1/2 RCH	837377
39	7 X 2-1/2 X 5/8 RCH	837385

Table 2-22: Hercules HSM Part Numbers

		Tubing Body		Cap	Socket HD	Mandrel	Nipple	Support Plates		Neoprene Packing Rings			Metal Packing Plates		Lower	Upper
		Size	Part No.	Part No.	Screw (6) Part No.		Part No.	Lower Part No.	Upper Part No.	Cable Size		Part No.	Hole Size		Part No.	Part No.
1		4-1/2	837393	837427	837435	2	837443	837468	837484	13/16		837500	15/16		837641	837781
2		"	"	"	"	"	"	"	"	15/16		837518	1–1/16		837658	837799
3		"	"	"	"	"	"	"	"	1–1/8		837526	1–1/4		837666	837807
4		"	"	"	"	"	"	"	"		7/16	837534		1/2	837674	837815
5		"	"	"	"	"	"	"	"		1/2	837542		9/16	837682	837823
6		"	"	"	"	"	"	"	"		5/8	837559		11/16	837690	837831
7		"	"	"	"	2-1/2	837450	837476	837492	13/16		837567	15/16		837708	837849
8		"	"	"	"	"	"	"	"	15/16		837575	1–1/6		837716	837856
9		"	"	"	"	"	"	"	"	1–1/8		837583	1–1/4		837724	837864
10		"	"	"	"	"	"	"	"		7/16	837591		1/2	837732	837872
11		"	"	"	"	"	"	"	"		1/2	837609		9/16	837740	837880
12		"	"	"	"	"	"	"	"		5/8	837617		11/16	837757	837898
13		5-1/2	837401	"	"	2	837443	837468	837484	13/16		837500	15/16		837641	837781
14		"	"	"	"	"	"	"	"	15/16		837518	1–1/16		837658	837799
15		"	"	"	"	"	"	"	"	1–1/8		837526	1–1/4		837666	837807
16		"	"	"	"	"	"	"	"	1–1/4		837625	1–3/8		837765	837906
17		"	"	"	"	"	"	"	"		7/16	837534		1/2	837674	837815
18		"	"	"	"	"	"	"	"		1/2	837542		9/16	837682	837823
19		"	"	"	"	"	"	"	"		5/8	837559		11/16	837690	837831
20		"	"	"	"	2-1/2	837450	837476	837492	13/16		837567	15/16		837708	837849
21		"	"	"	"	"	"	"	"	15/16		837575	1–1/16		837716	837856
22		"	"	"	"	"	"	"	"	1–1/8		837583	1–1/4		837724	837864
23		"	"	"	"	"	"	"	"		7/16	837591		1/2	837732	837872
24		"	"	"	"	"	"	"	"		1/2	837609		9/16	837740	837880
25		"	"	"	"	"	"	"	"		5/8	537617		11/16	837757	837898
26		7	837419	"	"	2	837443	837468	837484	13/16		837500	15/16		837641	837781
27		"	"	"	"	"	"	"	"	15/16		837518	1–1/16		837658	837799
28		"	"	"	"	"	"	"	"	1–1/8		837526	1–1/4		837666	837807
29		"	"	"	"	"	"	"	"	1–1/4		837625	1–3/8		837765	837906
30		"	"	"	"	"	"	"	"		7/16	837534		1/2	837674	837815
31		"	"	"	"	"	"	"	"		1/2	837542		9/16	837682	837823
32		"	"	"	"	"	"	"	"		5/8	837559		11/16	837690	837831
33		"	"	"	"	2-1/2	837450	837476	837492	13/16		837567	15/16		837708	837849

		Tubing Body		Cap	Socket HD	Mandrel	Nipple	Support Plates		Neoprene Packing Rings			Metal Packing Plates		Lower	Upper
		Size	Part No.	Part No.	Screw (6) Part No.			Lower Part No.	Upper Part No.	Cable Size		Part No.	Hole Size		Part No.	Part No.
34		"	"	"	"	"	"	"	"	15/16		837575	1–1/16		837716	837856
35		"	"	"	"	"	"	"	"	1–1/8		837583	1–1/4		837724	837864
36		"	"	"	"	"	"	"	"	1–1/4		837633	1–3/8		837773	837914
37		"	"	"	"	"	"	"	"		7/16	837591		1/2	837732	837872
38		"	"	"	"	"	"	"	"		1/2	837609		9/16	837740	837880
39			"	"	"	"	"	"	"		5/8	837617		11/16	837757	837898

Table 2-23 lists the proper wellhead part numbers for a given cable type. Find the cable type for your application, then under the same column, select the appropriate wellhead for the given Casing OD and Tubing size.

AL-ESP Field Service & Bypass Sys. \AL Engineering\InTouch ID 3255850\3.2\Release Date:20-Jul-2011\EDMS UID: 274783215\Produced: 07-Mar-2013 21:09:14

Private

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Table 2-23: HSM Wellhead Part Numbers

Casing OD Inch	Tubing OD	Jacket Dia.							
	Inch	1.188 in	0.594 in	1.125 in	0.469 in	0.938 in	0.5 in	0.844 in	0.438 in
	EUE Only	Round Cable	Flat Cable	Round Cable	Flat Cable	Round Cable	Flat Cable	Round Cable	Flat Cable
	2-7/8	837351		837344	837385	837336	837377	837328	837369
4–1/2	2–3/8			837021	837054	837013	837047	837005	837039
	2–7/8			837088	837112	837070	837104	837062	837096
5–1/2	2–3/8	837153		837146	837187	837138	837179	837120	837161
	2–7/8			837211	837245	837203	837237	837195	837229
7 837294		2–3/8	837286		837278	837310	837260	837302	837252
SOL <									

2.16 Hercules HHS Wellheads

A Hercules HHS wellhead is used when a bolted packing gland is preferred. The packing gland is split to facilitate installation after the tubing and cable have been run. All bodies are full opening, with heat-treated hinged slips for ease of installation and positive alignment.

Note
Hercules HHS wellhead pack-off procedures are provided in the *Cable* section.

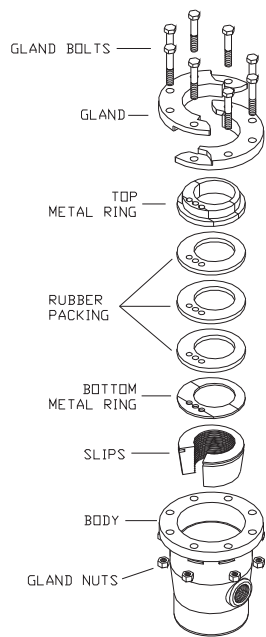


Figure 2-2: Hercules HHS Wellhead Drawing

2.16.1 HHS Wellhead Specifications

HHS wellhead specifications are shown in [Table 2-24](#).

Table 2-24: HHS Wellhead Specifications

Specification	4½ to 5½ in	7 in	8⅝ in	9⅝ in	10¾ in
working pressure, psi	1500	1500	1500	2000	2000
body test pressure, psi	3000	3000	3000	3000	3000
recommended slip load, lbf	80,000	80,000	100,000	100,000	100,000
maximum slip load, lbf	160,000	160,000	200,000	200,000	200,000

Specification	4½ to 5½ in	7 in	8⅝ in	9⅝ in	10¾ in
packing rating, psi	1500	1500	1500	1500	1500
packoff type	bolted packing gland				
inner string	2 in, 2.5 in	2 to 3 in	2 to 5.5 in	2 to 7 in	2 to 7 in
cap thread	n/a	n/a	n/a	n/a	n/a
bottom casing thread	8 rd	8 rd	8 rd	8 rd	8 rd
casing connection	female	female	female	female	female
minimum bore	4 to 4-15/16 in	6-7/16 in	8 in	8-7/8 in	9-15/16
side outlet size	2 in	2 in	2 in	2 in	2 in
material	ductile iron	ductile iron	steel	steel	steel
suspension	slip	slip	slip	slip	slip
height	14-1/2 in	14-1/2 in	13-1/2 in	16 in	16 in
weight, lbm	145	145	195	265	265
full opening	all sizes				

2.16.2

HHS Wellhead Application Guide

[Table 2-25](#) lists the proper wellhead part numbers for a given cable type. Find the cable type for your application, then under the same column, select the appropriate wellhead for the given casing OD and tubing size.

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Table 2-25: HHS Wellhead Application Guide

Casing OD Inch	Tubing OD Inch EUE Only	Jacket Dia.							
		1.188 in Round Cable	0.594 in Flat Cable	1.125 in Round Cable	0.469 in Flat Cable	0.938 in Round Cable	0.500 in Flat Cable	0.844 in Round Cable	0.438 in Flat Cable
4–1/2	2–1/16					528604		528471	
	2–3/8						529214	528422	529131
5–1/2	2–1/16							735159	
	2–3/8	528943	528380	528778	740951	528588	529222	528430	529149
	2–7/8		529479		529313	528620	529230	528489	529172
	3–1/2				528869			528513	
7 529156		2–3/8	528950	528398	528786	824839	528596	683037	528448
	2–7/8	528976		528844	529321	528646	529248	528497	529180
	3–1/2	528992		528877	529339	528679	529263	528521	529206
8–5/8	2–3/8			528794		316703		528455	529164
	2–7/8	528984		680835		528653	529255	528505	529198
	3–1/2	529008		528885		528687			
	4–1/2	529065		740555	734541	689802		528539	
	5–1/2	529081		686576		749523	529289		
9–5/8	2–7/8	685446		528851		528661			
	3–1/2	529016		528893	529370	733469		744888	741116
	4–1/2	529073		528919	743765	528729		528547	
	5–1/2	529099		528927	529404	528737			
6–5/8			826255						
10–3/4	2–3/8							741124	
	2–7/8	681783		737858					
	3–1/2	529040			731190	528695			
	4					834556			
	4–1/2	682625		688051			741827		
	5–1/2	529107		685552		528745	685545	746289	
	6–5/8	529115		528935					
	7	682682			529420				

Casing	Tubing OD	Jacket Dia.							
OD	Inch	1.188 in	0.594 in	1.125 in	0.469 in	0.938 in	0.500 in	0.844 in	0.438 in
Inch	EUE Only	Round Cable	Flat Cable	Round Cable	Flat Cable	Round Cable	Flat Cable	Round Cable	Flat Cable
		(EO) G4R 1 STR	(ETB0 F) G5F 2 COM	(PPEO) G5R 2 COM	(POTB) G4F 4SOL	(PPEO) G4R 4 SOL	(ELB) G4F 2 STR	(ETBO) G4R 6 SOL	(ELB) G5F 4 SOL
		(EO) G5R 1 COM	(EEF) GG5F 1 COM	(EER) G5R 2 COM		(ELBE) G4R 6 SOL	(ELB) GF5 2 COM	(PPEO) G4R 6 SOL	(POTB) G4F 6 SOL
		(EO) G5R 1 SOL	(ETBEF-300) G5F 2 COM	(PPEPE) O4R 2 STR		(EO) G4R 4 SOL	(ELTB) G4F 2 STR	(EO) G4R 6 SOL	(POF) G4F 6 SOL
		(ESO) G5R 2 COM	(POF) G4F 1 STR			(ETBO) G4R 4 SOL	(EEF) GG5F 4 SOL	(ETBO) G4R 6 SOL	(KEOTB) G4F 2 SOL
		(PPEO) G4R 1 STR	(POF) G5F 1 COM			(PPEPE) O4R 4 SOL	(POF) G4F 4 SOL	(PPEPE) O4R 6 SOL	(KEOTB) M4F 2 SOL
		(PPEO) G5R 1 COM	(POTB) G4F 1 STR			(POF) G5F 4 SOL		(KELB) G4F 2 SOL	
		(EER) G5R 1 COM	(POTB) G5F 1 COM			(POTB) G5F 4 SOL		(KELB) M4F 2 SOL	
		(ETBE-400) G5R 1 SOL							
		(EFE) G5R 2 COM							
		(SES0) G5R 2 COM							
		(SEFE) G5R 2 COM							
		(SETBO) G5R 1 SOL							
		(PPEPE) O4R 1 STR							
		(SETBE-400) G5R 1 SOL							

2.16.3 HHS Wellhead (4½-in Body)

Refer to [Table 2-26](#) for HHS wellhead (4½-in body) part numbers.

Table 2-26: HHS Wellhead (4½-in Body)

Parent	Body	Gland	Slip	Slip	RCH	FCH	Neoprene	Metal	Top	Bottom	Gland Bolts and Nuts		
Part No.	Part No.	Part No.	Size	Part No.	Hole Size	Hole Size	Ring Part No.	Ring Hole Size	Ring Part No.	Ring Part No.	Qty Req'd	Bolt Part No.	Nut Part No.
528414	525253	525220	1–1/2	833632	13/16		679399	15/16	679373	679381	8	824789	824359
732792	"	"	"	"	15/16		526384	1–1/16			"	"	"
733873	"	"	2	525410	3/4		526269	7/8	527283	525733	"	"	"
528422	"	"	"	"	13/16		526319	15/16	527309	525741	"	"	"
528570	"	"	"	"	15/16		526392	1–1/16	527333	525774	"	"	"
	"	"	"	"	1–1/8		526657	1–1/4	527408	525873	"	"	"
	"	"	"	"	1–1/4		526764	1–3/8	527499	525956	"	"	"
529131	"	"	"	"		7/16	526921	1/2	681403	683292	"	"	"
529214	"	"	"	"		1/2	526954	9/16	527580	526038	"	"	"
827006	"	"	"	"		5/8	527002	11/16	527606	526079	"	"	"
	"	"	"	"		11/16		13/16			"	"	"
	"	"	2-1/16	525444	3/4			7/8			"	"	"
528471	"	"	"	"	13/16		735175	15/16	735183	735167	"	"	"
528604	"	"	"	"	15/16		526418	1–1/16	527341	525782	"	"	"
528810	"	"	"	"	1–1/8		526665	1–1/4	527416	525881	"	"	"
	"	"	"	"	1–1/4			1–3/8			"	"	"
	"	"	"	"		7/16		1/2			"	"	"
	"	"	"	"		1/2		9/16			"	"	"
	"	"	"	"		5/8		11/16			"	"	"
	"	"	"	"		11/16		13/16			"	"	"
	"	"	2-1/2	525451	3/4		526277	7/8	527291	684118	"	"	"
	"	"	"	"	13/16		526343	15/16	527317	525758	"	"	"
528612	"	"	"	"	15/16		526434	1–1/16	527358	525790	"	"	"
	"	"	"	"	1–1/8		526673	1–1/4	527424	525899	"	"	"
	"	"	"	"	1–1/4		526772	1–3/8	527507	525964	"	"	"
	"	"	"	"		7/16	526939	1/2	527564	526020	"	"	"
746941	"	"	"	"		1/2	526962 9/16	9/16	527598	526046	"	"	"
	"	"	"	"		5/8	527010	11/16	527614	526087	"	"	"
	"	"	"	"		11/16	527101	13/16	527655	834184	"	"	"
	"	"	"	"		3/4	527101	13/16	527655	834184	"	"	"
	"	"	3	525493	13/16		526368	15/16	680298	680306	"	"	"
	"	"	"	"	15/16		526459	1–1/16	527366	525808	"	"	"

Parent	Body	Gland	Slip	Slip	RCH	FCH	Neoprene	Metal	Top	Bottom	Gland Bolts and Nuts		
Part No.	Part No.	Part No.	Size	Part No.	Hole Size	Hole Size	Ring Part No.	Ring Hole Size	Ring Part No.	Ring Part No.	Qty Req'd	Bolt Part No.	Nut Part No.
	"	"	"	"	1–1/8		526699	1–1/4	527457	525915	"	"	"
	"	"	"	"	1–1/4		526806	1–3/8	527515	525972	"	"	"
	"	"	"	"		7/16	526947	1/2	527572	681387	"	"	"
	"	"	"	"		1/2	526970	9/16	681411	526053	"	"	"
	"	"	"	"		5/8	527036	11/16	527622	526095	"	"	"
	"	"	"	"		11/16		13/16	681551	684209	"	"	"

2.16.4 HHS Wellhead (5½-in Body)

Refer to [Table 2-27](#) for HHS wellhead (5½-in body) part numbers.

Table 2-27: HHS Wellhead (5½-in Body)

Parent Part No.	Body Part No.	Gland Part No.	Slip Size	Slip Part No.	RCH Hole Size	FCH Hole Size	Neoprene Ring Part No.	Metal Ring Hole Size	Top Ring Part No.	Bottom Ring Part No.	Gland Bolts and Nuts		
											Qty Req'd	Bolt Part No.	Nut Part No.
741942	525261	525220	1–1/2	833632	13/16		679399	15/16	679373	679381	8	824789	824359
	"	"	"	"	15/16		526384	1–1/16			"	"	"
528380	"	"	2	525410	3/4		526269	7/8	527283	525733	"	"	"
528430	"	"	"	"	13/16		526319	15/16	527309	525741	"	"	"
528588	"	"	"	"	15/16		526392	1–1/16	527333	525774	"	"	"
528778	"	"	"	"	1–1/8		526657	1–1/4	527408	525873	"	"	"
528943	"	"	"	"	1–1/4		526764	1–3/8	527499	525956	"	"	"
529149	"	"	"	"		7/16	526921	1/2	681403	683292	"	"	"
529222	"	"	"	"		1/2	526954	9/16	527580	526038	"	"	"
740951	"	"	"	"		5/8	527002	11/16	527606	526079	"	"	"
	"	"	"	"		11/16		13/16			"	"	"
	"	"	2–1/16	525444	3/4			7/8			"	"	"
735159	"	"	"	"	13/16		735175	15/16	735183	735167	"	"	"
	"	"	"	"	15/16		526418	1–1/16	527341	525782	"	"	"
	"	"	"	"	1–1/8		526665	1–1/4	527416	525881	"	"	"
	"	"	"	"	1–1/4			1–3/8			"	"	"
	"	"	"	"		7/16		1/2			"	"	"
	"	"	"	"		1/2		9/16			"	"	"
	"	"	"	"		5/8		11/16			"	"	"
	"	"	"	"		11/16		13/16			"	"	"
	"	"	2–1/2	525451	3/4		526277	7/8	527291	684118	"	"	"
528489	"	"	"	"	13/16		526343	15/16	527317	525758	"	"	"
528620	"	"	"	"	15/16		526434	1–1/16	527358	525790	"	"	"
528828	"	"	"	"	1–1/8		526673	1–1/4	527424	525899	"	"	"
	"	"	"	"	1–1/4		526772	1–3/8	527507	525964	"	"	"
529172	"	"	"	"		7/16	526939	1/2	527564	526020	"	"	"
529230	"	"	"	"		1/2	526962	9/16	527598	526046	"	"	"
529313	"	"	"	"		5/8	527010	11/16	527614	526087	"	"	"
	"	"	"	"		11/16	527101	13/16	527655	834184	"	"	"
529479	"	"	"	"		3/4	527101	13/16	527655	834184	"	"	"
528513	"	"	3	525493	13/16		526368	15/16	680298	680306	"	"	"
	"	"	"	"	15/16		526459	1–1/16	527366	525808	"	"	"

Parent	Body	Gland	Slip	Slip	RCH	FCH	Neoprene	Metal	Top	Bottom	Gland Bolts and Nuts		
Part No.	Part No.	Part No.	Size	Part No.	Hole Size	Hole Size	Ring Part No.	Ring Hole Size	Ring Part No.	Ring Part No.	Qty Req'd	Bolt Part No.	Nut Part No.
528869	"	"	"	"	1–1/8		526699	1–1/4	527457	525915	"	"	"
	"	"	"	"	1–1/4		526806	1–3/8	527515	525972	"	"	"
	"	"	"	"		7/16	526947	1/2	527572	681387	"	"	"
	"	"	"	"		1/2	526970	9/16	681411	526053	"	"	"
	"	"	"	"		5/8	527036	11/16	527622	526095	"	"	"
	"	"	"	"		11/16		13/16	681551	684209	"	"	"

2.16.5 HHS Wellhead (6⁵/₈-in Body)

Refer to [Table 2-28](#) for HHS wellhead (6⁵/₈-in body) part numbers.

Table 2-28: HHS Wellhead (65⁄8-in Body)

Parent Part No.	Body Part No.	Gland Part No.	Slip Size	Slip Part No.	RCH Hole Size	FCH Hole Size	Neoprene Ring Part No.	Metal Ring Hole Size	Top Ring Part No.	Bottom Ring Part No.	Gland Bolts and Nuts		
											Qty Req'd	Bolt Part No.	Nut Part No.
	525279	525220	1–1/2	833632	13/16		679399	15/16	679373	679381	8	824789	824359
	"	"	"	"	15/16		526384	1–1/16			"	"	"
	"	"	2	525410	3/4		526269	7/8	527283	525733	"	"	"
	"	"	"	"	13/16		526319	15/16	527309	525741	"	"	"
	"	"	"	"	15/16		526392	1–1/16	527333	525774	"	"	"
	"	"	"	"	1–1/8		526657	1–1/4	527408	525873	"	"	"
	"	"	"	"	1–1/4		526764	1–3/8	527499	525956	"	"	"
	"	"	"	"		7/16	526921	1/2	681403	683292	"	"	"
824250	"	"	"	"		1/2	526954	9/16	527580	526038	"	"	"
	"	"	"	"		5/8	527002	11/16	527606	526079	"	"	"
	"	"	"	"		11/16		13/16			"	"	"
	"	"	2–1/16	525444	3/4			7/8			"	"	"
	"	"	"	"	13/16		735175	15/16	735183	735167	"	"	"
	"	"	"	"	15/16		526418	1–1/16	527341	525782	"	"	"
	"	"	"	"	1–1/8		526665	1–1/4	527416	525881	"	"	"
	"	"	"	"	1–1/4			1–3/8			"	"	"
	"	"	"	"		7/16		1/2			"	"	"
	"	"	"	"		1/2		9/16			"	"	"
	"	"	"	"		5/8		11/16			"	"	"
	"	"	"	"		11/16		13/16			"	"	"
	"	"	2-1/2	525451	3/4		526277	7/8	527291	684118	"	"	"
	"	"	"	"	13/16		526343	15/16	527317	525758	"	"	"
528638	"	"	"	"	15/16		526434	1–1/16	527358	525790	"	"	"
528836	"	"	"	"	1–1/8		526673	1–1/4	527424	525899	"	"	"
	"	"	"	"	1–1/4		526772	1–3/8	527507	525964	"	"	"
	"	"	"	"		7/16	526939	1/2	527564	526020	"	"	"
745422	"	"	"	"		1/2	526962	9/16	527598	526046	"	"	"
	"	"	"	"		5/8	527010	11/16	527614	526087	"	"	"
	"	"	"	"		11/16	527101	13/16	527655	834184	"	"	"
	"	"	"	"		3/4	527101	13/16	527655	834184	"	"	"
	"	"	3	525493	13/16		526368	15/16	680298	680306	"	"	"
683953	"	"	"	"	15/16		526459	1–1/16	527366	525808	"	"	"

Parent	Body	Gland	Slip	Slip	RCH	FCH	Neoprene	Metal	Top	Bottom	Gland Bolts and Nuts		
Part No.	Part No.	Part No.	Size	Part No.	Hole Size	Hole Size	Ring Part No.	Ring Hole Size	Ring Part No.	Ring Part No.	Qty Req'd	Bolt Part No.	Nut Part No.
	"	"	"	"	1–1/8		526699	1–1/4	527457	525915	"	"	"
	"	"	"	"	1–1/4		526806	1–3/8	527515	525972	"	"	"
	"	"	"	"		7/16	526947	1/2	527572	681387	"	"	"
	"	"	"	"		1/2	526970	9/16	681411	526053	"	"	"
	"	"	"	"		5/8	527036	11/16	527622	526095	"	"	"
	"	"	"	"		11/16		13/16	681551	684209	"	"	"

2.16.6 **HHS Wellhead (7-in Body)**

Follow this link or page down in the document for HHS wellhead (7-in body) part numbers.

Table 2-29: HHS Wellhead (7-in Body)

Parent Part No.	Body Part No.	Gland Part No.	Slip Size	Slip Part No.	RCH Hole Size	FCH Hole Size	Neoprene Ring Part No.	Metal Ring Hole Size	Top Ring Part No.	Bottom Ring Part No.	Gland Bolts and Nuts		
											Qty Req'd	Bolt Part No.	Nut Part No.
	525287	525220	1–1/2	833632	13/16		679399	15/16	679373	679381	8	824789	824359
	"	"	"	"	15/16		526384	1–1/16			"	"	"
528398	"	"	2	525410	3/4		526269	7/8	527283	525733	"	"	"
528448	"	"	"	"	13/16		526319	15/16	527309	525741	"	"	"
528596	"	"	"	"	15/16		526392	1–1/16	527333	525774	"	"	"
528786	"	"	"	"	1–1/8		526657	1–1/4	527408	525873	"	"	"
528950	"	"	"	"	1–1/4		526764	1–3/8	527499	525956	"	"	"
529156	"	"	"	"		7/16	526921	1/2	681403	683292	"	"	"
683037	"	"	"	"		1/2	526954	9/16	527580	526038	"	"	"
824839	"	"	"	"		5/8	527002	11/16	527606	526079	"	"	"
	"	"	"	"		11/16		13/16			"	"	"
	"	"	2–1/16	525444	3/4			7/8			"	"	"
	"	"	"	"	13/16		735175	15/16	735183	735167	"	"	"
	"	"	"	"	15/16		526418	1–1/16	527341	525782	"	"	"
	"	"	"	"	1–1/8		526665	1–1/4	527416	525881	"	"	"
	"	"	"	"	1–1/4			1–3/8			"	"	"
	"	"	"	"		7/16		1/2			"	"	"
	"	"	"	"		1/2		9/16			"	"	"
	"	"	"	"		5/8		11/16			"	"	"
	"	"	"	"		11/16		13/16			"	"	"
	"	"	2–1/2	525451	3/4		526277	7/8	527291	684118	"	"	"
528497	"	"	"	"	13/16		526343	15/16	527317	525758	"	"	"
528646	"	"	"	"	15/16		526434	1–1/16	527358	525790	"	"	"
528844	"	"	"	"	1–1/8		526673	1–1/4	527424	525899	"	"	"
528976	"	"	"	"	1–1/4		526772	1–3/8	527507	525964	"	"	"
529180	"	"	"	"		7/16	526939	1/2	527564	526020	"	"	"
529248	"	"	"	"		1/2	526962	9/16	527598	526046	"	"	"
529321	"	"	"	"		5/8	527010	11/16	527614	526087	"	"	"
529438	"	"	"	"		11/16	527101	13/16	527655	834184	"	"	"
	"	"	"	"		3/4	527101	13/16	527655	834184	"	"	"
528521	"	"	3	525493	13/16		526368	15/16	680298	680306	"	"	"
528679	"	"	"	"	15/16		526459	1–1/16	527366	525808	"	"	"

Parent	Body	Gland	Slip	Slip	RCH	FCH	Neoprene	Metal	Top	Bottom	Gland Bolts and Nuts		
Part No.	Part No.	Part No.	Size	Part No.	Hole Size	Hole Size	Ring Part No.	Ring Hole Size	Ring Part No.	Ring Part No.	Qty Req'd	Bolt Part No.	Nut Part No.
528877	"	"	"	"	1–1/8		526699	1–1/4	527457	525915	"	"	"
528992	"	"	"	"	1–1/4		526806	1–3/8	527515	525972	" " "		
529206	"	"	"	"		7/16	526947	1/2	527572	681387	"	"	"
529263	"	"	"	"		1/2	526970	9/16	681411	526053	"	"	"
529339	"	"	"	"		5/8	527036	11/16	527622	526095	"	"	"

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2.16.7 HHS Wellhead (8⁵/₈-in Body)

[Table 2-30](#) for HHS wellhead (8⁵/₈-in body) part numbers.

Table 2-30: HHS Wellhead (8⁵/₈-in Body)

Parent Part No.	Body Part No.	Gland Part No.	Slip Size	Slip Part No.	RCH Hole Size	FCH Hole Size	Neoprene Ring Part No.	Metal Ring Hole Size	Top Ring Part No.	Bottom Ring Part No.	Gland Bolts and Nuts		
											Qty Req'd	Bolt Part No.	Nut Part No.
528455	525311	525238	2	525428	13/16		526327	15/16	834275	834283	8	824789	824359
316703	"	"	"	"	15/16			1–1/16	681585	684142	"	"	"
528794	"	"	"	"	1–1/8		741140	1–1/4	834028	834010	"	"	"
	"	"	"	"	1–1/4						"	"	"
529164	"	"	"	"		7/16	681668	1/2	681650	681676	"	"	"
	"	"	"	"		1/2		9/16			"	"	"
	"	"	"	"		5/8		11/16		526079	"	"	"
528505	"	"	2–1/2	525477	13/16		526350	15/16	527325	525766	"	"	"
528653	"	"	"	"	15/16		526442	1–1/16	824433	824441	"	"	"
680835	"	"	"	"	1–1/8		526681	1–1/4	527432	525907	"	"	"
528984	"	"	"	"	1–1/4		526780	1–3/8	687376	687368	"	"	"
529198	"	"	"	"		7/16	741173	1/2	834093	834085	"	"	"
529255	"	"	"	"		1/2	733956	9/16	733931	733949	"	"	"
	"	"	"	"		5/8	733980	11/16	681635	681643	"	"	"
	"	"	3	525501	13/16			15/16	680298		"	"	"
528687	"	"	"	"	15/16		526467	1–1/16	527374	525816	"	"	"
528885	"	"	"	"	1–1/8		526707	1–1/4	527465	525923	"	"	"
529008	"	"	"	"	1–1/4		526814	1–3/8	527523	525980	"	"	"
	"	"	"	"		7/16		1/2			"	"	"
	"	"	"	"		1/2	526988	9/16			"	"	"
	"	"	"	"		5/8	527044	11/16	527630	526103	"	"	"
836528	"	"	"	"		21/32	836536	3/4	836544	836551	"	"	"
528539	"	"	4–1/2	525543	13/16		683920	15/16	682864	684134	"	"	"
689802	"	"	"	"	15/16		526491	1–1/16	742809	525824	"	"	"
740555	"	"	"	"	1–1/8		526715	1–1/4	834291	834309	"	"	"
529065	"	"	"	"	1–1/4		526848	1–3/8	527531	525998	"	"	"
	"	"	"	"		7/16		1/2			"	"	"
	"	"	"	"		1/2		9/16		741843	"	"	"
734541	"	"	"	"		5/8	741207	11/16	834218	834200	"	"	"
	"	"	5–1/2	525576	13/16			15/16			"	"	"
749523	"	"	"	"	15/16		526525	1–1/16	527382	745232	"	"	"
686576	"	"	"	"	1–1/8		526749	1–1/4	527481	525949	"	"	"

Parent Part No.	Body Part No.	Gland Part No.	Slip Size	Slip Part No.	RCH Hole Size	FCH Hole Size	Neoprene Ring Part No.	Metal Ring Hole Size	Top Ring Part No.	Bottom Ring Part No.	Gland Bolts and Nuts		
											Qty Req'd	Bolt Part No.	Nut Part No.
529081	"	"	"	"	1–1/4		526871	1–3/8	834051	684183	"	"	"
	"	"	"	"		7/16		1/2			"	"	"
529289	"	"	"	"		1/2	526996	9/16	834135	834127	"	"	"
	"	"	"	"		5/8	741256	11/16	829770		"	"	"

2.16.8 HHS Wellhead (9⁵/₈-in Body)

[Table 2-31](#) for HHS wellhead (9⁵/₈-in body) part numbers.

Table 2-31: HHS Wellhead (9⁵/₈-in Body)

Parent Part No.	Body Part No.	Gland Part No.	Slip Size	Slip Part No.	RCH Hole Size	FCH Hole Size	Neoprene Ring Part No.	Metal Ring Hole Size	Top Ring Part No.	Bottom Ring Part No.	Gland Bolts and Nuts		
											Qty Req'd	Bolt Part No.	Nut Part No.
	525329	525246	2	525436	13/16		741470	15/16	834234	834226	12	824789	824359
	"	"	"	"	1–1/8			1–1/4	684316	684308	"	"	"
528661	"	"	2–1/2	525485	15/16		741330	1–1/16	819920	819912	"	"	"
528851	"	"	"	"	1–1/8		834044	1–1/4	527440	834036	"	"	"
685446	"	"	"	"	1–1/4		678300	1–3/8	678318	678326	"	"	"
744888	"	"	3	525519	13/16		829069	15/16	741389	741371	"	"	"
733469	"	"	"	"	15/16		526475	1–1/16	819904	819896	"	"	"
528893	"	"	"	"	1–1/8		683342	1–1/4	683367	741512	"	"	"
529016	"	"	"	"	1–1/4		526822	1–3/8	819961	819953	"	"	"
741116	"	"	"	"		7/16	744144	1/2	744151	834242	"	"	"
	"	"	"	"		1/2		9/16			"	"	"
529370	"	"	"	"		5/8	527051	11/16	834168	834150	"	"	"
528547	"	"	4–1/2	525568	13/16		681072	15/16	681098	681080	"	"	"
528729	"	"	"	"	15/16		526517	1–1/16	681459	744318	"	"	"
528919	"	"	"	"	1–1/8		526723	1–1/4	527473	525931	"	"	"
529073	"	"	"	"	1–1/4		526863	1–3/8	681593	684175	"	"	"
	"	"	"	"		7/16		1/2			"	"	"
	"	"	"	"		1/2		9/16			"	"	"
743765	"	"	"	"		5/8	527069	11/16	731281	731273	"	"	"
	"	"	5–1/2	525584	13/16		741538	15/16	741561	741553	"	"	"
528737	"	"	"	"	15/16		824326	1–1/16	826743	824318	"	"	"
528927	"	"	"	"	1–1/8		526756	1–1/4	819946	819938	"	"	"
529099	"	"	"	"	1–1/4		526889	1–3/8	683391	683375	"	"	"
	"	"	"	"		7/16		1/2		741595	"	"	"
	"	"	"	"		1/2	686485	9/16	686493	834192	"	"	"
529404	"	"	"	"		5/8	527077	11/16	527648	526111	"	"	"
	"	"	6–5/8	741611	13/16			15/16			"	"	"
	"	"	"	"	15/16			1–1/16			"	"	"
826255	"	"	"	"	1–1/8		741637	1–1/4	741678	741652	"	"	"
	"	"	"	"	1–1/4		741660	1–3/8	820019	820001	"	"	"
	"	"	"	"		7/16		1/2	741702	741686	"	"	"
	"	"	"	"		1/2	741694	9/16			"	"	"

Parent Part No.	Body Part No.	Gland Part No.	Slip Size	Slip Part No.	RCH Hole Size	FCH Hole Size	Neoprene Ring Part No.	Metal Ring Hole Size	Top Ring Part No.	Bottom Ring Part No.	Gland Bolts and Nuts		
											Qty Req'd	Bolt Part No.	Nut Part No.
	"	"	"	"		5/8		11/16			"	"	"
	"	"	7	840280	1–1/8		835470	1–1/4	741736	741710	"	"	"
	"	"	"	"	1–1/4		741728	1–3/8	679217	678649	"	"	"
	"	"	"	681320		5/8	681338	11/16	737130	737148	"	"	"
	"	"	"	"		21/32		3/4	834259	834267	"	"	"
	"	"	"	"		11/16	686691	13/16	819995	819987	"	"	"

2.16.9 HHS Wellhead (10³/₄-in Body)

[Table 2-32](#) for HHS wellhead (10³/₄-in body) part numbers.

Table 2-32: HHS Wellhead (10¾-in Body)

Parent	Body	Gland	Slip	Slip	RCH	FCH	Neoprene	Metal	Top	Bottom	Gland Bolts and Nuts		
											Qty Req'd	Bolt Part No.	Nut Part No.
741124	525337	525246	2	525436	13/16		741470	15/16	834234	834226	12	824789	824359
	"	"	"	"	1–1/8			1–1/4	684316	684308	"	"	"
	"	"	2–1/2	525485	15/16		741330	1–1/16	819920	819912	"	"	"
737858	"	"	"	"	1–1/8		834044	1–1/4	527440	834036	"	"	"
681783	"	"	"	"	1–1/4		678300	1–3/8	678318	678326	"	"	"
	"	"	3	525519	13/16		829069	15/16	741389	741371	"	"	"
528695	"	"	"	"	15/16		526475	1–1/16	819904	819896	"	"	"
	"	"	"	"	1–1/8		683342	1–1/4	683367	741512	"	"	"
529040	"	"	"	"	1–1/4		526822	1–3/8	819961	819953	"	"	"
	"	"	"	"		8/16	744144	1/2	744151	834242	"	"	"
	"	"	"	"		1/2		9/16			"	"	"
731190	"	"	"	"		5/8	527051	11/16	834168	834150	"	"	"
	"	"	4–1/2	525568	13/16		681072	15/16	681098	681080	"	"	"
	"	"	"	"	15/16		526517	1–1/16	681459	744318	"	"	"
688051	"	"	"	"	1–1/8		526723	1–1/4	527473	525931	"	"	"
682625	"	"	"	"	1–1/4		526863	1–3/8	681593	684175	"	"	"
	"	"	"	"		8/16		1/2			"	"	"
741827	"	"	"	"		1/2		9/16			"	"	"
	"	"	"	"		5/8	527069	11/16	731281	731273	"	"	"
746289	"	"	5–1/2	525584	13/16		741538	15/16	741561	741553	"	"	"
528745	"	"	"	"	15/16		824326	1–1/16	826743	824318	"	"	"
685552	"	"	"	"	1–1/8		526756	1–1/4	819946	819938	" "	"	"
529107	"	"	"	"	1–1/4		526889	1–3/8	683391	683375	"	"	"
	"	"	"	"		8/16		1/2		741595	"	"	"
685545	"	"	"	"		1/2	686485	9/16	686493	834192	"	"	"
	"	"	"	"		5/8	527077	11/16	527648	526111	"	"	"
	"	"	6–5/8	741611	13/16			15/16			"	"	"
	"	"	"	"	15/16			1–1/16			"	"	"
528935	"	"	"	"	1–1/8		741637	1–1/4	741678	741652	"	"	"
529115	"	"	"	"	1–1/4		741660	1–3/8	820019	820001	"	"	"
	"	"	"	"		8/16		1/2	741702	741686	" "	"	"

Private

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Parent	Body	Gland	Slip	Slip	RCH	FCH	Neoprene	Metal	Top	Bottom	Gland Bolts and Nuts		
					Hole Size	Hole Size	Ring Part No.	Ring Hole Size	Ring Part No.	Ring Part No.	Qty Req'd	Bolt Part No.	Nut Part No.
Part No.	Part No.	Part No.	Size	Part No.									
	"	"	"	"		1/2	741694	9/16			"	"	"
	"	"	"	"		5/8		11/16			"	"	"
787564	"	"	7	840280	1–7/16		836569	1–1/16	836577	836585	"	"	"
835462	"	"	"	"	1–1/8		835470	1–1/4	741736	741710	"	"	"
682682	"	"	"	"	1–1/4		741728	1–3/8	679217	678649	"	"	"
529420	"	"	"	681320		8/16	681338	11/16	737130	737148	"	"	"
318485	"	"	"	"		21/32		3/4	834259	834267	"	"	"
529461	"	"	"	"		5/8	686691	13/16	819995	819987	"	"	"

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3 Technical and Engineering Information

3.1 Motor Resistance Tables and Winding Codes

The following tables show motor winding phase-to-phase resistance values at 68 degF (20 degC) and winding codes for various motors. Winding codes are not available for all listed motors. In some cases, different motors will have the same winding code indicating the stator winding as being the same, however, the motor's insulation system, bearings and mechanical configuration are different.

3.1.1 375–Series Motors

Motor resistances and winding codes for 375–Series Motors are shown in .

Table 3-1: Motor Resistances and Winding Codes, 375–Series Motors

Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
7.5	410	14	1.36	0050
7.5	480	11.5	2.15	0051
10.5	390	20.5	0.91	0070
10.5	680	12	2.94	0071
15	315	35.5	0.37	0100
15	400	28	0.59	0101
19.5	287	51	0.28	0130
19.5	387	38	0.48	0131
19.5	630	23	1.24	0132
22.5	330	51.5	0.33	0150
22.5	440	38.5	0.55	0151
22.5	730	23	1.42	0152
25.5	370	51	0.37	0170
25.5	500	37	0.63	0171
25.5	630	30	1.00	0172
25.5	760	25	1.61	0173

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3.1.2 456–Series Motors

Motor resistances and winding codes for 456–Series Motors are shown in .

Table 3-2: Motor Resistances and Winding Codes, 456–Series Motors

Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
10	435	12.5	1.22	
10	439	14	1.09	2020
12.5	425	22	1.22	
12.5	450	17.5	1.09	2020
15	434	21.5	0.63	2030
15	435	23	0.71	
15	655	14	1.52	2031
15	655	16	1.70	
18.75	420	30	0.71	
18.75	450	26.5	0.63	2030
18.75	635	20	1.70	
18.75	680	17.5	1.52	2031
20	450	28.5	0.54	
20	469	27.5	0.51	2040
20	750	17	1.68	
20	779	15	1.50	2041
25	410	39	0.37	
25	440	35.5	0.37	2050
25	450	35	0.54	
25	480	33	0.51	2040
25	675	23	1.23	
25	690	22	1.11	
25	730	22	1.68	
25	790	20	1.50	2041
25	895	18	1.50	2051
30	425	44.5	0.30	
30	440	45	0.30	2061
30	711	28	0.74	2060
30	750	25.5	1.01	
30	1260	15	3.14	

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Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
31.25	415	48	0.37	
31.25	450	44	0.37	2050
31.25	490	39	0.49	
31.25	705	27	1.11	
31.25	830	24	1.50	
31.25	910	22	1.50	2051
35	385	57	0.22	
35	675	33	0.66	
35	785	28	0.89	
37.5	415	61	0.30	
37.5	450	53	0.30	2061
37.5	725	33	0.74	2060
37.5	750	35	0.78	
37.75	1275	20	3.14	
40	430	59	0.25	
40	464	55	0.24	2080
40	770	33	0.75	
40	819	31	0.75	2081
40	880	29	1.02	
40	940	27.5	0.96	2082
40	1340	19	2.31	2083
50	440	72	0.25	
50	475	67	0.24	2080
50	675	47	0.49	
50	780	41	0.75	
50	815	39	0.70	
50	840	37.5	0.75	2081
50	885	35.5	0.70	2101
50	895	35	1.02	
50	955	33	0.92	
50	960	33	0.96	2082
50	1030	30.5	0.92	2102
50	1169	27.5	1.19	2103
50	1335	22.5	1.69	2104
50	1450	22	2.31	2083
50	1390	23	2.11	

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Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
60	640	59	0.37	
60	745	52	0.51	
60	810	47	0.58	
60	875	45.5	0.58	2121
60	970	39	0.83	
60	1330	29	1.50	2122
60	2145	18	3.39	2123
62.5	690	55	0.49	
62.5	740	53.5	0.49	2100
62.5	810	48	0.70	
62.5	900	44	0.70	2101
62.5	975	39	0.92	
62.5	1050	37.5	0.92	2102
62.5	1120	34	1.26	
62.5	1200	33	1.19	2103
62.5	1320	31	1.75	
62.5	1350	29.5	1.69	2104
62.5	1405	27	2.11	
70	540	82.5	0.24	
70	750	60	0.42	
70	805	55	0.40	2141
70	945	47	0.67	
70	1135	39	0.97	
70	1235	36	0.97	2143
70	1335	33	1.27	2144
70	2048	21.5	2.59	2145
75	660	72	0.37	
75	700	68	0.34	2120
75	830	57	0.58	
75	890	53.5	0.58	2121
75	1175	40	1.10	
75	1440	33	1.50	2122
75	2030	23	3.39	
75	2180	22	3.39	2123
80	635	80	0.27	
80	860	60	0.48	

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Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
80	920	59	0.45	2161
80	1085	46	0.77	
80	1160	47	0.77	2162
80	1310	39	1.10	
80	1345	39	1.10	2163
80	2382	22	2.95	2164
87.5	565	99	0.24	
87.5	610	91	0.24	2140
87.5	765	73	0.42	
87.5	825	67	0.40	2141
87.5	965	58	0.67	
87.5	1040	53	0.67	2142
87.5	1165	48	0.97	
87.5	1250	44.5	0.97	2143
87.5	1350	41	1.27	2144
87.5	2080	26.5	2.59	2145
90	710	81	0.30	
90	960	59	0.54	2180
90	1135	50	0.80	2181
90	1220	46	0.86	
90	1300	45	0.67	2242
90	1460	39	1.23	2182
90	1960	29	2.21	
100	650	97	0.27	
100	700	90.5	0.27	2160
100	790	80	0.33	
100	875	72	0.48	
100	920	70	0.47	
100	940	67.5	0.45	2161
100	1185	55	0.77	2162
100	1075	59	0.60	
100	1105	57	0.77	
100	1150	55	0.56	2200
100	1465	38	0.95	2201
100	1355	46	0.95	
100	1360	49	1.10	2163

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Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
100	2205	28.5	2.45	
100	2250	28	3.31	
100	2362	27	2.31	2202
100	2420	26	2.95	2164
110	935	74	0.40	2240
110	1190	60	0.66	
110	2380	30	2.69	
120	945	81	0.40	
120	1125	70	0.62	
120	1140	66.5	0.56	2241
120	1295	59	0.71	
120	1300	58	0.67	2242
120	2245	35	2.41	
120	2480	30.5	2.14	2243
125	810	98	0.33	
125	1095	72	0.60	
125	1180	67	0.56	2200
125	1480	55	0.95	2201
125	1380	57	0.95	
125	2240	35	2.45	
125	2400	33	2.31	2202
150	970	98	0.40	
150	1050	90.5	0.40	2240
150	1085	88	0.59	
150	1170	88.1	0.56	2241
150	1410	67.5	0.71	2242
150	2345	41	2.14	
150	2520	37.5	2.14	2243

3.1.3

456 Dominator Single-Section Motors

Motor winding phase-to-phase resistance values and winding codes for 456 Dominator single-section motors are shown in [Table 3-3](#).

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Table 3-3: Motor Resistances and Winding Codes, 456 Dominator Single-Section Motors

Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
50	960	32	1.30	4551
80	2150	23	3.59	4584
100	1345	45.5	1.05	4602
140	2420	35.5	2.49	4642
180	2420	46	1.85	4681

3.1.4

456 Dominator Tandem Motors

Motor winding phase-to-phase resistance values and winding codes for 456 Dominator tandem motors are shown in [Table 3-4](#).

Table 3-4: Motor Resistances and Winding Codes, 456 Dominator Tandem Motors

Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
24	439	35	0.654	4020
24	682	22.5	1.61	4021
36	415	55.5	0.343	4030
36	750	29.5	1.37	4031
36	901	25.5	1.78	4032
48	1039	29.5	1.76	4042
48	1363	22.5	2.91	4043
48	472	65	0.317	4044
48	877	35	1.18	4045
60	995	38.5	1.17	4051
60	1198	32	1.77	4052
60	1400	27.5	2.32	4053
60	894	43	0.933	4054
60	1097	35	1.44	4055
72	1437	32	2.09	4062
72	2288	20	5.94	4063
72	951	48.5	0.871	4064
72	2044	22.5	4.21	4065
84	968	55.5	0.737	4070

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Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
84	1394	38.5	1.59	4071
84	2102	25.5	3.82	4072
84	1252	43	1.27	4073
96	945	65	0.599	4082
96	1430	43	1.44	4083
96	2402	25.5	4.32	4084
108	881	78.5	0.496	4090
108	1427	48.5	1.27	4092
108	2520	27.5	4.02	4093
120	1181	65	0.741	4102
120	1586	48.5	1.41	4104
120	2598	29.5	4.12	4105
120	2194	35	2.76	4106
132	1076	78.5	0.6	4111
132	1299	65	0.811	4112
132	2635	32	3.71	4113
132	2413	35	3.03	4114
144	931	99	0.41	4120
144	1174	78.5	0.652	4121
144	2631	35	3.29	4123
144	2145	43	2.12	4124
144	2389	38.5	2.65	4125
156	1272	78.5	0.705	4131
156	1535	65	0.953	4132
156	2588	38.5	2.87	4133
168	1086	99	0.476	4140
168	2503	43	2.47	4142
180	2682	43	2.64	4150
192	1890	65	1.16	4160
192	2537	48.5	2.21	4161
204	2695	48.5	2.35	4170
204	2008	65	1.24	4171
216	1397	99	0.607	4180
216	2490	55.5	1.82	4181
216	2125	65	1.31	4182
216	1762	78.5	0.966	4183

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3.1.5 456–Series Hotline (HX) Motors

Motor resistances and winding codes for 456–series Hotline (Hx) motors are shown in .

Table 3-5: Motor Resistances and Winding Codes, 456–Series Hotline (Hx) Motors

Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
11.25	434	21	0.66	1530
30	985	21	1.37	1080
52.5	1400	29	1.25	1640
60	1150	55	0.56	1161
60	1160	47	0.76	1160
75	1150	46	0.57	1200
75	1400	38	0.93	1700
90	1135	48	0.63	1240
90	2400	28.5	2.17	1740

3.1.6 540–Series Motors

Motor resistances and winding codes for 540–Series Motors are shown in .

Table 3-6: Motor Resistances and Winding Codes, 540–Series Motors

Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
20	489	25.7	0.47	6020
20	778	16.2	1.25	6021
20	860	14.6	1.56	6022
20	1150	10.9	3.09	6023
25	516	30.5	0.47	6020
25	822	19.1	1.25	6021
25	909	17.3	1.56	6022
25	1215	12.9	3.09	6023
30	485	38.9	0.3	6030
30	795	23.8	0.83	6031
30	919	20.5	1.22	6032
30	1353	14.0	2.84	6033
37.5	513	46.0	0.3	6030

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Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
37.5	840	28.1	0.83	6031
37.5	971	24.3	1.22	6032
37.5	1430	16.5	2.84	6033
40	482	52.2	0.22	6040
40	812	31.0	0.61	6041
40	978	25.7	0.88	6042
40	1473	17.1	2.36	6043
50	509	61.8	0.22	6040
50	858	36.6	0.61	6041
50	1033	30.4	0.88	6042
50	1557	20.2	2.36	6043
50	499	63.1	0.18	6050
50	705	44.6	0.35	6051
50	809	38.9	0.47	6052
50	912	34.5	0.6	6053
50	1015	31.0	0.75	6054
50	1429	22.0	1.58	6055
60	475	79.5	0.14	6060
60	723	52.2	0.31	6061
60	971	38.9	0.56	6062
60	1095	34.5	0.71	6063
60	1219	31.0	0.88	6064
60	1466	25.8	1.27	6065
62.5	527	74.6	0.18	6050
62.5	745	52.7	0.35	6051
62.5	855	46.0	0.47	6052
62.5	964	40.8	0.6	6053
62.5	1073	36.6	0.75	6054
62.5	1510	26.0	1.58	6055
70	409	107.7	0.09	6070
70	843	52.3	0.36	6071
70	1132	38.9	0.65	6072
70	1422	31.0	1.08	6073
75	502	93.9	0.14	6060
75	764	61.7	0.31	6061
75	1026	46.0	0.56	6062

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Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
75	1157	40.8	0.71	6063
75	1288	36.6	0.88	6064
75	1550	30.4	1.27	6065
80	468	107.6	0.1	6080
80	798	63.1	0.27	6081
80	964	52.2	0.41	6082
80	1129	44.6	0.54	6083
80	1294	38.9	0.73	6084
80	2451	20.5	2.94	6085
87.5	433	127.1	0.09	6070
87.5	891	61.7	0.36	6071
87.5	1197	46.0	0.65	6072
87.5	1502	36.6	1.08	6073
100	494	127.3	0.1	6080
100	844	74.5	0.27	6081
100	1018	61.8	0.41	6082
100	1193	52.7	0.54	6083
100	1368	46.0	0.73	6084
100	2590	24.3	2.94	6085
100	1205	52.2	0.5	6100
100	1411	44.6	0.66	6101
100	2444	25.8	2.04	6102
120	702	107.6	0.15	6120
120	950	79.5	0.26	6121
120	1198	63.0	0.42	6122
120	1446	52.2	0.6	6123
120	2190	34.5	1.44	6124
120	2438	31.0	1.7	6125
125	1273	61.7	0.5	6100
125	1491	52.7	0.66	6101
125	2583	30.4	2.04	6102
130	1298	63.0	0.51	6130
130	2640	31.0	1.83	6131
130	1029	79.5	0.27	6132
130	2103	38.9	1.09	6133
140	819	107.6	0.17	6140

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Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
140	1108	79.5	0.31	6141
140	1398	63.0	0.48	6142
140	2265	38.9	1.25	6143
150	742	127.1	0.15	6120
150	1004	93.9	0.26	6121
150	1266	74.5	0.42	6122
150	1528	61.7	0.6	6123
150	2314	40.8	1.44	6124
150	2575	36.6	1.7	6125
150	1188	79.5	0.33	6150
150	2427	38.9	1.33	6151
160	936	107.6	0.19	6160
160	1267	79.5	0.35	6161
160	2258	44.6	1.04	6162
160	4240	23.8	3.91	6163
162.5	1371	74.5	0.51	6130
162.5	2790	36.6	1.83	6131
162.5	1087	94.0	0.27	6132
162.5	2223	46.0	1.09	6133
175	866	127.1	0.17	6140
175	1171	94.0	0.31	6141
175	1477	74.5	0.48	6142
175	2393	46.0	1.25	6143
180	1053	107.6	0.23	6180
180	1425	79.5	0.39	6181
180	2168	52.3	0.88	6182
180	2540	44.6	1.16	6183
187.5	1255	93.9	0.33	6150
187.5	2565	46.0	1.33	6151
200	989	127.1	0.19	6160
200	1339	93.9	0.35	6161
200	2386	52.7	1.04	6162
200	4480	28.1	3.91	6163
200	1170	107.6	0.24	6200
200	1470	85.0	0.43	6201
200	2410	52.2	0.98	6202

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Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
200	2822	44.6	1.29	6203
200	1996	63.1		6204
225	1113	127.1	0.23	6180
225	1506	93.9	0.39	6181
225	2292	61.7	0.88	6182
225	2685	52.7	1.16	6183
250	1237	127.1	0.24	6200
250	1375	111.3	0.43	6201
250	2547	61.7	0.98	6202
250	2982	52.7	1.29	6203
250	2110	74.5		6204

3.1.7 540–Series Hotline (HX) Motors

Motor resistances and winding codes for 540–series Hotline (Hx) motors are shown in .

Table 3-7: Motor Resistances and Winding Codes, 540–Series Hotline (Hx) Motors

Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
15	775	12	1.28	5520
30	1325	14	1.98	5540
37.5	975	24	0.74	5050
37.5	1375	17	1.77	5550
50	2000	15	2.84	5570
52.5	826	40	0.36	5070
52.5	2050	16	2.86	5570
60	2025	18	2.48	5080
60	2050	18	2.52	5580
67.5	2000	21	1.92	5590
70	1850	23	1.40	5600
75	1275	36	0.66	5100
75	1900	24	1.40	5600
90	1950	28	1.42	5620
100	2100	29	1.24	5640
105	2150	30	1.22	5640

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Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
120	1810	42	0.812	5660
135	950	91	0.21	5180
150	1445	65.5	.431	5201
150	2194	43	1.00	5200
150	2194	43	1.04	5700

3.1.8

562-Series Single-Section Motors

Motor resistances and winding codes for 562-series single-section motors are shown in .

Table 3-8: Motor Resistances and Winding Codes, 562-Series Single-Section Motors

Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
60	495	73.5	0.193	9408
60	850	43	0.625	9481
60	1205	30	1.33	9482
90	850	64	0.369	9490
90	1170	46.5	0.717	9491
90	1275	43	0.878	9494
90	1490	36.5	1.22	9492
90	2125	25.5	2.77	9493
120	850	85.5	0.283	9500
120	990	73.5	0.349	9501
120	1415	51.5	0.748	9502
120	1275	57	0.635	9503
120	2550	28.5	2.86	9504
120	4160	17	7.59	9505
150	885	102.5	0.229	9510
150	1415	64	0.581	9511
150	2655	34	2.31	9512
150	1240	73.5	0.48	9513
150	3190	28.5	3.50	9514
150	2305	39.5	1.59	9515
150	4160	21.5	5.88	9516
180	1275	85.5	0.41	9520

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Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
180	2550	43	1.64	9521
180	3190	34	2.37	9522
210	2480	51.5	1.25	9530
210	3720	34	3.16	9531
210	4160	34	3.96	9532
240	1415	102.5	0.35	9540
240	2550	57	1.20	9541
240	4250	34	3.58	9542
270	1275	128.5	0.25	9552
270	1595	102.5	0.40	9550
270	2550	64	1.01	9551
300	1415	128.5	0.28	9560
300	2480	73.5	0.92	9561
300	4250	43	2.65	9562
330	1560	128.5	0.30	9570
330	1950	102.5	0.48	9571
330	3895	51.5	1.92	9572
330	3505	57	1.63	9573
330	2725	73.5	0.90	9574
360	2550	85.5	0.79	9580
360	4250	51.5	2.09	9581
360	3400	64	1.32	9582
360	1700	128.5	0.33	9583
390	2300	102.5	0.56	9591
390	3685	64	1.43	9592
390	2765	85.5	0.85	9593
390	1840	128.5	0.36	9594
420	2480	102.5	0.61	9600
420	3965	64	1.54	9601
420	3470	73.5	1.13	9602
420	2975	85.5	0.92	9603
450	2125	128	0.41	9610
450	2655	102.5	0.65	9611
450	3720	73.5	1.36	9612
450	4250	64	1.64	9613

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3.1.9 562–Series Tandem Motors

Motor resistances and winding codes for 562–series tandem motors are shown in .

Table 3-9: Motor Resistances And Winding Codes, 562–Series Tandem Motors

Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
30	460	39.5	0.211	9630
30	745	24.5	0.612	9631
30	1170	15.5	1.80	9632
45	375	72	0.085	9010
45	430	63	0.218	9011
45	595	46	0.452	9012
45	700	39	0.601	9013
45	1140	23.5	1.74	9014
60	495	73.5	0.188	9350
60	850	43	0.617	9351
60	1205	30	1.32	9052
75	435	104	0.133	9020
75	615	73	0.250	9021
75	800	57	0.431	9022
75	980	46	0.667	9023
75	1255	35.5	1.07	9024
90	850	64	0.361	9360
90	1490	36.5	1.21	9361
90	1380	39.5	0.995	9362
90	2125	25.5	2.75	9363
90	1275	43	0.866	9364
90	450	120	1.10	9365
90	1170	46.3	0.353	9366
90	950	57	0.457	9367
105	485	130	0.104	9030
105	860	73	0.330	9031
105	1115	57	0.571	9032
105	1365	46	0.883	9033
120	850	85.5	0.275	9370

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Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
120	990	73.5	0.341	9371
120	1415	51.5	0.736	9372
120	1275	57	0.62	9373
120	2550	28.5	2.84	9374
120	4160	17	7.54	9375
135	1105	73	0.411	9040
135	1430	57	0.710	9041
135	1755	46	1.10	9042
135	460	177	0.076	9044
150	885	102	0.221	9380
150	1240	73.5	0.47	9383
150	1415	64	0.57	9381
150	2655	34	2.29	9382
150	3190	28.5	3.47	9384
150	2305	39.2	1.57	9385
150	4160	21.7	5.84	9386
150	985	91.5	0.299	9387
165	1350	73	0.492	9050
165	1745	57	0.849	9051
165	2140	46	1.31	9052
180	1275	85.5	0.40	9390
180	2550	43	1.61	9391
180	3190	34	2.70	9392
210	1240	102	0.30	9400
210	2490	51.5	1.23	9401
210	3720	34	3.12	9402
240	1415	102	0.34	9410
240	2550	57	1.18	9411
240	4250	34	3.54	9412
270	1275	128.5	0.24	9420
270	1595	102.5	0.38	9421
270	2545	64	0.98	9422
300	1415	128.5	0.266	9430
300	2480	73.5	0.90	9431
300	4250	43	2.61	9432
300	1775	102.5	0.424	9433

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Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
300	1235	147.5	0.243	9434
330	1360	147.5	0.266	9625
330	1560	128.5	0.29	9620
330	1950	102.5	0.46	9621
330	2730	73.5	0.88	9624
330	3505	57	1.60	9623
330	3895	51.5	1.89	9622
360	1700	129	0.32	9445
360	2125	102.5	0.50	9442
360	2550	85.5	0.77	9441
360	3400	64	1.30	9444
360	4250	51.5	2.05	9443
390	1385	171	0.20	9450
390	1840	129	0.34	9454
390	2080	114	0.51	9455
390	2300	102.5	0.55	9451
390	2765	85.5	0.83	9453
390	3685	64	1.40	9452
420	1970	128.5	0.37	9464
420	2480	102.5	0.59	9460
420	2975	85.5	0.89	9463
420	3470	73.5	1.10	9462
420	3965	64	1.50	9461
450	1860	147	0.36	9470
450	2125	128.5	0.39	9471
450	2655	102.5	0.63	9472
450	3180	86.7	0.95	8370
450	3720	73.5	1.33	9473
450	4250	64	1.61	9474

3.1.10

Enhanced 562 Dominator Motors

Enhanced 562 Dominator motor winding phase-to-phase resistance values are shown in [Table 3-10](#). Winding code cross references for the enhanced 562 Dominator motors are also shown.

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Table 3-10: Enhanced 562 Dominator Motor Winding Phase-To-Phase Resistance Values and Winding Code Cross References

New Base Nameplate Data			Phase-to-Phase Resistance (ohms)	Enhanced 562 Dominator Winding Code	Old 562 Dominator Winding Code
Horsepower	Voltage	Amperage			
30	460	39.5	0.505	F010	9630
30	747	24	1.3	F011	9631
30	1177	15.5	3.84	F012	9632
45	375	72	0.173	FA10	9010
45	430	63	0.236	FA11	9011
45	595	45.5	0.462	FA12	9012
45	704	38.5	0.691	FA13	9013
45	1143	23.5	1.78	FA14	9014
60	490	73.5	0.207	F020	9350
60	849	42.5	0.641	F021	9351
60	1207	30	1.37	F022	9352
75	439	103	0.135	FA20	9020
75	620	73	0.254	FA21	9021
75	802	56.5	0.438	FA22	9022
75	983	46	0.678	FA23	9023
75	1255	36	1.16	FA24	9024
90	842	64.5	0.393	F031	9360
90	1488	36.5	1.31	F036	9361
90	1380	39.5	1.15	F035	9362
90	2133	25.5	2.83	F037	9363
90	1273	42.5	0.89	F034	9364
90	412	132	0.096	F030	9365
90	1165	46.5	0.769	F033	9366
90	950	57	0.497	F032	9367
105	485	130	0.112	FA30	9030
105	865	73	0.334	FA31	9031
105	1118	56.5	0.578	FA32	9032
105	1371	46	0.894	FA33	9033
120	837	86.5	0.281	F040	9370
120	980	73.5	0.369	F041	9371
120	1410	51	0.796	F043	9372
120	1267	57	0.637	F042	9373
120	2558	28	2.9	F045	9374

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New Base Nameplate Data			Phase-to-Phase Resistance (ohms)	Enhanced 562 Dominator Winding Code	Old 562 Dominator Winding Code
Horsepower	Voltage	Amperage			
120	4279	17	9.37	F046	9375
120	2270	32	2.29	F044	not applicable
135	1110	73	0.415	FA42	9040
135	1435	56.5	0.717	FA43	9041
135	1760	46	1.11	FA44	9042
135	460	177	0.077	FA40	9044
135	948	85.5	0.316	FA41	not applicable
150	867	104	0.238	F050	9380
150	1404	64.5	0.613	F052	9381
150	2659	34	2.33	F054	9382
150	1225	73.5	0.506	F051	9383
150	3197	28	3.53	F055	9384
150	2300	39.5	1.79	F053	9385
150	4273	21	7.5	F056	9386
not applicable	not applicable	not applicable		not applicable	9387
165	1355	73.5	0.496	FA50	9050
165	1752	56.5	0.856	FA51	9051
165	2148	46	1.32	FA52	9052
180	1255	86.5	0.404	F061	9390
180	2545	42.5	1.64	F065	9391
180	3190	34	2.74	F066	9392
180	1900	57	0.915	F063	9395
180	1040	104	0.281	F060	not applicable
180	1685	64.5	0.723	F062	not applicable
180	2115	51	1.14	F064	not applicable
180	4051	26.5	5.53	F067	not applicable
210	1213	104	0.306	F070	9400
210	2468	51	1.25	F071	9401
210	3722	34	3.55	F072	9402
240	1386	104	0.367	F081	9410
240	2534	57	1.19	F083	9411
240	4255	34	4.02	F084	9412
240	1100	131	0.231	F080	9413
240	2247	64.5	0.943	F082	9414
270	1237	131	0.258	F090	9420

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New Base Nameplate Data			Phase-to-Phase Resistance (ohms)	Enhanced 562 Dominator Winding Code	Old 562 Dominator Winding Code
Horsepower	Voltage	Amperage			
270	1560	104	0.387	F091	9421
270	2528	64.5	1.05	F094	9422
270	1882	86.5	0.7	F092	9423
not applicable	not applicable	not applicable		not applicable	9425
270	2205	73.5	0.772	F093	not applicable
300	1374	131	0.284	F100	9430
300	2450	73.5	0.96	F103	9431
300	4242	42.5	3.14	F104	9432
300	1733	104	0.452	F101	9433
not applicable	not applicable	not applicable		not applicable	9434
300	2091	86.5	0.774	F102	9435
330	1512	131	0.311	F110	9620
330	1906	104	0.525	F111	9621
330	3878	51	2.13	F114	9622
330	3484	57	1.61	F113	9623
330	2695	73.5	0.933	F112	9624
not applicable	not applicable	not applicable		not applicable	9625
360	2510	86.5	0.773	F122	9441
360	2080	104	0.57	F121	9442
360	4230	51	2.32	F124	9443
360	3370	64.5	1.38	F123	9444
360	1649	131	0.338	F120	9445
390	1321	178	0.214	F130	9450
390	2253	104	0.581	F132	9451
390	3650	64.5	1.68	F134	9452
390	2718	86.5	0.834	F133	9453
390	1787	131	0.365	F131	9454
not applicable	not applicable	not applicable		not applicable	9455
420	2426	104	0.624	F141	9460
420	3932	64.5	1.8	F144	9461
420	3430	73.5	1.18	F143	9462
420	2928	86.5	0.896	F142	9463
420	1924	131	0.44	F140	9464

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New Base Nameplate Data			Phase-to-Phase Resistance (ohms)	Enhanced 562 Dominator Winding Code	Old 562 Dominator Winding Code
Horsepower	Voltage	Amperage			
450	3137	86.5	0.957	F153	8370
not applicable	not applicable	not applicable		not applicable	9470
450	2062	131	0.471	F151	9471
450	2600	104	0.666	F152	9472
450	3675	73.5	1.5	F154	9473
450	4213	64.5	1.92	F155	9474
450	1524	178	0.246	F150	9475

3.1.11

738 Dominator Motors

Motor resistances and winding codes for 738 Dominator motors are shown in [Table 3-11](#).

Table 3-11: Motor Resistances and Winding Codes, 738 Dominator

Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
200	1667	72.9	0.55	E040
250	2277	66.7	0.783	E050
250	1892	80.3	0.527	E051
250	3238	46.9	1.67	E052
250	4200	36.2	2.89	E053
300	4117	44.3	2.3	E060
300	2270	80.3	0.713	E061
300	3656	49.8	1.72	E062
350	2379	89.4	0.647	E070
350	3187	66.7	1.23	E071
350	4265	49.8	1.95	E072
400	2411	100.8	0.505	E080
400	3335	72.9	1.06	E081
400	4258	57.1	1.7	E082
450	2020	135.3	0.353	E090
450	2366	115.5	0.493	E091
450	3405	80.3	1	E092
450	4097	66.7	1.53	E093

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Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
500	1860	163.3	0.255	E100
500	2244	135.4	0.388	E101
500	3398	89.4	0.88	E103
500	4169	72.9	1.28	E104
550	2046	163.3	0.277	E110
550	2469	135.3	0.422	E111
550	3315	100.8	0.714	E112
550	4162	80.3	1.2	E113
600	1770	205.9	0.189	E120
600	2232	163.3	0.3	E121
600	3155	115.5	0.6	E123
600	3616	100.8	0.772	E124
650	1917	206	0.203	E130
650	2418	163.3	0.322	E131
650	3418	115.5	0.645	E132
650	3918	100.8	0.83	E133
700	2065	205.9	0.217	E140
700	2604	163.3	0.345	E141
700	3142	135.3	0.524	E142
700	4220	100.8	0.887	E144
750	2212	206	0.231	E150
750	3367	135.3	0.527	E151
750	3943	115.5	0.779	E153
800	2975	163.3	0.414	E160
800	3591	135.3	0.593	E161
800	2360	205.9	0.245	E163
850	3161	163.3	0.438	E170
850	3815	135.3	0.627	E171
850	2507	206	0.26	E172
900	4040	135.3	0.661	E180
900	3347	163.3	0.462	E181

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3.1.12 4–Pole 450/456–Series Motors

Motor resistances and winding codes for 4–pole 450/456–series motors are shown in .

Table 3-12: Motor Resistances and Winding Codes, 4–Pole 450/456–Series Motors

Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
20	1250	15.5	5.40	A100
30	1200	22.5	3.11	A150
40	1235	30.5	2.32	A200
50	1200	37.5	1.63	A240

3.1.13 4–Pole 540–Series Motors

Motor resistances and winding codes for 4–pole 540–series motors are shown in .

Table 3-13: Motor Resistances and Winding Codes, 4–Pole 540–Series Motors

Horsepower	Volts	Amps	Phase-to-Phase Resistance (ohms)	Winding Code
20	400	42	0.452	B060
20	1200	17	3.38	B061
40	400	82	0.209	B100
40	1275	27	2.05	B101
50	1275	34	1.70	B120
60	1325	47	1.18	B140
70	1535	42	1.49	B180
80	1260	59	.98	B200

3.2 Cable Resistance

Electric wire and cables are identified by numbered sizes that are derived by the cross-sectional area of the wire. This value is stated in circular mils. [Table 3-14](#) shows the cross-sectional area in circular mils, the resistance per thousand feet at 77 degF (25 degC) and the optimum current carrying capabilities for the common wire sizes used in submersible applications.

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Table 3-14: Cable Resistance

Size Number	Circular Mils	Ohms/1000 ft	Optimum Current (amperes)
6	26,240	0.41	55
4	41,740	0.258	70
2	66,360	0.163	90
1	83,690	0.129	110
1/0	105,600	0.102	200
2/0	133,100	0.080	300
4/0	211,600	0.050	400

Table 3-15 shows multiplier factors for conductor resistance.

Table 3-15: Multiplier Factors for Conductor Resistance

Conductor Temperature (degF [degC])	Resistance Multiplier Factor
122 to 131 [50 to 55]	1.13
140 to 149 [60 to 65]	1.17
158 to 167 [70 to 75]	1.21
176 to 185 [80 to 85]	1.25
194 to 203 [90 to 95]	1.28
212 to 221 [100 to 105]	1.32
230 to 239 [110 to 115]	1.36
248 to 257 [120 to 125]	1.40
266 to 275 [130 to 135]	1.44

3.3

Motor and Cable Resistance Calculation

Motor and cable resistance tables and the temperature multiplier factors for increasing temperatures are used to determine if the phase-to-phase resistance measurements of the downhole system are reasonable. The resistance measurement can also be useful for isolating the location of a fault.



Note

Resistance values for the motor and cable are provided in the *Motor Resistance Tables and Winding Codes* and *Cable Resistance* sections. Temperature multiplier values are provided in the *Cable Resistance* section.

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Note

For calculation purposes, the cable length doubles. The resistance value is the resistance down the cable, through the motor and back up the cable to the meter.

This is the calculated resistance. In practical application, measuring instruments are not capable of measurements to this accuracy, and splices may contribute to additional resistance. These calculations are used as a guideline. A resistance measurement considerably higher or lower than the calculated resistance indicates a potential problem. Resistance measurement interpretation is a function of experience and should be recorded for each installation.

Unbalanced phase-to-phase measurements indicate a problem and in the majority of cases are indicative of a motor problem.

One of the most important uses of the motor resistance tables is to ensure that the motor voltage listed on the motor nameplate is correct. Never assume the nameplate data is correct. Measure the motor phase-to-phase resistance and compare the measurement to data provided in the *Motor Resistance Tables and Winding Codes* section before installation of a unit.

Equation 3-1 is used to calculate the total phase-to-phase resistance.

Equation 3-1:

$$\text{total phase-to-phase resistance} = \text{motor resistance} + (\text{cable resistance} \times [\text{cable length} \times 2] \times \text{temperature multiplier})$$



Example

Example Data	
motor	456 series, 60 HP, 970 V, 39 A
cable	#4 copper conductor, 6000 ft (1829 m)
temperature	160 degF (70 degC)

Using Equation 3-1 —

$$\text{total phase-to-phase resistance} = 0.83 + (0.258 \text{ ohms} \times 12 \times 1.21) = 4.576 \text{ ohms}$$

3.4

Cable Temperature Calculation

The conductor temperature can be calculated using Equation 3-2. Conductor temperature can be used to select the appropriate cable type.

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**Note**

Significant cable life reduction may occur if cable is used at or near the maximum current rating.

Equation 3-2:

$$T_c = (a \times I^2) + T_{well}$$

where

Term	Definition
T_c	adjusted conductor temperature, degF
a	current carrying capacity factor, dimensionless
I	current, amperes
T_{well}	ambient well temperature, degF

[InTouch Content ID 3016052](#) contains an Excel spreadsheet that can be used to automatically generate an ESP power cable or MLE (cable only) Ampacity chart for a range of common cable types. Additional notes for MLE rating are included below.

MLE temperature rating: An MLE includes the flat cable length and the pothead that fits into the motor head. Typically the flat cable has a higher temperature rating than the pothead connection itself.

The attached spreadsheet (under [InTouch Content ID 3016052](#)) calculates the conductor temperature of the most common REDA cables, based on cable type, current, and downhole temperature. Note that for MLE's included in the spreadsheet, the calculation result DOES NOT include the pothead limitation. Be aware that this can be misleading.

The MLE temperature rating of the combined flat cable and pothead is based on the weaker component (usually pothead) and is found under the attributes in OneCAT details.

The use of the attached spreadsheet (under [InTouch Content ID 3016052](#)) is straight forward when checking power cable operating temperature under certain well conditions against published cable temperature rating. The rating is built into the spreadsheet.

However, to check the temperature limit of an MLE (combined flat cable and pothead) using the attached spreadsheet, make sure you do the following:

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1. Find the temperature rating of the MLE through OneCAT or GeMS.
2. Use the attached spreadsheet ([InTouch Content ID 3016052](#)) to calculate operating temperature (after entering your application data)
3. Compare the calculated operating temperature of selected cable to MLE rating determined in step -1- above. Calculated operating temperature must be below the MLE temperature rating for acceptable application.

DesignPro, in its current version (Jan 08) does not check selected MLE's against temperature rating. MLE rating check has to be done by the AE separately using the described procedure, above.

Future versions of DesignPro will include MLE rating calculation and comparison. This will be announced separately at the time of new version release.

3.4.1 Voltage Drop Calculation

Voltage drop is sometimes calculated using a temperature multiplier. This is not the best method because the voltage loss in the cable is not in phase with the voltage drop in the motor, but rather the current. To get the true voltage drop in the cable requires a power factor calculation that shows the voltage drop to be less than what would be predicted by making a resistance calculation. Ignoring the temperature multiplier results in a lower voltage drop calculation based on resistance only so this tends to offset the effect. Using a temperature multiplier and not accounting for power factor will result in an over-prediction of voltage loss and may provide more voltage at the motor than expected.

Cable voltage drop should be calculated using [Equation 3-3](#) and [Figure 3-1](#).

Equation 3-3:

$$\text{cable voltage drop} = \frac{(\text{pump depth} + 100 \text{ ft})(\text{voltage drop per } 1000 \text{ ft})}{1000}$$

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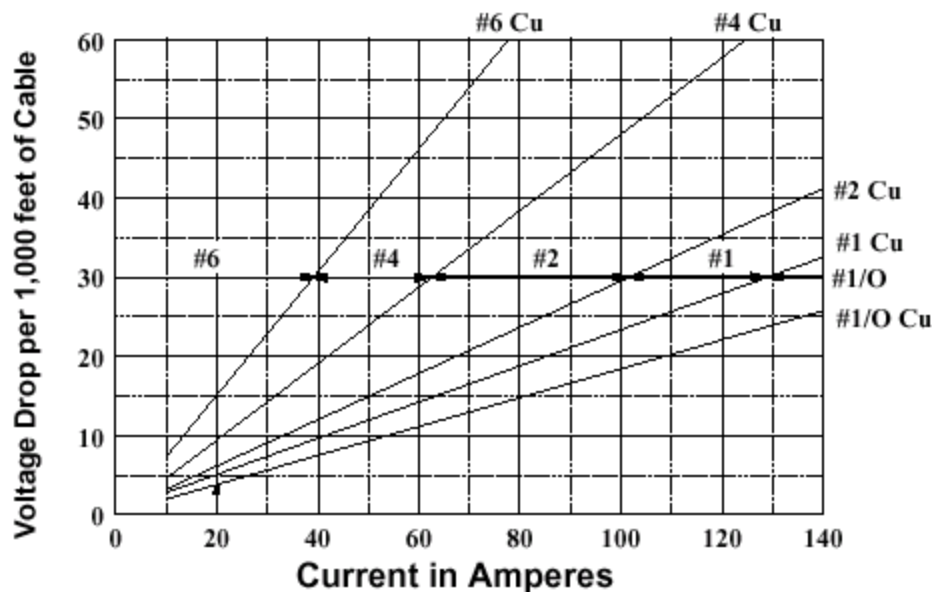


Figure 3-1: Cable voltage drop



Note

A horizontal line at 30 V/1000 ft is shown on [Figure 3-1](#). Typically, voltage drops in excess of this amount will not be economical due to the loss in efficiency.



Example

Calculate the cable voltage drop with a motor rated a 1130 V and 20 A with 7200 ft of #4 AWG copper cable in a well at 185 degF (85 degC).

Find the motor nameplate amperage on the X axis in the [Figure 3-1](#). Move vertically to the diagonal cable size line. Move horizontally to the left to find the voltage drop per 1000 feet on the Y axis.

Using [Equation 3-3](#), the cable voltage drop = $(7200 + 100) (9)/1000 = 66$ V.

3.5

Motor Cooling

[Figure 3-2](#) and [Figure 3-3](#) show the velocity of fluid moving passing various motors in various casing sizes at various flow rates. A general rule of thumb is to maintain a minimum flow rate of one ft/s (0.3 m/s) passing the motor. Fluid shrouds can be used to increase flow rate. There are factors other than fluid flow

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passing the motor that affect motor heating. An application engineer should review any design where flow rate passing the motor is less than one ft/s (0.3 m/s).

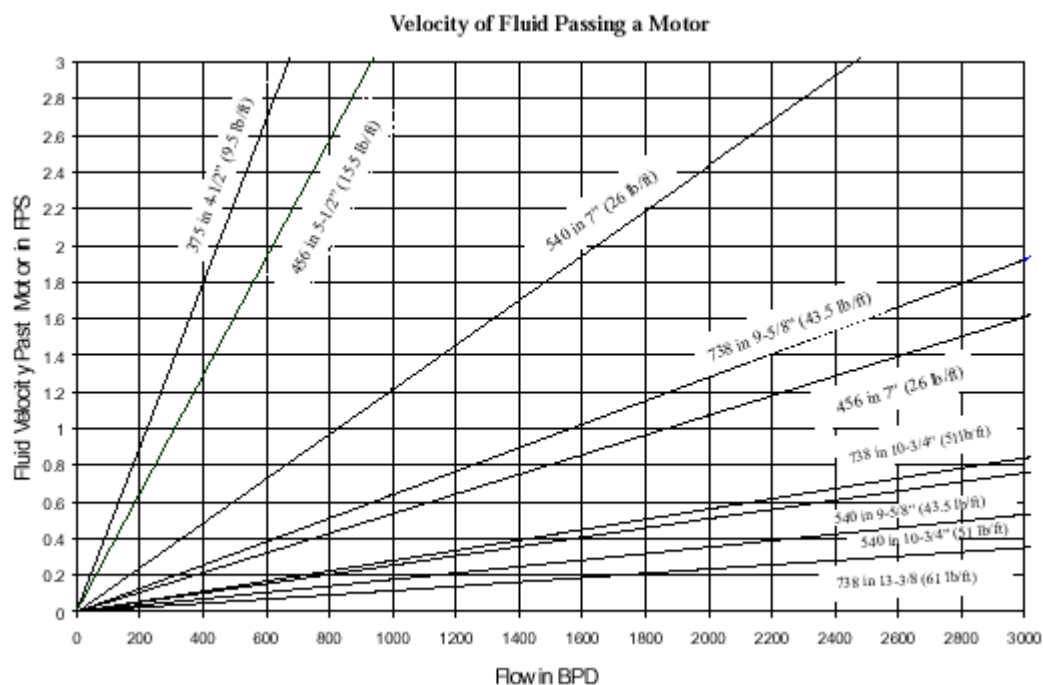


Figure 3-2: Velocity of fluid passing a motor, production rates less than 3000 bbl/day

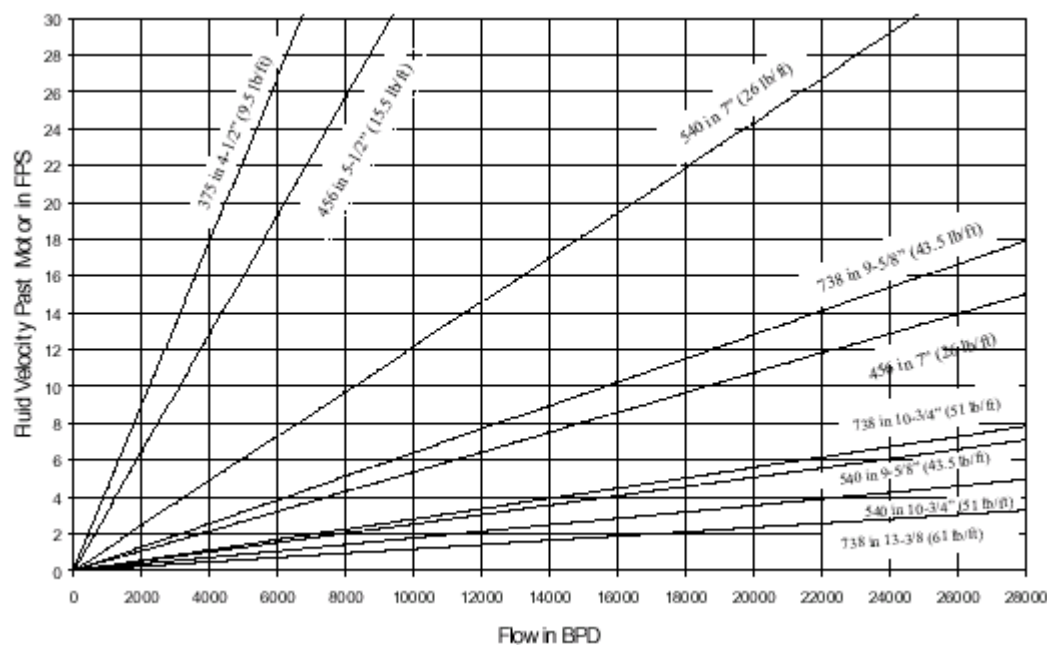


Figure 3-3: Velocity of fluid passing a motor, production rates more than 3000 bbl/day

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3.6 Voltage/Current Relationship of Motors

When designing or troubleshooting installations, it is useful to understand the voltage/current relationship of motors. This is especially true for deep wells. The voltage drop on initial startup should not be so great as to prevent the unit from starting. It is possible for a unit to start initially with no problems, then at a later time to encounter startup problems due to deposition buildup or wear in the unit.

The minimum voltage required to start a unit can vary due to different operating conditions. Schlumberger submersible motors require 2.5 to 3 times nameplate current or 50 to 55% of nameplate voltage at the terminals to start.

3.6.1 Voltage/Current Relationship

The following assumptions can be used to associate nameplate voltage to nameplate current.

- 100% nameplate volts gives 5 times nameplate current
- 80% nameplate volts gives 4 times nameplate current
- 60% nameplate volts gives 3 times nameplate current
- 40% nameplate volts gives 2 times nameplate current

3.6.2 Voltage Comparison Calculation

The following example is provided as a troubleshooting aid for analyzing applications. When sizing units for deep applications, the starting voltage percentage of the selected motor should be considered.

The calculations involved are based on assumptions and the assumptions are acceptable for practical use. The calculations are not intended to be mathematically exact, but an approximation that can be used.

The calculations and method are shown for the selection of the proper voltage motor for an application. This comparison of two motors of the same horsepower but different voltages illustrates how the available starting voltage at each motor can vary.

**Example**

Example Data	
depth	8,000 ft of 4 AWG cable
size of motor required	70 HP
casing size	5.5 in, 17 lbm/ft

3.6.2.1

Trial and Error Calculation Method

We find there are two motors available in the 456-series 70 HP size (980 V, 45 A and 1170 V, 38 A).

For our example determine the estimated starting voltage that will be present at the motor for the 980 V, 45 A motor.

1. The 4 AWG cable has 25 V/1000 ft of cable voltage loss ([Figure 3-1](#)) or a total of 200 V drop in the cable at motor nameplate amps. The required surface voltage would be $980 + 200 = 1180$ V.
2. Determine the percentage of full-load voltage that this motor will start at. This is done by using a 'rule of thumb' and by trial and error. The percentage can be arrived at quickly with experience.
 - a. Assume 80% voltage; the current is 4 times nameplate. The cable voltage drop is 4×200 V or 800 V. 1180 V – 800 V = 380 V. This is not 80%.
 - b. Assume 60% voltage; the current is 3 times 200 V or 600 V. 1180 V – 600 V = 580 V. This is not 60%. 580 V divided by 980 V is 49%; this is near percentage.
 - c. Assume 50% voltage; the current is 2.5 times 200 V or 500 V. 1180 V – 500 V = 680 V. 680 V ÷ 980 V = 58%. (Note: The percentage is between 50% and 60%)
 - d. Assume 55% voltage; the current is 2.75 times 200 V or 550 V. 1180 V – 550 V = 630 V. 630 V ÷ 980 V = 54%. The proper percentage that agrees with the current on start-up is 54%. This motor is marginal for this application.

Determine the estimated starting voltage that will be present at the motor for the 1170 V, 38 A unit.

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1. The 4 AWG cable has a voltage drop of 20 V/1000 ft (Figure 3-1) or 160 V drop in the cable at motor nameplate amps.
2. The required surface voltage is $1170\text{ V} + 160\text{ V} = 1330\text{ V}$.
 - a. Assume 80% voltage or 4 times nameplate current. The cable voltage drop is $4 \times 160\text{ V}$ or 640 V. $1330\text{ V} - 640\text{ V} = 690\text{ V}$. This is not 80%.
 - b. Assume 65% voltage or 3.25 times nameplate current. The cable voltage drop is $3.25 \times 160\text{ V}$ or 520 V. $1330\text{ V} - 520\text{ V} = 810\text{ V}$. $810\text{ V} \div 1330\text{ V}$ equals 61%, not 65%. The voltage percentage is approximately 63%. The 63% is much better (than the 980 V, 45 A motor) and this motor should have no start-up problems during its operation.

3.7

60 Hz Versus 50 Hz Operation

A Schlumberger submersible motor rotates at a nominal speed of 3500 rpm at 60 Hz. The motor rotates at 2915 rpm at 50 Hz. For each individual pump series, the diameter of the stages cannot be changed so the only variables available to modify performance within a series are the different types of stages, the number of stages and the speed they rotate. The rotational speed can only be changed by varying the frequency. When this speed change occurs, it changes the pump operation according to mathematical relationships known as affinity laws.

Affinity laws for a speed change



Note

These affinity laws apply only to pump behavior, not to well or system performance.

$$50\text{ Hz capacity flow} = 50 \div 60 \times 60\text{ Hz flow} = 0.833 \times 60\text{ Hz flow}$$

$$50\text{ Hz head} = (50 \div 60)^2 \times 60\text{ Hz head} = 0.694 \times 60\text{ Hz head}$$

$$50\text{ Hz hp pump load} = (50 \div 60)^3 \times 60\text{ Hz hp load} = 0.579 \times 60\text{ Hz hp pump load}$$

$$50\text{ Hz motor hp load} = 50 \div 60 \times 60\text{ Hz hp rating} = 0.833 \times 60\text{ Hz hp rating}$$

$$50\text{ Hz motor voltage} = 50 \div 60 \times 60\text{ Hz motor volts} = 0.833 \times 60\text{ Hz motor volts}$$

$$50\text{ Hz motor amps} = \text{same as } 60\text{ Hz motor amps}$$

$$50\text{ Hz pump efficiency} = \text{same as } 60\text{ Hz pump efficiency}$$

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3.8 Power Cable Banding

3.8.1 Banding Material

Various banding materials and their applications are shown in [Table 3-16](#).

Table 3-16: Banding Materials and Applications

Type	Code	Application
steel	black paint	normal installation where corrosion is not a problem
stainless steel	green paint	most oilwell applications with mild corrosive environments
Monel	red paint	Used in installations suspected of being highly corrosive with high concentration of H ₂ S
 Potential Severity:Major Potential Loss: Assets Danger Hazard Category:Machinery equipment hand tools Certain concentrations of H₂S will cause embrittlement and subsequent failure of stainless steel bands.		
 Potential Severity:Major Potential Loss: Assets Danger Hazard Category:Electrical If non-armored cable is used, cable clamp saddles must be used to prevent cable damage by the metal bands.		

3.8.2 Calculating Cable Band Length

[Equation 3-4](#) can be used to determine the length of bands required for a given tubing size, cable size, and type of band, (that is, continuous or individual).

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Equation 3-4:

$$\text{band length} = 3.14 (d_{\text{tubing}} + 0.5d_{\text{cable}}) + K$$

where

Term	Definition
d_{tubing}	tubing diameter, in or mm
d_{cable}	cable diameter, in or mm
K	5 in. (127 mm) for individual manual bands, 2 in. (51 mm) for continuous banding material

3.8.3

Determining the Quantity of Cable Bands

Cable clamps are available in various lengths to accommodate different equipment. A list of banding tools and materials is provided in the *Parts, Tools and Materials* section in the Appendix. [Table 3-17](#) shows guidelines for determining the quantity of required cable bands.

Table 3-17: Determining the Quantity of Cable Bands

75 bands/1000 ft (305 m) of tubing	2 bands/joint of tubing
110 bands/1000 ft (305 m) of tubing	3 bands/joint of tubing (recommended for heavier lead-jacketed cable)

3.9

Bolt Flange and Housing Strengths

The maximum pull is based on the smaller bolt capacity because no threaded joints are used.



Example

For a 675/738-series intake to a 540-series protector, the upper bolt pattern is 738-series protector and the lower bolt pattern is 540-series protector. The first series (675) is the equipment to be connected, the second series (738) is the upper connection bolt pattern type and the third series (540) is the lower connection bolt type.

[Table 3-18](#) shows the maximum pull based on bolt pattern and housing capacities for submersible pumps. [Table 3-19](#) shows the maximum pull based on bolt pattern and housing capacities for protectors. [Table 3-20](#) shows the maximum

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pull based on bolt pattern and housing capacities for motors. Table 3-21 shows the maximum pull based on bolt pattern and housing capacities for downhole monitoring equipment.

The ratings shown in the following tables are for new equipment being pulled evenly and straight. All fasteners must be torqued to specification. The lack of proper torque application to the bolted flanges can cause premature failure of the fasteners in high temperatures and/or straight pull conditions. The torsional effect of pulling on an assembly that is being axially loaded at the same time as a pulling action is occurring (for example, a unit assembly stuck in a deviated well) can greatly reduce the values shown in the following tables. The effect of setting off jars or other types of recovery hardware can cause parting at less than the values shown.

Table 3-18: Maximum Pull For Submersible Pumps

Series	Bolt Pattern	Maximum Pull (lbf) [Monel Bolt] [92,000 lbf/in ² Bolt Stress]	Maximum Pull (lbf) [Housing Capacity]	
			Vee Thread	Buttress Thread
A-338 (carbon steel)	(8) 5/16 in	56,500	33,350	42,400
A-338 (Redalloy)	(8) 5/16 in	56,500	66,200	79,400
D-387	(8) 5/16 in	56,500	not applicable	46,160
D-400 (carbon steel)	(8) 5/16 in	56,500	48,100	57,700
D-400 (Redalloy)	(8) 5/16 in	56,500	90,100	108,075
E-450	(6) 7/16 in	83,000	66,900	80,300
G-513 (carbon steel)	(11) 7/16 in	152,000	93,400	112,000
G-513 (Redalloy)	(11) 7/16 in	152,000	175,000	209,775
G-540	(11) 7/16 in	152,000	93,400	112,000
S-538 (carbon steel)	(11) 7/16 in	152,000	55,500	76,500
S-538 (Redalloy)	(11) 7/16 in	152,000	104,000	142,000
H-562 (carbon steel)	(12) 7/16 in	166,000	61,750	74,000
H-562 (Redalloy)	(12) 7/16 in	166,000	115,700	138,600
I-650	(14) 1/2 in	253,000	93,250	111,750

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Series	Bolt Pattern	Maximum Pull (lbf) [Monel Bolt] [92,000 lbf/in ² Bolt Stress]	Maximum Pull (lbf) [Housing Capacity]	
			Vee Thread	Buttress Thread
J-675 (carbon steel)	(14) 1/2 in	253,000	78,200	93,750
J-675 (Redalloy)	(14) 1/2 in	253,000	146,500	175,600
L-725	(14) 1/2 in	253,000	not applicable	102,225

Table 3-19: Maximum Pull For Protectors

Series	Bolt Pattern	Maximum Pull (lbf) [Monel Bolt] [92,000 lbf/in ² Bolt Stress]	Maximum Pull (lbf) [Vee Thread]	
			Carbon Steel	Redalloy
325/375	(6) 3/8 in	61,000	32,500	60,870
375	(6) 3/8 in	61,000	43,750	81,900
400/450	(6) 3/8 in	61,000	48,100	90,100
540	(11) 7/16 in	152,000	93,400	174,900
562	(11) 7/16 in	152,000	61,750	115,700
738	(13) 9/16 in	297,000	84,500	158,250
950/738	(12) 5/8 in	339,000	142,250	266,400

Table 3-20: Maximum Pull For Motors

Series	Bolt Pattern	Maximum Pull (lbf) [Monel Bolt] [92,000 lbf/in ² Bolt Stress]	Maximum Pull (lbf) [Vee Thread]	
			Carbon Steel	Redalloy
375	(6) 3/8 in	61,000	43,750	81,900
456	(6) 7/16 in	83,000	66,900	125,300
456RX	(8) 3/8 in	81,336	66,900	
540	(11) 7/16 in	152,000	93,400	174,900
562	(11) 7/16 in	152,000	61,750	115,700
738	(13) 9/16 in	297,000	84,500	158,200

Table 3-21: Maximum Pull For Downhole Monitoring Equipment

Series	Bolt Pattern	Maximum Pull (lbf) [Monel Bolt] [92,000 lbf/in² Bolt Stress]	Maximum Pull (lbf) [Vee Thread]	
			Carbon Steel	Redalloy
PSI				
all series	(6) 3/8 in	61,000	43,750	
DMT/DMST				

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Series	Bolt Pattern	Maximum Pull (lbf) [Monel Bolt] [92,000 lbf/in ² Bolt Stress]	Maximum Pull (lbf) [Vee Thread]	
			Carbon Steel	Redalloy
all series	(6) 3/8 in	61,000	43,500	
PumpWatcher				
all series	(6) 3/8 in	61,500	43,500	
Surveyor				
all series	(6) 3/8 in	61,500	43,500	
MultiSensor				
all series	(6) 3/8 in	61,500	43,500	

3.10

Pump Stage Information

Table 3-22 provides pump stage information for shafts. Table 3-23 provides flow rates and pressures for pump stages. The Schlumberger WCP Catalog lists current pump and stage types. All information in Table 3-22 and Table 3-23 is based on operation at 3500 rpm.

Table 3-22: Pump Stage Information, Shafts

Stage Type	Alias	Shaft Diameter (in)		Shaft Area (in²)	Max BHP Rating for Pump Shaft
338 Series - OD 3.375 in					
A400	A10	5/8	0.625	0.302	94
AN550	A14	5/8	0.625	0.302	94
AN900	A25	5/8	0.625	0.302	94
A1200		5/8	0.625	0.302	94
AN1200	A30	5/8	0.625	0.302	94
A1500		11/16	0.688	0.371	125
AN1500	A45	11/16	0.686	0.363	125
400 Series - OD 4.00 in					
DN280		1/2	0.500	0.196	44
D400	D12	5/8	0.625	0.302	94
DN440		5/8	0.625	0.302	94
DN450	D13	5/8	0.625	0.302	94
D475N		5/8	0.625	0.302	94
DN525		5/8	0.625	0.302	94
DN610		5/8	0.625	0.302	94
DN675		5/8	0.625	0.302	94

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Stage Type	Alias	Shaft Diameter (in)		Shaft Area (in ²)	Max BHP Rating for Pump Shaft
D725N		5/8	0.625	0.302	94
DN800		5/8	0.625	0.302	94
D950	D26	11/16	0.686	0.363	125
DN1000	D28	11/16	0.686	0.363	125
DN1100		11/16	0.686	0.363	125
DN1300	D31	11/16	0.686	0.363	125
D1350	D40	11/16	0.686	0.363	125
D1400		11/16	0.686	0.363	125
D1400N		11/16	0.686	0.363	125
DN1750	D51	11/16	0.686	0.363	125
DN1800		11/16	0.686	0.363	125
DN2150		11/16	0.686	0.363	125
DN3000	D82	7/8	0.875	0.595	256
DN3100		7/8	0.875	0.595	256
DN4000		7/8	0.875	0.595	256
D4300N		7/8	0.875	0.595	256
D5800N**		7/8	0.875	0.595	256
540 Series - OD 5.125 in					
GN1300		7/8	0.875	0.595	256
GN1600		7/8	0.875	0.595	256
G2000		7/8	0.875	0.595	256
GN2000	G52	7/8	0.875	0.595	256
GN2100		7/8	0.875	0.595	256
GN2500	G62	7/8	0.875	0.595	256
G2700	G75	7/8	0.875	0.595	256
G3100		7/8	0.875	0.595	256
GN3100	G59	7/8	0.875	0.595	256
GN3200		7/8	0.875	0.595	256
GN4000	G110	1	0.995	0.785	375
GN5200	G150	1	0.995	0.785	375
G5600		1	0.995	0.785	375
GN5600	G180	1	0.995	0.785	375
GN6200		1	0.995	0.785	375
GN7000	G220	1	0.995	0.785	375
GN10000(ES)	G325	1	0.995	0.785	375

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Stage Type	Alias	Shaft Diameter (in)		Shaft Area (in ²)	Max BHP Rating for Pump Shaft
GN10000	G325	1-3/16	1.188	1.107	637
538 Series - OD 5.375 in					
SN2600		7/8	0.875	0.595	256
SN3600		7/8	0.875	0.595	256
S5000N		1	1.000	0.785	375
SN8500		1	0.995	0.785	375
562 Series to OD 5.625 in					
HN10000		1	0.995	0.785	375
HN13000	H350	1	0.995	0.785	375
HN13500		1	0.995	0.785	375
HN15000		1	0.995	0.785	375
HN20000		1-3/16	1.188	1.107	637
HN21000		1-3/16	1.188	1.107	637
H21500N		1-3/16	1.188	1.107	637
H28000N		1-3/16	1.188	1.107	637
HN28000		1-3/16	1.188	1.107	637
675 Series - OD 6.75 in					
JN7000	J200	1-3/16	1.187	1.107	637
JN7500	I250	1-3/16	1.187	1.107	637
JN10000	I300	1-3/16	1.187	1.107	637
JN12000	J350	1-3/16	1.187	1.107	637
JN16000	J400	1-3/16	1.187	1.107	637
JN21000	J600	1-3/16	1.187	1.107	637
725 Series - OD 7.250 in					
L16000N		1-3/16	1.187	1.107	637
L43000N		1-1/2	1.500	1.767	1000
862 Series - OD 8.625 in					
M520A		1-3/16	1.187	1.107	637
M520A		1-3/8	1.375	1.485	840
M675A		1-3/16	1.187	1.107	637
M675A		1-3/8	1.375	1.485	840
950 Series - OD 9.50 in					
N1050A		1-1/2	1.500	1.767	1000
1000 Series - OD 10 in					
N1400NA		1-3/4	1.750	2.405	1500

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Stage Type	Alias	Shaft Diameter (in)		Shaft Area (in²)	Max BHP Rating for Pump Shaft
1125 Series - OD 11.25 in					
P2000A	P1000	1-11/16	1.686	2.233	1125
P2500A		1-11/16	1.686	2.233	1125
1325 Series - OD 13.25 in					
R1500A		1-3/4	1.750	2.405	1125
1450 Series - OD 14.50 in					
T4000A	T2000	1-15/16	1.937	2.947	1125

Table 3-23: Pump Stage Information, Flow Rates and Pressures

Stage Type	Alias	Recommended Capacity		B.E.P.				Shutoff Head (ft)
		(bbl/d)	(m³d)	Flow (bbl/d)	Head (ft)	Power (hp)	Eff%	
338 Series - OD 3.375 in								
A400	A10	200 to 500	32 to 80	396	16.3	0.102	46.3	21.6
AN550	A14	400 to 700	64 to 111	605	13.6	0.126	47.9	22.3
AN900	A25	550 to 1100	87 to 175	829	13.0	0.147	54.0	21.8
A1200		800 to 1650	127 to 262	1152	13.4	0.220	52.9	
AN1200	A30	800 to 1650	127 to 262	1131	13.6	0.215	52.8	21.0
A1500		1000 to 2000	159 to 318	1534	11.5	0.290	44.3	
AN1500	A45	1000 to 2000	159 to 318	1534	11.4	0.292	44.2	17.5
400 Series - OD 4.00 in								
DN280		100 to 500	16 to 80	378	13.6	0.083	50.1	21.5
D400	D12	200 to 550	32 to 87	412	23.7	0.134	53.4	29.6
DN440		100 to 550	16 to 87	417	19.8	0.125	48.6	28.5
DN450	D13	250 to 550	40 to 87	480	16.9	0.119	50.2	26.6
D475N*		200 to 625	32 to 99	474	23.5	0.162	50.8	32.4

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Stage Type	Alias	Recommended Capacity		B.E.P.				Shutoff Head (ft)
		(bbl/d)	(m³/d)	Flow (bbl/d)	Head (ft)	Power (hp)	Eff%	
DN525		300 to 625	48 to 99	531	16.9	0.126	52.6	26.9
DN610		350 to 760	56 to 121	612	25.4	0.197	58.2	32.2
DN675		320 to 830	52 to 135	683	23.4	0.196	60.1	33.2
D725N		350 to 925	56 to 147	719	23.5	0.207	60.2	32.5
DN800		500 to 950	80 to 151	709	25.1	0.221	59.1	32.2
D950	D26	600 to 1150	95 to 183	906	21.6	0.233	61.5	28.6
DN1000	D28	600 to 1250	111 to 199	917	23.4	0.267	59.1	29.0
DN1100		600 to 1350	95 to 215	1120	22.5	0.287	64.4	30.6
DN1300	D31	960 to 1640	153 to 261	1283	20.5	0.311	62.3	28.3
D1350	D40	900 to 1850	143 to 294	1336	20.9	0.336	61.0	28.0
D1400		900 to 1850	143 to 294	1389	24.6	0.393	64.2	30.5
D1400N		960 to 1700	153 to 271	1372	20.9	0.334	63.1	30.5
DN1750	D51	1200 to 2050	191 to 325	1799	18.6	0.361	68.5	30.5
DN1800		1200 to 2400	191 to 382	1857	18.2	0.334	74.5	31.7
DN2150		1300 to 2600	207 to 413	2048	23.4	0.600	58.7	30.6
DN3000	D82	2100 to 3700	334 to 588	2753	19.5	0.623	63.4	29.8
DN3100		2100 to 3900	334 to 620	3119	18.7	0.632	67.9	29.1
DN4000		3400 to 5200	541 to 827	4371	12.1	0.638	61.2	26.2
D4300N		3500 to 5400	557 to 859	4246	21.3	0.95	70.2	27.8
D5800N	7/8	4400 to 7000	700 to 1114	5642	22.4	1.33	70.1	29.9
540 Series - OD 5.125 in								

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Stage Type	Alias	Recommended Capacity		B.E.P.				Shutoff Head (ft)
		(bbl/d)	(m³/d)	Flow (bbl/d)	Head (ft)	Power (hp)	Eff%	
GN1300		800 to 1800	127 to 286	1407	38.7	0.685	58.5	52.7
GN1600		1000 to 2150	159 to 342	1704	43.0	0.902	59.7	59.5
G2000		1650 to 2500	262 to 398	2106	36.4	0.949	59.6	52.2
GN2000	G52	1650 to 2500	262 to 398	2106	36.4	0.949	59.6	52.2
GN2100		1650 to 2500	262 to 398	2204	41.7	1.067	63.3	54.5
GN2500	G62	1800 to 3100	286 to 493	2398	42.6	1.213	61.9	54.0
G2700	G75	2000 to 3400	318 to 541	2653	43.3	1.290	65.6	52.0
G3100		2200 to 3800	350 to 604	2684	43.7	1.473	58.6	54.2
GN3100	G59	2200 to 3800	350 to 604	2684	43.7	1.473	58.6	54.2
GN3200		2200 to 4100	350 to 652	3444	37.5	1.460	65.2	57.0
GN4000	G110	3200 to 4800	509 to 764	4056	33.5	1.472	67.9	53.0
GN5200	G150	3900 to 6600	620 to 1049	5511	28.3	1.740	66.1	47.3
G5600		4000 to 7500	636 to 1192	6155	27.6	1.787	69.9	45.0
GN5600	G180	4000 to 7500	636 to 1192	6155	27.6	1.787	69.9	45.0
G6200N		4500 to 8000	716 to 1273	6673	31.9	2.212	70.9	53.0
GN7000	G220	5000 to 9000	795 to 1431	7391	31.6	2.588	66.5	48.3
GN10000 (ES)	G325	7000 to 12,000	1113 to 1908	9685	30.8	3.327	66.0	43.6
GN10000	G325	7000 to 12,000	1113 to 1908	9685	30.8	3.327	66.0	43.6
538 Series - OD 5.375 in								
SN2600		1600 to 3200	254 to 509	2581	46.7	1.306	68.1	63.5
SN3600		2400 to 4600	382 to 731	3507	50.1	1.849	70.0	64.0

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Stage Type	Alias	Recommended Capacity		B.E.P.				Shutoff Head (ft)
		(bbl/d)	(m ³ d)	Flow (bbl/d)	Head (ft)	Power (hp)	Eff%	
S5000N		4000 to 5400	636 to 859	4730	41.53	2.04	70.6	
SN8500		6000 to 11,000	954 to 1749	8662	36.5	3.268	71.2	61.8
562 Series - OD 5.625 in								
HN10000		5000 to 18,000	795 to 2862	13,007	41.6	6.310	63.2	
HN13000	H350	9200 to 16,400	1463 to 2608	12556	43.9	6.426	63.1	67.5
HN13500				13004	41.6	6.312	63.2	66.2
HN15000		12,000 to 18,000	1908 to 2862	15177	42.1	6.933	67.9	68.7
HN20000				19119	28.1	6.501	60.9	53.3
HN21000		17,500 to 24,000	2782 to 3816	18759	56.1	11.001	70.4	61.0
HN21500		17,500 to 24,000	2782 to 3816	18759	56.1	11.001	70.4	61.0
H28000N		28,000 to 36,000	4451 to 5723	28982	47.9	15.01	68.14	54.0
HN28000		28,000 to 36,000	4451 to 5723	28982	47.9	15.01	68.14	54.0
675 Series - OD 6.75 in								
JN7000	J200	4500 to 9000	715 to 1431	7227	85.5	6.313	72	110
JN7500	I250	6500 to 9000	1034 to 1431	7697	49.5	3.638	77.1	78.1
JN10000	I300	8250 to 14,500	1312 to 2306	10690	57.2	5.843	77.1	75.5
JN12000	J350	8000 to 18,500	1272 to 2940	12652	84.6	9.972	79	109.9
JN16000	J400	12,800 to 19,500	2035 to 3101	16004	65.2	10.667	72.0	92.5
JN21000	J600	16,000 to 25,000	2544 to 3975	19387	54.6	11.349	68.6	97.0
725 Series - OD 7.250 in								

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Stage Type	Alias	Recommended Capacity		B.E.P.				Shutoff Head (ft)
		(bbl/d)	(m³/d)	Flow (bbl/d)	Head (ft)	Power (hp)	Eff%	
L16000N		11,000 to 20,000	1749 to 3180	15669	94.0	14.13	76.8	118
L43000N		36,000 to 54,000	5723 to 8585	45818	70.9	33.25	72.0	107
862 Series - OD 8.625 in				galUS/min				
M520A		12,000 to 24,000	1908 to 3816	544	105	18.60	77.3	146
M520A		12,000 to 24,000	1908 to 3816	544	105	18.60	77.3	146
M675A		19,000 to 32,500	3021 to 5168	755	133	29.26	73.5	150
M675A		19,000 to 32,500	3021 to 5168	755	133	29.26	73.5	150
950 Series - OD 9.50 in								
N1050A		24,000 to 43,000	3816 to 6837	34,595	156	55.7	71.3	215
1000 Series - OD 10 in								
N1400NA		35,000 to 64,000	5565 to 10,180	49,017	204	93.31	78.9	
1125 Series - OD 11.25 in								
P2000A	P1000	53,600 to 95,800	8523 to 15,233	79,420	189	152.7	72.5	299
P2500A		54,486 to 96,000	8663 to 15,262	83,841	205	159.0	79.4	318
1325 Series - OD 13.25 in								
R1500A		69,556 to 15,9992	11,060 to 25,440	3489	277	339.9	71.8	430
1450 Series - OD 14.50 in								
T4000A	T2000	90,436 to 18,7827	14,380 to 29,866	4438	356	479.5	83.2	466

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**Note**

* The use of D475N pump stages assembled with diffusers manufactured using worn tooling may result in pump performance problems. As a temporary measure, Singapore Product Engineering has manufactured new tooling. Diffusers manufactured with the temporary tooling yield lower brake horsepower problems than those manufactured with ratings of worn tooling. Diffusers made from temporary tooling may have brake horsepowers very close to API test limits. Diffusers manufactured from the temporary tooling are identified by a small slot on one of the diffuser ribs. [Figure 3-4](#) shows the slot location in a diffuser.



Figure 3-4: Slot Location in D475N Diffuser.

**Note**

** Use the correct impeller (p/n 1340306) and correct diffuser (p/n 1305820) when assembling D5800N pumps. Ensure quality control procedures are in place to prevent using the wrong impellers and diffusers when assembling pumps. Always review the Bill of Material to ensure that the proper parts are used to build the pump ordered.

3.11 Total Dynamic Head

One stage in a pump will move fluid a certain distance at a given flow rate. Numerous stages are required to pump fluid longer distances or at higher pressures. This total distance is expressed in head feet and is called total dynamic head (TDH). The TDH consists of three components

- vertical lift
- friction loss
- tubing discharge pressure.

3.11.1 Vertical Lift

Vertical lift is the distance from the producing fluid level to the surface. The fluid level above the pump can be obtained by either a 'fluid shot' or, if a downhole monitoring instrument is installed, the static bottomhole pressure can be converted to head feet using [Equation 3-5](#).

Equation 3-5:

$$\text{head feet} = \frac{p_{ws} \times 2.31}{sg}$$

where

Term	Definition
p_{ws}	static bottomhole pressure, lbf/in ²
sg	fluid specific gravity

3.11.2 Friction Loss

Additional pressure is required from the pump to overcome the friction losses caused by the fluid flowing inside the tubing. [Figure 3-5](#) shows the friction loss (water) due to flow in various pipe diameters.

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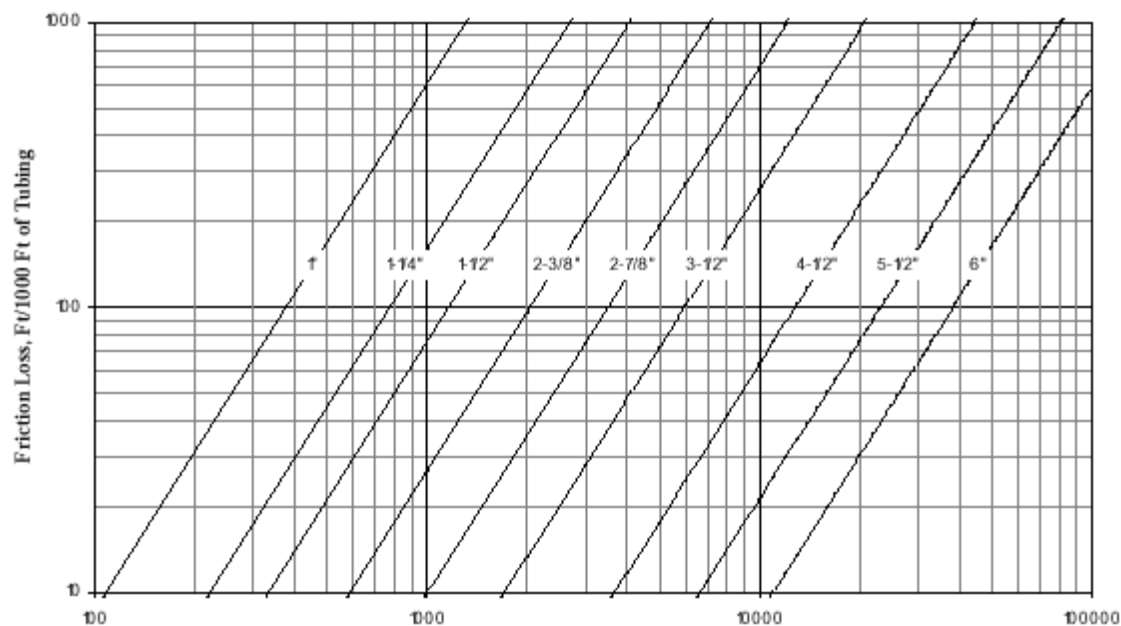


Figure 3-5: Friction loss due to flow (water)

3.11.3 Tubing Discharge Head

The pressure in the surface equipment must also be overcome by the pump. Since TDH is expressed in hd.ft, the system pressure can be converted from lbf/in² to hd.ft using the [Equation 3-6](#).

Equation 3-6:

$$\text{head feet} = \frac{\text{lbf/in}^2 \times 2.31}{\text{average specific gravity of the fluid}}$$

3.11.4 Total Dynamic Head Calculation

The TDH at a specific flow rate must be calculated to determine the number of stages required in a pump. The TDH can be calculated using [Equation 3-7](#).

Equation 3-7:

$$\text{TDH} = \text{vertical lift} + \text{friction loss} + \text{tubing discharge head}$$

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**Example****Calculating Total Dynamic Head**

Well data for the total dynamic head example is shown in [Table 3-24](#)

Table 3-24: Well Data For The Total Dynamic Head Example

Well Data	
flow rate	2200 bbl/d
producing fluid level	4200 ft
pump setting depth	4800 ft
average fluid specific gravity	0.93
tubing size	2.875 in
casing size	7 in
tubing discharge pressure	200 psi

1. Calculate vertical lift.

$$\text{vertical lift} = \text{pump setting depth} - \text{fluid level above the pump}$$

$$\text{vertical lift} = 4800 \text{ ft} - 600 \text{ ft} = 4200 \text{ ft}$$

2. Calculate friction loss.

$$\text{friction loss} = \text{friction loss per 1000 ft} \times \text{tubing length}$$

$$\text{friction loss} = (36/1000) \times 4800 \text{ ft} = 173 \text{ ft}$$

3. Calculate tubing discharge head.

$$\text{tubing discharge head} = \frac{\text{system pressure (lbf/in}^2\text{)} \times 2.31}{\text{average specific gravity of the fluid}}$$

$$\text{tubing discharge head} = (200 \times 2.31) \div 0.93 = 497 \text{ ft}$$

4. Calculate total dynamic head.

$$\text{Using the equation for calculating total dynamic head, } TDH = 4200 \text{ ft} + 173 \text{ ft} + 497 \text{ ft} = 4870 \text{ ft}$$

The total dynamic head in this example is 4870 ft.

3.11.5

Calculating Head Loss Across a Motor

Head loss across the motor should be considered when sizing a submersible pump. The head loss across the motor can be a significant factor. Heavy-wall casing can improve a desired velocity past the motor, but in many cases may

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cause adverse effects because of the minimal clearance between the motor and casing. This decreased clearance can cause a substantial pressure drop that the submersible pump has to overcome. Equation 3-8 can be used to calculate head loss across a motor.

Equation 3-8:

$$\text{head loss} = \frac{1.4 (\text{pump rate} \div 1000)^2}{(\text{casing ID} + \text{motor OD})^2 (\text{casing ID} - \text{motor OD})^3}$$

where

Term	Definition
head loss	feet of head loss per foot of motor length
pump rate	bbl/d

Add the total head loss required for the pump to overcome to the calculated total dynamic head and calculate the new number of stages and motor horsepower required. If the new pump requires a higher horsepower and longer motor, the procedure for calculating head loss across the motor should be repeated.

3.11.6

Fluid Time-To-Surface after Startup

The following procedure can be used to estimate the time required for fluid to reach the surface after startup. Two examples are provided. Both examples assume the pump is operating at B.E.P. (in the middle of the performance curve).

- In Example 1, the fluid level is unknown so the tubing length is used as a worst-case scenario.
- In Example 2, the static bottomhole pressure (p_{ws}) is known so the fluid level can be calculated. This will give a more accurate estimation of fluid time-to-surface.

Example 1 — Fluid level is unknown

Table 3-25: Data for Time-To-Surface after Startup Example 1

Data for Example 1	
pump	DN1750
tubulars	2.875-in tubing, 7300 ft

1. Calculate the tubing fill.

$$\text{tubing fill} = \text{bbl}/1000 \text{ ft} \times \text{tubing length}/1000$$

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$$\text{tubing fill} = 5.793 \times 7.3 = 42.3 \text{ bbl}$$

2. Calculate the pump flow rate (bbl/min).

$$\text{pump flow rate (bbl/min)} = \text{pump flow rate (bbl/d)} \div 1440$$

$$\text{pump flow rate (bbl/min)} = 1750 \div 1440 = 1.215 \text{ bbl/min}$$

3. Calculate the time to pump the fluid to the surface.

$$\text{fluid time to surface} = \text{tubing fill (bbl)} \div \text{pump flow rate (bbl/min)}$$

$$\text{fluid time to surface} = 42.3 \div 1.215 = 34.8 \text{ min}$$

Example 2 — Static BHP is known

Table 3-26: Data for Time-To-Surface after Startup Example 2

Data for Example 2	
pump	DN1750
tubulars	2.875-in tubing, 7300 ft
P_{ws}	800 lbf/in ²
sg	0.95

1. Calculate the tubing fill.

$$\text{tubing fill} = \text{bbl/1000 ft} \times \left[\frac{\text{tubing length} - \left(\frac{P_{ws} \times 2.31}{sg} \right)}{1000} \right]$$

$$\text{tubing fill} = 5.793 \times \left[\frac{7300 - \left(\frac{800 \times 2.31}{0.95} \right)}{1000} \right]$$

$$\text{tubing fill} = 31 \text{ bbl}$$

2. Calculate the pump flow rate (bbl/min).

$$\text{pump flow rate (bbl/min)} = \text{pump flow rate (bbl/d)} \div 1440$$

$$\text{pump flow rate (bbl/min)} = 1750 \div 1440 = 1.215 \text{ bbl/min}$$

3. Calculate the time to pump the fluid to the surface.

$$\text{fluid time to surface} = \text{tubing fill (bbl)} \div \text{pump flow rate (bbl/min)}$$

$$\text{fluid time to surface} = 31 \div 1.215 = 25.5 \text{ min}$$

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3.12

Calculation of Protector Thrust Loads

The following example shows how to determine the thrust load transferred to the protector thrust bearing when floater constructed pumps are used.

Thrust Load Calculation Example

Pump, Protector and Well Data	
pump	154—stage SN2600 FL-CT
protector	540—series LSL
BHST	200 degF
fluid sg	0.95

1. Calculate the total shutoff head.

$$\text{total shutoff head} = \text{shutoff head per stage} \times \text{number of stages}$$

$$\text{total shutoff head} = 63.5 \text{ ft} \times 154$$

$$\text{total shutoff head} = 9779 \text{ ft}$$



Note

Shutoff head per stage data is provided in the *Pump Stage Information* section.

2. Calculate the shutoff pressure.

$$\text{shutoff pressure} = (\text{total shutoff head} \times \text{fluid specific gravity}) \div 2.31$$

$$\text{shutoff pressure} = (9779 \text{ ft} \times 0.95) \div 2.31$$

$$\text{shutoff pressure} = 4022 \text{ lbf/in}^2$$



Note

The calculated shutoff pressure should be cross referenced with the values shown in [Table 3-27](#) to ensure pump housing burst pressures are not exceeded.

Table 3-27: Pump Housing Burst Limits

Allowable Burst Pressure Limits	
Pump Series	Pressure Limit (lbf/in ²)
338	5000
387	4000

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Allowable Burst Pressure Limits	
Pump Series	Pressure Limit (lbf/in ²)
400	5000
540	5000
538	5000
562	3000
650	3300
675	2550
725	2000
862	2000
950	2300
1000	3000
1125	3000
 Note These pressure limits can be increased 20% when buttress threads are used.	

3. Calculate the shutoff thrust.

$$\text{shutoff thrust} = \text{shutoff pressure} \times \text{shaft area}$$

$$\text{shutoff thrust} = 4022 \times 0.595 \text{ in}^2$$

$$\text{shutoff thrust} = 2393 \text{ lbf}$$

A 540 NTB bearing operating at 200 degF is rated to 2550 lbf of thrust and would be suitable for this application. (Table 3-28 shows protector thrust bearing ratings.)

If compression pumps are used, contact an application engineer to obtain the downthrust per stage type.

Table 3-28: Protector Thrust Bearing Ratings

Protector Series	Maximum Load (lbf at 60 Hz)	Temperature Rating for Maximum Load (degF)
325 NTB	900	250
325 KTB	2010	350
325 HL	2500	350
375 STB	995	190

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Protector Series	Maximum Load (lbf at 60 Hz)	Temperature Rating for Maximum Load (degF)
400 STB	1600	190
400 NTB	1600	250
400 KTB	3900	350
400 HL	8600	350
540 STB	2550	190
540 NTB	2550	250
540 KTB	9500	350
540 HL	12000	350
562 HL	17500	350
738 STD	3500	190
738 NTB	3500	250
738 KTB	8500	350
738 HL	20,000	350
950 STB	7850	190
950 HL	44,000	350
STB = REDA Babbitt Bearing NTB = REDA Bronze Bearing KTB = KMC Bronze Bearing HL = Glacier Tilt Pad		

3.13

Equipment Dimensions

Figure 3-6 shows a submersible pumping unit installed inside either casing, liner, or openhole and the related dimensions that are used to calculate the clearance available on the flat cable side.

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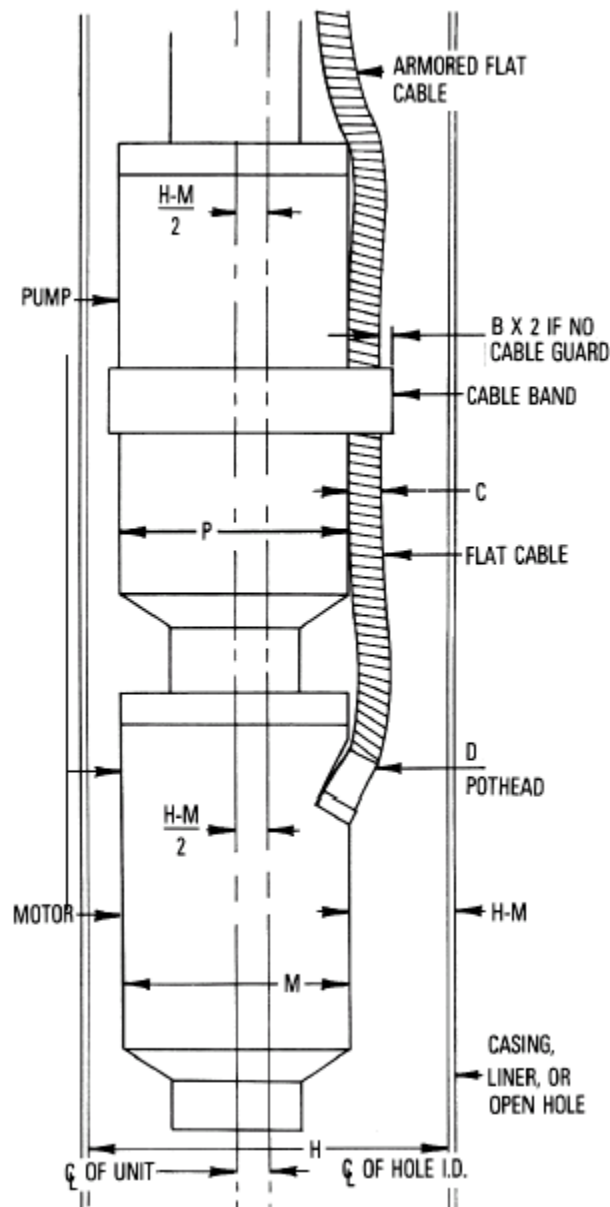


Figure 3-6: Downhole equipment dimensions

Equation 3-9 can be used to find the maximum available clearance or the maximum OD of the total unit.

Equation 3-9:

$$\text{pump clearance on flat cable side} = \frac{H}{2} - \left\{ \frac{P}{2} + C + B - \left(\frac{H - M}{2} \right) \right\}$$

where

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Term	Definition
H	ID of wellbore
M	largest diameter at any point on the motor
P	pump or protector diameter
C	armored flat cable thickness
B	band thickness

**Note**

End-view clearance drawings for specific configurations can be obtained from engineering.

[Figure 3-6](#) and assume that the protector used will be the same size as the pump and that no flat guards will be used. (Flat guards are not used in tight applications.) If flat guards are used, the flat guard dimension must be added. The protector OD must be used in place of the pump OD in the equation if a larger OD protector than pump OD is used.

**Note**

On all applications where the clearance is very close, a full gauge tool (slightly larger than equipment OD) should be run prior to the installation of equipment. This is to ensure that there are no burrs or tight spots in the wellbore. This operation will prevent many expensive mistakes when installing a submersible system in tight or heavy-wall casing.

Drift Diameter

The drift diameter is the minimum ID that the supplier (mill) must provide for API standard casing, drill pipe and tubing. The pipe ID should never be less than drift diameter if there is no casing damage and in many cases will be larger than drift. This is why it is important to run a gauge before installing equipment. Drift diameter standards (API) for casing are shown in [Table 3-29](#).

Table 3-29: Drift Diameter Standards for Casing

Casing Size	Drift Diameter
up to 8-5/8 in OD inclusive	1/8 in smaller than tabulated ID
over 8-5/8 in OD to 13-3/8 in OD inclusive	5/32 in smaller than tabulated ID
over 13-3/8 in OD	3/16 in smaller than tabulated ID

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eg**Example**

Well and Equipment Data for Downhole Equipment Clearance Example	
casing	5.5 in API 8rd, 14 lbm/ft, 5.012-in ID, 4.887 in drift diameter
pump	400 series (4.00-in OD)
motor	456 series (4.56-in OD)
band thickness	0.030 in
cable	3 kV high-temperature flat cable, 6 AWG conductor (0.505 in by 1.20 in) (See note below.)

**Note**

The thickness of flat cable can vary from a maximum to a minimum due to manufacturing tolerances. The maximum thickness for 6 AWG 3kV high-temperature flat cable is 0.55 in. The maximum thickness value should be used when calculating the maximum dimension of equipment. Cable dimensions and tolerances are provided in the *Cable Specifications* section in the *Cables* chapter.

Example Calculations

Using the [Equation 3-9](#) —

$$\frac{5.012}{2} - \left\{ \frac{4.00}{2} + 0.55 + 0.030 - \left(\frac{5.012 - 4.56}{2} \right) \right\}$$

$$\text{pump clearance on flat cable side} = 2.506 - 2 - 0.55 - 0.030 + 0.226$$

$$\text{pump clearance on flat cable side} = 0.152 \text{ in}$$

This maximum dimension (clearance) takes into consideration the entire unit with the centerline dimensions of both pump and motor. This indicates the unit will go into the wellbore.

The actual clearance at the pump and protector is $(5.012\text{--in pipe ID}) - (4.00\text{--in pump OD} + 0.55\text{--in flat cable} + .030\text{--in band}) = 0.432 \text{ in}$.

If flat guards are used, the OD is increased by approximately 0.16 in.

The OD at flanges and the motor head is reduced by passing flat through a slot cut out in equipment.

**Note**

On any installation where clearances are critical, check the placement of lockplates to ensure they do not increase the OD of the equipment. Equipment should also be calipered to obtain onsite dimensions.

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3.14

Oil Selection for Motors and Protectors

General guidelines for the selection of oil used in motors, protectors and any oil-filled motor accessories are provided in [Table 3-30](#). The proper oil will provide optimum lubrication, cooling, and motor efficiency for various downhole operating conditions.

Oil selection is based on the internal temperature of the operating motor. The same oil should be used in all motors, protectors and any oil-filled motor accessories in the same string of equipment. Recommended oil applications are shown in [Table 3-30](#).

Table 3-30: Recommended Oil Applications for Motors and Protectors

Motor Internal Temperature		Oil Selection	Application Notes	Containers	Part Numbers
< 270 degF	< 130 degC	Reda No. 3	cool well applications	5 gal	0979062
				15 gal	0979054
				55 gal	0976373
< 290 degF	< 145 degC	Reda No. 2	The oil will require heating during servicing when the surface temperature is below 32 degF (0 degC).	5 gal	0556548
				15 gal	0556530
				55 gal	0556498
< 360 degF	< 180 degC	Reda No. 5	Recommended for general applications and all applications that use 562-series motors. Should also be used as a minimum for Hotline applications.	5 gal	7004856
				15 gal	7004864
				55 gal	7004872
< 360 degF	> 180 degC	Reda No. 6	Under special requirements, this oil may be used down to 300 degF (150 degC) motor internal temperature. Motor efficiency may be up to 10% lower at startup due to high oil viscosity at bottomhole temperatures down to 150 degF (65 degC).	5 gal	7007248
				15 gal	7007230
				55 gal	7005556
		Reda No. 7	Reda No. 7 oil is used in the High-Temperature Extended Runlife Protector Reda No. 5 oil is used in the Abrasion-Resistant Extended Runlife Protector and the Corrosion-Resistant Extended Runlife Protector	5 gal	100021260
				15 gal	100021263
				55 gal	100021264

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3.14.1 Ordering

Because the selected oil is application dependant, the type of oil must be specified when placing an order. If the type of oil is not specified when the order is placed, motors and protectors will be shipped with the oil shown in [Table 3-31](#).

Table 3-31: Standard Oil Types

Motors	Protectors	Oil Type
562-series and Hotline	540-series and larger	Reda No. 5
all others	all others	Reda No. 2

3.14.2 Oil Compatibility

All Schlumberger motor or protector oils are compatible and may be mixed without chemical degradation. The resultant viscosity will be approximately equal to the proportional volume mix. Oils obtained from other sources (not Schlumberger approved) may not be compatible with the internal components of submersible equipment.

It is permissible to drain and refill motors, protectors and any oil-filled motor accessories with a different Schlumberger oil. Standard factory filling instructions and field service procedures apply.

For cold climate installations, the oil and/or the equipment may require heating to facilitate servicing.

3.14.3 Other considerations

Numerous commercial issues (for example, customer requirements for specific oils) may influence the selection of motor or protector oils. Contact InTouchSupport.com to confirm that a specific oil is appropriate for use in Schlumberger motors, protectors and any oil-filled motor accessories.

3.15 Dielectric Strength Test for Oil

The following procedure can be used to test the dielectric strength of motor oil and transformer oil. The applicable document is ASTM D877.

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3.15.1 Required Equipment

Equipment required to perform a dielectric test is

- Appropriate personal protective equipment.
- 1000 ml (1 qt) clean, dry containers
- Biddle/Foster Model OTS 60 auto, (p/n 99955-7) tester. The Biddle/Foster Model OTS 60 is the preferred equipment. If unavailable, a Hipotronics Model CC 60A, (p/n 56411-2) may be used. (A timer must be used with the Hipotronics tester.)
- Reda No. 2 oil



Note

All testing equipment must be kept clean and dry at all times.

3.15.2 Equipment Preparation

1. Check the test cup to ensure the gap setting between the electrodes is correct. A 0.100 in (+0.0005 in) gauge is supplied with the equipment.
2. Rinse the test cup with SC 140 solvent (p/n 73540-7) to remove any particulates.
3. Discard the solvent in the cup. Refill the cup with clean Reda No. 2 oil. Check the dielectric strength of the oil. A minimum dielectric strength of 25 kV must be achieved before the cup is ready for use. This may require that the cup be rinsed and filled several times with Reda No. 2 oil.
- 4.



Note

This procedure prepares the Biddle tester.

- a. Open the test chamber cover by turning the interlock knob $\frac{1}{4}$ turn counterclockwise.
- b. Place the test cup in the chamber so that the ends of the electrode supports rest on the cradle terminals at the top of the support horn.
- c. Close the chamber door and turn the interlock knob clockwise.
- d. Turn the power supply on and set the selector switch to **Manual**.

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- e. Press the **Start Test** button.
- f. Press and hold the **Raise** button until either the sample breaks down or the display shows 60 kV.
- g. Upon test completion, press the **Trip** button to reset the tester.

5.



Note

This procedure prepares the Hipotronics tester.

- a. Place the test cup in the holder.
- b. Close the door and turn the power switch on.
- c. Set the rate of rise to 3000 V/sec.
- d. Press the **Start** button.
- e. Upon test completion, press the **Reset** button.

3.15.3

Testing the Oil Sample

1. Carefully fill a 1000 ml (or 1 qt) container with the oil to be tested. If testing motor oil, sample the oil from the bottom of the motor.
2. Fill the clean test cup approximately one-third full with sample oil.
3. Gently swirl the cup so that the electrodes and surface area of the cup are well coated with oil. Discard the oil and refill the cup with sample oil. The sample of oil is now ready for testing.
- 4.



Note

This procedure uses the Biddle tester.

- a. Open the test chamber cover by turning the interlock knob $\frac{1}{4}$ turn counterclockwise.
- b. Place the test cup in the chamber so that the ends of the electrode supports rest on the cradle terminals at the top of the support horn.
- c. Close the chamber door and turn the interlock knob clockwise.
- d. Turn on the power.
- e. Set the selector switch to the ASTM D 877 test method.

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- f. Press the **Start Test** button.
- g. The Biddle tester will automatically finish the test procedure. After a two-minute and 20-second initial delay, the sample will be tested five times with one minute intervals between tests. After the tests are complete, the average of the five breakdown voltages will be displayed. Record this average value. Press the **Recall** button and record the five individual test values.

5.

**Note**

This procedure uses the Hipotronics tester.

- a. Set the rate of rise to 3000 V/sec.
- b. Place the test cup in the holder.
- c. Close the door and turn the power switch on.
- d. Using a timer, begin an initial time delay of 2 to 3 minutes.
- e. Press the **Start** button and record the dielectric strength.
- f. Press the **Reset** button and wait one minute before retesting the sample.
- g. A total of five tests values, at one-minute intervals, should be performed. Record the value from each test.

3.15.4

Interpretation of Results

ASTM D 877 requires that the five values be evaluated statistically in the following manner.

(high value - low value) multiplied by 3 = x

If **x** is lower than the second lowest value, then the average of the five values is statistically acceptable. The average of the values will be recorded as the final test result.

**Example**

The five recorded values are 32, 30, 28, 34 and 31.

$(34 - 28) \text{ multiplied by } 3 = 18$

Since 18 is less than the second lowest value (30), these five results are statistically acceptable. The average breakdown strength of 31 kV is the final test result.

If x is greater than the second lowest value, the five values are not statistically acceptable. For example, the five recorded values are 14, 25, 26, 27 and 27.

$(27 - 14) \text{ multiplied by } 3 = 39$

Since 39 is greater than the second lowest value (25), the five values are not statistically acceptable. At this point the oil in the test cup would be discarded. The cup would then be refilled with sample oil, and tested using the procedure appropriate to the tester. Assume the second test generated the following values; 27, 33, 35, 25 and 27. The average of the ten values (26.6 kV) would be recorded as the final test result.

3.15.5

Acceptance/Rejection Criteria for Motors

- The average dielectric strength of the oil must be greater than or equal to 15 kV.
- The visual appearance of the oil is indicative of the cleanliness, or quality workmanship, of the finished motor. The presence of metal particles and excessive discoloration and clarity of the oil will be considered as cause for rejection.
- The oil sample must pass both the dielectric strength test and visual appearance for the motor to be acceptable.

3.16

Dimensions and Capacities of Tubing and Casing

Dimensions and capacities of tubing and casing are provided in various field handbooks such as the Schlumberger *Field Data Handbook*. (The electronic version is called the *i-Handbook*.) The *Field Data Handbook* is available via <http://www.hub.slb.com/index.cfm?id=id159>. The *i-Handbook* is available via <http://www.hub.slb.com/index.cfm?id=id1384022>.

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3.17

Equipment Serial Numbering System

Equipment serial numbers consist of ten-digit alphanumeric characters. [Table 3-32](#) shows equipment serial number convention. [Table 3-33](#) shows product code nomenclature. [Table 3-34](#) shows product series nomenclature. [Table 3-35](#) shows manufacture designation.

Table 3-32: Equipment Serial Number Convention

Character	Designation
1	Product code
2	Series or type
3 or 3 and 4 ¹	Manufacture designation (1 or 2 characters)
4 or 5	Last digit of year manufactured (0 thru 9)
5 or 6	Month of manufacture (A through L)
6-10 or 7-11 ²	Last five or six digits of work order or a sequential number
11-15 or 12-16 ³	Running sequence number

¹ Effective August 2007, if manufacture designation contains two characters, the position of the characters following will shift accordingly, as indicated in the table.

² downhole monitoring equipment has six digits in this position

³ applicable only for WB2 and SD1

Table 3-33: Product Code Nomenclature

Equipment	New	Remanufactured	Used
motor	1	A	L
pump	2	B	M
protector	3	D	N
gas separator	4	E	O
switchboard	6	G	S
flat cable extension	7	H	T
cable	C	I	U
downhole monitoring equipment	P	J	V
intake	R	K	W
advanced gas handling system	8	Y	Z
oil and water separator	9	C	P
hydraulic release lower connector	H	not applicable	not applicable
ProMotor	G	7	Q

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Table 3-34: Product Series Nomenclature

Equipment	Designation	Series/Type
motor	A	375
	C	456
	D	540
	E	738
	H	562
	I	461 (117mm RUS)
pump	A	338
	B	338/375
	D	387
	E	387/375
	F	400
	H	405 (103mm RUS)
	1	400/540
	I	500, 540, 538/540
	2	540/738
	J	562, 562/540
	6	562/738
	N	675
	O	675/540
	P	675/738
	Q	725
	R	725/540, 725/738
	S	862
	T	862/738
	4	862/738/540
	U	950
	V	950/738
	Z	975/738
	W	1000/1000, 1000/738
	8	1125/950, 1125/738, 1325/950, 1450
	9	538

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Equipment	Designation	Series/Type
Poseidon	C	387/387, 387/375
	D	400
	E	400/375
	G	540
	9	538
	J	562, 562/540
	T	862/738
protector	A	325/375, 375/325/375
	B	375, 375/540
	C	400/456
	D	400/540
	E	456
	F	540
	G	738
	H	738/540
	I	405 (103mm RUS)
	J	738/562
	K	387/387, 387/456, 387/375
	L	950/738
	M	738/862
	R	540/562
	S	950
	T	400/400, 456/400
	U	400/375
Hotline Integrated Motor	H	562
gas separator	A	338/375, 387/375, 387
	B	400, 400/540, 400/375
	C	405 (103mm RUS)
	D	540
	F	675/540, 675/738
	G	562/540
	H	538/538, 538/540
oil and water separator	A	450, 456
	B	540
	C	738
	D	538/540
switchboard	0	all types

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Equipment	Designation	Series/Type
flat cable extension	3	375
	4	456
	5	540/562
	7	738
cable	0	all types
downhole monitoring equipment	A	375 PSI
	C	456 PSI
	D	540 PSI
	E	738 PSI
	H	562 PSI
	J	DMT MACH I
	K	DMT MACH II
	L	DMST multichannel (intake and discharge pressure, intake and motor temperature)
	M	DMST Multichannel (intake and discharge pressure, intake, winding and discharge temperature)
	S	Surveyor
intake	W	WB2
	V	SD1
	A	338/375, 387 (ARZ)
	B	400 (ARZ)
	C	400/540 (ARZ)
	D	540 (ARZ)
	E	538/540 (ARZ)
	F	675/738, 675/540 (ARZ)
	G	562/540 (ARZ)
	H	562/400
	I	405 (103mm RUS)
	J	540/400

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Equipment	Designation	Series/Type
bolt-on discharge head	J	338
	K	387
	L	400
	M	540
	N	562
	P	675
	Q	725
	R	862
advanced gas handling system	A	338
	C	387
	D	400
	E	405 (103mm RUS)
	G	540, 675/540
	S	538
	H	562/540, 675/562/540, 725/562/540, 862/562/540
lower connector	A	562 series
	B	540 series
ProMotor	C	456 series
	H	562 series

Table 3-35: Manufacture Designation

Designation	Location
A	Casper
B	Bartlesville
C	Garden Grove
D	Congo
E	Edmonton/Nisku
F	Field Trial
G	Gems/Tunisia
H	Argentina (Rincon Service Center)
I	China
J	Reda Free Zone (Egypt)
K	Lawrence Technology
L	Shreveport
M	Midland
N	Ecuador

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Designation	Location
O	Argentina (Comodoro Rivadavia Service Center)
P	Colombia
Q	Peru
R	Ras Al Khaimah
S	Singapore
T	Indonesia (Jakarta Service Center)
U	Indonesia (Pekanbaru Service Center)
V	Venezuela (Las Morochas Service Center)
W	Venezuela (Maturin Service Center)
X	Caltex
Y	Scotland
Z	PWR Mission/Brazil
AA	Macaé, Brazil
AB	La Yuca, Colombia
AC	Gabon
AD	Tripoli, Libya
AE	Okha, Russia
AF	Nizhnevartovsk, Russia
2	Kuwait
3	Yemen
4	Chad
5	Russia (Tyumen Product Center)
6	Russia (Muravlenko ART Center)
7	Venezuela (Barinas)
8	Russia (Nefteyugansk ART Center)
9	XPC (Xiqing Product Center)

3.18

Stator Codes

On the motor housing there is a field code number stenciled on the stator that provides specific information regarding the internal construction of the stator. This reference information is intended for use by field service and production personnel to verify the stator construction and winding type.

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Example

An example stenciled code is shown in [Figure 3-7](#)

F	-	90	-	10	-	135	-	4
1		2		3		4		5

Figure 3-7: Example stenciled field code on a stator

- The first position (alpha) indicates the housing type. The possible types are
 - C — carbon steel
 - F — ferric steel
- The second position (alpha or numeric) indicates the stator type as indicated by the part number description. Examples are 62 type, 87 type, 90 type, 91 type, 92 type and 93 type. This can also be two letters such as HX, PX, SX, MX.
- The third position (numeric) indicates the number of rotor positions inside the stator.
- The fourth position (numeric) indicates the voltage the part number specifies for the stator (three digits). Examples: 1350 V = 135, 900 V = 90.
- The fifth position (numeric) indicates the lamination type. The standard closed slot lamination is a 4.

3.19

Unit Conversion Tables

Conversion tables for units are provided in various field handbooks such as the Schlumberger *Field Data Handbook*. (The electronic version is called the *i-Handbook*.) The *Field Data Handbook* is available via <http://www.hub.slb.com/index.cfm?id=id159>. The *i-Handbook* is available via <http://www.hub.slb.com/index.cfm?id=id1384022>.

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3.20 Frequently Used Equations

3.20.1 Converting API Gravity (°API) and Specific Gravity (sg)

$$^{\circ}\text{API} = \frac{141.5}{\text{oil sg}} - 131.5$$

$$\text{oil sg} = \frac{141.5}{131.5 + ^{\circ}\text{API}}$$

3.20.2 Kilowatts, Kilovolt-Amperes, Horsepower and Amperes Calculations

Kilowatts Single Phase

$$\text{kilowatts} = \frac{\text{volts} \times \text{amps} \times \text{PF}}{1000}$$

Kilowatts Three Phase

$$\text{kilowatts} = \frac{1.73 \times \text{volts} \times \text{amps} \times \text{PF}}{1000}$$

Kilovolt-Amperes Single Phase

$$\text{kilovolt-ampere} = \frac{\text{volts} \times \text{amps}}{1000}$$

Kilovolt-Amperes Three Phase

$$\text{kilovolt-ampere} = \frac{1.73 \times \text{volts} \times \text{amps}}{1000}$$

Horsepower (Output) Single Phase

$$\text{horsepower (output)} = \frac{\text{volts} \times \text{amps} \times \text{EFF} \times \text{PF}}{746}$$

Horsepower (Output) Three Phase

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$$\text{horsepower (output)} = \frac{1.73 \times \text{volts} \times \text{amps} \times \text{EFF} \times \text{PF}}{746}$$

Amperes (when Horsepower is Known) Single Phase

$$\text{amperes} = \frac{\text{hp} \times 746}{\text{volts} \times \text{EFF} \times \text{PF}}$$

Amperes (when Horsepower is Known) Three Phase

$$\text{amperes} = \frac{\text{hp} \times 746}{1.73 \times \text{volts} \times \text{EFF} \times \text{PF}}$$

Amperes (when Kilowatts is Known) Single Phase

$$\text{amperes} = \frac{\text{kilowatts} \times 1000}{\text{volts} \times \text{PF}}$$

Amperes (when Kilowatts is Known) Three Phase

$$\text{amperes} = \frac{\text{kilowatts} \times 1000}{1.73 \times \text{volts} \times \text{PF}}$$



Note

Efficiency (EFF) is expressed as a decimal.



Note

Power Factor (PF) is expressed as a decimal.

3.20.3

Short-Circuit Current Calculation

$$I_{sc} = \frac{kVA_t \times 100}{\%Z_t \times 1.73 \times kV_{sec}}$$

where

Term	Definition
I_{sc}	short circuit current, ampere
kVA_t	transformer kilovolt-ampere

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$\%Z_t$	transformer impedance
kV_{sec}	maximum secondary voltage, volt

**Example****Short-Circuit Current Calculation Example**

Equipment Data	
transformer	225 kVA
impedance	3.7%
secondary voltage (maximum)	1600 V

$$I_{sc} = \frac{kVA_t \times 100}{\%Z_t \times 1.73 \times kV_{sec}}$$

$$I_{sc} = (225 \times 100) \div (3.7 \times 1.73 \times 1.6)$$

$$I_{sc} = 2197 \text{ A}$$

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4

Wellsite Procedures

**Note**

Refer to [Artificial Lift - REDA Maximus Reference Page ID 4057025](#) for REDA Maximus pertinent information.

4.1

Transport and Handling

**Note**

Always ship SPS equipment with the correct shipping caps in place, do not ship pieces of equipment bolted together.

4.1.1

Equipment Handling

Equipment transport and handling are important factors that influence the success of submergible operations. The nature of artificial lift equipment requires special handling procedures; procedures not normally used when handling and transporting many other types of pumping systems. The physical length of downhole components relative to their diameters can be large, especially for motors and pumps. For example, a 30–ft motor, 5.5 inches in diameter will deflect considerably if not transported and lifted properly. During the manufacturing process, there is considerable effort made to ensure that the stator is straight to within ± 0.002 in/ft. Allowing the motor to deflect excessively can result in catastrophic failure. For this reason, all downhole components should be transported in steel shipping boxes.

There are three areas of consideration; transportation, handling and placement. Although placement may not seem important, it has surprising influence on the entire process. Proper placement the first time eliminates the need for additional handling and reduces the risk of equipment damage.

4.1.2

Use of Steel Shipping Boxes

Handling Steel Shipping Boxes

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Steel shipping boxes should be used to transport motors, protectors, pumps, intakes, gas separators and pressure sensing instruments. These boxes are available in various lengths from 5.5 ft to 36.5 ft and various diameters from 6 to 12 inches (nominal diameter of component being shipped). One end of each box is colored to identify the head end of the equipment; the end that points towards the wellhead. Although metal shipping boxes provide some protection during handling and transport, they do not eliminate the need for special care and attention for downhole equipment.

Before equipment leaves the manufacturing facility, it is packed in the metal shipping box with rubber inserts to help absorb vibration and shock. At no time should the boxes be dragged, dropped or bounced to facilitate movement. In general, all equipment, (either in or out of shipping boxes) must be handled using a spreader bar or a lifting clamp. Never use a single winch line attached to the center of a component to lift it. Component damage, and serious personal injury can occur. Always use a spreader bar with chains or slings attached $\frac{1}{4}$ of the distance from each end. The sole use of a chain bridle or sling is not recommended.

A fork lift can be used but only when properly distributed support can be given to the box. For long boxes, the use of a fork lift may not be possible. In cases where a crane is being used, ensure that the box is supported securely in at least two places, approximately $\frac{1}{4}$ of the distance from each end.

Never use the shipping box handles for support. Handles are only designed to open, close and secure the box. They are not designed to support the weight of the box and the equipment inside. The use of a crane with a spreader bar and secured slings threaded through the handles and underneath the box is allowed only if the positions of the handles correspond with the $\frac{1}{4}$ length requirement. By threading through the handles and underneath the box, the handles are not supporting the weight of the box and the equipment inside it.

Shipping box lifting equipment and dimensions are shown in [Figure 4-1](#).

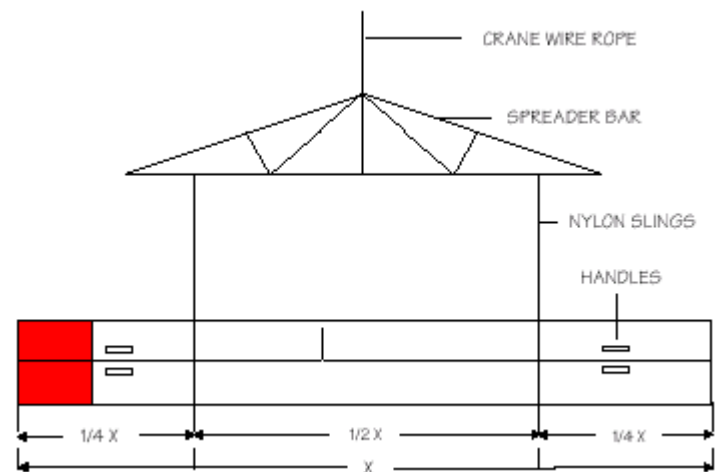


Figure 4-1: Lifting shipping boxes

Opening and Closing Steel Shipping Boxes

Steel shipping boxes should be opened and closed with a special tool. The tool has been developed to aid field service personnel in opening and closing the cover of a steel shipping box where mechanized lifting equipment is unavailable. The tool is suitable for all boxes including the H box design. When opening or closing a long and heavy box, two field service personnel located at both ends of the box should open and close the box using a coordinated sequence of movement.

The tool consists of two parts; the head (p/n 2010739) and extension bar (p/n 2010915).



Warning

Potential Severity: Serious
 Potential Loss: Personnel
 Hazard Category: Machinery equipment hand tools

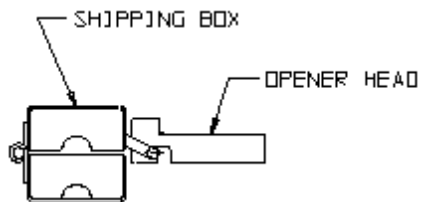
If a substitute extension bar is used, the front-end diameter must not be greater than 0.75 in. The extension bar and opener head must maintain a loose engagement to avoid personal injury during use.

Steel Shipping Box Opening Procedure

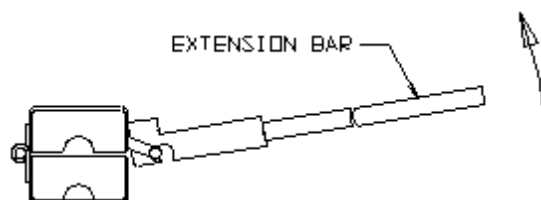
1. Engage the opener head on the shipping box handle.

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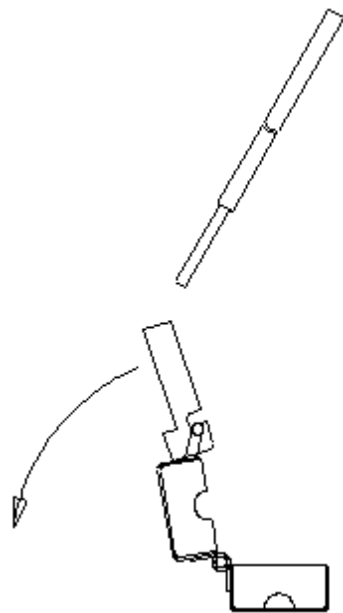
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2. Insert the extension bar into the opener head.



3. Swing the shipping box cover open. As the shipping box cover is swinging open, the extension bar will disengage from the opener head. Do not drop the extension bar.

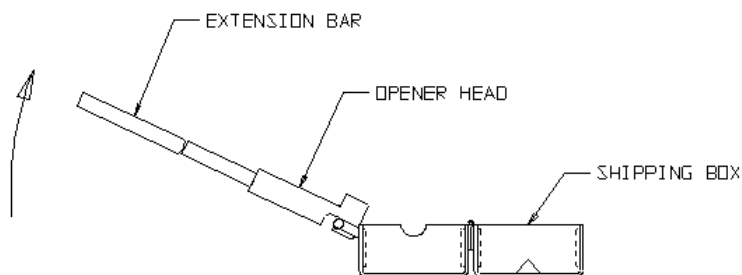


Steel Shipping Box Closing Procedure

1. Engage the opener head on the shipping box handle.

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2. Insert the extension bar into the opener head.
3. As the shipping box cover is swinging over its center of gravity, the cover will close and disengage the opener head that will then drop off of the extension bar. Do not to drop the extension bar.

Placing Equipment in Steel Shipping Boxes

Equipment should be placed in a steel shipping box with the top of equipment at the painted end of the box. Ensure shipping caps are installed before the equipment is placed in the box.

When loading equipment into a shipping box, be sure to use the correct two-piece rubber shipping blocks. Two blocks are used per box. Position the bottom half of each block at a distance equal to $\frac{1}{4}$ the component length from each end of the box.



Example

If a component is 28 ft long, the blocks would be placed 7 ft from the ends of each unit, leaving 14 ft of the component unsupported in the center. Do not use an additional block to support the middle of the unit.

Place the unit inside the box on the shipping blocks using an overhead hoist and a spreader bar. Center the unit as it is being lowered, leaving equal gap distances at each end between the unit and the shipping box. Use two, cut-to-length wooden blocks to fill these gaps between the unit and the shipping box at each end. Leave 0.125-inch clearance for expansion. Place the top half of the shipping blocks in the box directly above the bottom half. Carefully secure the box.

Shipping Box Transport

Shipping boxes should be placed on 4-inch by 4-inch wooden planks, 3 feet from each end and in the middle when being placed on a truck or on the ground. If the boxes will be stacked, planks should be placed between each layer. No more than one box should be hoisted in a single lift. When placed on a truck,

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no part of the box should extend beyond the end of the truck. Shipping boxes should be securely strapped to the deck of the truck at the location of the planks; shipping boxes should never move while being transported.

4.1.3 Cable

Handling

Power cable is typically shipped on large reels. The purpose of the reel is to allow easy cable feed into and out of the well and to protect the cable during transport and handling.

Transport

Cable should always be transported on cable reels. In the event that a power cable on a reel is being transported with the spliced flat cable extension on it, independently secure the extension and the power cable. A crane and spreader bar are recommended to lift cable reels. A forklift may be used. If a forklift is used, lift the reel through the spokes from the side, never against the cable. Reels should be secured to the truck when transported. Never allow the reel of cable to roll against objects that might crush or damage the cable or reel.

Placement

A section of heavy-walled pipe should be inserted through the reel to serve as an axle. Lift with a spreader bar attached to the axle. A short spacer should be placed between the reel and the stand on each side. (A thread protector, collar, Vic clamp or hammer union wing are some examples.) The cable must feed off to the wellhead from the top of the reel. Never handle a reel of cable by making a sling of wire line or chains that touch the cable. Protect the cable from being smashed or cut. The cable reel should be placed 75 to 100 feet from the wellhead and in the rig operator's direct line of sight.



Note

In an offshore environment this may not be feasible, however the spool should be placed as far back as possible.

On semi-submersible platforms or other floating rigs, two wheel and fork assemblies are used. This configuration is used to prevent the cable from being stretched and damaged when the rig compensators are in operation.

Cable reel lifting equipment is shown in [Figure 4-2](#).

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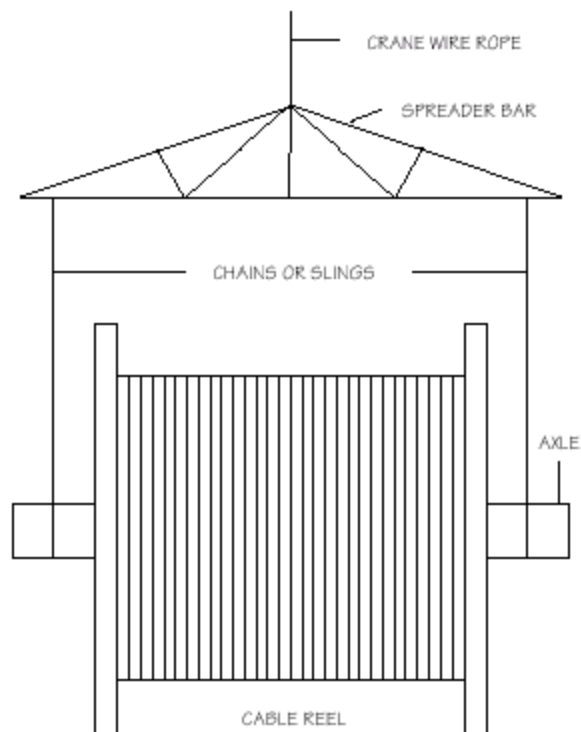


Figure 4-2: Cable reel lifting

A clear path should be provided for the cable to prevent it from being hooked or damaged during installation. Wooden planks should be used to keep the cable off the ground, preventing it from picking up any debris which may contaminate the well.

Cable Sheave Assembly

The cable sheave should be hung in the derrick, inline between the cable reel and wellhead and aligned with the wellhead to avoid any rubbing against tubing support.



Warning

Potential Severity: Serious
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

Ensure the cable sheave is attached securely with a chain and has a cable or chain safety backup.

The sheave should be no more than 30 feet above the ground in order to permit flexibility and avoid shock against the cable during running and pulling operations. During installation, the sheave should be supported approximately 10 feet above the ground to feed the flat cable extension (FCE) and splice

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through. After the FCE splice has been supported by cable bands, the sheave can be raised to the operational height of 30 feet. This procedure reduces stress on the pothead connection and FCE splice.

Cable running equipment and dimensions are shown in [Figure 4-3](#). The cable must be supported at point B during installation and pulling operations. The cable weight of distance A to B should not exceed 150 lbm.

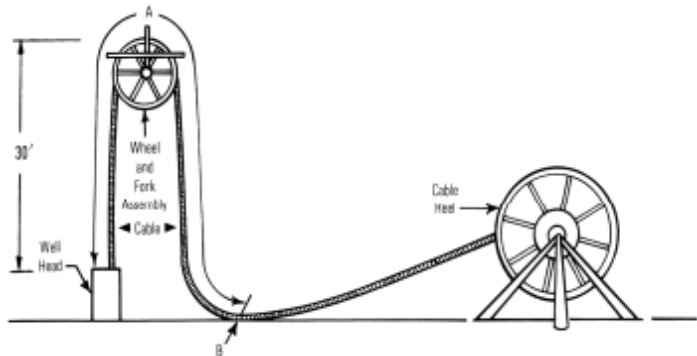


Figure 4-3: Cable running or pulling

4.1.4 Switchboard

The switchboard should be located approximately 50 to 100 feet from the well and close to the transformer bank or generator.



Potential Severity: Major
Potential Loss: Assets
Hazard Category: Electrical

The switchboard case should be grounded to the well casing.



Potential Severity: Major
Potential Loss: Assets
Hazard Category: Electrical

Extreme care should be used in handling the switchboard. When uncrating, pry bars should be used instead of a hammer. The interior of the switchboard must be kept free of moisture. Always examine internal components for possible shipping damage.

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4.1.5 Transformer or Generator

Transformers are provided with lifting hooks for loading, unloading and handling. A spreader bar and cables should be used to maintain the transformer in a vertical position to avoid damage. Always ensure the transformer cover is bolted in place before moving the transformer.

The transformer should be located approximately 50 to 100 feet from the well and close to the switchboard. The location of power lines, the access road and working clearance around the well should be considered when locating the surface electrical equipment.

4.1.6 Junction Box

The junction box is used for connecting the power cable and venting gas (in case gas has migrated up the power cable) between the well and switchboard. It should be in line with the switchboard if possible, and at a distance of approximately 25 feet (minimum 15 feet) from the well to permit a working area for the service rig. Junction box mounting and dimensions are shown in [Figure 4-4](#).



Potential Severity:	Catastrophic
Potential Loss:	Personnel
Hazard Category:	Electrical

Ensure the power is off before working at the junction box.

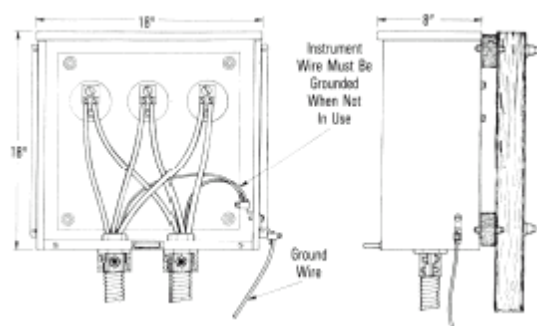


Figure 4-4: Junction box mounting

Junction box placement is shown in [Figure 4-5](#).

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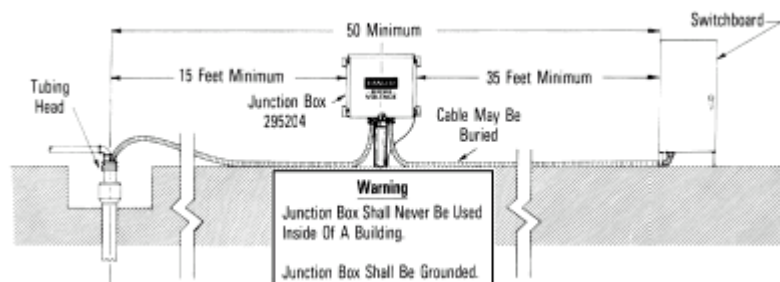


Figure 4-5: Junction box placement

4.1.7 Damaged Equipment

If a unit is suspected of being damaged or mistreated while being handled or transported, notify the responsible field service representative. The equipment will be inspected and repaired if necessary before it is placed in service. Pulling operations are time consuming, expensive and can be avoided.

4.2 Job Preparation

Prior to a new installation, check the condition of the casing. Run a gauge tool slightly larger than the OD of the equipment to ensure adequate clearance. If there was difficulty in pulling the pump from an existing well, investigate and determine the cause prior to running a new unit. Cable damage, missing bands, or tight spots may warrant the use of a scraper to clean the wellbore and eliminate potential running problems.

4.2.1 Checking Equipment

Check all materials and equipment to determine if everything necessary for installation is on location. This should be done immediately upon arrival at the wellsite. Check against the shipping order, quote and workover schedule in the following sequence.



Note

Refer to [InTouch content ID 5029387](#) for ESP Equipment Check lists

1. Remove box covers and record all information directly from nameplates on motors, pumps, gas separator, protector, flat cable extension, well cable, and switchboard.
2. Ensure that all other completion accessories are available at the wellsite and are functional.

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3. Ensure that backup equipment (when required) is at the wellsite and is functional.
4. Ensure that all installation consumables are available at the wellsite and are serviceable.
5. Ensure that all installation tooling is available at the wellsite and is functional.
6. Note the type of tubing support and confirm the proper size of casing, tubing, and cable pack-off.
7. Check the switchboard for proper fuses, potential transformer setup and CT ratio.
8. Ensure each coupling to verify that it is the correct size.
9. In the case of tandem pumps, determine which pump needs to be picked up and installed first.

**Note**

When installing tandem pumps ensure that the pump with a high-strength shaft (designated HSS on the nameplate) is installed closest to the motor.

10. Ensure the flat cable is the proper length and series for the motor.
11. Check the design of pump (head) against setting depth, rate, etc.
12. Check the power bank transformers for correct primary and secondary voltage ratings and kVA rating.

4.2.2

Preventing Galling of Flange Bolts

Installing Monel bolts in carbon steel heads can result in galling of the bolt and flange bolt hole threads. This can cause the bolt to bind in the bolt hole. The different hardnesses of the dissimilar materials are the root cause of this type of incident. If this happens, it may not be possible to back the bolt out to chase the threads with a thread tap. Often, the bolt breaks off in the flange. Application of an anti-seize compound (p/n 0849737) to Monel bolts will greatly reduce the incidence of galling and bolt breakage when bolts are threaded into carbon steel heads. Anti-seize compound should also be applied to carbon steel or Monel bolts being threaded into Redalloy heads.

**Note**

Some submersible pump components such as VGSA gas separators and ARS/DRS/GRS gas separators include Redalloy heads in the standard product. The Redalloy material may not be noted on the nameplate.

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The use of an air-operated tool to run bolts into dissimilar head material will greatly increase the likelihood of galling, even if an anti-seize compound is used.

If a bolt coated with anti-seize compound begins to bind while being threaded into the bolt hole, stop threading the bolt. Binding can occur with even miniscule debris in the bolt hole. Failure to stop threading the bolt into the bolt hole and continuing to force the bolt into the bolt hole will almost certainly result in bolt breakage and lost rig time. Slowly remove the bolt from the bolt hole and carefully chase the flange bolt hole threads with a thread tap. Clean any debris from the threads of the bolt. Reapply the anti-seize compound and carefully try again.

4.2.3 Motor Testing

Phase-To-Phase Measurements

The integrity of the motor's insulation should be tested prior to installation. Phase-to-phase measurements can be taken most accurately using a digital multimeter. Record the measurements and compare them to the data shown in the motor resistance tables in the *Technical and Engineering Information* chapter. Measurements must be balanced within 2% and the value cannot be less or more than 10% of the published value. Percentage unbalance can be calculated using [Equation 4-1](#).

Equation 4-1:

$$\% \text{ Unbalance} = \frac{\text{Maximum Deviation from Average Resistance}}{\text{Average Resistance}} \times 100$$

If the motor is an upper tandem or center tandem, a wye or star connection is required to take the measurements. Remove the lower shipping cap and interconnect all three male terminals using three jumper wires. The phase-to-phase measurements can now be taken from the top of the motor



Danger

Potential Severity:	Major
Potential Loss:	Assets
Hazard Category:	Electrical

Always perform a final visual inspection of the motor terminals for proper spacing in the slot and check for phase resistance after assembling the UMB. Refer to [Figure 4-6](#). For further details refer to [InTouch Content ID 3769116](#).

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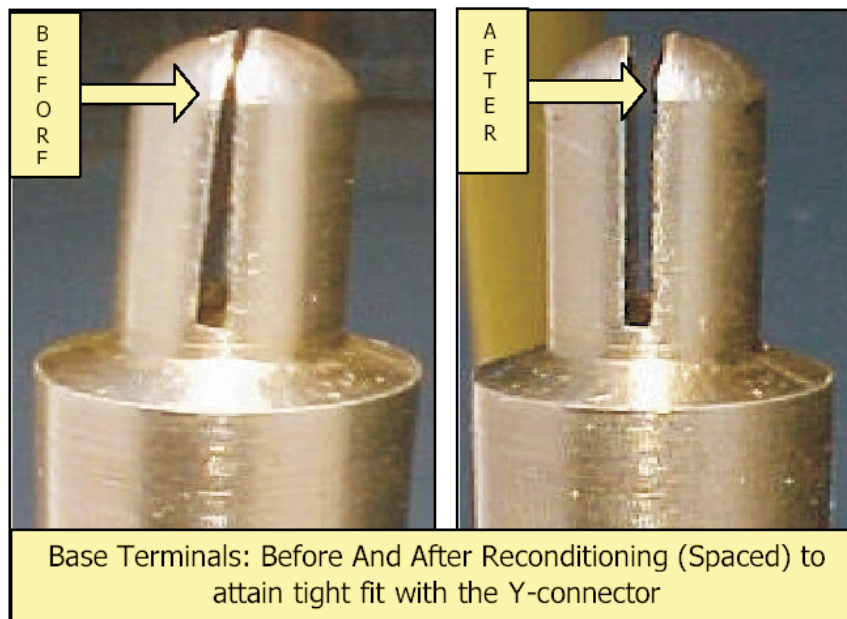


Figure 4-6: Base Terminals

Phase-To-Ground Measurements

The megger is used to check the motor insulation resistance to ground. The meter test lead marked **earth** is clipped onto a bolt head where a good ground can be established. The other probe lead is connected to each phase individually. Each phase on a new motor must measure a minimum of 1500 megohms to ground.



Danger

Potential Severity: Major
Potential Loss: Personnel
Hazard Category: Fire flammable

Never use a megger on the rig floor. A spark generated by a megger can ignite gasses emitted from the wellhead. Complete this test while the motor is in the shipping box. Additional megger tests can be completed safely at the cable reel once the cable is connected to the motor.



Danger

Potential Severity: Major
Potential Loss: Personnel
Hazard Category: Electrical

Always discharge each winding after the megger test. The residual high-voltage charge can shock personnel and/or damage other meters.

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Upper tandem motors require removal of the pothole cover. Each lead to ground is megged individually prior to being picked up.

Center tandem motors should be checked prior to being picked up. Each lead must be checked unless the center tandem motor has a universal base installed. If a center tandem motor has a **U** base or a PSI monitoring tool installed, megging a single lead is megging all three coils since the **Y** connection is closed.

Lower tandem motors are megger tested in the same manner as single-section motors and center tandem motors with the **Y** closed.

4.2.4 Cable Testing

Schlumberger cable is thoroughly tested before shipping. Ensure that no damage has occurred during transit.



Caution

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Any cable damage should be located and repaired prior to installation. Never take a chance that unrepaired damage will have no effect.

Cable testing can be performed using the following procedure.

1. Remove the shipping cap on the flat cable extension. Conductors at opposite ends must be clean, dry and clear of the cable reel and each other.
2. Using an ohmmeter, connect the ground lead to the cable armor and the probe to each individual lead. Record the measurements.
3. Using the ground lead and the probe lead, check phase A-B, B-C and A-C. Record the measurements. New cable should read near infinity both phase-to-ground and phase-to-phase. Depending on weather and humidity, measurements of only 1500 megohms are obtainable and considered normal.
4. Discharge each of the legs to ground after testing. A long piece of cable can hold a very high charge that can take over a minute to discharge.

4.2.5 Pothead Testing

The pothead should be pressure tested. [Table 4-1](#) shows part numbers for various pothead testing equipment.

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Table 4-1: Pothead Test Equipment Part Numbers

Motor Series	Pressure Test Cap	Gauge	Air Pump
375	305128	912923	557496
456	305136	912923	557496
540/562	305144	912923	557496
738	305151	912923	557496

Attach the pressure gauge and hand-operated air pump to the test cap. Submerge the pothead in water and apply pressure by operating the air pump. Test pressures and times are shown in [Table 4-2](#). If the pothead leaks, remove the test cap and discard the O-ring. Use a new O-ring and test again. If it continues to leak, reject the flat cable extension.

Table 4-2: Pothead Test Pressures and Times

Time	Pressure	Cable
5 min	0 psi ¹	used cable only
5 min	5 psi	new and used cable
5 min	40 psi	new and used cable

¹ Used cable that has just been pulled from a well may have gas trapped inside the armor and cable jacket. The gas may try to equalize through the pothead thus damaging the seal. This migration will cause a pressure increase that can be detected on the test gauge.

4.2.6

PSI Testing

Pressure Sensing Instrument (PSI) units should be tested prior to installation. Using a 1000 V megger, check the resistance of the unit to ground. If the unit is delatched, the measurement should be infinity. If the unit is latched (using a Triplet meter on the x1000 scale), the measurement should be between 35,000 ohms and 65,000 ohms depending on the ambient temperature. Cycling through the latch/delatch procedure and recording pressure and temperature measurements will ensure the integrity of the downhole unit.



Caution

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Do not use test equipment with output voltages greater than 1500 volts DC to test PSI units. This can damage the unit.

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4.2.7 Surveyor Testing

Surveyor field installation and testing procedures are provided in the [Surveyor Operation and Maintenance Manual InTouch Content ID 3473375](#).

4.2.8 Installation of Downhole Monitoring Equipment (DME)

1. Before removing the shipping cap and gasket ensure the unit is in a clean area.



Caution

Potential Severity: Light
Potential Loss: Assets, Personnel
Hazard Category: Electrical

Be careful do not to pull the three-phase terminal ring. Connected is a signal wire leading down to the instrument. Pulling the terminal ring may result in damage to the wire.



Caution

Potential Severity: Light
Potential Loss: Assets, Personnel
Hazard Category: Electrical

Use extreme caution when installing or removing adapters from the instrument. Pull the adapter straight away from its mating piece (along the tool longitudinal axis) if the adapter moves sideways and impacts a connector, it could damage the connector or wire.

2. The DME and the motor adapter can be connected to the motor and filled with oil from the sensor's base drain and fill valve. Once the motor cable is connected, the complete system (cable, motor, and DME) may be Megger tested from the free cable end.
3. Install a lifting clamp and lower the DME into the well until the clamp is about 18-in. from the top of the worktable.

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**Note**

Dedicated DME lifting clamps are not available. Also, there are no alignment pins, so possible damage may occur if tandem motor jacks are not used. Until DME lifting clamps are available, any lifting clamp with a smaller ID than the OD of the DME adapter will work (i.e., 400 series pump clamp or 375 series motor clamp). See [Figure 4-7](#).



Figure 4-7: Tandem Motor (lower end). *This photo shows the damage that can occur when tandem motor jacks are not used. If this equipment is damaged due to misalignment, failure of the DME will result due to well fluid entry.*

4. Tandem Motor jack assembly (P/N 100541677, 375/456/540/562 series motors or 44842, 738 series motors) is now slid into position on the worktable. Ensure that both hydraulic jacks are in their retracted position. Lower the DME down until the clamp rests in the cradle on top of the jacks.
5. Remove the shipping cap and slowly pump oil down the center of tool until full and all air is expelled.

**Note**

Use extreme caution to ensure no moisture or debris enters this area.

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6. Pick up the motor using procedure in section [4.5: Motor Installation](#) of this manual and remove the lower shipping cap.
7. Select the correct O-ring(s) to be used from ALFORM-D Connection Table. Remove the old O-ring. Lubricate the new O-ring with Dow Corning 111 (p/n 100386898) and carefully install to its correct position.

**Note**

Do not use metal tool when removing the O-ring from the flange groove. This can be the reason of equipment short runlife due to damage on the sealing surfaces.

A second backup O-ring is NOT required above the groove O-ring on 540 series and 562 series protector bases, between tandem motors and on PSI units.

-
8. Inspect the terminal pins and insulators to ensure they are not bent or broken. Have the rig operator lower the motor until it is approximately 4 - 6 inches above the DME. Then adjust both jacks simultaneously to slowly raise the DME and engage terminal pins into their respective motor terminals properly.

**Note**

Use caution not to damage the terminal insulators or O-rings.

-
9. Install the bolts and lock washers and torque to the required specifications listed in the connection table.
 10. Lower the motor(s) and DME into the well. Loosen the top shipping cap and place a lock washer between the cap and the motor head flange to provide a vent.
 11. Pull the motor(s) and DME all the way out of the well and fill the DME and motor using filling procedures in section [4.5.2: Motor Oiling Procedure \(Single-Section\)](#) of this manual, pump oil through both DME and motor(s).
 12. Phase-to-phase resistance checks should be made after each connection, keeping in mind the resistances are additive and should always be balanced.
 13. In the event that all motors cannot be pulled from the well because of limited (derrick) crown room, fill from the motor head of the lowest possible motor.
 14. Repeat the procedure in section [4.5.3: Tandem Motor Connection](#) for each CT motor that is added.

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4.2.9 Cold Weather Installations

Cold weather is defined as ambient temperatures less than 20 degF (-7 degC). Special procedures must be applied in cold-weather installations to prevent operating failures.

4.2.9.1 Equipment

Equipment considerations, best practices and the use of external heat are discussed in the following sections.

Equipment Considerations

- Ambient temperatures less than -45 degF (-43 degC) are considered too cold to properly assemble and install ESP equipment.
- Material selection should be based on bottomhole operating temperatures rather than surface temperatures. Lubricating oils and elastomers should be selected to ensure that optimum performance and run life is achieved for given well conditions. The installation procedures should ensure the selected equipment is installed and serviced properly.
- When installing equipment in cold temperatures (20 degF [-7 degC] or less), the equipment should be warmed before assembly to maximize run life. Oil in the motor and protector can become so viscous in very cold temperatures it cannot be pumped.
- When equipment is assembled, it must be serviced with oil before lowering into a well filled with kill fluids to warm it. The equipment will become contaminated with well fluids if not properly serviced.
- Working with O-rings in cold temperatures can be difficult. Problems have been observed with Aflas O-rings hanging up between flanges when assembling units in cold temperatures. Certain materials in O-rings become so rigid they can easily be pinched during assembly or break as they are installed. Minimum working temperatures for O-rings are shown in [Table 4-3](#).

Table 4-3: O-ring Minimum Working Temperatures

O-Ring Material	Minimum Working Temperature
Aflas	15 degF (-9 degC)
HSN	HSN -55 degF (-48 degC)

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Best Practices

Warming the equipment, oil and O-rings before assembly is strongly recommended. Transporting the equipment to the wellsite in a warm condition is recommended. (Overnight storage in a heated building will sufficiently warm the equipment.)

If the equipment temperature is less than 20 degF (-7 degC) and the equipment is submerged in well fluid, O-rings and mechanical seals can leak, contaminating the motor and protector with well fluids. If the equipment is placed in a well (full of fluid) to warm the equipment before the protector is completely serviced, the equipment will be contaminated. Temperature limitations for protector elastomers are shown in [Table 4-4](#).

Table 4-4: Protector Elastomer Minimum Working Temperatures

Protector Elastomer Material	Minimum Working Temperature
Aflas	15 degF (-9 degC)
HSN	HSN -55 degF (-48 degC)

Using External Heat



Danger

Potential Severity: Major
 Potential Loss: Personnel
 Hazard Category: Temperature

Always use caution when handling heated equipment.

The workover rig steam boiler (if available) or a portable steamer or warm-air blower should be used to provide heat. The steam or heat lines should be run to the equipment shipping boxes a minimum of 45 minutes prior to installation. Depending on the ambient temperature, heat source and heat capacity, longer times may be necessary. The warm air or steam line nozzle should be inserted into or connected to the shipping box container leaving container lids closed until the equipment is removed for installation. Enough heat should be applied to the equipment to ensure it stays warm while being serviced and installed. The object is to thoroughly heat the equipment so when the oil is applied during the top-off procedure, it will not be affected by the cold temperatures.

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**Tip**

The recommended warming temperature should range from 60 degF to 150 degF (16 degC to 66 degC). The best working temperature is 90 degF (32 degC).

Another warming method is to enclose the rig floor with a wind-proof material, usually canvas with a metal frame support. Warm air can be routed into the enclosure from the rig steam boiler or other source during the installation process to keep the equipment warm. This not only keeps the equipment warm, but also shields the equipment from undesirable elements (for example, blowing snow) during critical phases of the installation process.

Oil should be kept as warm as possible. Keep the oil can in a warm environment. The oil can should not be removed until it is time to service the equipment. Return the oil can to the warm environment to maintain the oil temperature.

O-rings can be warmed by putting them inside coverall pockets for an extended time (warming with body heat). An alternative is to keep them inside a warm vehicle until they are ready to be used. This practice should be implemented unless the materials in the O-rings can withstand cold temperatures at the surface as well as downhole temperatures, or until a better heating method is found. The ideal temperature should range from 60 degF to 150 degF (16 degC to 66 degC). Minimum working temperatures for O-rings are shown in [Table 4-3](#).

External Heating Sources Not Available

The use of external heating sources to warm downhole equipment is recommended, especially when performance-based contracts or alliances where runlife is part of the risk and reward criteria are involved. When ambient temperatures are less than the minimum requirements for fill oils and cable elastomers, alternate elastomer materials and blended oils have been substituted to permit servicing in cold temperatures. All efforts should be made to warm the equipment; material substitutions should be made only as a last resort.

**Note**

The following sections provide guidelines to heat the equipment by lowering the equipment into the well. Note that some form of equipment warm-up is still discussed, but the limitations are noted and those limitations should not be exceeded at any time.

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Shortcuts and the use of improper methods at time of installation can cause premature equipment failure. Sufficiently warming the equipment before installation is still important. Failure to do so can significantly reduce run life.

Guidelines for ESP installations when no external heating sources are available

- The fill oil should be warmed before the topping-off procedure is started. Oil temperature limitations are shown in [Table 4-5](#).

Table 4-5: Oil Temperature Limitations

Oil Type	Specific Gravity	Pour Point (Maximum)		Fill Temperature Without Heat ¹	
Reda No. 2	0.86 to 0.87	10 degF	-12.2 degC	20 degF	-6 degC
Reda No. 3	0.827	-90 degF	-68 degC	5 degF	-15 degC
Reda No. 5	0.834 to 0.844	-49 degF	-45 degC	23 degF	-5 degC
Reda No. 6	0.85 to 0.87	-44 degF	-42 degC	55 degF	13 degC

¹ Fill temperature is based on oil at 700 centistokes, the temperature at which the oil can be pumped.

- O-rings should be warmed by putting them inside coverall pockets for an extended time (warming with body heat) or by keeping them inside a warm environment until they are ready to be used.
- Some cables will require a warm-up procedure. [Table 4-6](#) shows temperature limits for cable materials.

Table 4-6: Cable Temperature Limitations

Material	Minimum Low Temperature (New)		Minimum Low Temperature (Used)	
Redalene insulation	-5 degF	-20 degC	5 degF	-15 degC
low-temperature nitrile jacket	-49 degF	-45 degC	-39 degF	-39.5 degC
Redahot nitrile jacket	-40 degF	-40 degC	-30 degF	-35 degC
EPDM jacket	-58 degF	-50 degC	-48 degF	-45 degC
EPDM insulation	-76 degF	-60 degC	-66 degF	-55 degC
lead	-148 degF	-100 degC	-138 degF	-95 degC

- Ensure the motor has a metal pothead cap and new O-ring before lowering into the well. Keep the motor in the well for a minimum of four hours to allow the equipment to warm.

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- If the well is full of kill fluid or is expected to flow while the equipment is being warmed, the protector must be completely serviced before lowering it into the well.
- If the motor has been placed in the well to warm, always check the protector for contamination by well fluids. Tie in the flat cable extension using a vertical tie-in; this puts less stress on the flat cable extension. Re-service the protector with warm oil.
- There have been problems resulting from assembling equipment, not servicing it with oil and lowering it into a well filled with kill fluid. The equipment always fills with well fluid in these circumstances.

Blended Oils

When the well BHT is greater than 175 degF (79 degC), the use of dual-base bushings in motors is required because a 50/50 blend of No. 3 and No. 5 oil at a concentration greater than 20% in No. 6 oil is too thin for the standard single-base bushing normally used in motors.

4.2.9.2

Cables

Cable considerations and best practices are discussed in the following sections.

Cable Considerations

Cable material selection should be based on bottomhole operating temperatures rather than surface temperatures. Elastomers should be selected to ensure optimum performance and run life for the well conditions. The installation procedures should ensure the selected cable equipment is handled and installed properly.

When installing cable strings and their various component parts in cold temperatures, the cable elastomer material must be heated to maximize run life.

Cable elastomers can become so rigid in cold temperatures that the cables can be pinched during assembly or break as they move over the cable wheel. There is no alternative but to heat the cable string before assembly and during installation when ambient temperatures are below the material useful temperature range. The cable should be transported to the wellsite in a warm condition if possible. Ideally, the cable should be stored overnight in a heated environment prior to running in the well.

Temperature limitations for cable materials are shown in [Table 4-6](#).

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Best Practices

- A heated cable spooling shack should be used for cabling operations when cold temperatures are expected.
- A workover rig steam boiler (if available) or a large-capacity portable steamer or warm-air blower should be used to provide heat. Another method is to enclose the cable spooler with wind-proof material (usually a canvas or tarp with a metal frame support). A warm-air vent from the rig steam boiler or warm-air unit is attached to the enclosure. This will keep the cable warm during the installation process. The steam line or heat line should be routed to the cable spooler reel as soon as it arrives on location. The cable should be minimally exposed to cold temperatures during installation on the surface. Temperature limitations for cable materials are shown in [Table 4-6](#).
- Never allow the cable to hang suspended overnight between the wellhead and the spooler in cold temperatures.
- Do not assemble the flat cable extension to the motor until the equipment is ready to run.
- For information on suspending the EPS Equipment in the well please see section [4.11.1: Suspending ESP String](#).

4.3

Lifting Clamp Installation

Refer to [InTouch Content ID 4557363](#), which specifies the procedures to load test and certify the mechanical lifting equipment, such as lifting clamp, lifting chain, centering hook and Motor tail clamp.

To choose the proper lifting clamps for your installation, please see section [2.4: Mechanical Lifting Equipment](#).



Note

Measure and record makeup length of pump/protector/motors.

A crane with a spreader bar and slings is used to lift equipment out of the shipping box and place equipment on wooden planks spaced evenly throughout the equipment length.

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When equipment is stabilized on the wooden planks and it is safe, install the lifting clamps complete with chains as per the approved procedure.

**Note**

Clamps must be clean and free from foreign material and visually inspect before each lift.

**Note**

Clamp bolts should be tightened according to [Table 2-4: Lifting Clamps](#) in section [2.4: Mechanical Lifting Equipment](#). Use torque wrench p/n 2003754 and sockets p/n 101168703 or p/n 101168675 depending on installation clamp bolt.

4.3.1**Motor Clamp Installation**

Choose the proper motor lifting clamp and install it so the shoulder of the clamp rests against the lip of the motor head to prevent the clamp from slipping. The cable slot in the clamp should be aligned with the pothole of the motor. This provides the maximum holding strength of the clamp. Tighten the bolts evenly to ensure equal spacing between the clamp faces.

If several tandem motors are being installed, place the clamp on the lower tandem motor first. On lower tandem and center tandem motors, it is important to align the cable slot on the motor clamp with the drain and fill valve to prevent it from being damaged.

4.3.2**Protector Clamp Installation**

The protector clamp is installed on the housing just below the top lock plate or lock weld in the case of a modular protector. Ensure the bolts are evenly tightened to ensure equal spacing between the clamp faces.

4.3.3**Pump Clamp Installation**

Pump clamps can be installed on the bottom two sections of the pump that will be picked up. If the pump clamp is a friction type, it is installed just under the top lock plate. If a shoulder type is used, slide the shoulder of the clamp into the groove located in the pump head.

A clamp is not normally used on the uppermost section of pump. The best method for picking up this section of pump is with a handling pup joint.

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The pup joint should be installed on the upper end of the pump and rig elevators used to pick up the pump if additional handling is necessary. This practice is best described in [Figure 4-8](#).

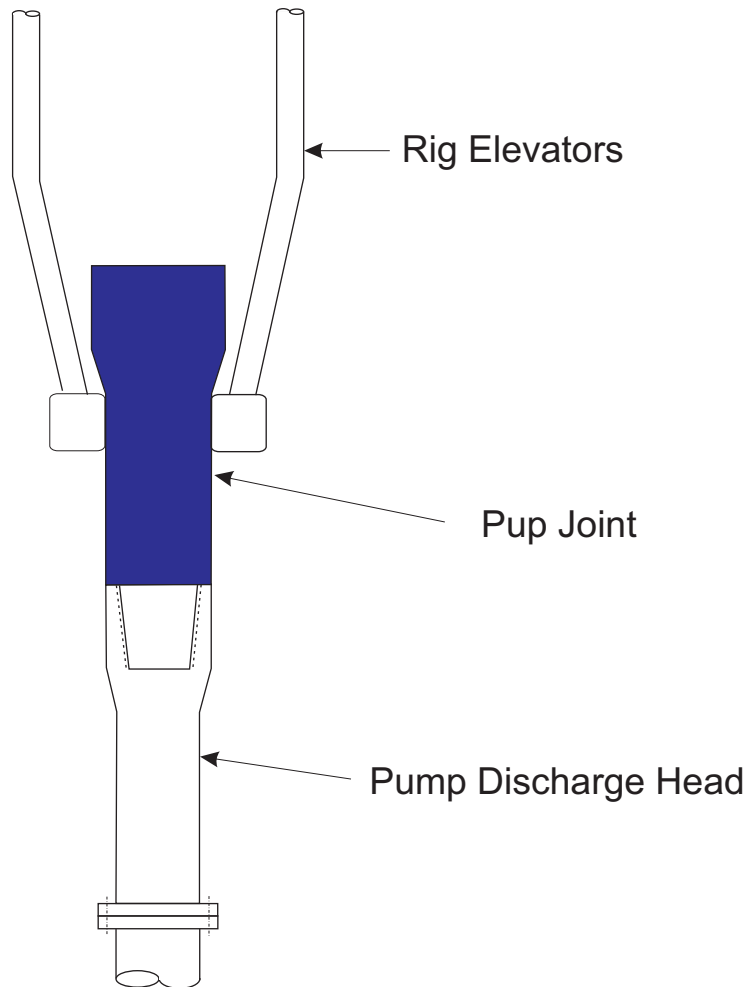


Figure 4-8: Lifting Collar Pup Joint

4.4 ESP Gauges

4.4.1 Product Description

4.4.1.1 Overview

The Phoenix systems are installed on downhole ESP (electric submersible pump) pumping equipment and use the ESP power cable for communication and power. The downhole sensors measure pump and reservoir parameters to give the operator information to better optimize the ESP system.

The Phoenix Select Lite, Standard, and Advanced systems consist of up to three components: Base Unit and either 0, 1, or 2 Remote Sensors. The Phoenix Select Base Unit is installed on the base of the ESP motor and measures intake pressure, intake temperature, tri-axial vibration, motor winding temperature, and current leakage of the ESP electrical system. The Phoenix Select Remote Sensors can be installed at the pump intake and pump discharge. The remote sensors measure intake or discharge pressure, intake or discharge temperature, and tri-axial vibration. The Phoenix Select Remote Sensors communicate to the Phoenix Select Base Unit over a single electrical conductor inside a 1/4-in diameter armored line. Communication to surface is achieved through a connection to the neutral point of the lower ESP motor by the Phoenix Select Base Unit, and consequently, the ESP cable. The Phoenix Select Base Units retransmit the remote sensor information to the surface as required. The Phoenix Select gauge operates at higher temperatures than the previous XT offering, 150 degC (302 degF) vs 125 degC (257 degF).

The Phoenix XT has been the standard ESP gauge in the market for many years. The XT is fully qualified for 125 degC and uses Analog Telemetry to transmit information to the surface panel. It is compatible with all UniConn/PIC equipment as well as with legacy panels like ISP and ISU.

The Phoenix xt150 is based on proven MultiSensor and Phoenix XT monitoring technology for downhole ESP equipment. Improved with microelectronic components, and digital telemetry to ensure that the most reliable, faster and accurate information is provided to analyze artificial lift performance. The Phoenix xt150 operates at 150 degC (302 degF). Phoenix XT and xt150 are offered as a single pressure Type 0 gauge and a dual pressure Type 1 gauge. It measures intake pressure, discharge pressure (Type1 only), intake temperature, motor winding or motor oil temperature, single axis vibration and current leakage

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of the ESP electrical system. The Phoenix xt150 is only compatible with the UniConn and PICv2 surface acquisition system. The Phoenix xt150 offers the following improvements compared to the analogue version (Phoenix XT):

- Higher temperature rating, 150 degC (302 degF) versus 125 degC (257 degF)
- Higher pressure accuracy, 0.1% full scale versus 0.2% full scale
- Digital telemetry that eliminates analogue data uncertainties

The Phoenix UltraLite system is a base unit only. The UltraLite gauge uses plastic surface mount components to reduce cost, but limits the operating temperature to 100 degC (212 degF) maximum. The UltraLite supports a single pressure transducer that measures the motor oil pressure and temperature, which is assumed equal to annulus pressure and temperature, respectively. Unlike the xt150 and Select systems, there is no external thermocouple to measure motor winding temperature. The UltraLite also measures single axis vibration and current leakage.

The Phoenix UltraLite, xt150, and Select systems transmit the acquired measurements over the ESP power cable using digital telemetry. This digital telemetry offers several advantages over the analog telemetry used in the older MultiSensor and XT gauges:

- The data is digitized downhole so that the transmitter and receiver accuracy and performance have no bearing on the accuracy of the measurement. Noise and interference in the communication system cannot cause data to become inaccurate.
- The telemetry protocol uses error checking on the data received at the surface to ensure that the data interpreted by the surface equipment is the same as the data measured downhole.

Refer to [Table 4-7: Phoenix Gauge Comparison](#) for a list of parameters monitored by the Phoenix gauges.

Table 4-7: Phoenix Gauge Comparison

Measurements	Ultralite	XT/xt150		Select		
		Type 0	Type 1	Lite	Standard	Advanced
Motor oil pressure	✓					
Motor oil temperature	✓					
Current leakage	✓	✓	✓	✓	✓	✓
Annulus pressure @ motor		✓	✓	✓	✓	✓
Annulus temperature @ motor		✓	✓	✓	✓	✓
Motor winding temperature		✓	✓	✓	✓	✓
Motor vibration, single axis (X axis)	✓	✓	✓			
Motor vibration, tri-axial (X, Y, Z axes)				✓	✓	✓
Discharge pressure			✓		✓	✓
Discharge temperature					✓	✓
Discharge vibration, tri-axial					✓	✓
Pump intake pressure						✓
Pump intake temperature						✓
Pump intake vibration, tri-axial						✓

4.4.1.2

Surface Acquisition System Compatibility

The Phoenix XT, Select Lite, Select Standard and Select Advanced systems are supported by the UniConn/Phoenix Interface Card (PIC) and legacy ISP/ISU surface acquisition systems. However, the Phoenix UltraLite and the Phoenix xt150 systems are only supported by the UniConn/PIC (version 2). An earlier version of the Phoenix UltraLite system developed under rapid response uses an obsolete telemetry system and is supported by both the UniConn/PIC and legacy ISP/ISU surface acquisition systems.

**Note**

Obsolete UltraLite Part Number	Old Description	Current UltraLite Part Number	Description
100225679	UltraLite 13Cr	100670906	Base Gauge, Ultralite, 13Cr, Vit (A346)
100276501	UltraLite Carbon Steel	100670910	Base Gauge, Ultralite, Carbon steel, Vit (A346)

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4.4.2 Specifications

4.4.2.1 Phoenix UltraLite Gauge

The Phoenix UltraLite is installed at the base of motor and has no remote sensor connections.

Table 4-8: Phoenix UltraLite Measurements

Parameter	Metrology Range	Accuracy Full Scale	Resolution
Motor Oil Pressure	0 to 5,000 psig	±10 psi	0.1 psi
Motor Oil Temperature	0 to 100 degC	±2 degC	0.5 degC
Single Axis Vibration	0 to 30 g	±1 g	0.1 g
Current Leakage	0 to 25 mA	±0.05 mA	0.001 mA

Table 4-9: Phoenix UltraLite Base Gauge Operational/Environmental

Description	Specification	Notes
External Pressure	5,000 psi gauge	
Operating Temperature	100 degC standard	
Survival Temperature	125 degC for 1 hour	Sensor not calibrated to operate above 100 degC
Elastomers	Viton 90 Hard	
Metallurgy	13 chrome steel Carbon steel	

The Phoenix UltraLite gauges are either installed integral to the motor when the motor is manufactured or bolted on at the base of the motor like the other Phoenix gauges.

The Phoenix UltraLite bolt-on gauge weighs approximately 44 lbm (20 kg).

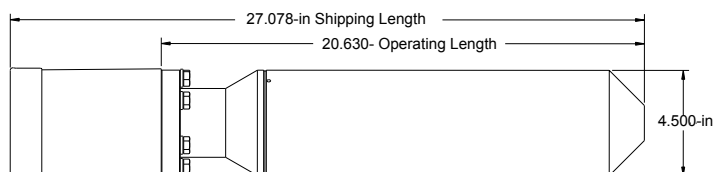


Figure 4-9: Phoenix UltraLite Gauge

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4.4.2.2 Phoenix XT Base Gauge

The Phoenix XT base unit is installed at the base of motor and has one optional hydraulic discharge connection. Refer to OneCAT for various sizes of discharge subs.

Table 4-10: Phoenix XT Measurement Specifications

Parameter	Metrology Range	Accuracy Full Scale	Resolution
Intake Pressure ¹	0 to 5,800 psig	±10 psi	1 psi
Discharge Pressure ¹	0 to 5,800 psig	±10 psi	1 psi
Intake Temperature	0 to 125 degC	±2 degC	0.1 degC
Motor Winding/Oil Temperature	0 to 325 degC	±4 degC	0.1 degC
Vibration	0 to 30 g	±1 g	0.1 g
Current Leakage	0 to 25 mA	±0.05 mA	0.001 mA

¹ Calibrated pressure range.

Table 4-11: Phoenix XT Base Gauge Operational/Environmental

Description	Specification
External Pressure	6,500 psi absolute
Operating Temperature	125 degC standard
Shock/Vibration	250 g shock / 40 g vibration
Metallurgy	13% Chrome Steel
Elastomers	Viton and Aflas

The Phoenix XT base unit weights approximately 55.1 lbm (25 kg).

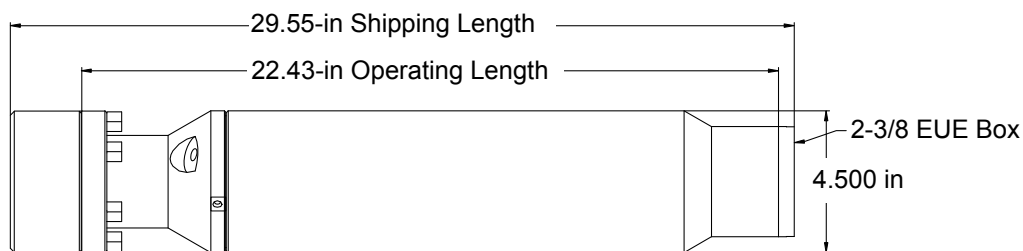


Figure 4-10: Phoenix XT Gauge

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4.4.2.3 Phoenix xt150 Base Gauge

The Phoenix xt150 base unit is installed at the base of motor and has one optional hydraulic discharge connection. Refer to OneCAT for various sizes of discharge subs.

Table 4-12: xt150 Measurement Specifications

Measurement	Range	Accuracy	Resolution	Drift	Update Rate
Intake Pressure ¹	0 to 5,800 psi	± 5 psi	0.1 psi	5 psi / year	4 s
Discharge Pressure ¹	0 to 5,800 psi	± 5 psi	0.1 psi	5 psi / year	4 s
Intake Temperature	0 to 150 degC	± 2 degC	0.1 degC	n/a	4 s / 8 s
Motor Temperature	0 to 409 degC	±4 degC	0.1 degC	n/a	36 s
Vibration	0 to 30 g	1 g	0.1 g	n/a	Variable
Current Leakage	0 to 25 mA	0.05 mA	0.001 mA	n/a	Variable

¹ Calibrated pressure range.

Table 4-13: xt150 Base Gauge Operational/Environmental

Parameter	Value
Maximum Operating Pressure	6,500 psi
Maximum Operating Temperature	150 degC
Metallurgy	13 Chrome
Elastomers	Aflas and Viton
Insulation Test Rating	5,000 V

The Phoenix xt150 base unit weights approximately 55.1 lbm (25 kg).

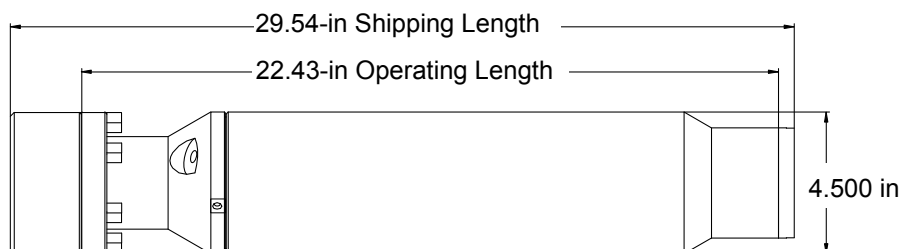


Figure 4-11: Phoenix xt150 Gauge

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4.4.2.4 Phoenix Select

Select Base Gauge

The Phoenix Select base unit is installed at the base of motor and has two remote sensor connections.

Table 4-14: Select Base Gauge Measurements

Parameter	Metrology Range	Accuracy Full Scale	Resolution
Pressure	0 to 5,000 psig	±5 psi	0.1 psi
Temperature	0 to 150 degC	±2 degC	0.1 degC
Motor Winding/Oil Temp	0 to 409 degC	±4 degC	0.1 degC
Tri-Axial Vibration	0 to 30 g	±1 g	0.1 g
Current Leakage	0 to 25 mA	±0.05 mA	0.001 mA

Table 4-15: Select Base Unit Operational/Environmental

Description	Specification
External Pressure	5,000 psi gauge
Operating Temperature	150 degC standard
Metallurgy	13% Chrome Steel
Elastomers	Viton or Aflas

The Phoenix Select base unit weights approximately 55.1 lbm (25 kg).

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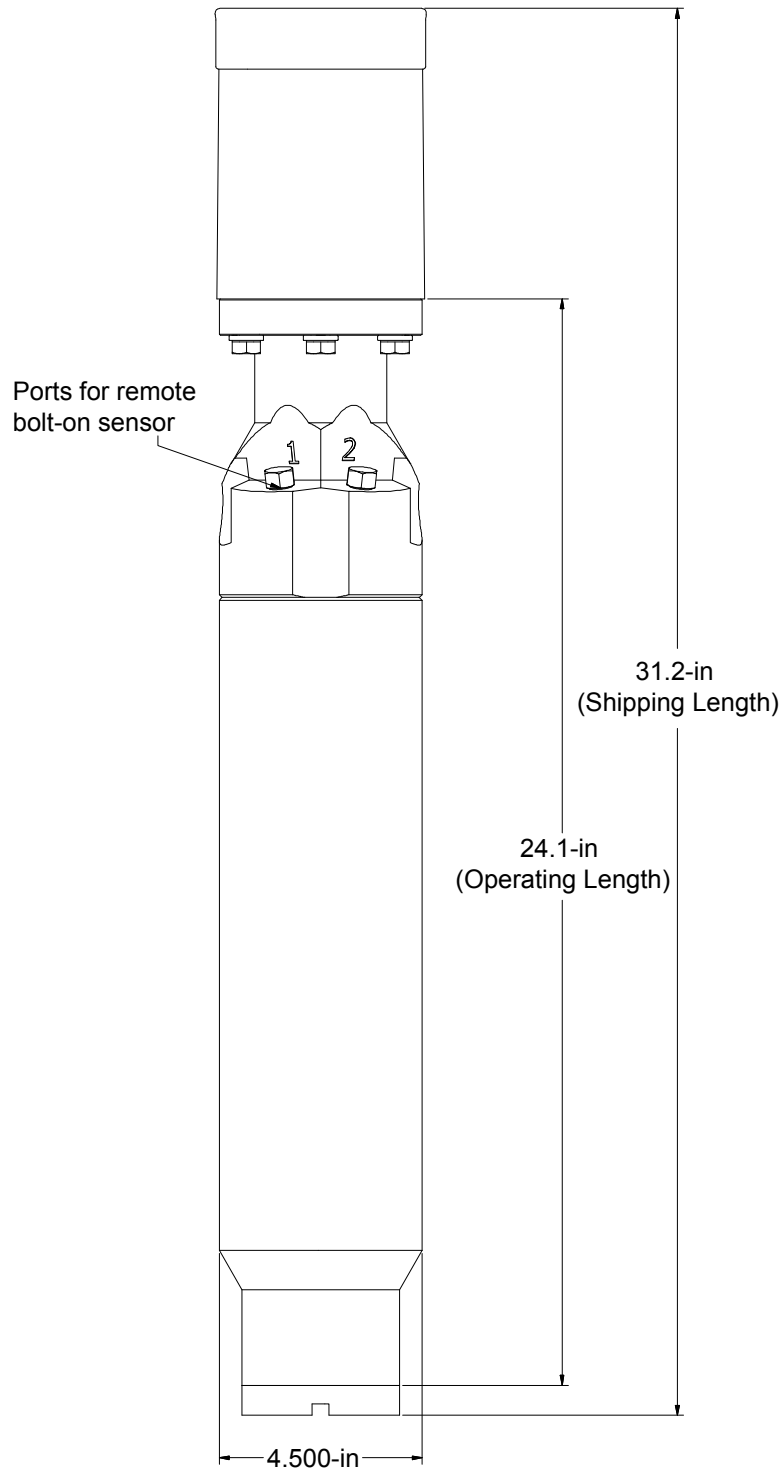


Figure 4-12: Select Base Unit Dimensions

The Phoenix Select 375 base unit is installed at the base of motor and does not support any remote sensor connections.

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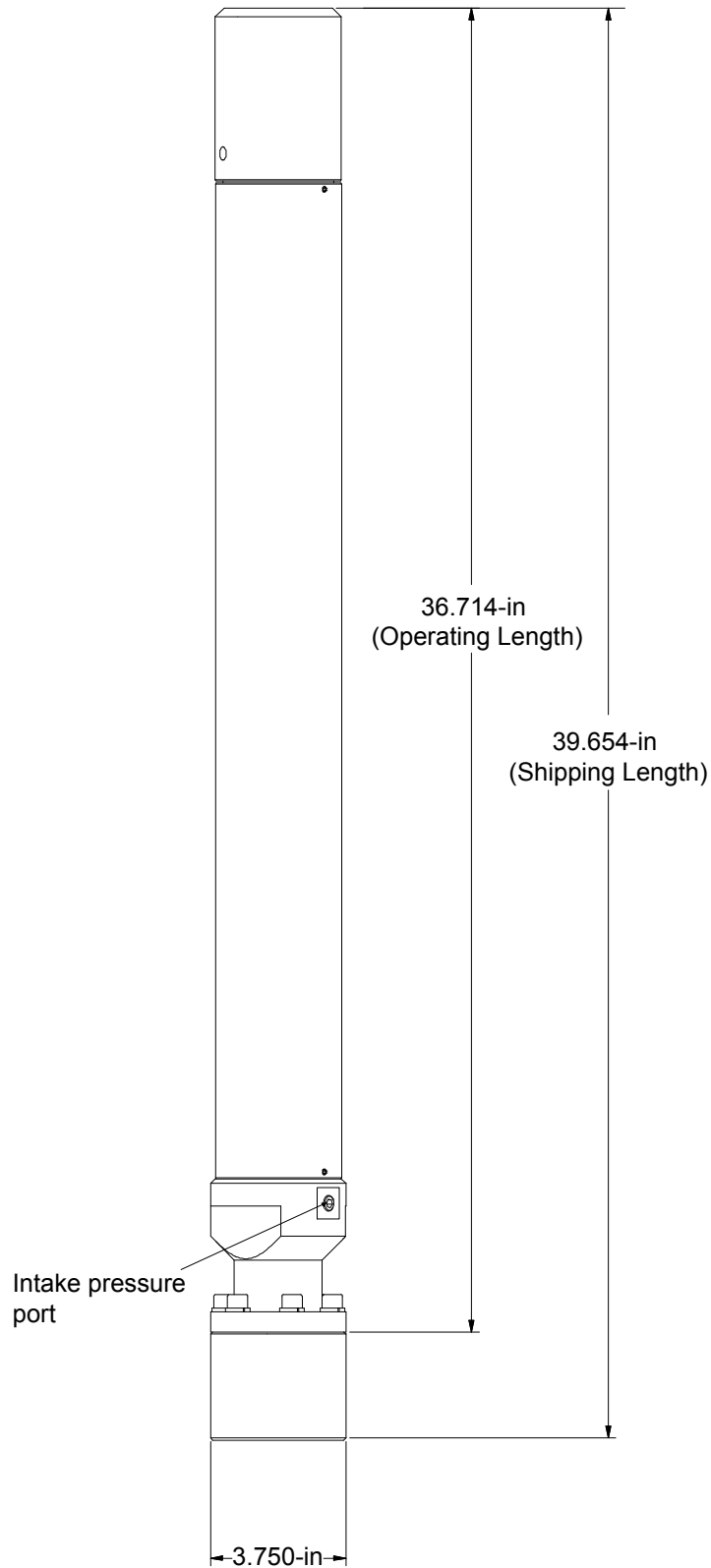


Figure 4-13: 375 Select Base Unit Dimensions

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The Phoenix Select 375 base unit weights approximately 48.5 lbm (22 kg).

Select Remote Sensors (Discharge and Intake)

The Phoenix Select remote sensor is ordered according to the pump size and connects to the remote link connection on a base gauge.

Table 4-16: Select Remote Sensor Measurements

Parameter	Metrology Range	Accuracy, Full Scale	Resolution
Pressure	0 to 5,000 psig	±5 psi	0.1 psi
Temperature	0 to 150 degC	±2 degC	0.1 degC
Tri-Axial Vibration	0 to 30 g	±1 g	0.1 g

Table 4-17: Select Remote Sensor Operational and Environmental Specifications

Description	Specification
External Pressure	5,000 psi gauge
Operating Temperature	150 degC standard
Shock/Vibration	250 g shock / 40g vibration
Metallurgy	Stainless Steel AISI 630 (17-4PH)
Maximum distance from Base Unit	100 m

Refer to [Figure 4-14](#) for the Phoenix Select remote sensor dimensions for a sensor that can be installed on a 562 pump.

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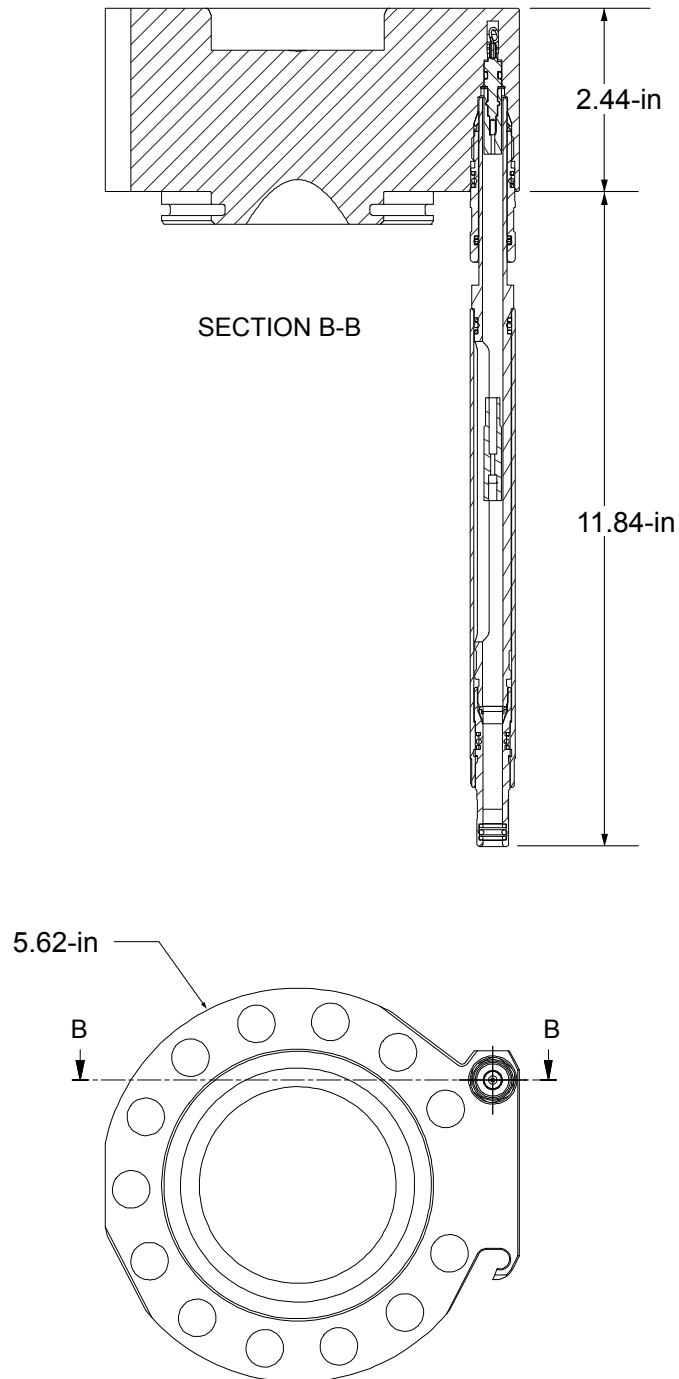


Figure 4-14: 562 Remote Sensor Dimensions.



Note

See [Table 4-18: Select Remote Sensor Dimensions](#) for running ODs.

[Figure 4-15](#) shows the configuration of the Phoenix Select Advanced.

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**Note**

In order for the remote sensor to be installed at the pump intake, a special Phoenix Select Extended Coupling is required to take into account the additional length of the remote sensor.

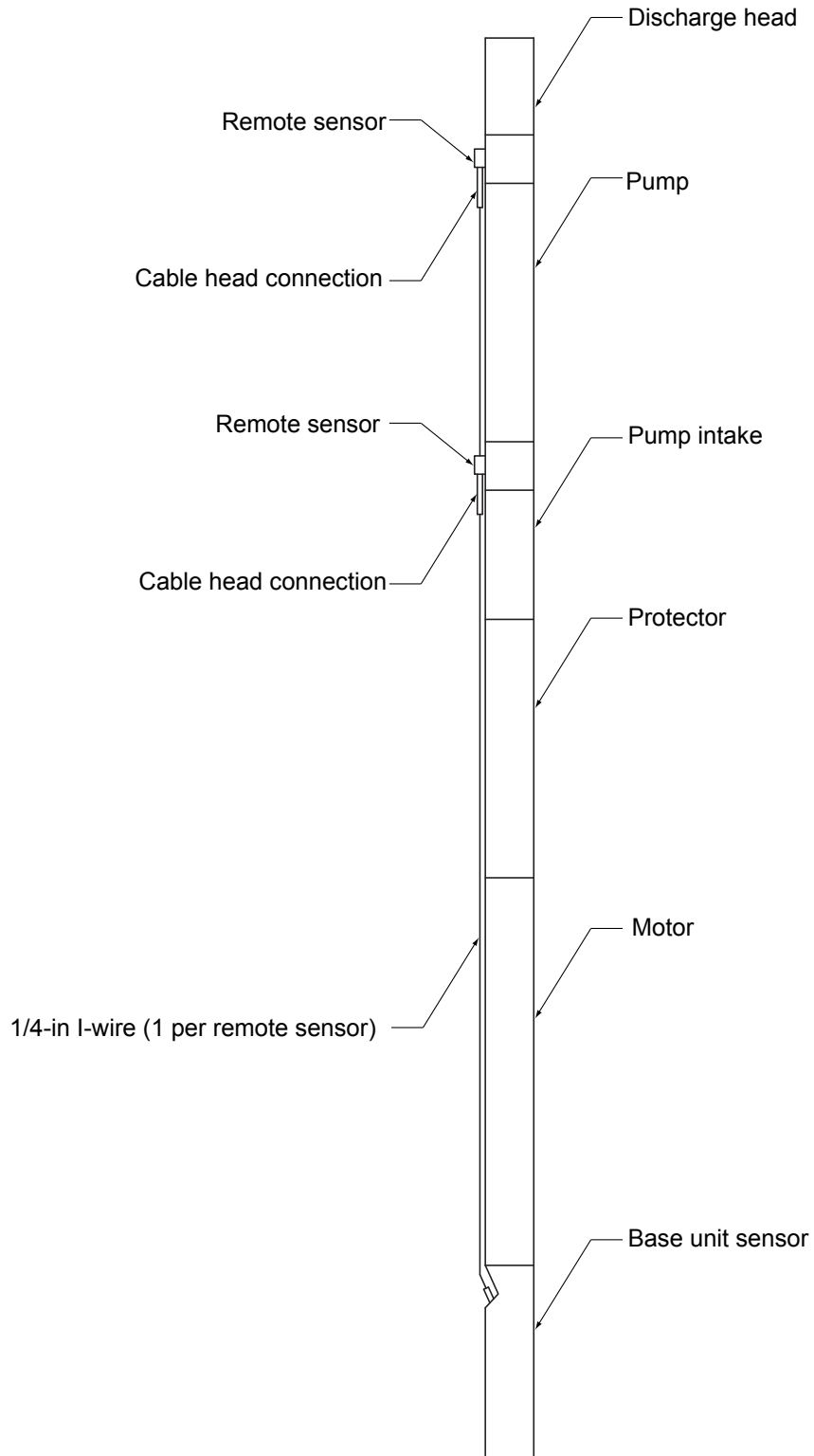


Figure 4-15: Phoenix Select Advanced with Remote Sensor Configuration

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Maximum Running OD of Remote Sensors

The Phoenix Select remote sensor is sized according to the pump and location it is to be installed. The following table defines the maximum running OD for all configurations.



Note

For the same remote sensor, the running OD will differ if installed at the pump intake or discharge.

The running OD dimensions include the cable protector clamp.

Table 4-18: Select Remote Sensor Dimensions

Part Number	Description	Operating Length (in)	Approximate Shipping Weight
100260203 or 100260204	400 SERIES	2.44	13.23 lbm (6 kg)
100252035 or 100260311	538/540 SERIES		17.64 lbm (8 kg)
100260206 or 100260207	562 SERIES		19.84 lbm (9 kg)
100260208 or 100260209	675 SERIES		26.45 lbm (12 kg)

Table 4-19: Select ESP Remote Sensor Diameters and Minimum Casing Sizes

ESP Remote Sensor	Part Numbers	Motor Series	Discharge Remote Sensor		Intake Remote Sensor	
			Running Diameter	Minimum Casing Size and Standard Drift	Effective Running Diameter	Minimum Casing Size and Standard Drift
Phoenix Select Remote Sensor, Reda 400	100260203 100260204	456	5.000-in	6 5/8-in x 35 lb/ft Drift 5.450-in Dia	5.164	6 5/8-in x 35 lb/ft Drift 5.450-in Dia
		540, 562			Motor Diameter Critical	As Required For Motor

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ESP Remote Sensor	Part Numbers	Motor Series	Discharge Remote Sensor		Intake Remote Sensor	
			Running Diameter	Minimum Casing Size and Standard Drift	Effective Running Diameter	Minimum Casing Size and Standard Drift
Phoenix Select Remote Sensor, Reda 538/540	100252035 100260311	540	6.066-in (With 540 Pump 5.25-in dia)	7-in x 26 lb/ft Drift 6.151-in Dia	6.160-in	7-in x 23 lb/ft Drift 6.241-in Dia
		562			6.254-in	7-in x 20 lb/ft Drift 6.331-in Dia
		540	6.130-in (With 538 Pump 5.38-in dia)	7-in x 26 lb/ft Drift 6.151-in Dia	6.160-in	7-in x 23 lb/ft Drift 6.241-in Dia
		562			6.254-in	7-in x 20 lb/ft Drift 6.331-in Dia
Phoenix Select Remote Sensor, Reda 562	100260206 100260207	562	6.545-in	8 5/8-in x 52lb/ft Drift 7.310-in Dia	6.545-in	8 5/8-in x 52lb/ft Drift 7.310-in Dia
Phoenix Select Remote Sensor, Reda 675	100260208 100260209	562	7.700-in	8 5/8-in x 36lb/ft Drift 7.700-in Dia	7.700-in	8 5/8-in x 36lb/ft Drift 7.700-in Dia
		738			7.990-in	9 5/8-in x 70.30 lb/ft Drift 8.001-in Dia

**Note**

The above table does not take into account any protectolisers, clamps or fittings which tend to offset or centralize the ESP String or parts of it. If such equipment is used in a completion a design review must be carried out to ensure the system is within drift of the smallest casing diameter into which it will be run, and that no excess deflection of the ESP string is imposed. An external drift of the ESP system or parts of the system is recommended prior to installation.

Refer to the best practice at InTouch Content ID 4594994 for the latest version of this information.

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4.4.3 Wellsite Procedures

This section covers the wellsite installation procedures for base unit gauge configurations (Ultralite, XT Type 0, xt150 Type 0 and Select Lite). Refer to the appropriate operation manual for more advanced gauge configurations (XT Type 1, xt150 Type 1, Select Standard, Select Advanced, etc.) installation procedures.

4.4.3.1 Confirm Firmware Version

The Phoenix Select Lite, Standard and Advanced systems are supported by the UniConn/Phoenix Interface Card (PIC) and legacy ISP/ISU surface acquisition systems.

The XT gauge is compatible with PICv1 (firmware ver. 2.1 or later) and PICv2 (firmware ver. 1.15 or later). The XT gauge is supported by the UniConn/Phoenix Interface Card (PIC) and legacy ISP/ISU surface acquisition systems.

The xt150 gauge is only compatible with surface equipment (UniConn) equipped with a PICv2. The firmware version of the PICv2 must be 1.20 or newer and the UniConn firmware needs to be version 1.404r1 or newer for the tool to work properly.

The UltraLite gauge is only compatible with surface equipment (UniConn) equipped with a PICv2. The firmware version of the PICv2 must be 1.15 or newer and the UniConn firmware needs to be version 1.404r1 or newer for the tool to work properly.



Note

The UniConn/PICv2 (with latest firmware) is recommended for all Phoenix gauge applications.

It is always a good idea to check with InTouch for the latest firmware releases and alerts. See [InTouch content 4456403](#) for the PICv2 and [InTouch content 3985891](#) for the UniConn. Refer to the UniConn manual to see how to install the PICv2, configure, and check the firmware versions.

4.4.3.2 Pre-Installation Checks

Before running the gauge into the well, several checks should be made to determine that it is functioning properly. Perform the following checks before running the downhole tool into the well:

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1. Inspect the housing for physical damage or imperfections.
2. If the top connection of the downhole tool is protected with a shipping/handling cap secured by six bolts to the lower flange. Remove the handling cap and ensure that the two O-rings are new. If they are not, replace them with new O-rings before connecting to the ESP motor.

**Warning**

Potential Severity: Serious
 Potential Loss: Personnel
 Hazard Category: Electrical

Caution should be used when using a megommeter. All safety guidelines as outlined in the operation manual for the meter, and Schlumberger policies must be followed.

3. Inspect the pin to make sure it is not bent or damaged. Test the electrical integrity of the feedthru with a megohmmeter (Megger). The gauge should be tested up to 2.5 kV. (see Megger procedure or the Gauge User Manual in InTouch)
4. Before connecting the gauge to an ESP motor, it should be first connected to the surface equipment to verify that the gauge is functioning properly. Power up the Surface equipment and ensure that the readings are being displayed. If errors are reported refer to the Troubleshooting section in the UniConn manual. Be sure to resolve all faults before proceeding with installation of the gauge in the well.
5. Values displayed should represent ambient conditions. Depending on accuracy of each tool, those values could be different. Refer to the User Manual Specifications section for nominal values (pressure, temperature and vibration) and the Functional Test section for the testing procedure.

**Note**

It is optional (but strongly recommended) to apply pressure and verify proper function of discharge port (XT and xt150 only). Check value read at surface panel is ± 10 psi of applied pressure.

**Note**

Vibration refresh period for digital gauges could be 30 seconds or more. The first vibration measurement is acquired within the first few seconds of applying power to the gauge, after which no measurement is taken until

the next sample has been transmitted to the surface gear. There is also an averaging setting on UniConn and the time needed to display a change in Vibration could be even higher.

Also note if the temperature at the wellhead is significantly below operating temperature of the tool (< 0 degC) the tool should be warmed up or the readings could be out of specification.

6. Disconnect the gauge from the surface equipment.

4.4.3.3

Connecting the Downhole Gauge

The gauge must be connected to the ESP system. Connection can be done either with an I-wire or if necessary, without an I-wire through the use of an adapter. Refer to [Figure 4-16](#) for an example of an I-wire type connection.

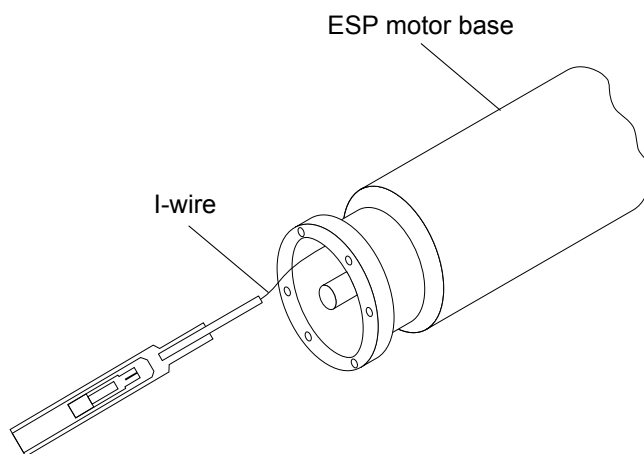


Figure 4-16: I-wire Connection

Refer to [Figure 4-17](#) for ESP application without neutral point or I-wire for an example of an ESP system that does not have an electrical neutral point or I-wire available. An adapter is available for applications such as this.

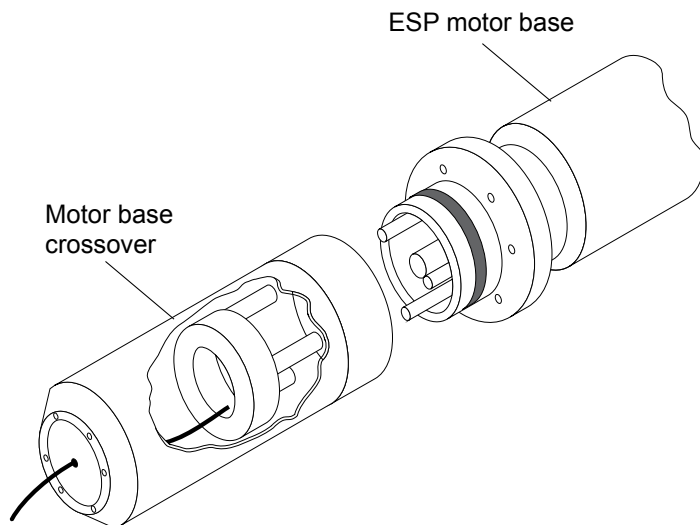


Figure 4-17: ESP application without neutral point or I-wire

I-wire Connection

Perform the following steps when installing a gauge on the lower end of an ESP assembly.

1. Remove the lower handling cap from the ESP motor.
2. Locate the I-wire.



Note

If the motor has three-phase pin connectors, the motor base adapter should be bolted to the ESP motor base. Locate the I-wire in the motor base adapter

3. A boot connector and splice kit is supplied with every gauge assembly. The boot connector is composed of three parts; a crimp connector, a Teflon sleeve and a boot. Refer to [Figure 4-18](#) for how the boot connector and splice kit are designed to go together.

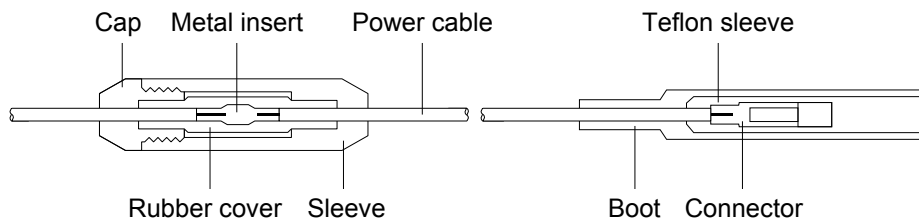


Figure 4-18: Crimping and splicing connector, sleeve, and boot

4. Slip the boot and the Teflon sleeve over the I-wire and far enough down the wire to be out of the way for now.

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**Note**

A splice kit is supplied, in case the I-wire is not long enough to install the boot connector and plug onto the feedthru. Install the splice kit if necessary.

5. Strip approximately 6 mm (~1/4 in) of insulation from the end of the I-wire.
6. Set the DMC crimping tool for #12 AWG. Crimp the crimp connector onto the wire.
7. Slip the Teflon sleeve over the crimp connector and the boot over the Teflon sleeve.

Connecting Gauge to ESP Motor

The gauge can be connected to the motor in the horizontal position before picking the motor up in the rig or with the ESP string hanging vertical in the rig. Both options are described in the following sections.

Horizontal Gauge Installation (Before Lifting)

The gauge can be connected to the motor in the horizontal position before picking the motor up (vertical) in the rig. To connect the gauge to the ESP string, perform the following steps:

**Note**

Remove any gaskets remaining on the motor from the shipping cap. New O-rings should be installed on the gauge flange.

1. The gauge should be appropriately oriented (rotated about the tool axis).
2. Align the gauge with the ESP string and move it within a couple of inches from the bottom of the motor. Remove the dummy boot from the feedthrough.

**Note**

Care should be taken not to damage the feedthrough.

3. Plug the previously installed boot connector on the I-wire firmly onto the feedthrough on the bottom face of the ESP motor.
4. When the boot connector is securely fastened to the feedthrough, the gauge should be pushed home into the motor flange. Be very careful not to snag the I-wire.

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5. Install and tighten all flange bolts to 30 ft.lbf. The recommended the torque wrench setting is 26 ft.lbf, as per ED-119.

**Danger**

Potential Severity: Major
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

Do not allow the lower end of the gauge to drag the ground, the V-door, or along the rig floor when hoisting the ESP string up into the derrick or mast. Dragging the tool can cause damage to the joint and damage the internal parts of the tool.

Vertical Gauge Installation (After Lifting)

The gauge may be connected to the lower end of the ESP motor with the ESP string hanging vertical in the rig. To attach the gauge to the motor while vertical, perform the following steps:



Note

Remove any gaskets remaining on the motor from the shipping cap. New O-rings should be installed on the gauge flange.

1. Lift the gauge onto the rig floor with the transit cap in place, ensuring that the cavity in the upper head is covered to prevent fluid and loose matter from falling into the cavity.
2. Orient the gauge.
3. Pick up (with derrick hoist) the ESP string so that the motor is far enough above the gauge to remove the boot protector cap from the feedthrough.
4. Plug the boot connector (already installed on the I-wire) directly onto the feedthrough.
5. When the boot connector is securely pushed onto the feedthrough and lift the gauge to the ESP motor to fully engage the upper flange of the gauge .
6. Lower the ESP and gauge to a solid work surface so that the bolts can be safely installed.
7. Install and tighten all flange bolts to 30 ft.lbf. The recommended the torque wrench setting is 26 ft.lbf, as per ED-119.

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4.4.3.4 Functional Testing

Before running into the well, the gauge should be checked to make sure nothing was damaged or disturbed while connecting the gauge to the ESP string. To check this; perform the following:

1. Connect the surface equipment to the pothead of the motor. Connect the positive (+ or LINE) lead to a motor phase. If this is not permitted, then connect it to the cable once connected. The Negative (or CHASIS) lead should be connected to ESP housing or ESP cable armor.
2. Power up the surface equipment. All diagnostic and parameter readings should be recorded. If errors are reported refer to the Troubleshooting section in the UniConn manual ([InTouch ID 3953183](#)). Be sure to resolve all faults before proceeding with installation of the gauge in the well.
3. Verify all readings are within specifications. Refer to the User Manual Specifications section for nominal values (pressure, temperature and vibration).

The following nominal values are shown as an example:

Pressure – 15 psi $\pm$ 10 psi

Temperature within 3.6 degF (2 degC) of surface temperature

Vibration < 0.5 g when not moving and responding to movement

4. Perform megohmmeter (megger) test on both the ESP motor and the gauge to a value of 2.5 kV.



Note

The positive (+) lead of the Megger should always be connected to ground (earth) when attached to a Phoenix Select. Never spark (arc) discharge voltage to ground. Always discharge voltage through the Megger.

Refer to the Megger Test Procedure in the specific gauge operations manual for details about checking insulation resistance.

4.4.3.5 Pressure Discharge Installation

This section describes the discharge line installation process for the Multisensor, Phoenix XT and Phoenix xt150 gauges.

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Discharge Line Options

When preparing Type-1 gauge systems, a pressure discharge line (1/4-in tubing) must be connected between the pump discharge and the gauge. To connect the pressure discharge line, perform the following. Refer to [Figure 4-19: Gauge With Discharge Pressure Line](#) for a diagram.

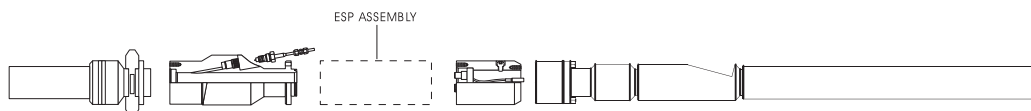


Figure 4-19: Gauge With Discharge Pressure Line

There are now two variations of discharge pressure line and three variations of Discharge Head (port) that may be encountered in the field. [Table 4-20: Identifying Discharge Line Installation Procedure](#) shows how each one can be utilized during the installation depending on what version of the Line and Discharge Head that is available. In all cases it is assumed that the downhole tool is a gauge.

Table 4-20: Identifying Discharge Line Installation Procedure

Discharge Line	Pressure Discharge Port		
	Large Port	NPT Bulkhead	Jam Nut
Old Line (A046002-405)	Standard Install See section 4.4.3.5 Large Port Installation with Old Discharge Line	Non Standard See section 4.4.3.5 NPT Bulkhead Installation with Old Discharge Line	Non Standard See section 4.4.3.5 Jam Nut Installation with Old Discharge Line
New Line (100120402)	Do Not Use See section 4.4.3.5 Large Port Installation with New Discharge Line	Non Standard See section 4.4.3.5 NPT Bulkhead Installation with New Discharge Line	Standard Install See section 4.4.3.5 Jam Nut Installation with New Discharge Line

Identifying the Discharge Head Port

The old Discharge Head port that is most common on earlier Discharge Subs is shown in the following figure. This is called the large port.

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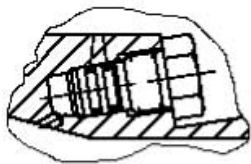


Figure 4-20: Large Discharge Port

Another old version, which is less common is shown in [Figure 4-21: NPT Bulkhead Discharge Port](#) and will be supplied complete with a Swagelok type fitting.

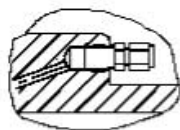


Figure 4-21: NPT Bulkhead Discharge Port

The new Discharge Head port that will feature on all discharge Head's on the future is shown in the following figure.

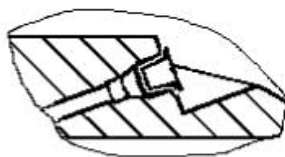


Figure 4-22: New Jam Nut Discharge Port

Identifying the Discharge Lines

The old Discharge Line is as shown in the [Figure 4-23: Old MultiSensor Discharge Pressure Line](#).

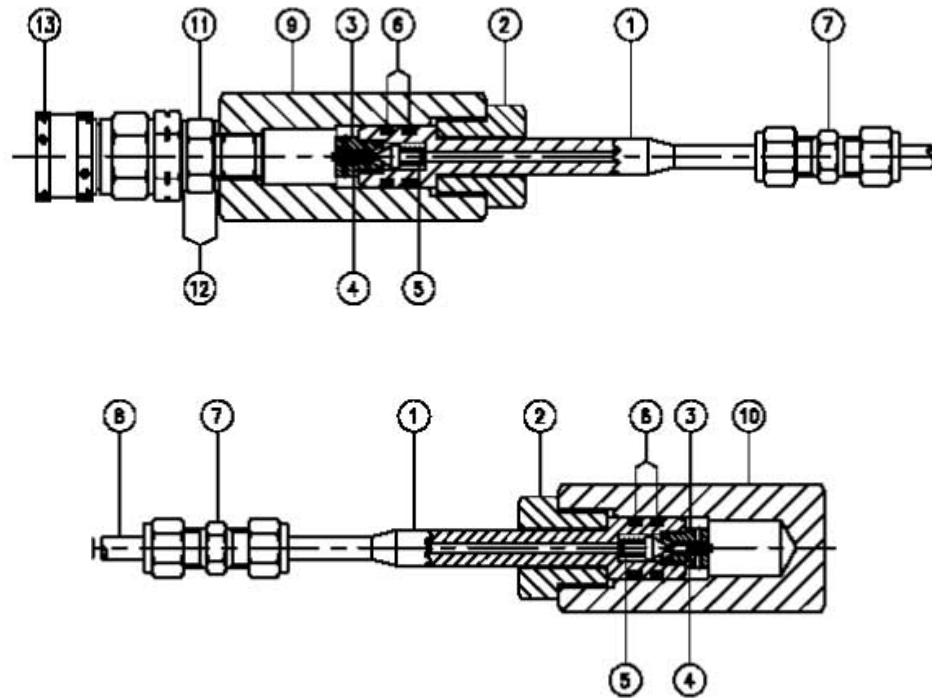


Figure 4-23: Old MultiSensor Discharge Pressure Line

The new Discharge Line and details of the jam nut and ferrules are as shown in the [Figure 4-24: New Gauge Discharge Pressure Line](#).

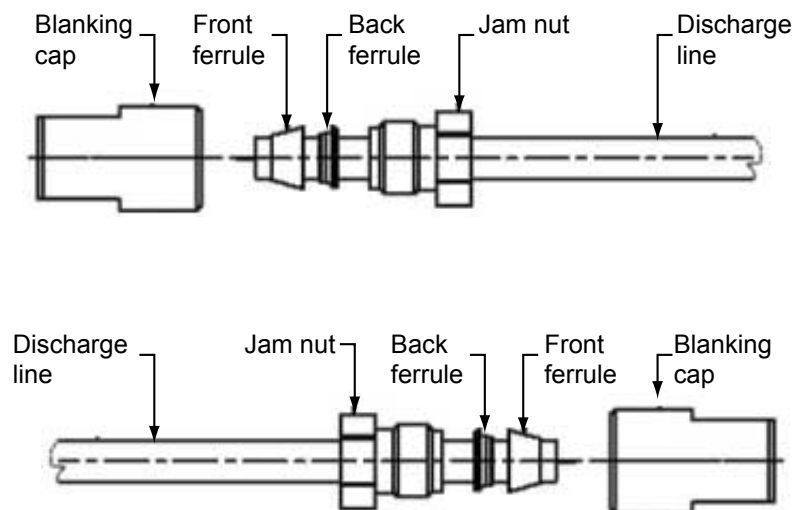


Figure 4-24: New Gauge Discharge Pressure Line

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Installation Procedures

Verify Port Depth

When using Reda discharge pressure sub, the port depth must be verified prior to the jam nut installation and cutting the discharge line. The port depth for should be between 2.3-in and 2.4-in.



Note

If the port depth is between 1.0-in to 1.4-in, the control line must be cut close to the MIN line on discharge pressure sub. Do not exceed the measured port length.

1. Verify the depth of the port on the discharge pressure sub.
 - a. Insert a short piece of control line in the port.
 - b. Mark the control line at the discharge pressure sub.
 - c. Remove the control line and measure the distance from the inserted end to the mark.



Figure 4-25: Measure Port Depth

2. Hold the discharge line against the discharge pressure sub and mark the cable.
 - If the port depth is 2.3-in to 2.4-in, mark the control line below MAX line and above MIN line on discharge pressure sub.

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Figure 4-26: Standard Port Depth Between 2.3-in and 2.4-in

- If the port depth is 1.0-in to 1.4-in, mark the control line at or close to MIN line on discharge pressure sub.



Figure 4-27: Port Depth Between 1.0-in and 1.4-in

3. Cut the control line at the mark.

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Jam Nut Installation with New Discharge Line

**Note**

The discharge line has been filled with fluid of a known density so that error correction for the length of the line can be made when the data is being analyzed. It is important to minimize the loss of hydraulic fluid during this process.

1. Refer to [Figure 4-24: New Gauge Discharge Pressure Line](#). Remove the blanking cap (1) from one end of the line and replace with the Discharge Line Adaptor. Attach the other end of the Adaptor to a hand pump and close the bleed valve on the pump.
2. Remove the blanking cap on the other end of the line and make up to the gauge discharge port. Tighten the connection to 15 ft-lbf (equivalent to 1-1/4 turns).

**Note**

Before installing the discharge line inspect the ferrules on the end of tubing. If they are damaged or scratched cut off the end of the tubing and replace them with new parts.

3. With the hand pump, apply pressure (3,000 psi) to the hydraulic oil. Hold pressure at 3,000 psi for 10 min. Watch the fitting and tubing for pressure drop-off to ensure no leaks are present.
4. Once the tubing and Swagelok fitting pass the hydrostatic pressure test, position the 1/4-in discharge line so that it can be banded to the ESP string or is contained in the ESP neck clamps.

**Note**

Position the 1/4-in discharge line as close as possible along the OD of the gauge and the ESP string to avoid snagging when the string is run into the well.

5. The discharge line will be terminated at the discharge pressure sub.
 - a. Cut the discharge line to length. Refer to [4.4.3.5 Verify Port Depth](#) for details.
 - b. De-burr the discharge line edges.
 - c. Insert the precut discharge line into the discharge port.

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Figure 4-28: Install Discharge Line

- d. Screw the Swagelok fitting into the discharge pressure sub port.
- e. Tighten the jam nut connection to 15 ft-lbf (equivalent to 1-1/4 turns). Refer to [Figure 4-24: New Gauge Discharge Pressure Line](#) for details of the jam nut and ferrules.



Figure 4-29: Tighten Connection

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Jam Nut Installation with Old Discharge Line



Note

The has been filled with fluid of a known density so that error correction for the length of the line can be made when the data is being analyzed. It is important to minimize the loss of hydraulic fluid during this process.

Refer to [Figure 4-23: Old MultiSensor Discharge Pressure Line](#) for the following procedures.

1. Cut the lower connection off using a tubing cutter above the lower tube coupling (7).
2. De-burr the edges.
3. Slide the jam nut, back ferrule and front ferrule onto the tube and make up to the discharge port on the gauge.
4. Tighten the connection to 15 ft-lbf (equivalent to 1-1/4 turns). Refer to [Figure 4-24: New Gauge Discharge Pressure Line](#) for details of the jam nut and ferrules.
5. Attach the other end of the Adaptor to a hand pump. If fittings are not readily available, cut off the upper connection completely and make up the tube to the Discharge Line Adapter.



Note

To avoid fluid loss always ensure that at least one end of the line is sealed off when cutting the tube.

6. With the hand pump, apply pressure (3,000 psi) to the hydraulic oil. Hold pressure at 3,000 psi for 10 min. Watch the fitting and tubing for pressure drop-off to ensure no leaks are present.
7. Once the tubing and Swagelok fitting pass the hydrostatic pressure test, position the 1/4-in discharge line so that it can be banded to the ESP string or is contained in the ESP neck clamps.



Note

Position the 1/4-in discharge line as close as possible along the OD of the gauge and the ESP string to avoid snagging when the string is run into the well.

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8. The discharge line will be terminated at the discharge pressure sub.
 - a. Cut the discharge line to length. Refer to [4.4.3.5 Verify Port Depth](#) for details.
 - b. De-burr the discharge line edges.
 - c. Insert the precut discharge line into the discharge port.



Figure 4-30: Install Discharge Line

- d. Screw the Swagelok fitting into the discharge pressure sub port.
 - e. Tighten the jam nut connection to 15 ft-lbf (equivalent to 1-1/4 turns). Refer to [Figure 4-24: New Gauge Discharge Pressure Line](#) for details of the jam nut and ferrules.

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Figure 4-31: Install Connection

NPT Bulkhead Installation with Old Discharge Line



Note

The has been filled with fluid of a known density so that error correction for the length of the line can be made when the data is being analyzed. It is important to minimize the loss of hydraulic fluid during this process.

Refer to [Figure 4-23: Old MultiSensor Discharge Pressure Line](#) for the following procedure.

1. Cut the lower connection off using a tubing cutter above the lower tube coupling (7).
2. De-burr the edges.
3. Slide the jam nut, back ferrule and front ferrule onto the tube and make up to the discharge port on the gauge.
4. Tighten the connection to 15 ft-lbf (equivalent to 1-1/4 turns). Refer to [Figure 4-24: New Gauge Discharge Pressure Line](#) for details of the jam nut and ferrules.
5. Attach the other end of the Adaptor to a hand pump. If fittings are not readily available, cut off the upper connection completely and make up the tube to the Discharge Line Adapter.

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**Note**

To avoid fluid loss always ensure that at least one end of the line is sealed off when cutting the tube.

6. With the hand pump, apply pressure (3,000 psi) to the hydraulic oil. Hold pressure at 3,000 psi for 10 minutes. Watch the fitting and tubing for pressure drop-off to ensure no leaks are present.
 7. Once the tubing and Swagelok fitting pass the hydrostatic pressure test, position the 1/4-in discharge line so that it can be banded to the ESP string or is contained in the ESP neck clamps.
-

**Note**

Position the 1/4-in discharge line as close as possible along the OD of the gauge and the ESP string to avoid snagging when the string is run into the well.

8. The discharge line will be terminated at the discharge pressure sub.
 - a. Cut the discharge line to length.
 - b. De-burr the edges.
 - c. Slide the tube into the loosened Swagelok fitting on the bulkhead fitting.
 - d. Tighten the connection to 15 ft-lbf (equivalent to 1-1/4 turns).
-

NPT Bulkhead Installation with New Discharge Line

**Note**

The has been filled with fluid of a known density so that error correction for the length of the line can be made when the data is being analyzed. It is important to minimize the loss of hydraulic fluid during this process.

Refer to [Figure 4-24: New Gauge Discharge Pressure Line](#) for the following procedure.

1. Remove the blanking cap (1) from one end of the line and replace with the Discharge Line Adaptor.
2. Attach the other end of the Adaptor to a hand pump and close the bleed valve on the pump.

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3. Remove the blanking cap on the other end of the line and make up to the gauge discharge port.
4. Tighten the connection to 15 ft-lbf (equivalent to 1-1/4 turns).

**Note**

Before installing the discharge line inspect the ferrules on the end of discharge line. If they are damaged or scratched cut off the end of the discharge line and replace them with new parts.

5. With the hand pump, apply pressure (3,000 psi) to the hydraulic oil. Hold pressure at 3,000 psi for 10 minutes. Watch the fitting and tubing for pressure drop-off to ensure no leaks are present.
6. Once the tubing and Swagelok fitting pass the hydrostatic pressure test, position the 1/4-in discharge line so that it can be banded to the ESP string or is contained in the ESP neck clamps.

**Note**

Position the 1/4-in discharge line as close as possible along the OD of the gauge and the ESP string to avoid snagging when the string is run into the well.

7. The discharge line will be terminated at the discharge pressure sub.
 - a. Cut the discharge line to length.
 - b. De-burr the edges.
 - c. Slide the tube into the loosened Swagelok fitting on the bulkhead fitting.
 - d. Tighten the connection to 15 ft-lbf (equivalent to 1-1/4 turns).

Large Port Installation with Old Discharge Line

**Note**

The has been filled with fluid of a known density so that error correction for the length of the line can be made when the data is being analyzed. It is important to minimize the loss of hydraulic fluid during this process.

Refer to [Figure 4-23: Old MultiSensor Discharge Pressure Line](#) for the following procedure.

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1. Cut the lower connection off using a tubing cutter above the lower tube coupling (7).
2. De-burr the edges.
3. Slide the jam nut, back ferrule and front ferrule onto the tube and make up to the discharge port on the gauge.
4. Tighten the connection to 15 ft-lbf (equivalent to 1-1/4 turns). Refer to [Figure 4-24: New Gauge Discharge Pressure Line](#) for details of the jam nut and ferrules.
5. Attach the other end of the Adaptor to a hand pump. If fittings are not readily available, cut off the upper connection completely and make up the tube to the Discharge Line Adapter.

**Note**

To avoid fluid loss always ensure that at least one end of the line is sealed off when cutting the tube.

6. With the hand pump, apply pressure (3,000 psi) to the hydraulic oil. Hold pressure at 3,000 psi for 10 min. Watch the fitting and tubing for pressure drop-off to ensure no leaks are present.
7. Once the tubing and Swagelok fitting pass the hydrostatic pressure test, position the 1/4-in discharge line so that it can be banded to the ESP string or is contained in the ESP neck clamps.

**Note**

Position the 1/4-in discharge line as close as possible along the OD of the gauge and the ESP string to avoid snagging when the string is run into the well.

8. The discharge line will be terminated at the discharge pressure sub.
 - a. Make up the top end of the connection (cut off in step 5) to the discharge sub.
 - b. Cut the discharge line to length.
 - c. De-burr the edges and install a Swagelok fitting on the end of the discharge line.
 - d. Screw the Swagelok fitting into the discharge pressure sub port and tighten the connection to 15 ft-lbf (equivalent to 1-1/4 turns).

Large Port Installation with New Discharge Line

This installation is not supported, as the fittings required to make up the discharge port are not supplied with the new discharge line. If fittings are available by other means (top part shown in [Figure 4-23: Old MultiSensor Discharge Pressure Line](#)) then proceed as per section 4.4.3.5 Large Port Installation with Old Discharge Line.

4.4.3.6 Select Remote Sensor Installation Procedure

The Phoenix Select remote sensors connect directly to the base gauge using a single conductor cable that provides the conduit for both power and digital telemetry. Refer to the Phoenix Select ESP Operations Manual ([InTouch ID 4399658](#)) for the remote sensor installation procedure.

4.4.4 Troubleshooting

Refer to the troubleshooting section in the system acquisition manual if errors are reported by the surface equipment.

- Integrated Surface Panel (ISP) Operation Manual ([InTouch ID 3957143](#))
- UniConn Reference Guide ([InTouch ID 3953183](#))

4.5 Motor Installation



Note: Maximus information

For Maximus, refer to [Appendix C: Maximus](#).

Before picking up the first section of the motor into a vertical position, a tail clamp should be positioned approximately one-third of the way up from the bottom. This provides support when the motor is being hoisted. A crane should be used in conjunction with the winch line on the rig to safely hoist the equipment to a point where the chain assembly on the motor clamp can be secured into the hook assembly in the elevators. As the blocks are raised, the crane helps support the motor's weight to prevent it from bending. Turn the elevators so that the weight is not against the latch. [Figure 4-32](#) shows a motor being lifted with a rig winch and crane.

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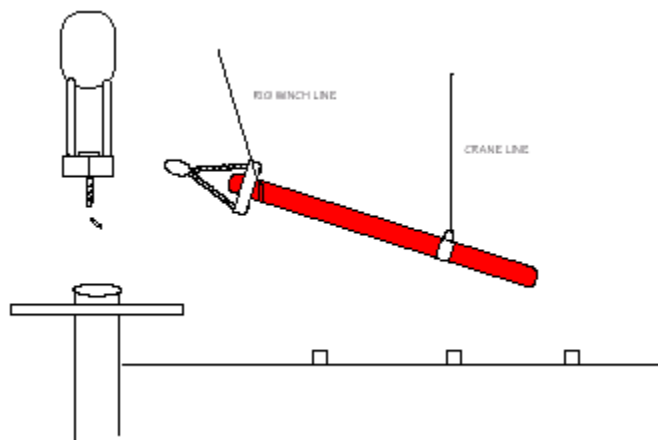


Figure 4-32: Lifting the motor with a rig winch and crane

4.5.1 Alternate Lifting Method

If a crane is not available, an alternate lifting method can be used to hoist a motor into position. The method uses a winch line and tag line. The tag line is used to control movement towards the rig floor. A tail clamp is positioned one-third of the way down from the top for the winch line. The winch line and traveling blocks are operated by one person. The motor is lifted using the winch line until near vertical and then the load is transferred onto the main blocks keeping the winch line slack. Another winch line or catline should be attached to the tag line to prevent contact with the rig floor or other obstructions. The lifting procedure is shown in [Figure 4-33](#).



Note

The position of the tail clamp is important. Positioning the clamp one-third of the way down from the top places the least amount of stress on the motor, preventing it from bending.

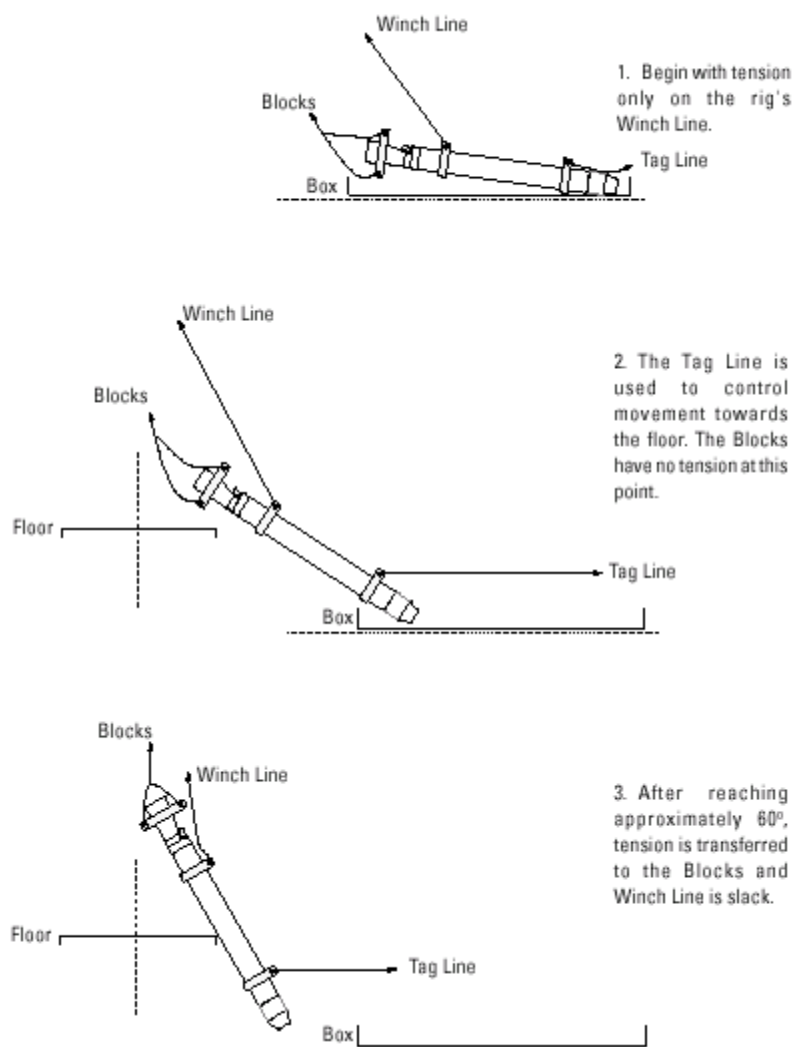


Figure 4-33: Lifting the motor with a rig winch and tag line

4.5.2

Motor Oiling Procedure (Single-Section)

The motor is ready for servicing once it is in a vertical position over the well.



Note

The 456 Dominator Motor has a short Drain and Fill Valve. Alemite adaptors produced prior to 2009 may not be suitable for filling these motors, please see [InTouch Content 4366376](#).

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Proper installation of plugs is vital to ensuring the Total-Run-Life (TRL) of the ESP system is achieved. Ensure the vent plugs (standard, short and long) and drain-and-fill valves used are the latest revision per the OneCAT bill of materials.



Figure 4-34: Standard (Vent and Drain) Plug



Figure 4-35: Short and Long Vent Plugs



Figure 4-36: Drain and Fill Valve

When a plug or valve is removed and installed the following steps must be taken:

1. Remove the plug or valve. If removing the plug, use a 3/4-in wrench to backup the valve to prevent loosening the valve.
2. Remove the old lead gasket. Do not reuse lead gaskets.
3. Install a new lead gasket. Do not install two lead gaskets.
4. Install the plug (with lead gasket) or valve (with lead gasket). Torque plugs to 15 ft-lbf and valves to 35 ft-lbf.



Caution

Vent plug and drain-and-fill valve

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head, before and after servicing.

1. At the bottom of the motor locate the drain-and-fill valve. Tighten this assembly to 35 ft-lbf.
2. Lower the motor into the well. Remove the top shipping cap, coupling and the Allen screw (shaft plug) from the center of the shaft.
3. Reinstall the shipping cap with a lock washer between the cap and motor head flange to provide a vent during the oil filling procedure.

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4. Lift the motor out of the well. Remove the vent plug and lead gasket from the drain-and-fill valve at the base of the motor.

**Note**

When removing the vent plug, use a 3/4-in wrench to back-up the drain-and-fill valve to prevent it from loosening.

5. Select the appropriate oil for the application and connect the oil filling pump, hose and adapter assembly to the oil can.

**Note**

Purge the air from the oil filling hose with oil before filling the motor.

6. Install the adapter and slowly pump (45 rev/min) oil into the motor until it flows out of the motor head. Wait for five minutes to allow air bubbles to migrate out the top of the motor and then pump again until oil flows. Continue this process until it takes only one turn of the oil pump to cause oil to flow from the top of the motor.
7. Remove the adapter and hose from the motor and install the vent plug with a new lead gasket (p/n 15008) into the drain-and-fill valve. Torque the vent plug to 15 ft-lbf.
8. Lower the motor into the well until the clamp rests on top of the worktable. Remove the motor shipping cap and slowly pump oil down the center of the shaft to ensure all air is expelled from the motor. Rotating the shaft rapidly with a spline wrench will help displace any air.

**Note**

Ensure no moisture or debris enters the motor.

9. Replace the shaft plug and tighten. Be sure the plug is not protruding past the end of the shaft when tight.
10. Replace the coupling. Ensure the shaft rotates smoothly. Servicing is now complete.
11. Check all drain-and-fill valves and vent plugs to insure that they are torqued to specification.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

4.5.3 Tandem Motor Connection

The motor is ready for servicing once it is in a vertical position over the well.

1. At the bottom of the motor locate the drain-and-fill valve. Tighten this assembly to 35 ft-lbf.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head, before and after servicing.

2. Lower the motor into the well. Loosen the top shipping cap and place a lock washer between the cap and motor head flange to provide a vent during the oil filling procedure.
3. Lift the motor out of the well. Remove the vent plug and lead gasket from the drain-and-fill valve at the base of the motor.

**Note**

When removing the vent plug, use a 3/4-in wrench to back-up the drain-and-fill valve to prevent it from loosening.

4. Select the appropriate oil for the application and connect the oil filling pump, hose and adapter assembly to the oil can.

**Note**

Purge the air from the oil filling hose with oil before filling the motor.

5. Install the adapter and slowly pump (45 rev/min) oil into the motor until it flows out of the motor head. Wait for five minutes to allow air bubbles to migrate out the top of the motor and then pump again until oil flows. Continue this process until it takes only one turn of the oil pump to cause oil to flow from the top of the motor.
6. Remove the adapter and hose from the motor and install the vent plug with a new lead gasket (p/n 15008) into the drain-and-fill valve. Torque the vent plug to 15 ft-lbf.

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7. Lower the motor into the well until the clamp is approximately 18 in from the top of the worktable.
8. Slide the motor jack assembly (p/n 100541677, 375/456/540/562-series motors or p/n 44842, 738-series motors) into position on the worktable. Ensure that both hydraulic jacks are in their retracted position.
9. Lower the motor until the clamp rests in the cradle on top of the jacks.
10. Remove the motor shipping cap and slowly pump oil down the center of the shaft to ensure all air is expelled from the motor. Rotating the shaft rapidly with a spline wrench will help displace any air.

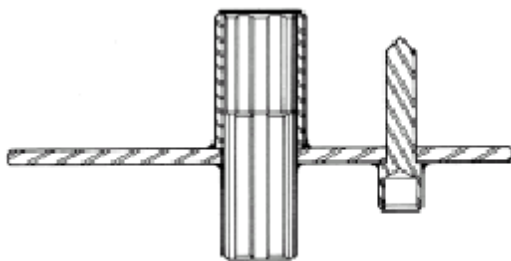
**Note**

Ensure no moisture or debris enters the motor.

11. Select the spline alignment tool from [Table 4-21](#). A spline alignment tool is shown in [Figure 4-37](#).

Table 4-21: Spline Alignment Tool Part Numbers

Motor Series	Shaft size	Part Number
375	0.68 in	44115
456	1.00 in	44271
456	1.12 in	455360
540	1.12 in	44297
540	1.18 in	44263
540	1.37 in	320937 (most common)
562	1.37 in	1079920
738	1.50 in	44289
738	2.00 in	2006419

**Figure 4-37: Spline alignment tool****Private**

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12. Place the alignment tool on the motor coupling and turn the motor shaft until the sleeve slides down over the guide pin on the motor head. (There is an alignment groove machined into the motor head and base indicating the location of the guide pin.) [Figure 4-38](#) shows the alignment tool in place.

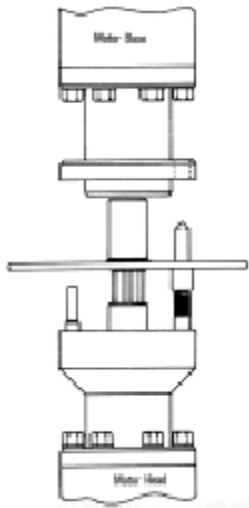


Figure 4-38: Spline alignment tool in place for tandem motor connections

13. Pick up the next motor and remove the lower shipping cap. Place the alignment tool on the upper motor shaft and turn until the pin of the alignment tool slips into the alignment hole on the motor base. The splines on the two shafts are now mated to ensure proper engagement of the motor coupling.
14. Remove the old O-ring from motor base.



Caution

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

Do not use a metal tool when removing the O-ring from the flange groove. Damaged sealing surfaces can result in short equipment runlife.

15. Lubricate a new O-ring with Dow Corning 111 (p/n 100386898). Carefully install the O-ring into its correct position. Refer to the ALFORM-D connection tables ([InTouch ID: 3255850](#)) for a complete list of O-ring part numbers, coupling and bolt torque information.



Note

For 540 series and 562 series protector bases, a second backup O-ring is NOT required above the O-ring groove between tandem motors and on PSI units.

16. Inspect the terminal pins and insulators to ensure they are not bent or broken. Have the rig operator lower the top motor until it is approximately 4 to 6 inches above the lower motor. Then pump both jacks simultaneously to slowly raise the lower motor and engage the coupling, making sure the alignment pin slides into the respective hole and that the motor terminals are properly engaged.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Use care not to damage the terminal insulators or O-rings.

17. Install the bolts and lock washers. Torque to the specifications listed in the connection table.
18. Lower both motors into the well. Loosen the top shipping cap and place a lock washer between the cap and the motor head flange to provide a vent.
19. Pull the motors completely out of the well and repeat the oil filling procedures; this time filling both motors. In the event that all motors cannot be pulled from the well because of limited crown room, fill from the motor head of the lowest possible motor. Replace the vent plug with a new lead gasket (p/n 15008) and ensure all other plugs are tight (15 ft-lbf).
20. Phase-to-phase resistance checks should be made after each connection. Resistances are additive and should always be balanced.
21. Check all drain-and-fill valves and vent plugs to insure that they are torqued to specification.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

22. Repeat the above procedures for each motor that is added.

4.5.4

Upper Tandem Motor Oiling Procedure

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

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Before servicing the upper tandem motor, remove the allen screw (shaft plug) located in the center of the shaft. Service the motor with oil, replace the shaft plug and tighten.



Note

Ensure the plug is not protruding past the end of the shaft when tight.

Replace the coupling and check to ensure the shaft rotates smoothly. Service is now complete.

4.6 **Protector Assembly**



Note: Maximus information

For Maximus, refer to [Appendix C: Maximus](#).

4.6.1 **Protector Installation**

Protectors are lifted using the same methods used to lift motors. Motor lifting procedures are provided in the *Motor Installation* section.

- 1. Lift the protector to a position two to three feet above the motor.
- 2. Remove the lower shipping cap and gasket.
- 3. Remove the old O-ring from protector base.



Caution

Potential Severity: Light
Potential Loss: Assets
Hazard Category: Machinery equipment hand tools

Do not use a metal tool when removing the O-ring from the flange groove. Damaged sealing surfaces can result in short equipment runlife.

- 4. Lubricate a new O-ring with Dow Corning 111 (p/n 100386898). Refer to the ALFORM-D connection tables ([InTouch ID: 3255850](#)) for a complete list of O-ring part numbers, coupling and bolt torque information.

**Note**

Do not use metal tool when removing the O-ring from the flange groove. This can be the reason of equipment short runlife due to damage on the sealing surfaces.

A second backup O-ring is NOT required above the groove O-ring on 540 series and 562 series protector bases, between tandem motors and on PSI units.

Always test couplings on a test shaft prior to be sent to the field. Also, once crew is on the rig site, test couplings on the equipment that is going to be used in the installation. Do not rely on part numbers only.

5. Use the spline wrench to ensure the motor and protector turn free and smooth.
6. Place the coupling on the motor shaft and have the rig operator slowly lower the protector until the shafts are engaged.
7. Turn the protector so that the flat cable slots on the protector flange and motor flange line up with each other. Lower the protector until the flanges meet, making sure not to damage the O-ring. Install the bolts and torque to specification.
8. Pick up the assembly, remove the motor clamp and lower the assembly until the weight rests on the protector clamp.

**Note****Installation Caution, All Series Protectors to All Series Motors**

If the upper snap-ring (coupling adapter retainer) is left in place on the protector's lower shaft spline, there is a possibility of a tolerance stack-up which can cause a protector problem, motor problem or both. Observe the following procedures when installing all series protectors to all series motors.

When a Universal Coupling Adapter is on the Protector Lower Shaft Splines

1. Remove the retaining snap-ring from below adapter.
2. Remove the adapter from protector shaft.
3. Remove the upper retaining snap-ring from protector shaft.

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When a Universal Coupling Adapter is not on the Protector Shaft

1. Feel the lower protector shaft to determine if either snap-ring is in its groove. If either is present, remove and discard.
2. Proceed with installation procedure.

In either case, the protector lower shaft spline must not have any snap rings installed when connecting to a 562 motor. Leaving the snapping installed can force the protector in upthrust and the motor in downthrust, resulting in a failure.

4.6.2 Tandem Protector Assembly

When installing tandem protectors remove the spline adapter from the shaft of the upper protector. Remove the adapter with an ice pick.

4.6.3 Oil filling procedures

Protectors are filled with oil prior to shipment from the manufacturing facility. Typically, some oil is lost during field installation and refilling the protector is required. Oil refilling procedures at the wellsite are provided in [4.9: Oil Filling Procedures for Protectors](#).

4.7 Pump Shimming and Installation

Procedures and specifications for field spacing (shimming) submersible pumps are provided in this document. Select the proper thrust setting tools and shims based on the series of pump equipment and shaft sizes and then follow applicable procedures.

1. Always check the equipment first and make a note of all the nameplate data.
2. Always make sure the equipment on location is the equipment that has been designed for this particular well.
3. Always remove BOH to replace O-ring before RIH, then install BOH back using new bolts. Refer to the ALFORM-D connection tables ([InTouch ID: 3255850](#)) for a complete list of O-ring part numbers, coupling and bolt torque information.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

Do not use a metal tool when removing the O-ring from the flange groove. Damaged sealing surfaces can result in short equipment runlife.

4. In large pumps (series 675 and higher) top shipping cap should be removing the first on the rig floor to purge the vacuum for easy removing bottom shipping cap in further. Then install top shipping cap on place to follow installation procedure.

On pumps built in compression construction, the shafts must be matched to the protector so that the thrust developed by the pump can be transferred to the protector thrust bearing.

High BHT applications, those above 250 degF with multiple tandem pumps, should be evaluated for possible excessive shaft growth. Please submit downhole conditions and pump configurations to InTouch for these cases.

**Warning**

Potential Severity: Serious
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

The lower part of string should be covered while shimming.

4.7.1

Tool Requirements

Proper shaft spacing requires the use of thrust setting tools, commonly referred to as H-tools, for the specific pump series and configuration. Tool descriptions and part numbers are shown in [Table 4-22](#).

Table 4-22: Thrust Setting Tools

Thrust Setting Tool Description	Part Number
338 pump to 338 pump or 338 pump to 338 intake or LT Pump	1089044
338, 387, or 400 intake to 325, 375, or 400 protector	1088996
387 pump to 387 pump or 387 pump to 387 intake	1089036
400 pump to 400 pump or 400 pump to 400 intake or DN5–21 AGH	1089028
400 pump (CT) to D20–60 VGSA	1089028
D4300N pump to pump or D20-60 VGSA or D20-60 AGH (400 series)	2005330
D4300N pump to pump or D20-60 VGSA or D20-60 AGH (387 series)	100024881

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Thrust Setting Tool Description	Part Number
D5800N pump to pump or D20-60 VGSA or D20-60 AGH (400 series)	2005330
D5800N pump to pump or D20-60 VGSA or D20-60 AGH (387 series)	100024881
D8-42 Poseidon (Extended Shaft) to D20-60 VGSA (400 series)	2005330
D8-42 Poseidon (Extended Shaft) to D20-60 VGSA (387 series)	100024881
540 pump to 540 pump or 540 pump to 540 intake or 562/540 pump to 540 protector	1083930
562 pump to 562 pump or 562 pump to H100–250 AGH	1083948
650 pump to 650 pump	48504
675 pump to 675 pump or 675 pump to 675/562 AGH	1050400
725 pump to 725 pump or 725 pump to 725/562 AGH	1320045
862 pump to 862 pump or 862 pump to 862/562 pump	1089432
950 pump to 950 pump or 950 protector	354597
650, 675, 725, 862, 950, 1000 and 1125 LT and integral intake pumps with 738 flange to 738 protector	46672
D20–60 VGSA to D20–60 AGH (CT)	2005330

4.7.2

Material Requirements

Shim material required to properly match shafts is shown in [Table 4-23](#).

Table 4-23: Shim Material

Shim, 0.550-in Diameter, to Fit 0.62–, 0.68– and 0.87-in Shafts	
Shim Thickness	Part Number
1/200-in thick (0.005-in)	7005218
1/64-in thick	1146497
1/32-in thick	1146505
1/16-in thick	1146513
Shim, 0.812-in Diameter, to Fit 1.00– inch and 1.187-in Shafts	
Shim Thickness	Part Number
1/200-in thick (0.005-in)	7005192
1/64-in thick	614461
1/32-in thick	614479
1/16-in thick	614487
Shim, 1.161-in Diameter, to Fit 1.37-in and 1.50-in Shafts	
Shim Thickness	Part Number
1/200-in thick (0.005-in)	7005200
1/64-in thick	1146521
1/32-in thick	1146539

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1/16-in thick

1146547

4.7.3

Spacing Requirements

Table 4-24 shows spacing requirements by component series and type.

Table 4-24: Spacing Requirements by Component Series and Type

Series	Type	Spacing Requirement
338	floaters	no requirement
338	bottom floater/compression	set flush
338	compression ring	flush to 1/64-in lift
387	floaters	no requirement
387	bottom floater/compression	set flush
387	compression ring	flush to 1/64-in lift
400	D5–21 AGH	flush to 1/64-in lift
400	D20–60 AGH	flush to 1/64-in lift
400	floaters	no requirement
400	bottom floater/compression	set flush
400	compression ring or compression	flush to 1/64-in lift
540	GN20–40 AGH (compression ring — compression)	flush to 1/64-in lift
540	GN40–80 AGH (compression ring — compression)	flush to 1/64-in lift
540	floaters	no requirement
540	bottom floater/compression	set flush
540	compression ring	flush to 1/64-in lift
538	floaters	no requirement
538	compression ring	flush to 1/64-in lift
562	H100–250 AGH with 562-series pumps containing HN10000, HN13000, HN13500, HN15000, HN21000 and HN28000 pump stages	to protector, flush to 1/64-in lift to pump, flush
562	H100–250 AGH with a 562-series pump containing H15500H and H21500N pump stages	to protector, flush to 1/64-in lift to pump, 1/16-in lift
562	compression except H15500N and H21500N	flush to 1/64-in lift
562	compression ring except H15500N and H21500N	flush to 1/64-in lift
562	compression, H15500N and H21500N pump stages	1/32-in lift

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Series	Type	Spacing Requirement
562	compression ring, H15500N and H21500N pump stages	1/32-in lift
562	compression, H22500N	1/16-in lift
650 and 675	compression	1/32-in lift
675	H100–250 AGH with a 675–series pump	to protector, flush to 1/64-in lift to pump, 1/16-in lift
725	compression	1/16-in lift
725	H100–250 AGH with a 725–series pump containing LN16000 pump stages	to protector, flush to 1/64-in lift to pump, 3/32-in lift
862	compression	1/16-in lift
862	H100–250 AGH with a 862–series pump	to protector, flush to 1/64-in lift to pump, 3/32-in lift
862	M90-470 Poseidon	to protector, flush to 1/64-in lift to pump, 1/16-in lift
950	compression	1/16-in lift
1000	compression	1/16-in lift
1125	compression	1/16-in to 3/32-in lift
all	hydraulic balance	set flush


Note
In 338–, 387–, 400–, 540– and 538–series pumps, compression pumps (typically available only in housing sizes less than 50 housing) are effectively the same as bottom floater pumps with the exception that the entire impeller stack is compressed. These pumps are not intended to transfer thrust to the protector thrust bearing.


Note
Compression or compression ring advanced gas handlers should be shimmed to the pump requirements shown in this table according to series.

4.7.4

Mixing Pumps

If compression type pumps and floater type pumps are being run in the same string the compression type pumps must be placed at the bottom of the string. Once all of the pumps are bolted together they must be turned using the correct tool to make sure the complete assembly rotates.

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If a floater type AGH (advanced gas handler) is being used with compression pumps, install the motor and protector as normal then install the intake, then the AGH. No shimming is required up to this point. Install the compression pumps per the standard procedure.

For tandem pump installations with high-strength shafts, ensure the pump with the high-strength shaft is installed first, next to the intake/protector.

4.7.5 **Factory Shimmed Equipment**

When installing factory filled or shimmed equipment it is not necessary to shim the equipment in the Field. Always make sure the complete assembly rotates smoothly.

4.7.5.1 **Mixing Factory Shimmed with non-Factory Shimmed Equipment**

When mixing factory shimmed and non-factory shimmed pumps, the factory shimmed pump section should be installed as the lowest pump in the string. That is next to the intake/protector.

If factory shimmed pumps and floater type pumps are being run in the same string the factory shimmed pumps must be placed at the bottom of the string. Once all the pumps are bolted together they must be turned using the correct tool to make sure the complete assemble rotates.

Use factory shimmed coupling between factory shimmed and standard equipment, apply the standard shimming procedure and standard spacing requirements. Standard couplings can be used between standard pumps.



Note

Factory shims sometimes get stuck in a shaft; confirm all of these shims are removed before assembling.



Note

Check for smooth, free rotation of each individual section and overall assembly after spacing.

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4.7.6

Shimming Procedure for Pumps (338–Series Through 562–Series Pumps Except 562–Series Pumps Containing H15500N and H21500N Stages)

**Note**

The following procedure is used to shim pumps (338–series through 562–series pumps except 562–series pumps containing H15500N and H21500N stages).

1. If a bolt-on intake or gas separator will be used, remove the shipping caps and assemble the intake/gas separator to the protector prior to beginning shaft matching. It is not necessary to shim between these units and the protector because the shaft in an intake or gas separator will sit down on the protector shaft.

**Note**

Verify that the bolt-on intake/gas separator, protector shafts and coupling all touch when bolted together. Be sure that there is some lift of the intake/gas separator upper shaft end when the equipment is assembled. If there is not sufficient shaft play for the shafts to touch and there is a gap between the shafts and coupling, install sufficient shims to fill the gap up to 1/64-in lift.

2. Pick up the intake/protector assembly with the proper lifting clamp and proceed with the protector-to-motor assembly, flat cable tie-in and protector service procedure.
3. Lower the assembled and serviced motor/protector/intake section into the well until the lifting clamp is resting on the wellhead.
4. Attach the appropriate lifting clamp to the lower pump section. Remove the shipping caps and check the shaft for free rotation and endplay. Tap the shaft into the full-down position with a rubber mallet or wood block before matching.

**Note**

It is advisable to record the shaft measurement at the bottom end with the shaft in the full-down position. This is to verify that the matching process is started from the full-down position after the pump is lifted vertical. This should be done on the pipe deck prior to starting the assembly.

5. Lift the pump to the vertical position approximately two to three feet above the intake head. After the pump is lifted, recheck the lower shaft setting to make sure the matching process is started from the full-down position.
6. Attach the proper thrust setting tool to the head of the intake with the coupling installed as shown in Figure 4-39. Loosen the jam nut on the center bolt and turn the bolt down until it touches the divider in the coupling. Tighten the bolt to 60 lbf in (5 lbf ft). **Do not overtighten!** Tighten the jam nut while holding a backup on the center bolt. Remove the thrust setting tool.

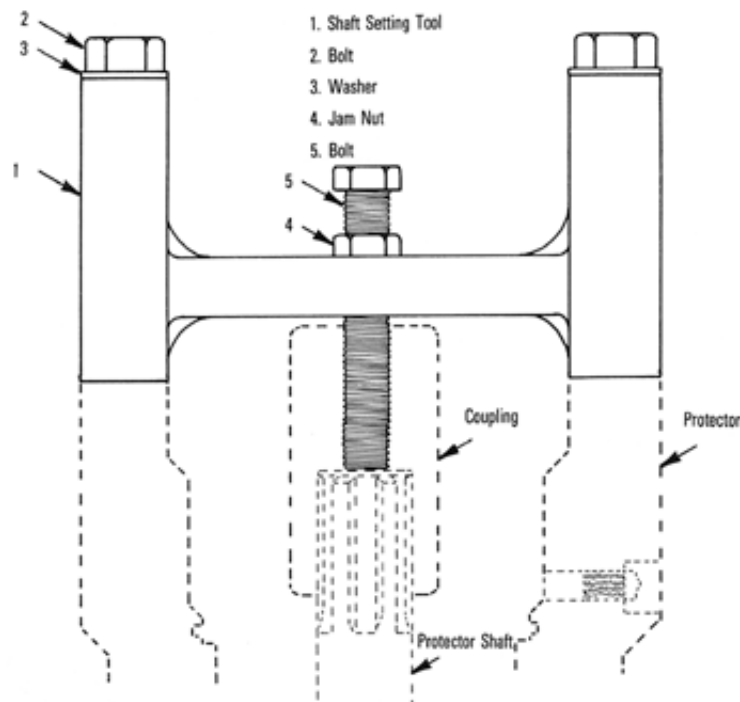


Figure 4-39: Thrust setting tool on the top end of the protector

7. Install the thrust setting tool on the pump base as shown in Figure 4-40. When doing so, reverse the attaching bolts, not the tool. The center bolt head must remain next to the bottom of the pump shaft. Using shims, fill space A (see Figure 4-40) between the pump shaft and the bolt head with the appropriate amount of shims to fill, but not lift the shaft.



Note

Start with 1/16-in shims and graduate down in thickness until no 1/200-in shims will fit into the gap. Once this is done, there will be less than 1/200-in gap remaining.

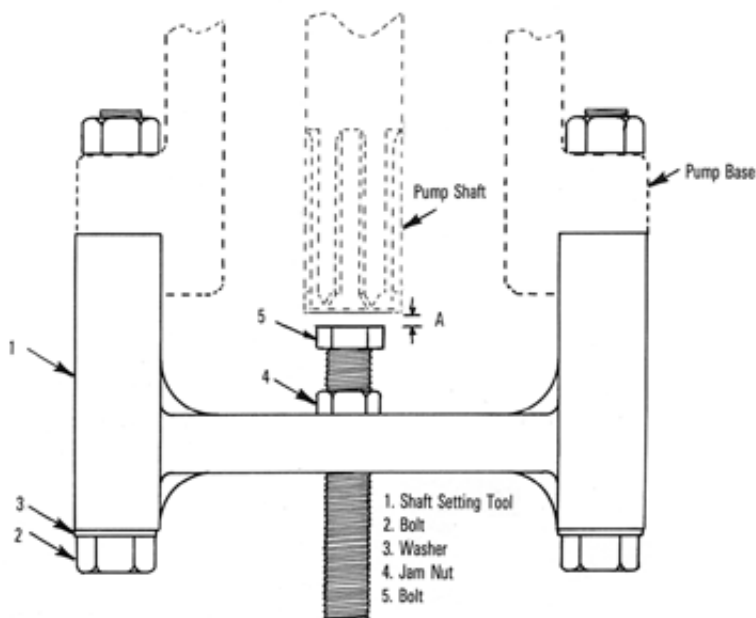


Figure 4-40: Thrust setting tool on the pump base

8. Remove the shims from space **A** and add a 1/64-in shim from the shim stack. This will satisfy the spacing requirement shown in [Table 4-24](#). Install the shim stack into the intake coupling. Remove the thrust setting tool from the bottom of the pump and install the pump on the intake. Tighten bolts in accordance with torque specifications.
9. Repeat steps 4, 5, 6, 7 and 8 for each additional tandem pump.



Note

- 1) Steps 4, 5, 6, 7 and 8 may also be used for matching integral-intake lower-tandem (LT) pumps to protectors. The proper thrust setting tool will be required to match from the LT base to the protector.
- 2) Check for smooth, free rotation of each individual section and overall assembly after spacing.

4.7.7

Shimming Procedure for Pumps (338-Series, 387-Series and 400-Series Pumps)



Note

Use this shimming procedure when measurements must be taken and available space restricts placing shims in the gap.

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1. If a bolt-on intake or gas separator will be used, remove the shipping caps and assemble the intake/gas separator to the protector prior to beginning shaft matching. It is not necessary to shim between these units and the protector because the shaft in an intake or gas separator will sit down on the protector shaft.

**Note**

Verify that the bolt-on intake/gas separator, protector shafts and coupling all touch when bolted together. Be sure that there is some lift of the intake/gas separator upper shaft end when the equipment is assembled. If there is not sufficient shaft play for the shafts to touch and there is a gap between the shafts and coupling, install sufficient shims to fill the gap up to 1/64-in lift.

2. Pick up the intake/protector assembly with the proper lifting clamp and proceed with the protector-to-motor assembly, flat cable tie-in and protector service procedure.
3. Lower the assembled and serviced motor/protector/intake section into the well until the lifting clamp is resting on the wellhead.
4. Attach the appropriate lifting clamp to the lower pump section (or AGH if being used). Remove the shipping caps and check the shaft for free rotation and endplay. Tap the shaft into the full-down position with a rubber mallet or wood block before matching.

**Note**

It is advisable to record the shaft measurement at the bottom end with the shaft in the full-down position. This is to verify that the matching process is started from the full-down position after the pump is lifted vertical. This should be done on the pipe deck prior to starting the assembly.

5. Lift the pump/AGH to the vertical position approximately two to three feet above the intake head. After the pump is lifted, recheck the lower shaft setting to ensure the matching process is started from the full-down position.
6. Attach the proper thrust setting tool to the head of the intake with the coupling installed as shown in [Figure 4-41](#). Loosen the jam nut on the center bolt and turn the bolt down until it touches the divider in the coupling. Tighten the bolt to 60 lbf-in (5 lbf-ft). **Do not overtighten!** Tighten the jam nut while holding a backup on the center bolt. Remove the thrust setting tool.

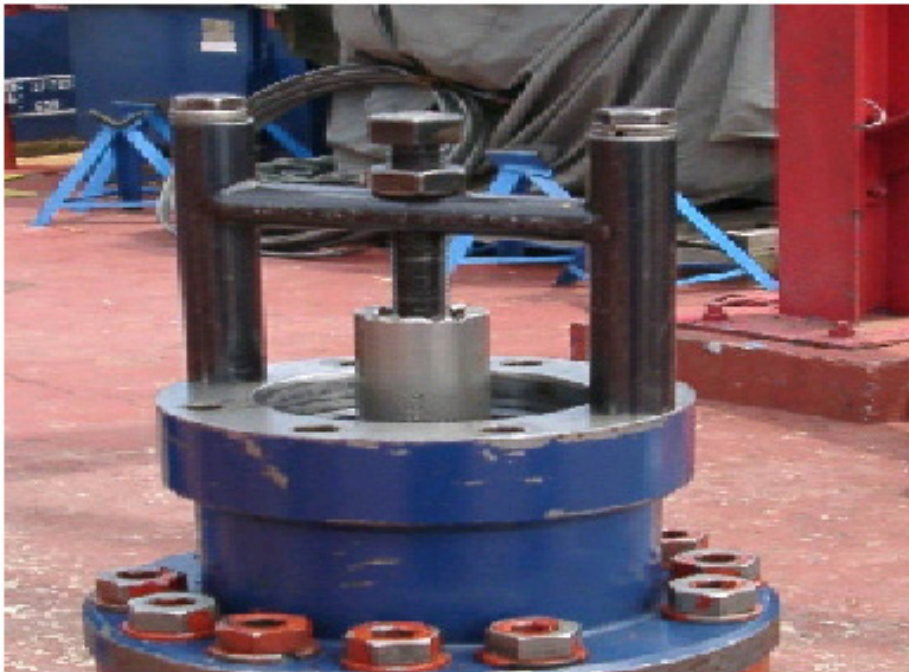


Figure 4-41: Shimming tool positioned on top of the intake adaptor

7. Remove the shimming tool from the flange. Measure the distance the center bolt is protruding from the horizontal part of the H (Figure 4-42). Ideally, this should be done with a vernier gauge. This is dimension **A**.



Figure 4-42: Measuring distance A on shimming tool

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8. Slacken the jam nut.
9. Install the thrust setting tool on the pump base as shown in Figure 4-43. When doing so, reverse the attaching bolts, not the tool. The center bolt head must remain next to the bottom of the pump/AGH shaft.

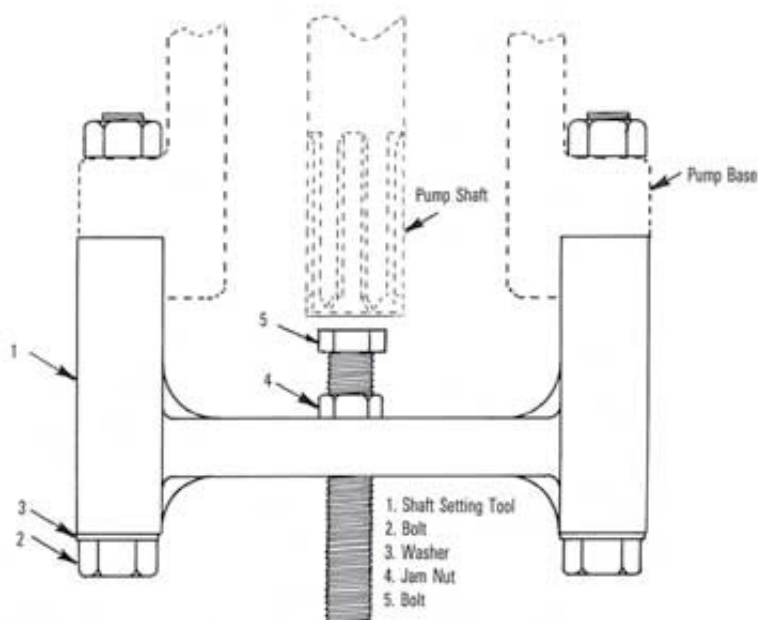


Figure 4-43: Thrust setting tool on the pump base

10. Using your fingers, adjust the center bolt until it touches the shaft above (do not lift the shaft) and measure the distance between the center bolt and the horizontal. This is dimension **B**.



Note

Dimension B must be less than dimension A.



Figure 4-44: Measuring Dimension B

11. Remove the thrust setting tool from the bottom of the pump.
12. Determine dimension **C** ($C = A - B$). Dimension **C** is the difference between dimension **A** and dimension **B** or the gap between the coupling divider and the shaft.
13. Set up value **C** on the caliper. Using shims, fill up the space between the caliper legs with the appropriate amount of shims to fill the gap. Refer to [Figure 4-45](#).



Note

Start with 1/16-in shims and graduate down in thickness until no 1/200-in shims will fit into the gap. Once this is done, there will be less than 1/200-in gap remaining.

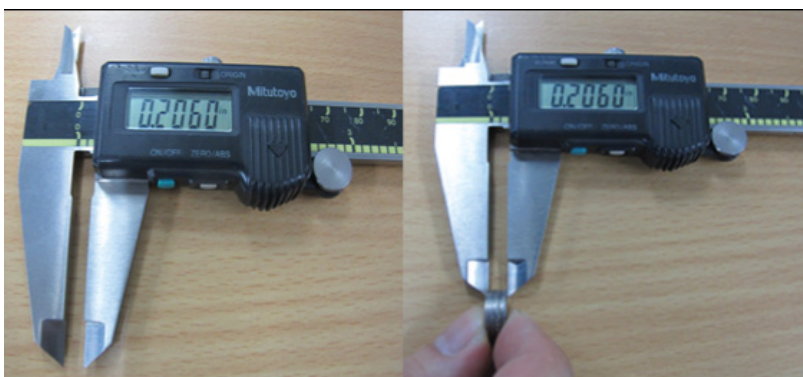


Figure 4-45: Setting up caliper to reflect dimension C

14. Remove the shims from the caliper and add a 1/64-in to this shim stack. This will satisfy the spacing requirement shown in [Table 4-24](#).

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15. Install the shim stack on the intake coupling and subsequently install the pump onto the intake. Tighten the bolts in accordance with the torque specifications.
16. Repeat steps 4 to 15 for each additional tandem pump.

**Note**

Check for smooth, free rotation of each individual section and overall assembly after spacing.

4.7.8

Shimming Procedure for Pumps Requiring Lift (562-Series Pumps Containing H15500N and H21500N Stages and 650-Series Through 1125-Series Pumps)

**Note**

The following procedure is used to shim pumps requiring lift (562-series pumps containing H15500N and HN21500N stages and 650-series through 1125-series pumps). The shafts of these pumps must be matched for lifting, instead of with a gap which would allow the shaft to drop as it transfers load to the protector thrust bearing. This done because larger pumps do not have a downthrust washer for the impellers to ride on.

Procedures for determining the clearance between the protector shaft and pump shaft and setting thrust are:

1. If a bolt-on intake or gas separator will be used, remove the shipping caps and assemble the intake/gas separator to the protector prior to beginning shaft matching.
-

**Note**

Verify that the bolt-on intake/gas separator, protector shafts and coupling all touch when bolted together. Be sure that there is some lift of the intake/gas separator upper shaft end when the equipment is assembled. If there is not sufficient shaft play for the shafts to touch and there is a gap between the shafts and coupling, install sufficient shims to fill the gap up to 1/64-in lift.

2. Pick up the intake/protector assembly with the proper lifting clamp and proceed with the protector-to-motor assembly, flat cable tie-in and protector service procedure.

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3. Lower the assembled and serviced motor/protector/intake section into the well until the lifting clamp is resting on the wellhead.
4. Attach the appropriate lifting clamp to the lower pump section. Remove the shipping caps and check the shaft for free rotation and endplay. Tap the shaft into the full-down position with a rubber mallet or wood block before matching.



Note

It is advisable to record the shaft measurement at the bottom end with the shaft in the full-down position. This is to verify that the matching process is started from the full-down position after the pump is lifted vertical.

5. Lift the pump to the vertical position approximately two to three feet above the intake head. After the pump is lifted, recheck the lower shaft setting to ensure the matching process is started from the full-down position.
6. Attach the proper thrust setting tool to the head of the intake/protector with the coupling installed as shown in [Figure 4-41](#). Loosen the jam nut on the center bolt and turn the bolt down until it touches the divider in the coupling. Tighten the bolt to 60 lbf-in (5 lbf-ft). **Do not overtighten!** Tighten the jam nut while holding a backup on the center bolt. Remove the thrust setting tool.
7. Install the thrust setting tool on the pump base as shown in [Figure 4-46](#). When doing so, flip the attaching bolts, not the tool. The center bolt head must remain next to the bottom of the pump shaft. Using shims, fill space **A** (see [Figure 4-46](#)) between the pump shaft and the bolt head with the appropriate amount of shims to fill the space and place the shims in the coupling.

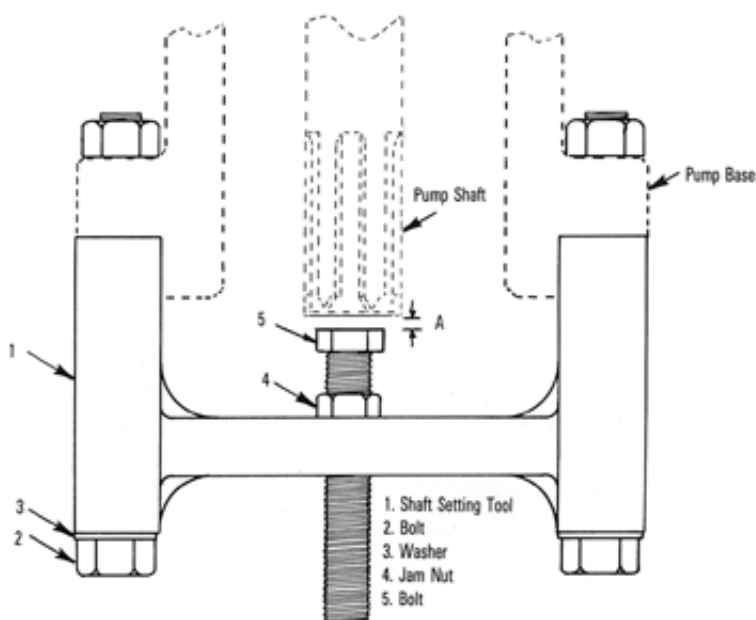


Figure 4-46: Thrust setting tool on the pump base

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8. Remove the thrust setting tool. Select additional shims to achieve the correct shaft lift. The correct shaft lift is determined by the pump series. (Specifications are provided in [Table 4-24](#).) Place these additional shims in the coupling and bolt the pump to the intake/protector. Torque to specifications.
9. Set the thrust on additional tandem pumps using procedures shown in steps 3, 4, 5, 6, 7 and 8.

**Note**

Each additional tandem pump must be lifted the same amount as the lower pump (that is, shims must be added as in Step 8 at each tandem joint) to ensure the complete rotating assembly is lifted the amount specified in [Table 4-24](#).

4.8

Flat Cable Attachment to the Motor

**Note: Maximus information**

For Maximus, refer to [Appendix C: Maximus](#).

Use the following procedure to prepare the motor for cable attachment.

1. Raise the assembly until the motor head is approximately five feet above the rig floor.
2. Feed the flat cable through the cable sheave. Using a winch line, hoist the sheave to a height ten feet above the rig floor. Do not bend the cable excessively when feeding it through the sheave. The flat cable splice should be supported by a person when it is being pulled from the spool to prevent it from being stretched.

**Note**

Hold the cable, do NOT pull on the pothead

3. Temporarily band the flat cable to the protector so that the pothead is just above the motor terminal cover. The numbers on the pothead should be facing out, otherwise it will not connect to the motor.
4. Remove the pothead cover and check the motor phase-to-phase resistance. (Phase-to-phase resistance values are shown in the *Technical and Engineering Information* section.)

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**Note**

Resistances are additive when tandem motors are connected.

**Note**

The phase-to-ground motor resistance will be measured at the cable reel after the flat cable is connected to the motor.

5. Remove the shipping cap from the flat cable extension and the terminal cover from the motor head.

**Note**

Ensure this area remains free of moisture and debris.

6. The O-ring on the pothead must be changed before the terminals can be connected. Lubricate the new O-ring with Dow Corning 111 (p/n 100386898). Refer to the ALFORM-D connection tables ([InTouch ID: 3255850](#)) for a complete list of O-ring part numbers.

**Caution**

Potential Severity:	Light
Potential Loss:	Assets
Hazard Category:	Machinery equipment hand tools

Do not use a metal tool when removing the O-ring from the flange groove. Damaged sealing surfaces can result in short equipment runlife.

4.8.1

Phasing the Power Cable**Note**

Phasing of the motor and cable can also be done with a Motor Rotation Indicator. Since models might differ please read the instructions accompanying the device.

**Hint**

Reda equipment shafts should turn counter clockwise when viewed from the top pump down.

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Achieving the correct rotation on the first start begins when the Flat Cable Extension (FCE) is attached to the motor. All motors have their terminal leads identified A, B and C at the pothole. When the power supply phasing corresponds with the motor phasing, the pump will rotate in the correct direction. Prior to making connections, the following must be determined.

- the way the incoming power is phased
- after the FCE is connected to the motor, which leads of the power cable correspond with the motor terminals A, B and C

After these are determined, the cable identified as phase A can be connected to the power supply phase A, phase B to B and C to C.

1. Determine which phase of the FCE is the easiest to connect to the motor lead marked A. Connect a jumper wire between ground and the phase identified as the one which will be connected to motor lead A. A good grounding location is any bolt on the motor. Using an ohm meter, go to the cable reel and connect one test lead to ground (the cable reel or armor). Use the other test lead to determine which of the three conductors is grounded. Mark this phase A.
2. Repeat the Step 1 procedure for B and C phases as the FCE is connected to the motor. When this is done, the three phases of cable will be identified at the top end of the spool.
3. If a field attachable connector is installed at the wellhead, care must be taken to ensure the rotation is unchanged. The procedures above can be used to track the phasing through the wellhead penetrator and upper pigtail to the junction box. The upper pigtail conductors should be marked using different color electrical tape to identify which is phase A, B and C.

Integrity of the complete string and phasing accuracy of the cable can be checked using the Motor Rotation Indicator. After the ESP has been assembled, connect the Motor Rotation Indicator terminals to the cable conductors on the cable reel. Rotate the shaft rapidly with a spline wrench in counterclockwise direction. Determine phase sequence and mark conductors with different color pieces of tapes to identify each phase A, B and C. In cases where the indicator shows no sign of rotation, indicate the that connection was missed or there is problem with the cable.

4.8.2 Flat Cable Tie-In

1. Using motor lead pliers and a small hammer, grip the motor lead marked A just below the terminal. Connect this lead to the corresponding pothead terminal.

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**Note**

Do not pull more than 2.5 inches of lead from the pothole when making this connection. Do not damage the insulation.

2. Using approximately three feet of Teflon® tape, start at the bottom and apply two half-lap layers of tape to the connection. Repeat this taping procedure for the other leads.

**Note**

For motors with terminal voltages greater than 2500 V, three (four if space permits) half-wrap layers should be applied to each lead.

3. Bind the three leads together using four half-wrap layers and double knot Teflon tape to prevent it coming loose.

**Note**

Do not use cotton tape. Cotton tape (p/n 521203) is obsolete. Teflon tape should be used to protect the connection in place of the cotton tape. Three or four layers, wrapped over all three conductors and knotted with two half-hitches has been used with success in high temperature applications.

4. Ensure the pothole is full of oil.
5. Carefully push the leads into the motor head. Ensure there is no foreign material between the flanges.
6. Tighten the pothead flange. To ensure a good connection, the pothead flange must be drawn down evenly, alternating from bolt-to-bolt a few threads at a time. Torque to specification. (Torque specifications are shown in the *Technical and Engineering Information* section.)

4.8.3

Non-Maximus Plug-In Pothead Installation**Note: Maximus information**

For Maximus, refer to [Appendix C: Maximus](#).

A redesign of the older X-type plug-in potheads and flat cable assemblies has been completed for 456– and 540–series motors. No parts are interchangeable between the older X-type and the new–design flat cables or pothead assemblies.

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The new design uses the same installation O-rings as the tape-in pothead. Equipment tests and assembly procedures for the plug-in design are the same as tests and procedures for the tape-in design.

1. Remove the shipping cap from the flat cable extension.


Caution

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

Do not use a metal tool when removing the O-ring from the flange groove. Damaged sealing surfaces can result in short equipment runlife.

2. Look inside the pothead flange to determine if a flat-black colored filler washer is present. Potheads may be shipped with the filler washer loose in the protective cap or installed in the pothead cavity. If a filler washer is not installed, the light gray color of the pothead sealing block will be observed.
3. If a filler washer is present, do not permanently remove it. Apply a coating of Dow #4 grease and install it into the pothead cavity. Push it into the cavity; it will be pushed to the proper location when the pothead is plugged to the motor terminal block.


Note

Do not install two filler washers. The installation of two filler washer may result in future disassembly problems when the pothead is to be unplugged from the motor.

4. Purge air and connect an oil-fill hose to the motor head fill valve. Pump slowly until oil comes out around the pothead block. Clean any residual contamination from the flat cable pothead.
5. Fill the pothead area with the correct oil, wet pothead terminals and insulator with oil, align terminals, then carefully push the pothead assembly into the motor pothead by hand. Install retaining nuts, tighten down evenly and torque to recommended specifications. (Torque specifications are shown in the *Technical and Engineering Information* section.)

4.9

Oil Filling Procedures for Protectors

Protectors are filled with oil prior to shipment from the manufacturing facility. Oil is lost from the protector during field assembly, therefore, some refilling is required. Oil filling procedures for the various protectors are provided in the following sections.

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**Note**

When using protectors with sand avoidance head, be sure that all plugs in the sand avoidance head are removed from the drain hole.

Proper installation of plugs is vital to ensuring the Total-Run-Life (TRL) of the ESP system is achieved. Ensure the vent plugs (standard, short and long) and drain-and-fill valves used are the latest revision per the OneCAT bill of materials.



Figure 4-47: Standard (Vent and Drain) Plug



Figure 4-48: Short and Long Vent Plugs



Figure 4-49: Drain and Fill Valve

When a plug or valve is removed and installed the following steps must be taken:

1. Remove the plug or valve. If removing the plug, use a 3/4-in wrench to backup the valve to prevent loosening the valve.
2. Remove the old lead gasket. Do not reuse lead gaskets.
3. Install a new lead gasket. Do not install two lead gaskets.
4. Install the plug (with lead gasket) or valve (with lead gasket). Torque plugs to 15 ft-lbf and valves to 35 ft-lbf.

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4.9.1

66L Type (except 738, see note below)**Note**

This protector is different from 738 Type 66. Use proper procedures for the correct series protectors.

738 protector part numbers 0332841, W332841, 0662015 are to be serviced per the procedure in [4.9.1: 66L Type \(except 738, see note below\)](#).

**Caution****Vent plug and drain-and-fill valve**

Potential Severity: Light

Potential Loss: Assets

Hazard Category: Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

1. Starting from the top of the protector using a 5/16-in allen wrench, remove the plug from vent 2. The number of each vent is stamped on the housing next to the corresponding hole.
2. Remove and discard the lead gasket that accompanies each plug. A new lead gasket installed over a used gasket can result in a leak. The vent plugs should be kept together where they will not be misplaced or forgotten.
3. The next vent down is stamped "Drain." It should not be removed but checked to ensure that it is tight. All vent plugs are torqued to 15 ft-lbf.
4. Below the drain valve is vent 1. Remove the plug from vent 1.
5. In the base of the protector is a drain-and-fill valve. Tighten this valve using a 3/4-in socket to 35 ft-lbf and tighten the vent plug that is in the center of the valve using the 5/16-in allen wrench to 15 ft-lbf.

**Note**

When removing the vent plug, use a 3/4-in wrench to backup the drain-and-fill valve to prevent it from loosening.

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6. All protectors are serviced from the drain-and-fill valve located in the motor head. Locate this valve and tighten it using the 3/4-in socket to 35 ft-lbf.
7. Remove the vent plug and lead gasket from the center of the drain-and-fill valve.
8. Purge the air out of the oil pump and hose/adaptor assembly, then screw the adaptor into the drain-and-fill valve.

**Note**

Vent plugs are torqued to 15 ft-lbf.

9. Connect the oil hose to the adaptor and begin pumping oil into the motor head.
10. Turn the pump handle slowly at a rate of 45 rpm (do not exceed 40 psi) to allow the air to migrate up and out of vent 1. As oil starts to flow from vent 1, it will still contain some air bubbles. Continue to pump until the oil is bubble free.
11. Reinstall the vent plug with a new lead gasket (p/n 15008). Torque vent plug to 15 ft-lbf.
12. Continue pumping until oil flows from vent 2. Again, ensure that the oil is bubble-free before installing the vent plug with a new lead gasket.
13. Continue pumping until oil flows from the weep hole located in the base of the intake.
14. Remove the adaptor from the drain-and-fill valve in the motor head.
15. Install the vent plug with a new lead gasket and torque 15 ft-lbf. The protector is now serviced and is ready to install in the well.
16. Check all other drain-and-fill valves and vent plugs to insure that they are torqued to specification.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

4.9.2 738 Type 66 (see note below)



Note

This protector is different from 66L Type. Use proper procedures for the correct series protectors.



Note

738 protector part numbers 0332841, W332841, 0662015 are to be serviced per the procedure in [4.9.1: 66L Type \(except 738, see note below\)](#)



Caution

Vent plug and drain-and-fill valve

Potential Severity:	Light
Potential Loss:	Assets
Hazard Category:	Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

These protectors are identified by a lower housing that is roughly 9 inches shorter than the standard 66L lower housing. Additionally, it has three vent plugs, #3, #4, and #5, that the standard 66L does not have.



Note

Attempts to service this type of protector in the "normal" manner will result in excessive oil pump back pressure (from restricted flow) after the #4 vent plug has been replaced.

1. Mount the protector and make a conventional oil fill connection at the motor head drain-and-fill valve.



Note

When removing the vent plug, use a 3/4-in wrench to backup the drain-and-fill valve to prevent it from loosening.

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2. Remove vent plug #1 and begin pumping to evacuate all air from the coupling area and lower housing.

**Note**

Vent plugs are torqued to 15 ft-lbf.

3. Pump until bubble-free oil flows from #1 vent. Stop pumping and install the #1 vent plug using new lead gasket.
 4. Remove vent plug #4 and begin pumping. Continue until bubble-free oil flows from the #4 vent. Stop pumping and install the #4 vent plug using a new lead gasket.
 5. Remove the oil pumping adaptor from the motor head drain-and-fill. Install the vent plug using new lead gasket.
 6. Remove vent plug #2 and vent plug #3. Install the oil pump adaptor in vent #3. Lower the protector, attach the oil pump and pump until bubble-free oil flows from vent #2.
 7. Remove the adaptor, Reinstall vent plug #2 and vent plug #3 using new lead gaskets. This completes the oil fill procedure for new units.
 8. Check all other drain-and-fill valves and vent plugs to insure that they are torqued to specification.
-

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

Servicing the Protector for Rerun 540, 738 Type 66

The practice of rerunning protectors is not recommended by engineering. If the client insists, the protector should be serviced in a vertical position during the pulling operation.

**Note**

Reservicing presupposes the protector to be in excellent mechanical condition, internally and externally.

1. Remove the drain-and-fill valve. Check the protector fluid for contamination (i.e., well fluid).
 - If well fluid is present, the protector should NOT be rerun.

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- If the oil is clear, reinstall the drain-and-fill valve with a new lead gasket (p/n 23903). Remove the vent plug, install the oil filling adapter in preparation for refilling.

**Note**

Vent plugs are torqued to 15 ft-lbf.

2. Remove vent plug #5 and check for well fluid. Well fluid at this point is indicative of upper seal leakage. If there is no water present and oil is clear, reinstall the vent plug using a new lead gasket.
 3. Remove vent plug #1 and vent plug #4.
 4. Begin pumping and continue until bubble-free oil flows from vent #1.
 5. Reinstall vent plug #1 using a new lead gasket.
 6. Resume pumping until bubble-free oil flows from vent #4.
 7. Reinstall vent plug #4 using a new lead gasket.
 8. Remove the oil pump adaptor and reinstall the drain-and-fill vent plug using a new lead gasket.
 9. Remove vent plug #3.
 10. Install the oil pump adaptor and pump until clear oil is being expelled from vent tube in the protector head.
 11. Remove the adaptor and reinstall vent plug #3 using a new lead gasket.
 12. Clean all well fluid from the ID of protector head and fill with clean oil.
 13. Install shipping caps and gaskets. This completes the reservice procedure.
 14. Check all other drain-and-fill valves and vent plugs to insure that they are torqued to specification.
-

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

4.9.3

PFSB (Pre-Filled Single Bag) Type

The following procedure is for the single and double bag 540– and 400–series pre-filled protectors. Filling techniques are the same as the 66L, but the design utilizes a short and long vent plug in conjunction with an inner valve.

**Caution****Vent plug and drain-and-fill valve**

Potential Severity: Light

Potential Loss: Assets

Hazard Category: Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

1. Beginning from the top, torque the vent plug in PF#3 to 15 ft-lbf.
2. Remove the vent plug and lead gasket from PF#4. This vent plug is longer than the rest and should be kept isolated from the others. [Figure 4-50](#) shows the various styles of vent plugs.



Standard Plug
0023879
(carbon steel),
0170803
(Monel)



Long Plug
1020221
(Monel)



Short plug
0876821
(Monel)

Figure 4-50: Vent plug identification

3. In the lower section of the protector there are two vents (PF#1 and PF#2) and a drain-and-fill valve (D). Remove the vent plug and lead gasket from PF#1. This is a short plug used to prevent the shipping valve (located behind the plug) from opening.

**Note**

When removing the vent plug, use a 3/4-in wrench to backup the drain-and-fill valve to prevent it from loosening.

4. Locate the drain-and-fill valve in the motor head and tighten. Remove the vent plug from the center of the valve.

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5. Purge the air out of the oil pump and hose/adaptor assembly, then screw the adaptor into the drain-and-fill valve.
6. Connect the oil hose to the adaptor and begin pumping oil into the motor head.

**Note**

Vent plugs are torqued to 15 ft-lbf.

7. When bubble-free oil is flowing from vent PF#1, replace the vent plug with the long plug originally taken from PF#4 with a new lead gasket (p/n 15008).
-

**Note**

Failure to install the long plug in PF#1 will render the protector useless.

The long plug can be identified by a slot across its head that resembles a screwdriver slot. (See [Figure 4-50](#).) This long plug opens the shipping valve, allowing communication with the inside of the bag. [Figure 4-51](#) shows the location of the long plug.

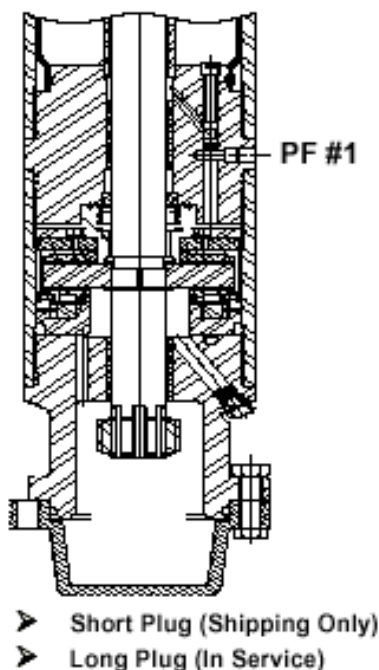


Figure 4-51: PF #1 valve location

8. Continue to pump through the motor head until bubble-free oil flows from PF#4.

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9. Install the short vent plug in PF#4 (originally taken from PF#1) with a new lead gasket. Ensure only one lead gasket is installed and that the plug is not over tightened.
10. Continue to pump until oil flows from the weep hole located in the base of the pump.
11. Remove the adapter from the drain-and-fill valve in the motor head.
12. Install the vent plug with a new lead gasket and torque 15 ft-lbf. Servicing is now complete.
13. Check all other drain-and-fill valves and vent plugs to insure that they are torqued to specification.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

4.9.4 Tandem Protector Filling Instructions (66L on Top, PFSB on Bottom)

**Note**

Do not confuse this protector with 540 series 66L.

**Caution****Vent plug and drain-and-fill valve**

Potential Severity: Light

Potential Loss: Assets

Hazard Category: Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

1. Remove the vent plug and lead gasket from vent 2 of the 66L protector.
2. Remove the vent plug and lead gasket from vent 1 of the 66L protector.
3. Remove the vent plug and lead gasket from the drain-and-fill valve located in the base of the 66L protector.

**Note**

When removing the vent plug, use a 3/4-in wrench to backup the drain-and-fill valve to prevent it from loosening.

4. Remove the long vent plug and lead gasket from PF#4 of the bag protector.
5. Remove the short vent plug and lead gasket from PF#1 of the bag protector.
6. Remove the vent plug and lead gasket from the drain-and-fill valve located in the motor head.

**Note**

When removing the vent plug, use a 3/4-in wrench to backup the drain-and-fill valve to prevent it from loosening.

7. Purge the air from pump and hose then connect the adapter to the motor drain-and-fill valve.

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8. Slowly pump oil until it flows from PF#1 in the PFSB. Continue pumping until all air bubbles have been expelled.

**Note**

Vent plugs are torqued to 15 ft-lbf.

9. Install the long vent plug with a new lead gasket (p/n 15008) in vent PF#1.
10. Continue pumping until bubble-free oil flows from PF#4 of the lower protector.
11. Install the short vent plug with a new lead gasket in vent PF#4.
12. Continue pumping until bubble-free oil flows from vent 1 of the 66L protector.
13. Lower the assembly into the well until the base of the 66L is slightly above floor level.
14. Connect the adapter to the drain-and-fill valve located in the base of the 66L protector.
15. Pump until bubble-free oil flows from vent 1 of the 66L protector.
16. Install the vent plug and new lead gasket in vent 1.
17. Continue pumping until bubble-free oil flows from vent 2 of the 66L protector.
18. Install the vent plug and new lead gasket in vent 2.
19. Continue pumping until bubble-free oil flows from the weep hole located in the base of the intake.
20. Remove the hose and adapter from the drain-and-fill valve.
21. Install the vent plug with a new lead gasket.
22. Check all other drain-and-fill valves and vent plugs to insure that they are torqued to specification.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

4.9.5 Tandem protector filling instructions (PFSB on top, 66L on bottom)

**Note**

Do not confuse this protector with 540 series 66L.

**Caution****Vent plug and drain-and-fill valve**

Potential Severity:	Light
Potential Loss:	Assets
Hazard Category:	Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

1. Starting from the top of the protector 66L using a 5/16-in allen wrench, remove the plug from vent 2. The number of each vent is stamped on the housing next to the corresponding hole.
2. Remove and discard the lead gasket that accompanies each plug.

**Note**

A new lead gasket installed over a used washer can result in a leak. The vent plugs should be kept together where they will not be misplaced or forgotten.

3. The next vent down is stamped "Drain." It should not be removed but checked to ensure that it is tight. All Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

**Note**

When removing the vent plug, use a 3/4-in wrench to backup the drain-and-fill valve to prevent it from loosening.

4. Below the drain valve is vent 1. Remove the plug from vent 1.
5. In the base of the protector is a drain-and-fill valve. Tighten this valve using a 3/4-in socket to 35 ft-lbf.

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6. Tighten the vent plug that is in the center of the valve using the 5/16-in allen wrench to 15 ft-lbf.
7. All protectors are serviced from the drain-and-fill valve located in the motor head. Locate this valve and tighten it using the 3/4-in socket to 35 ft-lbf. Remove the vent plug and lead gasket from the center of the drain-and-fill valve. Purge the air out of the oil pump and hose/adaptor assembly, then screw the adaptor into the drain-and-fill valve in the motor head.

**Note**

Vent plugs are torqued to 15 ft-lbf.

8. Connect the oil hose to the adaptor and begin pumping oil into the motor head.
9. Turn the pump handle slowly at a rate of 45 rpm (do not exceed 40 psi) to allow the air to migrate up and out of vent 1. As oil starts to flow from vent 1, it will still contain some air bubbles. Continue to pump until the oil is bubble free.
10. Reinstall the vent plug with a new lead gasket (p/n 15008). Torque vent plug to 15 ft-lbf.
11. Continue pumping until bubble-free oil flows from vent 2.
12. Reinstall the vent plug with a new lead gasket.
13. On the top of PFSB, torque the vent plug in PF#3 to 15 ft-lbf.
14. Remove the vent plug and lead gasket from PF#4.

**Note**

This vent plug is longer than the rest and should be kept isolated from the others. The following figure shows the various styles of vent plugs.

15. In the lower section of the PFSB protector are two vents (PF#1 and PF#2) and a drain-and-fill valve (D). Remove the vent plug and lead gasket from PF#1. This is a short plug used to prevent the shipping valve (located behind the plug) from opening.
16. Continue pumping until bubble-free oil flows from vent PF#1.
17. Replace the vent plug with the long plug originally taken from PF#4 and a new lead gasket (p/n 15008).

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**Note**

Failure to install the long plug in PF#1 will render the protector useless.

The long plug can be identified by a slot across its head that resembles a screwdriver slot. This long plug opens the shipping valve, allowing communication with the inside of the bag. See the following figure.

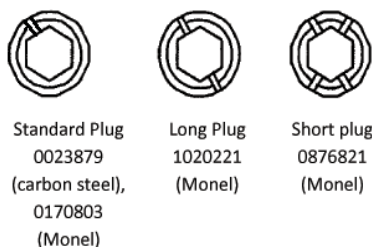


Figure 4-52: Vent plug identification

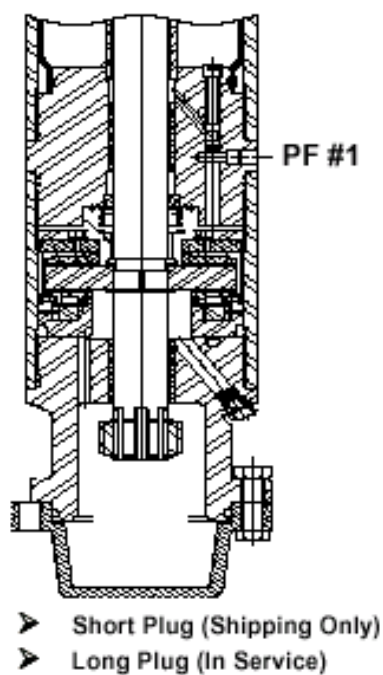


Figure 4-53: PF #1 valve location

**Note**

Vent plugs are torqued to 15 ft-lbf.

18. Continue to pump through the motor head until bubble-free oil flows from PF#4.

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19. Install the short vent plug in PF#4 (originally taken from PF#1) with a new lead gasket. Ensure only one lead gasket is installed and that the plug is not over tightened.
20. Continue pumping until bubble-free oil flows from the weep hole located in the base of the intake.
21. Remove the adapter from the drain-and-fill valve in the motor head. install the vent plug with a new lead gasket. Servicing is now complete.
22. Check all other drain-and-fill valves and vent plugs to insure that they are torqued to specification.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

4.9.6 950/738 Series Pre-Filled Bag Protector

Location of Vent Plugs

Vent plugs PF#1, drain-and-fill, and #5 are located in the lower section of the protector, approximately 10 inches above the base. Vent plugs #2, #6, and PF#7 are located in the middle section of the protector. Vent plugs #3 and PF#4 are located in the top portion of the protector. All vent plug locations are marked.



Caution

Vent plug and drain-and-fill valve

Potential Severity: Light
Potential Loss: Assets
Hazard Category: Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

Filling Procedure

1. Remove the long vent plugs and lead gaskets from vents #3 and PF#4 in the top of the protector.



Note

Long plugs can be identified by a slot across the head resembling a screwdriver slot.



Note

When removing the vent plug, use a 3/4-in wrench to backup the drain-and-fill valve to prevent it from loosening.

2. Raise the protector until filling from the motor head drain-and-fill valve is possible.
3. Install an oil pump at the motor head drain-and-fill valve.
4. Remove the vent plug and lead gasket from vent #5.
5. Start pumping slowly until bubble-free oil flows from vent #5.

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**Note**

Vent plugs are torqued to 15 ft-lbf.

6. Reinstall vent #5 using a new lead gasket.
 7. Remove the short plug and lead gasket from vent PF#1 and replace with one of the long vent plugs removed from vent #3 or PF#4 using a new lead gasket.
 8. Remove vent plugs and lead gaskets from vents #2 or #6 in the middle section of the protector.
 9. Pump slowly until bubble-free oil flows from either vent #2 or #6.
 10. Reinstall the vent plug using a new lead gasket at the vent with oil flow.
 11. Continue pumping until bubble-free oil flows from remaining open vent.
 12. Reinstall the vent plug using new lead gasket.
 13. Remove the short vent plug and lead gasket from vent PF#7 and replace with the remaining long vent plug and new lead gasket.
 14. Continue pumping slowly until bubble-free oil flows from vent #3. Reinstall the vent plug using new lead gasket.
-

**Note**

Short vent plugs are used in vents #3 and PF#4. Be careful not to strip threads on short plugs.

15. Continue pumping until bubble-free oil flows from vent PF#4. Reinstall the vent plug using new lead gasket.
-

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

4.9.7 738/562 and 738/738 Protector BPBSL

Figure 4-54 shows a 738 BPBSL protector.

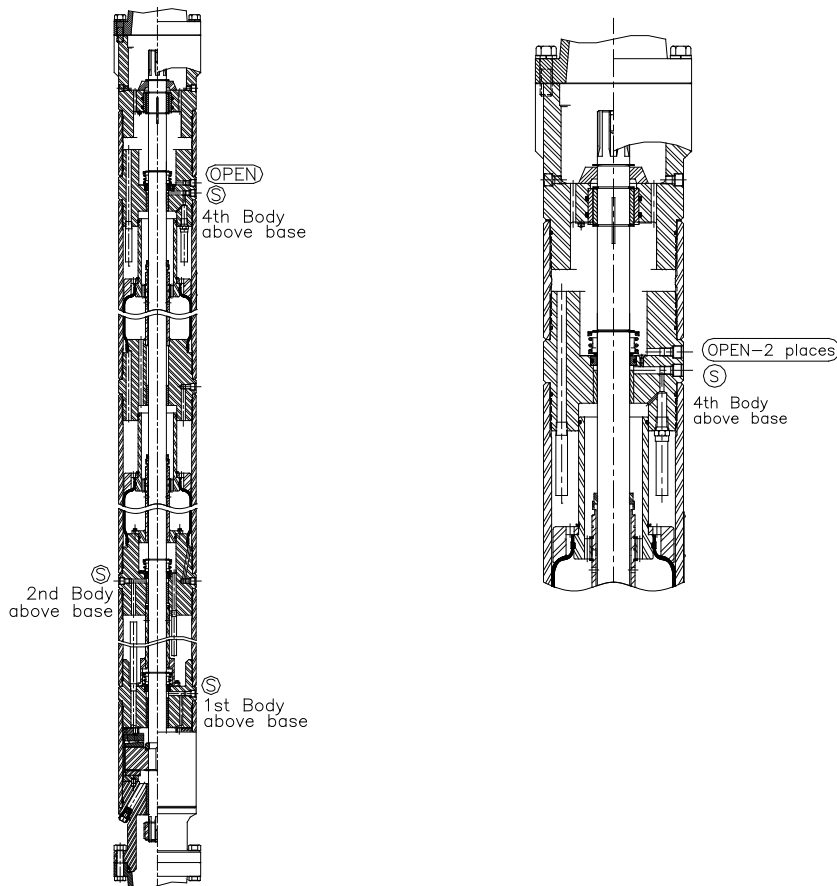


Figure 4-54: 738 BPBSL protector



Caution

Vent plug and drain-and-fill valve

Potential Severity:	Light
Potential Loss:	Assets
Hazard Category:	Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

1. Remove the shipping caps and install the protector between the motor and intake. Connect the cable to the motor.

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**Note**

When removing the vent plug, use a 3/4-in wrench to backup the drain-and-fill valve to prevent it from loosening.

2. Remove the vent plug from the drain-and-fill valve in the motor head and connect the oil fill pump and adapter.
 3. Remove the lower “S” plug from the thrust bearing center body (the first body above the base).
-

**Note**

Vent plugs are torqued to 15 ft-lbf.

4. Slowly pump motor oil into the head of the motor until bubble-free oil flows from the open “S” hole. Reinstall the standard plug using a new lead gasket.
5. Remove the standard plug from the “S” hole located in the second body above the base.
6. Continue to pump motor oil into the head of the motor until bubble-free oil flows from the open “S” hole. Reinstall the standard plug using a new lead gasket.
7. There are two ports marked “OPEN” in the top body (4th body above the base). The two “OPEN” ports are 180 degrees apart in the body. These “OPEN” ports should remain open during installation into the well. The OPEN ports are NOT in the head of the protector.
8. Remove the “S” and “OPEN” plugs from the highest center body (fourth body above the base).
9. Continue pumping motor oil through the motor head until bubble-free oil flows from the “S” port.
10. Reinstall the standard plug using a new lead gasket in the “S” port.
11. Continue pumping until bubble-free oil flows from the “OPEN” holes.
12. Confirm that the two plugs in the head of the protector are open.
13. Check all other drain-and-fill valves and vent plugs to insure that they are torqued to specification.

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**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

**Note**

The 738/562 or 738/738 BPBSL protector is not the same as a 540 or 456 modular protector. There is no valve in the base, and no PF plug used to actuate a valve, therefore installing a long plug is not necessary.

**Note**

For tandem 738 BPBSL protectors ensure that the “OPEN” holes of the lower protector are closed with standard plugs and new lead gaskets.

4.9.8 Modular Protector BSL

Figure 4-55 shows a BSL modular protector.

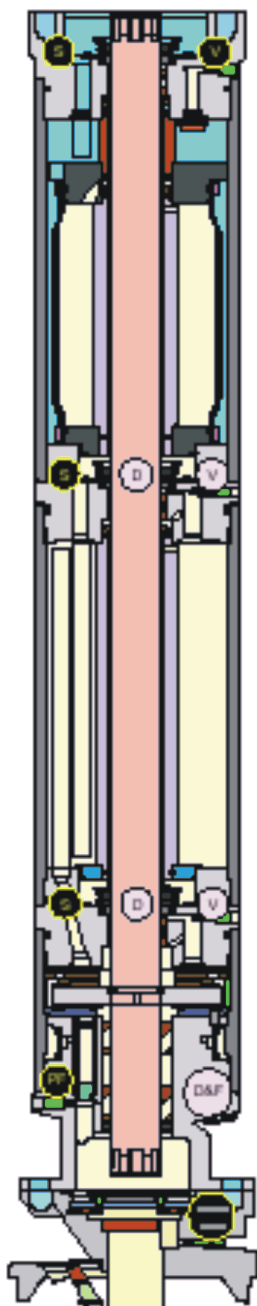


Figure 4-55: BSL modular protector. Arrow indicates the location of the long plug

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**Caution****Vent plug and drain-and-fill valve**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

1. Remove the shipping caps and install the protector between the motor and intake. Connect the cable to the motor.
2. Remove the long plug from the “S” hole in the protector head. The long plug can be identified by a slot across its head that resembles a screwdriver slot.
3. Remove the standard plug from the “V” hole in the protector head.
4. Remove the standard plug from “S” hole located above the lower labyrinth chamber.
5. Remove the standard plug from the “S” hole located above the thrust chamber.
6. Remove the short plug from the “PF” hole in the protector base.

**Note**

Use a 3/4-in wrench to backup the valve to prevent it from loosening when removing the vent plug.

7. Remove the standard plug from the motor head drain-and-fill valve.
8. Install the oil pump adapter, purge any air from the oil pump/hose assembly, and connect.
9. Slowly pump motor oil through the motor head until bubble-free oil flows from the “PF” hole in the protector base. Install the long plug with a new lead gasket.

**Note**

Vent plugs are torqued to 15 ft-lbf.

10. Continue pumping until bubble-free oil flows from the “S” hole on top of the thrust chamber. Install a standard plug with a new lead gasket.

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11. Continue pumping until bubble-free oil flows from the “S” hole on top of the lower labyrinth chamber. Install a standard plug with a new lead gasket.
12. Continue pumping until bubble-free oil flows from the “V” hole in the protector head. Install a standard plug with a new lead gasket.
13. Continue pumping until bubble-free oil flows from the “S” hole in the protector head. Install the short plug with a new lead gasket.
14. Continue pumping until bubble-free oil flows from the weep hole located in the intake base.
15. Remove the oil fill pump and adapter from the motor head and install a standard plug with a new lead gasket.
16. Check all other drain-and-fill valves and vent plugs to insure that they are torqued to specification.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

**Caution****Vent plug and drain-and-fill valve**

Potential Severity: Light
Potential Loss: Assets
Hazard Category: Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

1. Remove the shipping caps and install the protector between the motor and intake. Connect the cable to the motor.
2. Remove the long plug from the “S” hole in the protector head. The long plug can be identified by a slot across its head that resembles a screwdriver slot.
3. Remove the standard plugs from the “S” and “V” holes located on top of the lower bag chamber.
4. Remove the standard plug from the “S” hole located above the thrust bearing chamber.
5. Remove the short plug from the “PF” hole in the protector base.

**Note**

Use a 3/4-in wrench to backup the valve to prevent it from loosening when removing the vent plug.

6. Remove the standard plug from the motor head drain-and-fill valve.
7. Install the oil pump adapter, purge any air from the oil pump/hose assembly, and connect.
8. Slowly pump motor oil through the motor head until bubble-free oil flows from the “PF” hole in the protector base. Install the long plug with a new lead gasket.

**Note**

Vent plugs are torqued to 15 ft-lbf.

9. Continue pumping until bubble-free oil flows from the “S” hole above the thrust bearing chamber. Install a standard plug with a new lead gasket.
10. Continue pumping until bubble-free oil flows from the “V” hole located on top of the lower bag chamber.

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11. Install a standard plug with a new lead gasket.
12. Continue pumping until bubble-free oil flows from the "S" hole located on top of the lower bag chamber. Install a standard plug with a new lead gasket.
13. Continue pumping until bubble-free oil flows from the "S" hole located in the protector head. Install the short plug with a new lead gasket.
14. Continue pumping until bubble-free oil flows from the weep hole in the intake base.
15. Remove the oil fill pump and adapter from the motor head and install a standard plug with a new lead gasket.
16. Check all other drain-and-fill valves and vent plugs to insure that they are torqued to specification.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

4.9.10 Modular Protector BSBSL

Figure 4-57 shows a BSBSL modular protector.

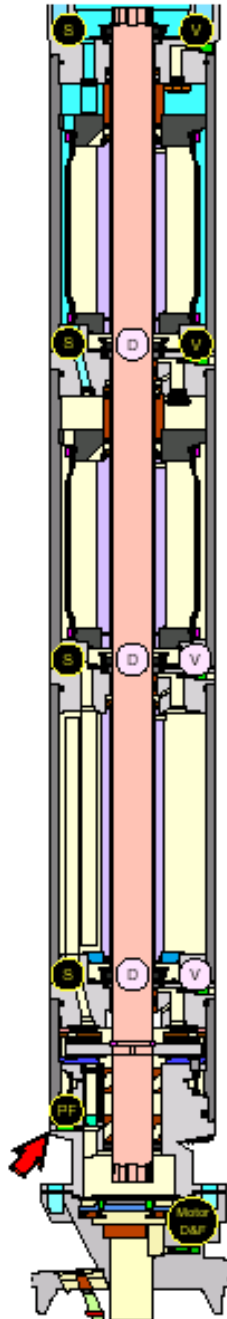


Figure 4-57: BSBSL modular protector. Arrow indicates the location of the long plug

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**Caution****Vent plug and drain-and-fill valve**

Potential Severity: Light

Potential Loss: Assets

Hazard Category: Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

1. Remove the shipping caps and install the protector between the motor and intake. Connect the cable to the motor.
2. Remove the long plug from the “S” hole in the protector head. The long plug can be identified by a slot across its head that resembles a screwdriver slot.
3. Remove the standard plug from the “V” hole in protector head.
4. Remove the standard plugs from the “S” and “V” holes located above the middle bag chamber.
5. Remove the standard plug from the “S” hole located above the lower labyrinth chamber.
6. Remove the standard plug from the “S” hole located above the thrust chamber.
7. Remove the short plug from the “PF” hole in the protector base.

**Note**

Use a 3/4-in wrench to backup the valve to prevent it from loosening when removing the vent plug.

8. Remove the standard plug from the motor head drain-and-fill valve.
9. Install the oil pump adapter, purge any air from the oil pump/hose assembly, and connect.
10. Slowly pump motor oil through the motor head until bubble-free oil flows from the “PF” hole in the protector base. Install the long plug with a new lead gasket.

**Note**

Vent plugs are torqued to 15 ft-lbf.

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11. Continue pumping until bubble-free oil flows from the “S” hole on top of the thrust chamber. Install a standard plug using a new lead gasket.
12. Continue pumping until bubble-free oil flows from the “S” hole on top of the lower labyrinth chamber. Install a standard plug using a new lead gasket.
13. Continue pumping until bubble-free oil flows from the “S” and “V” holes on top of the middle bag chamber. Install standard plugs with new lead gaskets.
14. Continue pumping until bubble-free oil flows from the “V” hole in the protector head. Install a standard plug with a new lead gasket.
15. Continue pumping until bubble-free oil flows from the “S” hole in the protector head. Install the short plug with a new lead gasket.
16. Continue pumping until bubble-free oil flows from the weep hole in the intake base.
17. Remove the oil fill pump and adapter from the motor head and install a standard plug with a new lead gasket.
18. Check all other drain-and-fill valves and vent plugs to insure that they are torqued to specification.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

4.9.11 Modular Protector LSLSL

Figure 4-58 shows a LSLSL modular protector.

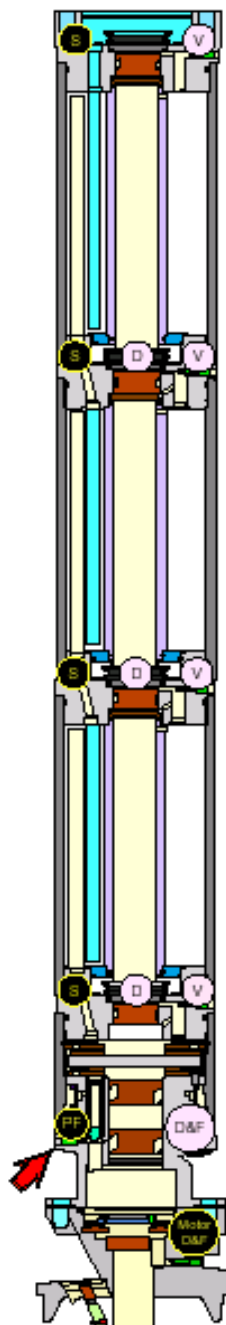


Figure 4-58: LSLSL modular protector. Arrow indicates the location of the long plug

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**Caution****Vent plug and drain-and-fill valve**

Potential Severity: Light

Potential Loss: Assets

Hazard Category: Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

1. Remove the shipping caps and install the protector between the motor and intake. Connect the cable to the motor.
2. Remove the long plug from the “S” hole in the protector head. The long plug can be identified by a slot across its head that resembles a screwdriver slot.
3. Remove the standard plug from the “S” hole located above the middle labyrinth chamber.
4. Remove the standard plug from the “S” hole located above the lower labyrinth chamber.
5. Remove the standard plug from the “S” hole located above the thrust bearing chamber.
6. Remove the short plug from the “PF” hole in the protector base.

**Note**

Use a 3/4-in wrench to backup the valve to prevent it from loosening when removing the vent plug.

7. Remove the standard plug from the motor head drain-and-fill valve.
8. Install the oil pump adapter, purge any air from the oil pump/hose assembly, and connect.
9. Slowly pump motor oil through the motor head until bubble-free oil flows from the “PF” hole in the protector base. Install the long plug with a new lead gasket.

**Note**

Vent plugs are torqued to 15 ft-lbf.

10. Continue pumping until bubble-free oil flows from the “S” hole above the thrust bearing chamber. Install a standard plug with a new lead gasket.

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11. Continue pumping until bubble-free oil flows from the “S” hole located on top of the lower labyrinth chamber. Install a standard plug with a new lead gasket.
12. Continue pumping until bubble-free oil flows from the “S” hole above the middle labyrinth chamber. Install a standard plug with a new lead gasket.
13. Continue pumping until bubble-free oil flows from the “S” hole located in protector head. Install the short plug with a new lead gasket.
14. Continue pumping until bubble-free oil flows from the weep hole in the intake base.
15. Remove the oil fill pump and adapter from the motor head and install a standard plug with a new lead gasket.
16. Check all other drain-and-fill valves and vent plugs to insure that they are torqued to specification.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

**Caution****Vent plug and drain-and-fill valve**

Potential Severity: Light

Potential Loss: Assets

Hazard Category: Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

1. Remove the shipping caps and install the protector between the motor and intake. Connect the cable to the motor.
2. Remove the long plug from the “S” hole in the protector head. The long plug can be identified by a slot across its head that resembles a screwdriver slot.
3. Remove the standard plug from the “S” hole located above the middle labyrinth chamber.
4. Remove standard plugs from the “S” and “V” holes located above the lower bag chamber.
5. Remove the standard plug from the “S” hole located above the thrust bearing chamber.
6. Remove the short plug from the “PF” hole in the protector base.

**Note**

Use a 3/4-in wrench to backup the valve to prevent it from loosening when removing the vent plug.

7. Remove the standard plug from the motor head drain-and-fill valve.
8. Install the oil pump adapter, purge any air from the oil pump/hose assembly, and connect.
9. Slowly pump motor oil through the motor head until bubble-free oil flows from the “PF” hole in protector base. Install the long plug with a new lead gasket.

**Note**

Vent plugs are torqued to 15 ft-lbf.

10. Continue pumping until bubble-free oil flows from the “S” hole above the thrust bearing chamber. Install a standard plug with a new lead gasket.

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11. Continue pumping until bubble-free oil flows from the “V” hole located on top of the lower bag chamber. Install a standard plug with a new lead gasket.
12. Continue pumping until bubble-free oil flows from the “S” hole located on top of the lower bag chamber. Install a standard plug with a new lead gasket.
13. Continue pumping until bubble-free oil flows from the “S” hole above the middle labyrinth chamber. Install a standard plug with a new lead gasket.
14. Continue pumping until bubble-free oil flows from the “S” hole located in the protector head. Install the short plug with a new lead gasket.
15. Continue pumping until bubble-free oil flows from the weep hole in the intake base.
16. Remove the oil fill pump and adapter from the motor head and install a standard plug with a new lead gasket.
17. Check all other drain-and-fill valves and vent plugs to insure that they are torqued to specification.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

4.9.13

Modular Protector LSBPB

Figure 4-60 shows a LSBPB modular protector.

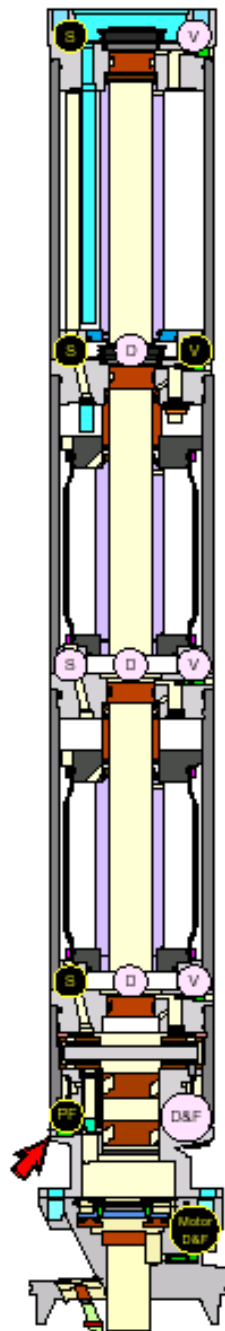


Figure 4-60: LSBPB modular protector. Arrow indicates the location of the long plug

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**Caution****Vent plug and drain-and-fill valve**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

1. Remove the shipping caps and install the protector between the motor and intake. Connect the cable to the motor.
2. Remove the long plug from the “S” hole in the protector head. The long plug can be identified by a slot across its head that resembles a screwdriver slot.
3. Remove the standard plugs from the “S” and “V” holes located above the middle bag chamber.
4. Remove the standard plug from the “S” hole located above the thrust bearing chamber.
5. Remove the short plug from the “PF” hole in protector base.

**Note**

Use a 3/4-in wrench to backup the valve to prevent it from loosening when removing the vent plug.

6. Remove the standard plug from the motor head drain-and-fill valve.
7. Install the oil pump adapter, purge any air from the oil pump/hose assembly, and connect.
8. Slowly pump motor oil through the motor head until bubble-free oil flows from the “PF” hole in the protector base. Install the long plug with a new lead gasket.

**Note**

Vent plugs are torqued to 15 ft-lbf.

9. Continue pumping until bubble-free oil flows from the “S” hole above the thrust bearing chamber. Install a standard plug with a new lead gasket.
10. Continue pumping until bubble-free oil flows from the “V” hole located on top of the middle bag chamber. Install a standard plug with a new lead gasket.

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11. Continue pumping until bubble-free oil flows from the “S” hole located on top of the middle bag chamber. Install a standard plug with a new lead gasket.
12. Continue pumping until bubble-free oil flows from the “S” hole located in the protector head. Install the short plug with a new lead gasket.
13. Continue pumping until bubble-free oil flows from the weep hole in the intake base.
14. Remove the oil fill pump and adapter from the motor head and install a standard plug with a new lead gasket.
15. Check all other drain-and-fill valves and vent plugs to insure that they are torqued to specification.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

4.9.14

Modular Protector BPBSL

Figure 4-61 shows a BPBSL modular protector.

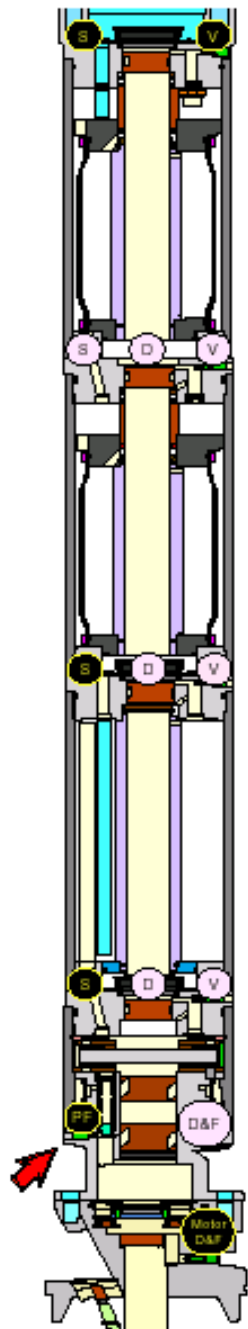


Figure 4-61: BPBSL modular protector. Arrow indicates the location of the long plug

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**Caution****Vent plug and drain-and-fill valve**

Potential Severity: Light

Potential Loss: Assets

Hazard Category: Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

1. Remove the shipping caps and install the protector between the motor and intake. Connect the cable to the motor.
2. Remove the long plug from the “S” hole in the protector head. The long plug can be identified by a slot across its head that resembles a screwdriver slot.
3. Remove the standard plug from the “V” hole in the protector head.
4. Remove the standard plug from the “S” hole located above the lower labyrinth chamber.
5. Remove the standard plug from the “S” hole located above the thrust chamber.
6. Remove the short plug from the “PF” hole in the protector base.

**Note**

Use a 3/4-in wrench to backup the valve to prevent it from loosening when removing the vent plug.

7. Remove the standard plug from the motor head drain-and-fill valve.
8. Install the oil pump adapter, purge any air from the oil pump/hose assembly, and connect.
9. Slowly pump motor oil through the motor head until bubble-free oil flows from the “PF” hole in the protector base. Install the long plug with a new lead gasket.

**Note**

Vent plugs are torqued to 15 ft-lbf.

10. Continue pumping until bubble-free oil flows from the “S” hole above the thrust chamber. Install a standard plug using a new lead gasket.

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11. Continue pumping until bubble-free oil flows from the “S” hole above the lower labyrinth chamber. Install a standard plug using a new lead gasket.
12. Continue pumping until bubble-free oil flows from the “V” hole in the protector head. Install a standard plug using a new lead gasket.
13. Continue pumping until bubble-free oil flows from the “S” hole in the protector head. Install the short plug with a new lead gasket.
14. Continue pumping until bubble-free oil flows from the weep hole in the intake base.
15. Remove the oil fill pump and adapter from the motor head and install a standard plug with a new lead gasket.
16. Check all other drain-and-fill valves and vent plugs to insure that they are torqued to specification.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

4.9.15

Modular Protector LSBPBSL

Figure 4-62 shows a LSBPBSL modular protector.

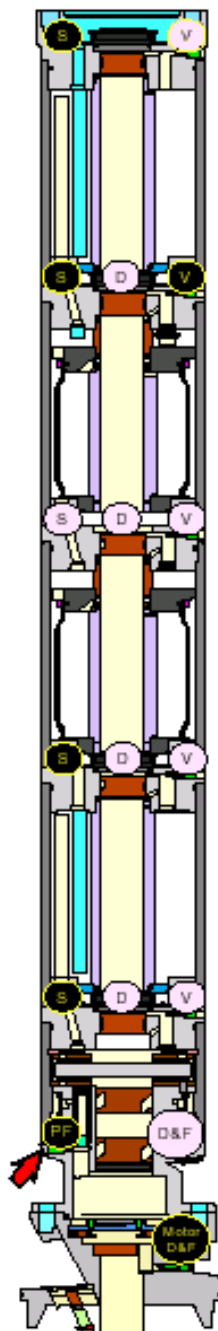


Figure 4-62: LSBPBSL modular protector. Arrow indicates the location of the long plug

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**Caution****Vent plug and drain-and-fill valve**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

1. Remove the shipping caps and install the protector between the motor and intake. Connect the cable to the motor.
2. Remove the long plug from the “S” hole in the protector head. The long plug can be identified by a slot across its head that resembles a screwdriver slot.
3. Remove the standard plugs from the “S” and “V” holes located above the top bag chamber.
4. Remove the standard plug from the “S” hole located above the lower labyrinth chamber.
5. Remove the standard plug from the “S” hole located above the thrust bearing chamber.
6. Remove the short plug from the “PF” hole in the protector base.

**Note**

Use a 3/4-in wrench to backup the valve to prevent it from loosening when removing the vent plug.

7. Remove the standard plug from the motor head drain-and-fill valve.
8. Install the oil pump adapter, purge any air from the oil pump/hose assembly, and connect.
9. Slowly pump motor oil through the motor head until bubble-free oil flows from the “PF” hole in the protector base. Install the long plug with a new lead gasket.

**Note**

Vent plugs are torqued to 15 ft-lbf.

10. Continue pumping until bubble-free oil flows from the “S” hole above the thrust bearing chamber. Install a standard plug with a new lead gasket.

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11. Continue pumping until bubble-free oil flows from the “S” hole located above the lower labyrinth chamber. Install a standard plug with a new lead gasket.
12. Continue pumping until bubble-free oil flows from the “V” hole located above the top bag chamber. Install a standard plug with a new lead gasket.
13. Continue pumping until bubble-free oil flows from the “S” hole located above the top bag chamber. Install a standard plug with a new lead gasket.
14. Continue pumping until bubble-free oil flows from the “S” hole located in the protector head. Install the short plug with a new lead gasket.
15. Continue pumping until bubble-free oil flows from the weep hole in the intake base.
16. Remove the oil fill pump and adapter from the motor head and install a standard plug with a new lead gasket.
17. Check all other drain-and-fill valves and vent plugs to insure that they are torqued to specification.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

4.9.16

Modular Protector BPBSBPB

Figure 4-63 shows a BPBSBPB modular protector.

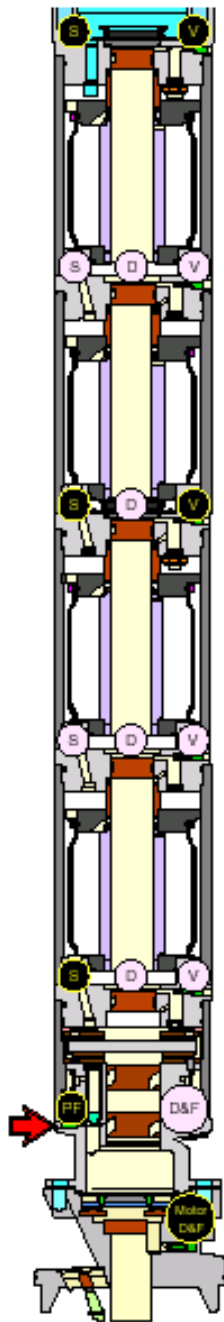


Figure 4-63: BPBSBPB modular protector. Arrow indicates the location of the long plug

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**Caution****Vent plug and drain-and-fill valve**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

1. Remove the shipping caps and install the protector between the motor and intake. Connect the cable to the motor.
2. Remove the long plug from the “S” hole in the protector head. The long plug can be identified by a slot across its head that resembles a screwdriver slot.
3. Remove the standard plug from the “V” hole in protector head.
4. Remove the standard plugs from the “S” and “V” holes located above the third bag chamber from the top.
5. Remove the standard plug from the “S” hole located above the thrust chamber.
6. Remove the short plug from the “PF” hole in protector base.

**Note**

Use a 3/4-in wrench to backup the valve to prevent it from loosening when removing the vent plug.

7. Remove the standard plug from the motor head drain-and-fill valve.
8. Install the oil pump adapter, purge any air from the oil pump/hose assembly, and connect.
9. Slowly pump motor oil through the motor head until bubble-free oil flows from the “PF” hole in the protector base. Install the long plug with a new lead gasket.

**Note**

Vent plugs are torqued to 15 ft-lbf.

10. Continue pumping until bubble-free oil flows from the “S” hole above the thrust chamber. Install a standard plug with a new lead gasket.

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11. Continue pumping until bubble-free oil flows from the “S” and “V” holes above the third bag chamber. Install standard plugs with new lead gaskets.
12. Continue pumping until bubble-free oil flows from the “V” hole in the protector head. Install a standard plug with a new lead gasket.
13. Continue pumping until bubble-free oil flows from the “S” hole in the protector head. Install the short plug with a new lead gasket.
14. Continue pumping until bubble-free oil flows from the weep hole in the intake base.
15. Remove the oil fill pump and adapter from the motor head and install a standard plug with a new lead gasket.
16. Check all other drain-and-fill valves and vent plugs to insure that they are torqued to specification.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

4.9.17

LSMPM Extended Runlife Protector oil fill procedure

A LSMPM Extended Runlife Protector is shown in [Figure 4-64](#).

**Caution****Vent plug and drain-and-fill valve**

Potential Severity:	Light
Potential Loss:	Assets
Hazard Category:	Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

1. Remove the shipping caps and the four 1/8-in NTP plugs in the protector head.

**Note**

If this protector will be installed below another protector, the four NPT plugs in the head of the lower protector should not be removed.

2. Install the protector between the motor and intake.
3. Connect the power cable to the motor.
4. Remove the long plug from the S port in the protector top seal body. (The long plug should have a screwdriver slot).
5. Remove the standard plugs from the S and V ports located above the middle bellows chamber.
6. Remove the standard plug from the S port located above the thrust bearing chamber.
7. Remove the short plug from the PF port in the protector base.

**Note**

Use a 3/4-in wrench to back up the valve to prevent it from loosening while removing the vent plug.

8. Remove the standard plug from the motor head D and F valve.
9. Install the oil-fill adapter, purge air from oil pump/hose assembly and connect the hose.

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10. Slowly pump the specified motor oil through motor head until bubble-free oil flows from the PF port in the protector base. Install a long plug with a new lead gasket.

**Note**

Vent plugs are torqued to 15 ft-lbf.

11. Continue pumping until bubble-free oil flows from the S port above the thrust bearing chamber. Install a standard plug with a new lead gasket.
12. Continue pumping until bubble-free oil flows from the V port located on top of the middle bellows chamber. Install a standard plug with a new lead gasket.
13. Continue pumping until bubble-free oil flows from the S port located on top of the middle bellows chamber. Install a standard plug with a new lead gasket.
14. Continue pumping until bubble-free oil flows from the S port located in protector upper seal body. Install a short plug with a new lead gasket.
15. Continue pumping until bubble-free oil flows from the NPT ports in the protector head.
16. Remove the filling pump/adaptor from the motor head and install a standard plug with a new lead gasket.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

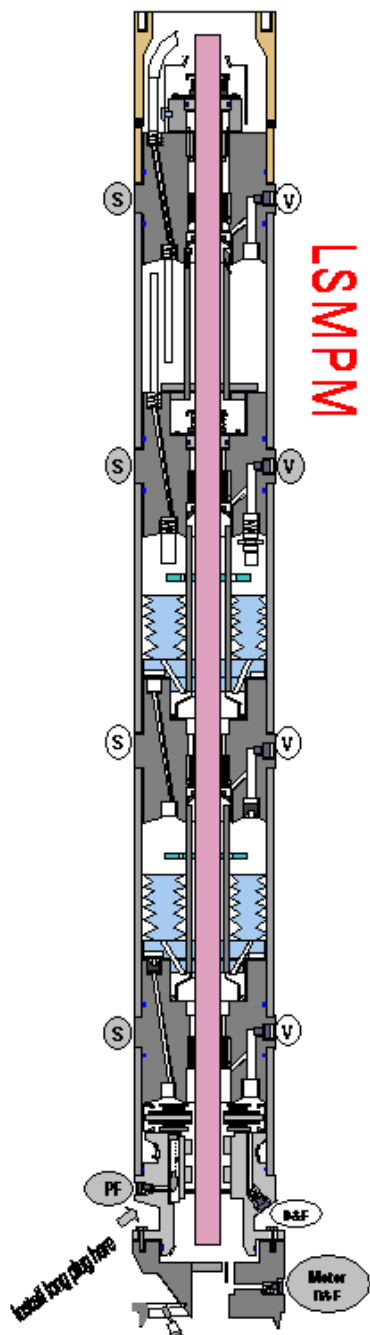


Figure 4-64: LSMPM Extended Runlife Protector.

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4.9.18

LSMPMPM Extended Runlife Protector oil fill procedure

A LSMPMPM Extended Runlife Protector is shown in [Figure 4-65](#).



Caution

Vent plug and drain-and-fill valve

Potential Severity:	Light
Potential Loss:	Assets
Hazard Category:	Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

1. Remove the shipping caps and the four 1/8-in NTP plugs in the protector head.



Note

If this protector will be installed below another protector, the four NPT plugs in the head of the lower protector should not be removed.

2. Install the protector between the motor and intake.
3. Connect the power cable to the motor.
4. Remove the long plug from the S port in protector top seal body. (The long plug should have a screwdriver slot).
5. Remove the standard plugs from the S and V ports located below the top chamber.
6. Remove the standard plug from the S port located above the thrust bearing chamber.
7. Remove the short plug from the PF port in the protector base.



Note

Use a 3/4-in wrench to back up the valve to prevent it from loosening while removing the vent plug.

8. Remove the standard plug from the motor head D and F valve.

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9. Install the oil-fill adapter, purge air from oil pump/hose assembly and connect the hose.
10. Slowly pump the specified motor oil through motor head until bubble-free oil flows from the PF port in the protector base. Install a long plug with a new lead gasket.

**Note**

Vent plugs are torqued to 15 ft-lbf.

11. Continue pumping until bubble-free oil flows from the S port above the thrust bearing chamber. Install a standard plug with a new lead gasket.
12. Continue pumping until bubble-free oil flows from the V port located below the top chamber. Install a standard plug with a new lead gasket.
13. Continue pumping until bubble-free oil flows from the S port located below the top chamber. Install a standard plug with a new lead gasket.
14. Continue pumping until bubble-free oil flows from the S port located in the protector top seal body. Install a short plug with a new lead gasket.
15. Continue pumping until bubble-free oil flows from the NPT ports in the protector head.
16. Remove the filling pump/adapter from the motor head and install a standard plug with a new lead gasket.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

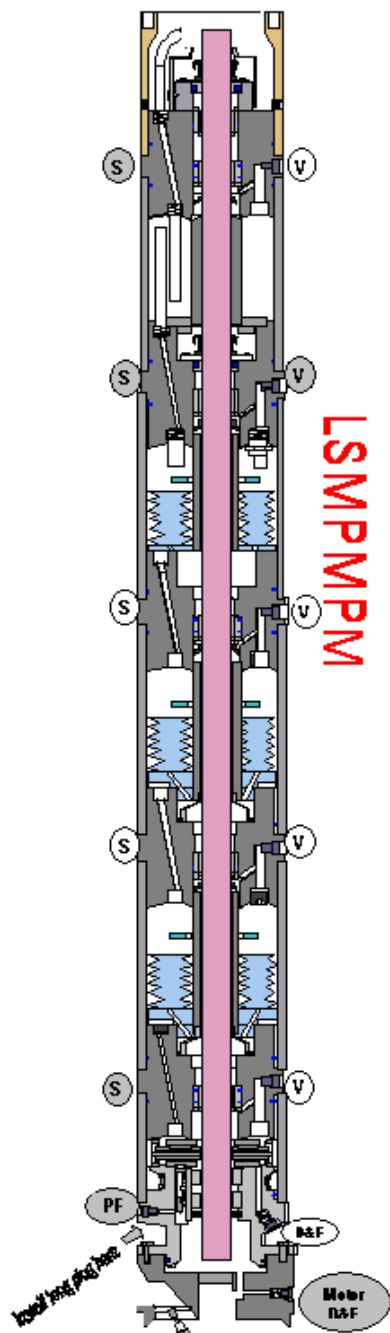


Figure 4-65: LSMPMPM Extended Runlife Protector.

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4.9.19

LSBPB Extended Runlife Protector oil fill procedure

An LSBPB Extended Runlife Protector is shown in [Figure 4-66](#).

**Caution****Vent plug and drain-and-fill valve**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

1. Remove the shipping caps and the four 1/8-in NTP plugs in the protector head.

**Note**

If this protector will be installed below another protector, the four NPT plugs in the head of the lower protector should not be removed.

2. Install the protector between the motor and intake.
3. Connect the power cable to the motor.
4. Remove the long plug from the S port in protector top seal body. (The long plug should have a screwdriver slot).
5. Remove the standard plugs from the S and V ports located above the middle bag chamber.
6. Remove the standard plug from the S port located above the thrust bearing chamber.
7. Remove the short plug from the PF port in the protector base.

**Note**

Use a 3/4-in wrench to back up the valve to prevent it from loosening while removing the vent plug.

8. Remove the standard plug from the motor head D and F valve.
9. Install the oil-fill adapter, purge air from oil pump/hose assembly and connect the hose.

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10. Slowly pump the specified motor oil through motor head until bubble-free oil flows from the PF port in the protector base. Install a long plug with a new lead gasket.

**Note**

Vent plugs are torqued to 15 ft-lbf.

11. Continue pumping until bubble-free oil flows from the S port above the thrust bearing chamber. Install a standard plug with a new lead gasket.
12. Continue pumping until bubble-free oil flows from the V port located on top of the middle bag chamber. Install a standard plug with a new lead gasket.
13. Continue pumping until bubble-free oil flows from the S port located on top of the middle bag chamber. Install a standard plug with a new lead gasket.
14. Continue pumping until bubble-free oil flows from the S port located in the protector upper seal body. Install a short plug with a new lead gasket.
15. Continue pumping until bubble-free oil flows from the NPT ports in the protector head.
16. Remove the filling pump/adaptor from the motor head and install a standard plug with a new lead gasket.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

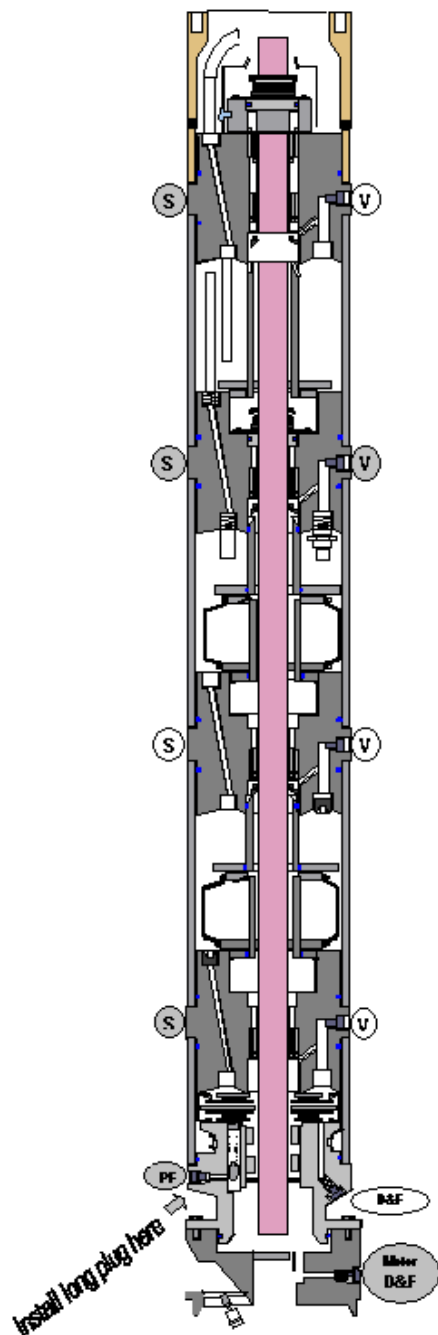


Figure 4-66: LSBPB Extended Runlife Protector.

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4.9.20

BSMPM Extended Runlife Protector oil fill procedure

An BSMPM Extended Runlife Protector is shown in [Figure 4-67](#).



Caution

Vent plug and drain-and-fill valve

Potential Severity:	Light
Potential Loss:	Assets
Hazard Category:	Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

1. Remove the shipping caps and the four 1/8-in NTP plugs in the protector head.



Note

If this protector will be installed below another protector, the four NPT plugs in the head of the lower protector should not be removed.

2. Install the protector between the motor and intake.
3. Connect the power cable to the motor.
4. Remove the long plug from the S port in protector top seal body. (The long plug should have a screwdriver slot).
5. Remove the standard plug from the V port in protector top seal body.
6. Remove the standard plugs from the S and V ports located above the middle chamber.
7. Remove the standard plug from the S port located above the thrust chamber.
8. Remove the short plug from the PF port in the protector base.



Note

Use a 3/4-in wrench to back up the valve to prevent it from loosening while removing the vent plug.

9. Remove the standard plug from the motor head D and F valve.

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10. Install the oil-fill adapter, purge air from oil pump/hose assembly and connect the hose.
11. Slowly pump the specified motor oil through motor head until bubble-free oil flows from the PF port in the protector base. Install a long plug with a new lead gasket.

**Note**

Vent plugs are torqued to 15 ft-lbf.

12. Continue pumping until bubble-free oil flows from the S port above the thrust chamber. Install a standard plug with a new lead gasket.
13. Continue pumping until bubble-free oil flows from S and V ports above the middle chamber. Install standard plugs with new lead gaskets.
14. Continue pumping until bubble-free oil flows from the V port located in the protector top seal body . Install a standard plug with a new lead gasket.
15. Continue pumping until bubble-free oil flows from the S port located in the protector top seal body. Install a short plug with a new lead gasket.
16. Continue pumping until bubble-free oil flows from the NPT ports in the protector head.
17. Remove the filling pump/adapter from the motor head and install a standard plug with a new lead gasket.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

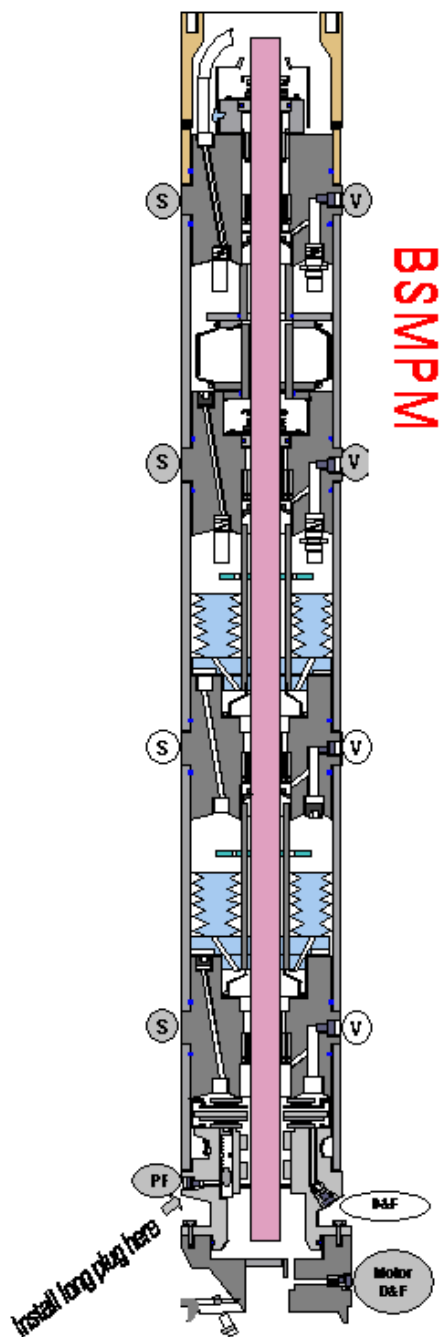


Figure 4-67: BSMPM Extended Runlife Protector.

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4.9.21

BSM Extended Runlife Protector oil fill procedure

A BSM Extended Runlife Protector is shown in [Figure 4-68](#).

**Caution****Vent plug and drain-and-fill valve**

Potential Severity: Light

Potential Loss: Assets

Hazard Category: Machinery equipment hand tools

Ensure the vent plugs and drain-and-fill valves used are the latest revision per the OneCAT bill of materials. Make sure that a calibrated torque wrench is used to install plugs. Vent plugs should be torqued to 15 ft-lbf and the drain-and-fill valves should be torqued to 35 ft-lbf.

1. Remove the shipping caps and the four 1/8-in NTP plugs in the protector head.

**Note**

If this protector will be installed below another protector, the four NPT plugs in the head of the lower protector should not be removed.

2. Install the protector between the motor and intake.
3. Connect the power cable to the motor.
4. Remove the long plug from the S port in protector top seal body. (The long plug should have a screwdriver slot).
5. Remove the standard plugs from the S and V ports located above the lower chamber.
6. Remove the standard plug from the S port located above the thrust chamber.
7. Remove the short plug from the PF port in the protector base.

**Note**

Use a 3/4-in wrench to back up the valve to prevent it from loosening while removing the vent plug.

8. Remove the standard plug from the motor head D and F valve.
9. Install the oil-fill adapter, purge air from oil pump/hose assembly and connect the hose.

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10. Slowly pump the specified motor oil through motor head until bubble-free oil flows from the PF port in the protector base. Install a long plug with a new lead gasket.

**Note**

Vent plugs are torqued to 15 ft-lbf.

11. Continue pumping until bubble-free oil flows from the S port above the thrust chamber. Install a standard plug with a new lead gasket.
12. Continue pumping until bubble-free oil flows from the S and V ports located above the middle chamber. Install standard plugs with a new lead gaskets.
13. Continue pumping until bubble-free oil flows from the V port located in the protector top seal body. Install a standard plug with a new lead gasket.
14. Continue pumping until bubble-free oil flows from the S port located in the protector top seal body. Install a short plug with a new lead gasket.
15. Continue pumping until bubble-free oil flows from the NPT ports in the protector head.
16. Remove the filling pump/adaptor from the motor head and install a standard plug with a new lead gasket.

**Note**

Confirm that all vent plug(s) and drain-and-fill valve(s) are flush to or recessed into the equipment base, body and/or head after servicing. Vent plugs are torqued to 15 ft-lbf and drain-and-fill valves are torqued to 35 ft-lbf.

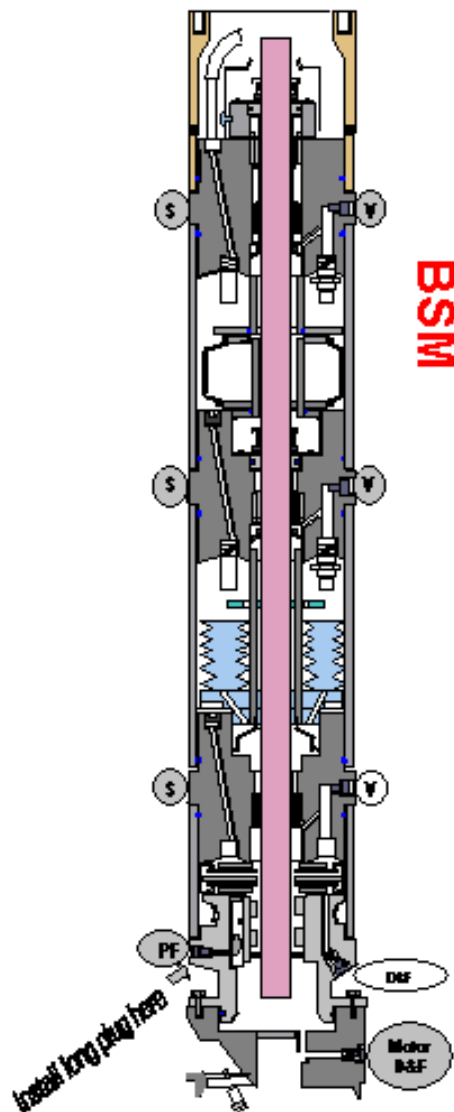


Figure 4-68: BSM Extended Runlife Protector.

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4.10

Power Cable Banding

**Note**

Signode air bander repair manual can be found on [InTouch Content ID 3465535](#).

Inspect the condition of the rig equipment to ensure that it is in good working order. Tubing slips should be clean and sharp. The backups on the power tongs should be in good condition to prevent the tubing from rotating when making up a joint. Inspect the backup dies for buildup of paraffin that would allow the tubing to slip. Any questionable equipment should be reported to rig supervisor before proceeding with the installation.

**Note**

Many power cables have been damaged because the backups were not turned over or slipped while making up the first joint.

The rig must be centered over the hole to provide adequate clearance for the cable and cable bands. This alignment should be checked periodically when running the equipment in the hole.

Instruct the person operating the automatic cable spooler to ensure the cable feeds off the reel at an even speed and tension. The practice of allowing the cable tension, as the unit is being lowered, to furnish spooling power should not be tolerated. The cable should always be spooled off the top of the reel.

54 in (1.4 m) cable sheaves are recommended because larger cable sheaves reduce stress to the cable. Once the flat cable extension (FCE) splice is banded to the tubing, the cable sheave can be raised to its running height of not more than 30 ft (9 m) above the ground. It should be secured with its load sling and a safety sling attached to a separate support.

To avoid damage to the flat cable make sure that it does not pass over any of the lock plates on the equipment. Once the FCE (Flat Cable Extension) is installed ensure it does not pass over a lock plate on the equipment as this will increase the OD of the equipment or this will reduce the available clearance and possibly cause a premature failure.

Pneumatic banding is preferred to hand banding since tension applied to the band can be controlled with far greater precision. Bands must be placed so that the buckle is located on the right hand side of the cable in the void between the cable and the tubing. This will prevent the buckle from being pushed into

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the cable and damaging it when the band is being tightened. Cable bands should be tight enough to distort the armor but not crush it. If a band is loose, remove and replace it. When placing bands on the flat cable extensions, make sure the cable fits against the flat spots on the head and base of each piece of equipment. The flat cable extension splice must be a minimum of 5 ft (1.5 m) above the pump discharge head and not across a tubing collar. Bands must be installed above and below all splices to prevent any cable load from being transmitted to the splice.

Typically, two bands per tubing joint is the minimum requirement for supporting the power cable. One band should be placed 4 ft (1.2 m) above the tubing collar and one in the middle of the joint. Consideration should be given to installing several bands (10 per joint) to the first two joints. This practice provides additional cable support in the area of “tubing cut off” that may occur during a fishing operation.

4.10.1

Installation of Cable Protectolisers with Non-Pre-Engaged Fasteners

The following procedure should be used to install Lasalle cable protectolisers.



Note

Use a standard pneumatic installation kit for this procedure.



Note

All protectolisers are supplied in the closed position.

1. Undo the saddle clamp fastener at the top of the protectoliser to enable access for the cable.
2. Undo the protectoliser cap-screw fastener using the impact wrench (supplied). This will allow the collar to be opened.
3. Move the protectoliser up to the pump component flange neck and position the cable in the slot.
4. Close the protectoliser around the pump component flange neck and engage the cap-screw fastener by at least two to three full threads into the protectoliser body using the T-bar supplied with the installation kit. (If any resistance is met, then the cap-screw fastener is cross-threaded. Any further tightening will result in the protectoliser threads being damaged and the protectoliser having to be replaced.)

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5. Tighten the cap-screw fastener using the impact wrench until the unit stalls (between five to eight seconds).

**Note**

The collar/body cap-screw fasteners are torqued to 30 lbf-ft. The saddle clamp/body cap-screw fasteners (if used) are torqued to 20 lbf-ft.

6. The cap-screw fastener torque applied using the impact wrench should be periodically confirmed using the torque wrench supplied with the installation kit.

4.10.2

Removal Of Cable Protectolisers With Non-Pre-Engaged Fasteners

The following procedure should be used to remove Lasalle cable protectolisers.

**Note**

Use a standard pneumatic installation kit for this removal procedure.

1. Ensure the hexagonal bit of the impact wrench is fully inserted into the cap-screw head before activating the impact wrench.
2. Activate the impact wrench and fully remove the cap-screw from the saddle clamp and remove the cable.
3. Activate the impact wrench and fully remove the cap-screw from the protectoliser body.
4. Remove the cable protectoliser from the pump string.
5. Using the T-bar (provided in the installation kit), engage the cap-screw fasteners by at least two to three full threads into the protectoliser body and saddle clamp. If any resistance is encountered, the cap-screw fastener is cross-threaded and any further tightening will result in the protectoliser threads being damaged and the protectoliser having to be replaced.
6. Tighten the cap-screw fasteners using the impact wrench (provide in the installation kit).
7. The protectoliser can now be stored for future use.

4.10.3

Installation of All-Cast Cable Protectors

The following procedure is used to install Lasalle all-cast cable protectors.

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**Note**

Use a standard pneumatic installation kit for this procedure.

**Note**

All protectors are supplied in the closed position.

1. Disengage the protector collar locking mechanism using the impact wrench. This involves ensuring the cap-screw fastener is fully retracted from the toggle mechanism. Rotate the toggle 90 degrees then remove the toggle from the locating lug of the protector body.
 2. Move the protector up to the tubing ensuring the cables/control lines are correctly positioned in the appropriate slots.
 3. Close the protector around the tubing ensuring the toggle is secured within the locating lug of the protector body.
 4. Torque the cap-screw fasteners to 30 lbf-ft using the impact wrench from the installation kit until the unit stalls (between 5 to 8 seconds).
-

**Note**

The cap-screw fastener torque applied using the impact wrench should be periodically confirmed using the torque wrench supplied with the installation kit. The torque applied on the first 10 protectors should be confirmed followed by 10% of the remainder. A change of installation personnel should also initiate a confirmation of torque values.

4.10.4

Removal of All-Cast Cable Protectors

The following procedure is used to remove Lasalle all-cast cable protectors.

**Note**

Use a standard pneumatic installation kit for this removal procedure.

1. Ensure the hexagonal bit of the torque wrench or T-bar is fully inserted into the cap-screw.
2. Release the tension of the fastener a few turns prior to fully unscrewing the fastener with the impact wrench.
3. Remove the cable protector from the tubing.
4. Using the T-bar engage the cap-screw fastener into the protector body.

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5. Fully tighten the cap-screw fastener using the impact wrench from the installation kit.
6. The protector can now be stored for future use.

4.11

Lowering Equipment into the Well

This practice will ensure that the motor is physically connected to the rest of the equipment. There is a tool (**drawing 100075454**) available that will allow the complete string of equipment to be rotated.

Once the last pump assembled to the string the complete assembly should be rotated using the correct tool, while this is being done a rotation meter should be connected to the motor leads to determine that the motor is turning. This can also be used to determine phase rotation. If a rotation meter is not available a Fluke or Triplet may be used to detect the EMF.

When running the unit into the well, the rig operator must be aware of the cable at all times. Maintain a running speed of approximately 1000 ft/hr (305 m/hr) to allow the spooler to provide minimum cable tension. All stops and starts must be slow and smooth to prevent the tubing from bouncing. Bouncing will damage the cable and protector seals.

4.11.1

Suspending ESP String

It is not recommended to interrupt installation and leave equipment hanging in the well or derrick overnight. There are, however, cases where this is unavoidable. In such cases the following practices are recommended.

Suspending Equipment Above Ground

Submersible equipment is designed to be hung and will not support its own weight. A string may be hung (not racked) in the derrick provided it is hung by equipment that is rated and certified for the specific use of lifting ESP equipment. To minimize hazards, it must also be secured in a way to eliminate horizontal movement.

Suspending Equipment in Well Prior to Run-In-Hole (RIH)

Job timing must be taken into consideration before starting to assemble equipment. If equipment cannot be run immediately after assembly, the equipment may need to be hung in the well hole and the hole sealed for control. This most often occurs on unlit land rigs which operate only during the day

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light hours. BOPs and HyDrills cannot fully seal around cable and may cause damage; therefore, the MLE should not be installed. In this case, the following procedure must be followed.

Assemble the equipment string according to the procedures in this manual, including servicing the motor. Do not connect the MLE to the motor. Completely service the protector as per procedure. Equipment can then be hung in the well and the BOP can seal around the tubing. When the equipment is retrieved, remove the long plug in the PF port before attaching the MLE (this is to reduce the amount of oil lost from the pothead.) Attach the MLE to the pothead and service the protector as per procedure, replacing the long plug when appropriate. Ensure that the oil is being pumped in through the D&F Valve in the motor head. Continue running equipment in hole as per normal operating procedure.

There have been problems resulting from assembling equipment, not servicing it with oil and then lowering it into a well full of kill fluids. The equipment will always fill with well fluid in these circumstances resulting in premature failure.

Suspending Equipment During Run-In-Hole (RIH)

If during the run it is decided the HyDrill must be closed around the cable then the closing pressure should be limited to 500 lbf/in² (3447 kPa) to minimize cable damage. Always electrically test and examine the cable for damage prior to continuing the run.

There is no way to fully secure the well around a cable. Closing the HyDrill around the cable as described above will minimize the leakage area but will not eliminate it. To fully secure the well the cable would have to be cut and the HyDrill closed around the tubing above the cable. Then the cable would need to be repaired (spliced) when the running operation continued.

4.11.2

Check Valve

A check valve may be installed at the pump discharge when installing a pump in a well where no gas problem exists. In gassy wells, the check valve should be installed one to six joints above the pump depending on the amount of gas. This practice provides a compression chamber for the purpose of avoiding a gas lock in the pump.

4.11.3

Bleeder Valve

The bleeder valve is installed one to two joints above the check valve. Be sure to check the bleeder plug for tightness (torque 30 lbf).

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4.11.4 **Checking Cable**

Take cable readings approximately every 1000 ft (305 m). Phase-to-phase continuity tests should indicate a balanced reading that will increase as the cable is subjected to increased well temperatures. Phase-to-ground insulation tests should show a high megohm reading (greater than 1500 megohms) at the beginning and slowly drop off as the unit is lowered into the well. After the BOPs have been removed, a final cable check should be made to ensure the integrity of the power cable and motor.

4.11.5 **Pump Setting Depth**

The pump setting depth should be calculated to ensure that the motor bottom is above the perforations. If the pump will be set in the perforated area, a motor shroud must be installed to provide adequate motor cooling.

4.11.6 **Clearance Check Before Running in the Wellbore**

The field service technician should verify that the wellhead is full gauge and the blocks are centered over the wellbore. Often, the BOP stack will be installed prior to the technicians arrival at the wellsite, making visual verification difficult. If this is the case, the following procedure should be used to confirm proper alignment and adequate clearance for the cable to pass through unobstructed.

1. After installing the initial joint of tubing, it should be pulled back up and inspected to ensure there is no evidence of the cable rubbing or damage to the cable bands. Any interference, rubbing or snagging of the bands, clamps or cable must be investigated and corrected before continuing run-in-hole operations.
2. Perform clearance checks at regular intervals during run-in-hole operations to confirm the rig has not settled or shifted or other clearance problems have not developed. These checks can prevent loss of the cable and downhole equipment in the well.
3. The practice of pulling-out-of-hole one or two tubing joints should be performed several times during run-in-hole operations. This will help identify any settling or shifting of the rig, or if conditions in the wellhead or BOP change.

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4.11.7 Landing the Tubing Hanger

The tubing hanger should be landed carefully to prevent cable damage. Because a tubing hanger is often offset, it should be landed with the cable facing in the same direction as it was pulled. This will provide a much easier task of pipe fitting the wellhead.

4.12 Pulling Equipment from the Well

Equipment removed from a well is valuable equipment and should be treated appropriately. Avoid any rough handling. Pulling speeds should not exceed 1000 ft/hr (305 m/hr) in gassy wells. This is to allow time for gas to migrate out of the cable at a slower rate, thus reducing the chance of damage. Cable bands should be cut off rather than pried or torn off. Tearing off a band can cause the band to cut into the cable.



Note

Submersible equipment must not be racked in the derrick. It is designed to be hung and will not support its own weight. Equipment may be hung in the derrick provided it is hung by equipment that is rated and certified for the specific use of lifting ESP equipment.

4.12.1 Annulus Blowdown Requirements

Gas or vapor may be present in ESP power cables and penetrators even if the annular fluid is thought not to contain gas. This gas or vapor may or may not originate from the produced fluids and can be air, which has been entrained within the cable layers. Relieving the pressure is required to ensure that the ESP power cable is not damaged due to the effects of explosive decompression.

The field service technician shall ensure the client is aware of the proper implementation of the blowdown procedure, thereby ensuring the integrity of the power cable. If the annulus is not blown down, the risk of shortened runlife, increased expenditure or loss of revenue is enhanced.

Definitions

Term	Definition
blowdown rate	This rate of pressure relief is set to allow a controlled expansion of any gas trapped inside the cable and to give it time to start to migrate out of the cable. The recommended blowdown rate is 50 psi/min.

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step blowdown	The pressure drop allowed on each blowdown stage. Fifty percent maximum step blowdown is recommended to prevent any trapped gas from expanding to a volume large enough to damage the cable before it is allowed to migrate out of the cable.
rest period	The time between each step blowdown. Once any trapped gas has expanded to its maximum volume during the 50% maximum step blowdown it is essential that time is given for this expanded gas to migrate out of the cable before proceeding to the next blowdown stage. The recommended rest period is 30 minutes.
final threshold	The final pressure after the last blowdown stage. Although the potential expansion is at its greatest, once below this threshold, the mechanical integrity of the cable is sufficient to contain the pressure until it dissipates normally. The recommended final threshold is 500 psi or less.
agreed minimum pressure	This can be atmospheric in most cases but there may be variances. For example, a minimum positive pressure is maintained in an installation where the thermal dynamics may induce an undesirable vacuum in the annulus when the well is shut in.

Blowdown Procedure

The blowdown procedure can be used to blow down the intertubular void containing the submersible motor power cable and may be used in all situations and well conditions, independent of actual annulus or wellbore fluid, where a packer or other pressure-containment equipment separates the annulus from the process.

1. Record the starting pressure (that is, the maximum test pressure).
2. Blow down the annulus at a maximum of 50 psi/min. Continue until the annulus pressure is half of the starting pressure.
3. Shut in the annulus for 30 minutes. At the end of the 30 minute period, record the pressure.
4. Blow down the annulus at a maximum of 50 psi/min. Continue until the annulus pressure is half of the pressure recorded in the previous step (25% of the starting pressure).
5. Shut in the annulus for 30 minutes. At the end of the 30 minute period, record the pressure.
6. Repeat step 4 and step 5 until 500 psi or less has been shut in for the 30-minute period.
7. Blow down the annulus at a maximum of 50 psi/min to an agreed minimum pressure.

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**Example**

Example 1, maximum annulus test pressure = 4000 psi, minimum pressure = 50 psi

Total Time	Action Time	Annulus Pressure	Action
0 min	0 min	4000 psi	Record the starting pressure.
40 min	40 min	2000 psi	Blow down at 50 psi/min to 50 % of the starting pressure.
70 min	30 min	2000 psi	Shut in the annulus for 30 min, then record the new pressure.
90 min	20 min	1000 psi	Blow down at 50 psi/min to 50 % of the latest recorded pressure.
120 min	30 min	1000 psi	Shut in the annulus for 30 min, then record the new pressure.
130 min	10 min	500 psi	Blow down at 50 psi/min to 50 % of the latest recorded pressure.
160 min	30 min	500 psi	Shut in the annulus for 30 min, then record the new pressure.
170 min	9 min	50 psi	Blow down at 50 psi/min to the minimum pressure (50 psi).

**Example****Example 2**

If the annulus pressure starts at 0 psi, then increases to 800 psi when a tubing test is applied, the tubing must be bled back until the annulus pressure has dropped back at a maximum of 50 psi/min to 50 % of the increase (400 psi). It must be held here for 30 minutes, then bled back to the original conditions. Irrespective of the tubing pressures and volumes, the blowdown must be controlled with respect to the annulus pressure.

4.12.2

Information Discovery After a 'Short-Run' Event

Typically, the first person to come in contact with a 'short-run' is the Field Service Technician. For the purpose of discussion, a short run will be one less than twelve months from the last date of installation. It is the desire of all involved parties to determine the cause of the failure and correct the problem.

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**Warning**

Potential Severity: Serious
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

Do not teardown or inspect equipment in the field. All Schlumberger and third party equipment needs to be shipped back to a certified facility. If equipment is taken apart in the field valuable information can be lost. Failures need to be analyzed in a controlled area and done by qualified personnel. Failure to follow this rule can lead to additional cost to your customer and to Schlumberger.

Following is a listing of some conditions worth noting on the pull report whenever a short-run failure has occurred.

Heat Indicators

- brittle, hardened, or cracked O-rings
- melted solder on motor leads (terminals or leads may pull off)
- blistered paint
- melted lead (flat cable extension)

Housings

- evidence of corrosion, erosion, abrasives, asphaltene, scale, etc.
- burn marks or holes in the housings
- signs of erosion inside the heads and bases of pumps (this is more noticeable on pump bases, particularly near the O-ring groove)

Shafts

- Are the splines twisted? Which piece? Head or base end?
- 'Turns Excessively Easy' Is it broken? Unless the piece in question is a 62D or S-type motor, it is relatively easy to determine if a shaft is broken.

Motors

- phase-to-phase and phase-to-ground readings

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- fluids in motor(s), type (oil, water, other)
 - color (clear, discolored, black)
 - bronze filings
 - location (top or bottom of motor)

**Note**

Do not state that a motor contained well fluid. Well fluid is a generic term and may be misleading to the client and investigators.

- Does the motor turn freely, rough, seized, or with great effort? The longer (heavier) the motor is the harder it will be to break it free.
- Are all vent plugs in place?

Protectors

- Types of fluids in each chamber. Fluid samples should be taken while the protector is still vertical immediately after being pulled from the well.
- Does there seem to be abnormal bushing wear?
- Do you see anything unusual when looking into the top of the protector? Is there any "debris" present? If so, note it and leave it in the protector. If you don't know what the debris is don't speculate.
- Are the vent plugs where they're supposed to be? Make sure and note that the long and short plugs have been put in the "shipping" vent holes.

Intakes

- Is the intake section plugged? If so with what? Is the screen in good condition or does it appear to be collapsed?
- Is the intake screen corroded?
- Is any foreign material visible inside the intake?
- Does there seem to be excessive bushing wear?

Pumps

Pumps can be subjected to abuse from abrasives, misapplication, corrosion, etc. If you think there is a problem affecting the operation of the pump first look for the obvious. If there is nothing clearly wrong, you may want to draw some comparisons. How does this pump differ from a new one? Was there any foreign material in the tubing adapter or check valve? Does the intake offer any clues?

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Cable

- physical condition — burned, jacket blown out, corroded, twisted, armor pulled apart
- resistance readings abnormal
- condition of flat cable extension. If there is a good reason to suspect the motor is burned, get the client's approval to cut the flat cable extension a foot above the pothead and leave this attached to the motor. This may help to determine the cause of motor failure. If the client will not allow this, install a clean pothole cap with a new O-ring. Also look in the pothole and see if there's any sign of water entry such as water tracks or stains.

Switchboard

- What does the recording amp chart show? Was the unit pumping off? Does the chart show erratic operation? Power spikes? Fluctuating load?
- What were the underload and overload shutdowns set at? Do these setpoints make sense?
- If the motor controller is a K095, download the data base and alarm logger history. Print this out and submit it with the pulling report.
- Are there any fuses blown?

General

Did the client mention anything unusual such as power disruptions, someone closing a valve or having difficulty keeping the unit running? If anyone makes statements or comments that you think may be important later, write down their name for future reference. Don't speculate. Make every possible effort to provide as much data as possible but don't comment on things that are not visible such as internal conditions.

4.12.3

Storing Equipment for Future Use

If the possibility exists that the equipment will be reused or stored for future use, the protector should be drained and refilled with new oil. The pump should be flushed out, preferably with fresh water, then oiled and the caps installed.

The protector should not be reused unless the client will assume responsibility. If the protector will be stored or reused, it should be checked while still in the vertical position to determine whether well fluid has entered the lower section. The *Protector Assembly* section provides servicing instructions for protectors.

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**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

If a protector is placed in a horizontal position without first being drained, water will flow to the other section. If stored, this can cause a seal to malfunction on the next installation, resulting in a burned motor.

If the oil in the motor contains water or is contaminated, it should be flushed out with clean oil and capped. Water can enter the motor only after passing through the protector (assuming the cable connection at the motor head and the connection between the motor and protector are secure). Since the motor is in a vertical position, any water will be in the bottom and can be checked by removing the plug in the motor base. If no water is present but the oil is dirty, the oil should be replaced.

Shipping caps should be installed where applicable to prevent the leakage of oil or the entry of moisture or dirt.

The cable should be properly spooled onto a reel, sprayed with oil and stored out the sun.

The switchboard should be kept upright unless the oil has been removed from the dashpots.

Minor items are the check and bleeder valves. A coat of oil should preserve them in good condition.

The same precautions used in handling new equipment applies to used equipment. To avoid high repair costs due to mishandling, all precautions handling the equipment must be observed.

4.12.4

Checking Equipment for Immediate Rerun

In addition to the items outlined above, the following guidelines will be useful in determining whether equipment is suitable for immediate rerun.

Motors

- The motor phase-to-ground resistance should be a minimum of 1000 megohms with the megger and be balanced phase-to-phase.
- The motor should be drained, flushed and refilled with oil. Note the condition of the oil being drained from the motor. If brass fillings are present, bearing wear is likely which can lead to mechanical failure. After flushing and

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refilling, check phase-to-ground resistance again. If measurements have not improved, the risk of running the motor is high. (The phase-to-ground resistance of a new motor will measure more than 2000+ megohms)

- Visually check the entire unit. Inspect the housing for corrosion. Inspect the pothead for damage. Inspect the leads for any insulation deterioration. Check the terminals for tightness and good joints. Inspect the top of the motor for cuttings of any type and inspect the flange and sealing area for any washout or damage.
- The motor should turn smoothly with no abnormal drag.
- Check tandem units individually.
- If flat cable is rerun, it should be pressure tested and inspected for any deterioration.

Protectors

It is not recommended to rerun a protector. If the client wants to rerun a protector, then the following steps should be observed.



Note

Always keep the protector in a vertical position.

- Check flange and sealing areas at top and bottom. Remove the O-ring on the bottom and physically check the ID and OD for any damage.
 - Drain all sections of the protector, flush and reservice from the motor head drain and refill with oil. Use new lead gaskets at all vent plugs and use new O-rings. The *Protector Assembly* section provides servicing instructions for protectors.
 - The protector should rotate freely with no abnormal drag.
-



Note

If there is water in the lower section of the protector, it should not be rerun.



Note

Do not rerun bag-type protectors. There is not a good method to check for water inside the bags.

Pump and Intake

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Deciding to rerun a pump is the most difficult decision. Many factors must be considered.

- The pump should freely rotate with no drag.
- The shaft should not have any excess side-to-side movement.
- Inspect the discharge, lower pump flange and the intake flanges for corrosion and erosion. All sealing surfaces should be carefully inspected. Carefully inspect the threads on the discharge head. This is an area where corrosion and erosion often occurs.
- Check tandem units individually.

In addition to the preceding checks, the following should be considered.

- history of scale in the well, previous runs, etc.
- number of days run and type of fluid pumped (abrasive or relatively clean)
- barrels per day being pumped at time of unit being pulled (pump or well conditions)
- Why was well pulled? Cable failure, low production, tubing leak, well stimulation, etc.?
- Check for any indication of excessive heat. Excess heat is a sign of operating in downthrust, lack of fluid flow, or both.

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Cables

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5 Cables

5.1 Product Description

5.1.1 ESP Power Cable

The ESP power cable incorporates materials with high-integrity insulation and armor systems. Various grades of armor ranging from standard galvanized steel to Monel® alloys are available. Cables are made of either a modified variant of ethylene propylene diene methylene (EPDM) or an enhanced polypropylene formulation. The polypropylene formulation supports a high temperature rating of up to 121 degC [250 degF] without compromising mechanical and insulating properties. A basic configuration of the main components covers a wide range of ESP applications and is suitable for most well conditions. Each standard cable can be customized to suit the specific requirements of a given well, including temperature and pressure ratings, corrosive properties, and gas/oil ratios.

5.1.2 Motorlead Extension (MLE)

The motor lead extension (MLE) designs provide the optimum combination of installation clearance and downhole performance. Polyimide is used as the primary insulation to significantly reduce cable dimensions. For extreme high temperature or specialty applications, PEEK is utilized as the insulation material. The same patented and proprietary materials used in Schlumberger power cables are used in the MLE designs as secondary insulations and jackets to maximize system reliability. The MLE designs support temperature ratings of 121 degC [250 degF] and higher.

5.2 Health, Safety and Environment (HSE)

5.2.1 Storage and Handling

5.2.1.1 MLE Storage

Motorlead Extensions should always be stored in a dry environment, indoors with the shipping cap on and O-rings installed. Any pothead with a lead gasket should also be stored with the provided lead gasket, O-ring and ship cap. The MLE should be left in the same coiled configuration for storage until ready for install. The cable portion coming out of the pothead flange needs to be straight at all times – any side to side or excessive bending is a risk to cause conductor movement which has potential to break the bond between the conductor to sealing mechanism. Any MLE that is received with a desiccant pack(s) in the shipping cap should be stored with the desiccant packs remaining until ready for installation. The desiccant pack(s) should be blue. For packs that have turned pink in color refer to section [5.6.1: Electrical Problems](#).

Once an MLE is spliced to the cable, it is recommended to still store this indoors, with O-rings and gaskets installed and the ship cap secured.

5.2.1.2 Cable Handling

Power cable is typically shipped on large reels. The purpose of the reel is to allow easy cable feed into and out of the well and to protect the cable during transport and handling.

Cable should always be transported on cable reels. In the event that a power cable on a reel is being transported with the spliced flat cable extension on it, independently secure the extension and the power cable. A crane and spreader bar are recommended to lift cable reels. A forklift may be used. If a forklift is used, lift the reel through the spokes from the side, never against the cable. Reels should be secured to the truck when transported. Never allow the reel of cable to roll against objects that might crush or damage the cable or reel.

**Warning**

Potential Severity: Serious
Potential Loss: Assets, Personnel
Hazard Category: Machinery equipment hand tools

Always ensure that the weight of the load is less than the load rating of the lifting equipment. Never exceed the load rating of hoists, cranes, forklifts, spreader bars, chains, slings or other lifting equipment. Refer to the Artificial Lift Mechanical Lifting Standard ([InTouch ID 4557363](#)) for more information regarding lifting.

Prior to moving the cable ensure that the path the move will take is clear. Warn all personnel in the area that the move is about to take place.

Placement

A section of heavy-walled pipe, which has been approved for the purpose, should be inserted through the reel to serve as an axle. Lift with a spreader bar attached to the axle. A short spacer should be placed between the reel and the stand on each side.

Never handle a reel of cable by making a sling of wire line or chains that touch the cable. Protect the cable from being smashed or cut.

**Caution**

Potential Severity: Light
Potential Loss: Assets
Hazard Category: Machinery equipment hand tools

Always use a spreader bar when lifting cable reels. Never lift a reel of cable with a sling of wire line or chains.

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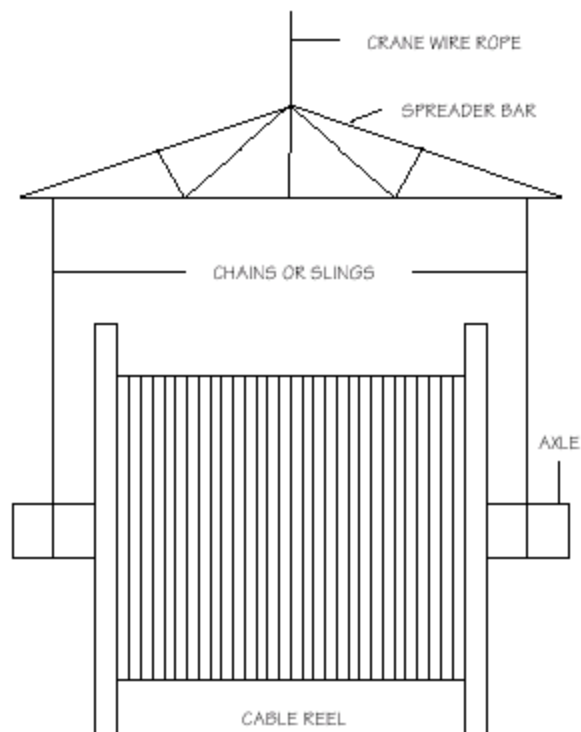


Figure 5-1: Cable reel lifting

A clear path should be provided for the cable to prevent it from being hooked or damaged during installation. Wooden planks should be used to keep the cable off the ground, preventing it from picking up any debris which may contaminate the well.

5.3 Specifications

5.3.1 Cables

There are three RedaMAX power cable families which are offered. They are RedaMAX250, RedaMax400 and Redalead. The cable specifications can be found in [OneCAT](#).

Table 5-1: ESP Power Cable Specifications

Type	Profile	Maximum Temperature	Insulation	Barrier	Jacketing
RedaMax 250 PPEO	Round	96 degC (205 degF)	Polypropylene	None	Nitrile

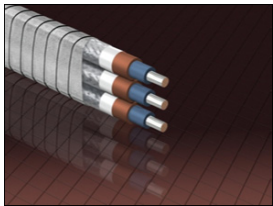
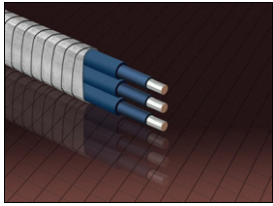
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Type	Profile	Maximum Temperature	Insulation	Barrier	Jacketing
RedaMax 250 PP	Flat	121 degC (250 degF)	Polypropylene	None	Polypropylene
RedaMax 250 PTB	Flat	121 degC (250 degF)	Polypropylene	Fluoropolymer Tape + Braid	None
RedaMax 250 POTB	Flat	121 degC (250 degF)	Polypropylene	Fluoropolymer Tape + Braid	Nitrile
RedaMax 400 EE	Round	204 degC (400 degF)	EPDM	None	EPDM
RedaMax 400 ETBE	Flat & Round	204 degC (400 degF)	EPDM	Fluoropolymer Tape + Braid	EPDM
Redalead ELB	Flat	232 degC (450 degF)	EPDM	Lead + Braid	None
Redalead ELBE	Round	232 degC (450 degF)	EPDM	Lead + Braid	EPDM
Redalead EHLTB	Flat	232 degC (450 degF)	EPDM	Heavy Lead + Fluoropolymer Tape	None
Materials and specifications are subject to change without notice.					

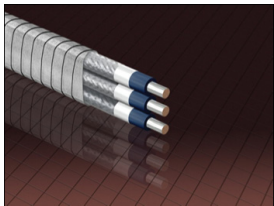
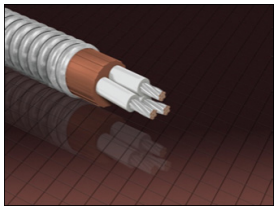
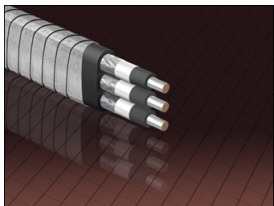
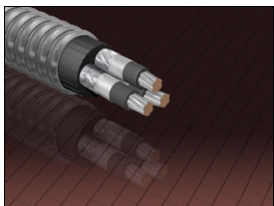
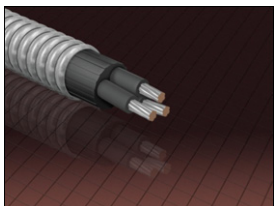
5.3.1.1 RedaMAX Cable

Table 5-2: RedaMAX Power Cable

Cable	Profile	Insulation	Barrier	Jacketing
 Figure 5-2: RedaMAX250 POTB	Flat	Polypropylene	Fluoropolymer Tape	Nitrile
	The POTB cable is designed primarily for oil well applications with moderate gas/oil ratio and moderate oil cuts.			
 Figure 5-3: RedaMAX250 PP	Flat	Polypropylene	None	Polypropylene
	The PP cable is designed primarily for oil well applications with medium-temperature, non-aggressive wells.			

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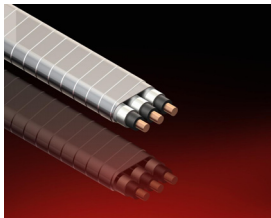
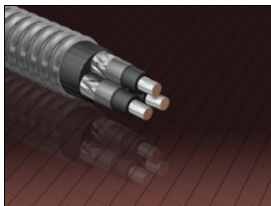
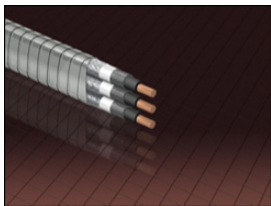
Cable	Profile	Insulation	Barrier	Jacketing
 Figure 5-4: RedaMAX250 PTB	Flat	Polypropylene	Fluoropolymer Tape	None
The PTB cable is designed primarily for oil well applications with medium-temperature , mildly aggressive wells.				
 Figure 5-5: RedaMAX250 PPEO	Round	Polypropylene	None	Nitrile
The PPEO cable is designed primarily for oil well applications with low-temperature and low levels of gas.				
 Figure 5-6: RedaMAX400 ETBE (flat)	Flat	EPDM	Fluoropolymer Tape	EPDM
The ETBE cable is designed primarily for oil well applications with high-temperature, moderate gassy wells.				
 Figure 5-7: RedaMAX400 ETBE (round)	Round	EPDM	Fluoropolymer Tape	EPDM
The ETBE cable is designed primarily for oil well applications with high-temperature, moderate gassy wells.				
 Figure 5-8: RedaMAX400 EE	Round	EPDM	None	EPDM
The EE cable is designed primarily for oil well applications with high-temperature, low GOR.				

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5.3.1.2 Redalead Cable

Table 5-3: Redalead Power Cable

Cable	Profile	Insulation	Barrier	Jacketing
 Figure 5-9: Redalead ELB	Flat	EPDM	Lead	EPDM
	The ELB cable is designed primarily for oil well applications with high-temperature, gassy and/or corrosive wells.			
 Figure 5-10: Redalead ELBE	Round	EPDM	Lead	EPDM
	The ELBE cable is designed primarily for oil well applications with high-temperature, gassy and/or corrosive wells.			
 Figure 5-11: Redalead EHLTB	Flat	EPDM	Heavy Lead + Fluoropolymer Tape	None
	The EHLTB cable is designed primarily for oil well applications with high-temperature, high gassy and/or high corrosive wells.			

5.3.2 Motor Lead Extensions

There are several MLE (motor lead extension) designs offered. Motor lead extension designs are shown in [Table 5-6](#). Physical properties for motor lead extensions are shown in [Table 5-5](#). Specifications for motor lead extensions can be found in [OneCAT](#). Standard MLE configurations can be found in [Table 5-6](#). Other designs are available depending on the options selected.

Table 5-4: Motor Lead Extensions (MLE) Specifications

ESP System	MLE Type	Pothead Connection	Maximum Temperature	Insulation	Barrier
Dominator & Maximus – 456 / 562 series	KEOTB	Tape & Plug-In	121 degC (250 degF)	Polyimide/ EPDM	Fluoropolymer Tape + Nitrile

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ESP System	MLE Type	Pothed Connection	Maximum Temperature	Insulation	Barrier
Dominator – 375 / 456	KELTB-LP	Tape & Plug-in	149 degC (300 degF)	Polyimide/ EPDM	Lead + Fluoropolymer Tape
Dominator & Maximus – 375 / 456 / 562 / 738	KELB	Tape-in (All)	149 degC (300 degF)	Polyimide/ EPDM	Lead
		Plug-In (456, 562)	204 degC (400 degF)		
Maximus – 562 / 738 series (Trident)	PEEKLB	Plug-in	149 degC (300 degF)	PEEK	Lead
			204 degC (400 degF)		
Dominator – 456 / 562 (Hotline I)	KEHLB	Tape-In	232 degC (450 degF)	Polyimide/	Heavy Lead
562 series Hotline550	PEEKLB	Tape-In	260 degC (500 degF)	PEEK	Lead
562 series Hotline SA3	XHLTB	Plug-In	288 degC (550 degF)	High Temp Polyimide	Heavy Lead + Fluoropolymer Tape
Materials and specifications are subject to change without notice.					

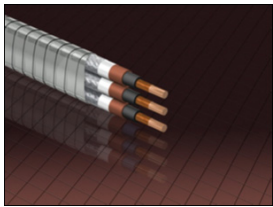
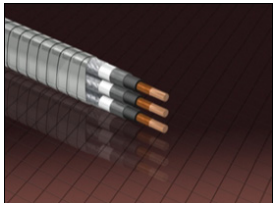
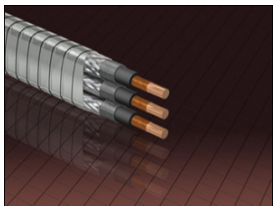
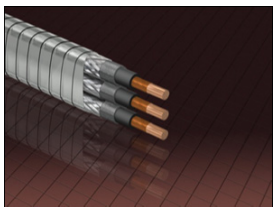
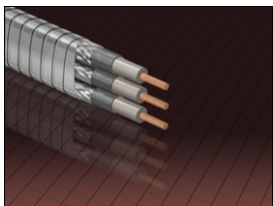
Table 5-5: Physical Properties of Motor Lead Extensions

Temperature range	The maximum conductor temperature range is from 250 to 550 degF (121 to 288 degC). The temperature rating is based on the selected options.
Conductor	The conductors are fully annealed, high conductivity copper.
Insulation	The KELB, KEOTB and KELTB insulation is made of dual layers of Polyimide tape. The secondary insulation is EPDM (ethylene propylene diene methylene). The PEEKLB insulation is made of extruded PEEK (polyetheretherketone).
Barrier	Lead and fluorocarbon tape barriers prevent chemical attack and gas decompression failures for superior performance in hot, gassy environments.
Jacket	A proprietary nitrile rubber formulation with exceptional physical properties and oil resistance is used in the KEOTB design.
Armor	Galvanized steel and Monel are standard.

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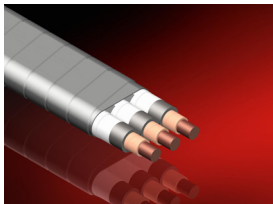
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Table 5-6: Motor Lead Extensions

MLE	Profile	Insulation	Barrier	Jacketing
 Figure 5-12: KEOTB	Flat	Polyimide/ EPDM	Fluoropolymer Tape	Nitrile
	Dominator & Maximus – 456 / 562 series Tape & Plug-In			
 Figure 5-13: KELTB	Flat	Polyimide/ EPDM	Lead + Fluoropolymer Tape	None
	Dominator – 375 / 456 Tape & Plug-in			
 Figure 5-14: KELB	Flat	Polyimide/ EPDM	Lead	None
	Dominator & Maximus – 375 / 456 / 562 / 738 Tape & Plug-In (456, 562)			
 Figure 5-15: KEHLB	Flat	Polyimide/ EPDM	Heavy Lead	None
	Dominator – 456 / 562 (Hotline I) Tape-In			
 Figure 5-16: PEEKLB	Flat	PEEK	Lead	None
	Maximus – 562 / 738 series (Trident), Plug-in 562 series Hotline550, Tape-In			

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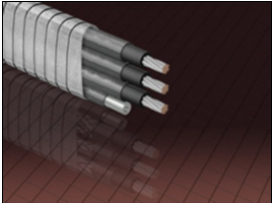
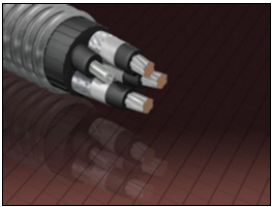
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MLE	Profile	Insulation	Barrier	Jacketing
 Figure 5-17: XHLTB	Flat	High Temp Polyimide	Heavy Lead + Fluoropolymer Tape	None
	562 series Hotline SA3 Plug-in			

5.3.3 Injection Tubes

Cables with integral stainless steel tubing(s) are available. This tubing is used for chemical injection or downhole control applications. Combining the power cable and injection tube into one package simplifies workover operations and prevents damage to the injection tube. Flat and round cables with integral injection tubes are shown in [Figure 5-18](#) and [Figure 5-19](#).

Table 5-7: Cable with Integral Tube Specifications

	Profile	Maximum Temperature	Insulation	Barrier	Jacketing
 Figure 5-18: Flat cable with integral tube	flat	Refer to cable specifications			
	Flat cable with an integral stainless steel tube.				
 Figure 5-19: Round cable with integral tube	round	Refer to cable specifications			
	Round cable with an integral stainless steel tube.				

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5.4 Technical and Engineering Information

5.4.1 Cable Nomenclature

Nomenclature uniquely identifies the components of cable construction. The designations indicate the materials selected, starting with the insulation and ending with the armor. Nomenclature for cables is shown in [Table 5-8](#).

Table 5-8: Cable and MLE Nomenclature

Component	Symbol	Description
conductor	#/1	# AWG/solid conductor
	#/7	# AWG/stranded (round) conductor
	#/7 C/S	compacted strand conductor
insulation	PPE, P	polypropylene copolymer
	E	EPDM (ethylene propylene diene methylene) rubber
	PEEK	PEEK
	K	polyimide film
	X	high-temperature polyimide
barrier	TB	high-temperature tape with braid
	L	lead
	HL	heavy lead
jacket	PE	HDPE (high-density polyethylene)
	P	polypropylene copolymer
	O	oil-resistant nitrile rubber compound (four temperature ratings).
	E	EPDM rubber compound
armor	G	galvanized steel
	HG	heavy-galvanized steel
	GG	double-galvanized steel
	SS	stainless steel
	M	monel
voltage rating	3	3 kV cable voltage rating
	4	4 kV cable voltage rating
	5	5 kV cable voltage rating
	8	8 kV cable voltage rating

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Component	Symbol	Description
geometry	F	flat cable configuration
	R	round cable configuration
	S	single cable configuration

Examples

ETBE SS5R

Term	Definition
E	EPDM insulation
TB	tape and braid
E	EPDM jacket
F	flat profile armor
SS	stainless steel armor
5	5kV rating
R	round cable

POTB G3F

Term	Definition
P	polypropylene insulation
O	oil-resistant nitrile jacket
TB	tape and braid
G	galvanized steel armor
3	3 kV rating
F	flat cable

PPEO GG5R

Term	Definition
PPE	polypropylene insulation
O	oil-resistant nitrile jacket
GG	double-galvanized armor
5	5kV rating
R	round cable

ELB M4F

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Term	Definition
E	EPDM insulation
L	lead barrier
B	braid
M	monel armor
4	4kV rating
F	flat cable

5.4.2

Cable Construction Materials

Schlumberger uses a modular cable concept. A cable is considered a customized assembly of components (conductor, insulation, etc.). The selection of these components is based on a thorough analysis of the client's well conditions and expected performance. Through the proper selection of materials, the cable design can be precisely tailored to provide best-in-class application performance. The focus is on optimizing total value (price and performance) to the client.

Commonly used materials for various cable components are:

Conductor

- stranded
- compacted strand
- solid

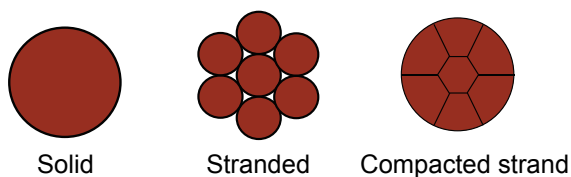


Figure 5-20: Cable conductor

Insulation

- polypropylene copolymer
- EPDM (ethylene propylene diene methylene) rubber

Barrier

- high-temperature tape
- lead

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Jacket

- HDPE (high-density polyethylene)
- oil-resistant nitrile rubber formation (four temperature ratings)
- EPDM rubber formation



Figure 5-21: Cable jacket

Armor

- galvanized steel (0.020 in, 0.025 in or 0.034 in thick strips available)
- stainless steel
- monel

Depending on the type of cable and the application, the armor can be formed and wrapped in several different profiles:

- Flat profile (used for flat cables)
- Flat interlocked profile (used when a lower profile is required on round cable)
- Round profile (typically used on round cables)

The option of double armored layers is also available.

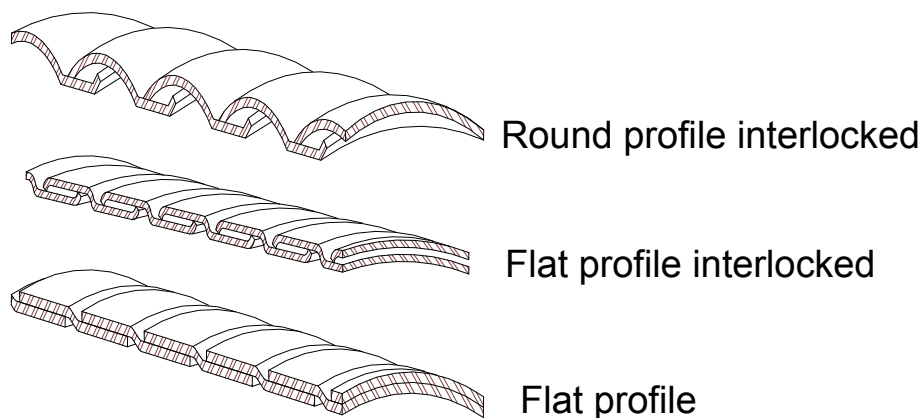


Figure 5-22: Cable armor

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Outer Jacket

For seafloor, low temperature environments, a PVC (poly-vinyl-chloride) extruded jacket can be applied on the exterior of the metallic armor layer.

5.5 Wellsite Procedures

5.5.1 Overview



Note

In cold climates splicing tapes should be kept warm before and during splicing a cable. This can be done by keeping the tapes in a pocket. Tapes should be kept at 25 degC or 77 degF.

The following guidelines and procedures are used to ensure maximum-performance cable splices. Different procedures are used based on the cable type and configuration. Do not deviate from these procedures without engineering approval.

Rig personnel, contract consultants and clients are often critical of the time required to make a splice. There are few clients that cannot afford the extra time required to do a quality job. Do not give in to pressure placed on you by personnel who are unfamiliar with the importance of following proper cable splicing procedures. A successful cable splice starts with the following procedure.

1. Plan ahead. Make arrangements to arrive on location early if you know there is a cable splice required before running in the hole.
2. If a lower pigtail will be installed after the unit is on bottom, have the leads cut and prepped ahead of time. Have the splicing table, tools and materials ready by the time the end of the reel is reached.
3. Energize the soldering iron before the armor is applied so that it's ready when needed.



Note

It is recommended that there is no less than 500 ft between splices. We do not have a recommendation for the maximum number of splices allowed in a cable run; however, multiple splices can severely affect run life.

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5.5.2 Cable splicing

5.5.2.1 Termination Procedures

[Power Cable Termination for Splicing \(100285583\)](#)

[Flat Cable Extensions Terminations for Splicing \(100285587\)](#)

[Splice Code And Nicopress Part Numbers \(100650432\)](#)

5.5.2.2 MLE Splicing Procedures

[MLE to Round Profile Power Cable Splice \(100716529\)](#)

[MLE to Flat Profile Power Cable Splice \(100357294\)](#)

[4/1 XTHLB \(or XHLTB\) MLE to 4/1 EHLTB Cable Splice \(100979795\)](#)

[Hotline II \(Hotline550\) MLE and Power Cable Splice \(100389530\)](#)

[Non-Lead Splice \(100795611\)](#)

[5 Conductor Flat Splice \(100357298\)](#)

5.5.2.3 Splicing Procedure, Round Cable to Round Cable

1. Position the ends of the two power cables on the splice bench and secure the clamps onto the armor. Do not overtighten the clamps.
2. Prepare the cable ends. [Figure 5-23](#) shows dimensions for round cables without barriers. [Figure 5-24](#) shows dimensions for round cables with a taped or extruded barrier between the insulation and jacket. Carefully remove the armor to the dimension shown in the figures without cutting into the jacket. Remove any burrs or sharp edges on the cut and bend the armor to fit snugly around the jacket.

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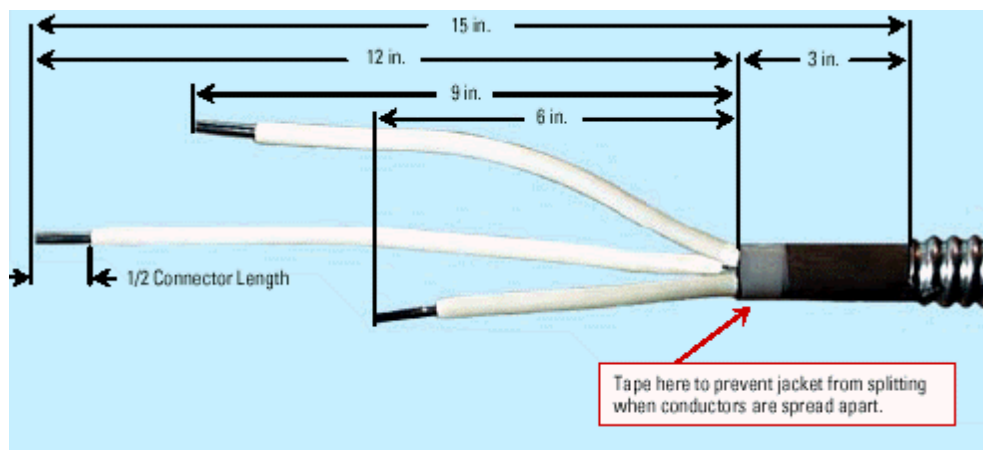


Figure 5-23: Conductor preparation for cables without barriers

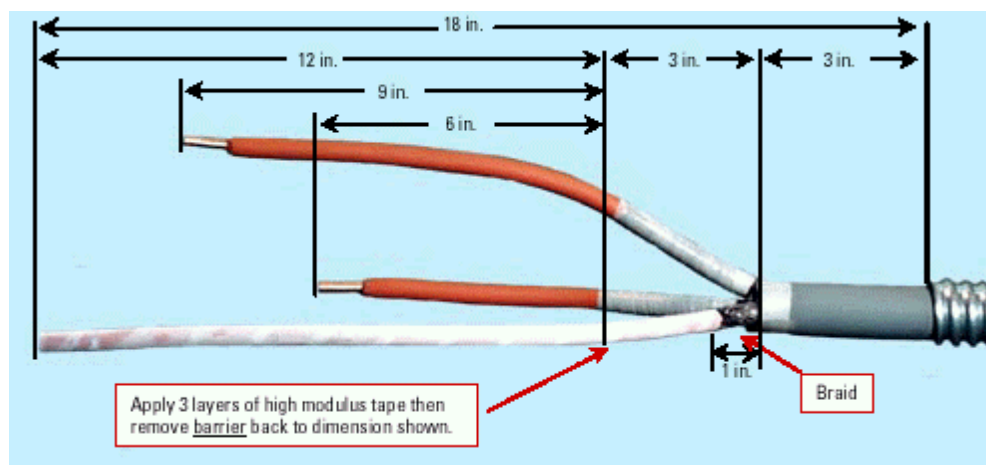


Figure 5-24: Conductor preparation for cables with barriers

3. Measure the jacket back to the dimension shown in the preceding figures. Slightly score the jacket with a knife at this location. Using a new piece of 50-grit emery cloth, abrade the jacket until it is smooth from the mark back to the armor.



Caution

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Never reuse emery cloth when abrading different materials such as copper, insulation or jackets. This will embed copper into the insulation and degrade the integrity of the splice.

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4. Apply three tight layers of high-modulus tape on the jacket at the insulation interface. The tape prevents the jacket from splitting when the conductors are separated during the splicing procedure. It will be removed later in the procedure once the conductors are pulled back together.
5. Remove the jacket back to the dimension shown in the preceding figures using extreme care to not damage the insulation. There must be no cuts in the insulation beyond the bare conductor area.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Cuts in the conductor insulation are not permitted past the bare conductor area. If the insulation is damaged when removing the jacket, remove the damaged area and start a new splice.

6. Remove any braid to the dimension shown. Clean the conductors using CRC (contact cleaner) or a naphtha-based solvent and a clean rag. Spread the conductors apart slightly using caution not to bend more than is necessary to apply tape. Apply three half-lapped layers of high-modulus tape to each conductor beginning against the outer jacket and extending 3 in (76 mm) toward the cut ends of the conductors. Apply the three layers tightly and continuously with no ridges at the transition areas.
7. If the cable uses barriers, remove any taped or extruded barrier back to the high-modulus tape. (See the preceding figure.)
8. Cut the conductors with a hacksaw back to lengths shown in the preceding figures.
9. Obtain the appropriate splicing sleeves. Descriptions and part numbers for the sleeves are shown in the *Splicing Sleeves* table in the *Parts, Tools and Materials* section. Remove the insulation to the dimension shown in the preceding figures.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Be careful not to nick or ring cut the copper when removing the insulation. This can create a hot spot which can result in a failure.

10. Ensure the sleeve fits all the way on and snugly against the insulation. If too much insulation was removed, use a file to trim the copper back to the proper dimension.

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11. Remove filler from the outside diameter of any stranded conductors.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Do not remove the alloy coating covering the copper conductor.

12. Prepare the end of the cable to be spliced. Spread the conductors apart slightly and place the cable ends together such that one conductor on the second cable butts against the shortest conductor on the first cable. Mark the other two conductors and cut where indicated. (See the preceding figures.) After cutting, check to ensure all three conductors butt up to each other when placed tightly together.
13. Remove the insulation from the ends of the second cable. Taper the insulation ends that butt up against the splicing sleeve using 100-grit emery cloth. Clean the exposed conductors with a clean cloth that has been moistened with CRC (contact cleaner). Allow to completely dry.

**Note**

5000-volt cable with EPDM-type insulation does not require tapered ends.

14. Use a megger to test both cables to ensure electrical integrity before proceeding.
15. Install the splicing sleeves over the conductors. Do not remove the abrasive material from the inside of the sleeves. Its purpose is to puncture the surface of the copper allowing a connection with very low resistance. Some sleeves also contain an antioxidant to prevent corrosion. Do not remove the antioxidant.
16. Use a crimping tool (nicopress tool) to place crimps in the sleeve. Refer to [Figure 5-25](#). Place a crimp at position **A**. Rotate the crimping tool 90 degrees and place a crimp at position **B**. Rotate the crimping tool 90 degrees and place a crimp at position **C**.

Place a crimp at position **D**. Rotate the crimping tool 90 degrees and place a crimp at position **E**. Rotate the crimping tool 90 degrees and place a crimp at position **F**.

**Note**

Crimp **A** and crimp **D** should be at the very end of the sleeve so there is no flare.

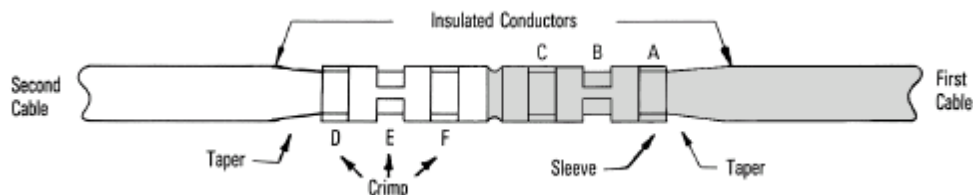


Figure 5-25: Splicing sleeve installation

17. Check the outside diameter of the pressed crimps with crimping gauge. The gauge must pass over the pressed crimp without binding.
18. Repeat the splicing sleeve installation procedure for the remaining conductors.



Caution

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Improper installation of splicing sleeves can result in splice failure.

19. Use a strip of 100-grit emery cloth on each sleeve to remove any burrs or sharp edges. Discard the emery cloth and clean the splice with CRC (contact cleaner) or naphtha-based solvent.
20. Use a new strip of 100-grit emery cloth to roughen all exposed conductor insulation.



Note

Do not use emery cloth on the insulation after it has been used on the splicing sleeves or a lead jacket.

21. Clean exposed insulation and splicing sleeves by wiping with a clean cloth that has been moistened with CRC or a naphtha-based solvent. Allow to dry completely.
22. Cover each splicing sleeve with five layers of high-temperature tape. Start the first layer of tape on the insulation 0.5 in (13 mm) from one end of the sleeve. Continue over the sleeve with half-lapped wraps until 0.5 in (13 mm) of the insulation is covered beyond the other end of the sleeve. Wrap the second layer in the opposite direction filling the void area in the taper flush with the nicopress sleeve. Extend each succeeding layer of tape 0.5 in (13 mm) beyond the ends of the previous layer. The tape must be wound continuously in a smooth spiral with no gaps or wrinkles.

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23. Cover the exposed area of each splice with six layers of high-modulus tape. Start the first layer of tape on the insulation one inch beyond the ends of the high-temperature tape. The tape should be held taut when applying and snugly wrapped around the insulation in a smooth spiral with no gaps or wrinkles. Continue over the sleeve with half-lapped wraps until one inch of insulation is covered beyond the other end of where the high-temperature tape finished. Cut the tape and begin a second layer wrapped in the opposite direction. Extend each successive layer 0.5 in (13 mm) out from the previous, cutting the tape at the end of each layer until the final layer covers the full length of the conductor and butts up against the end of the jacket.

**Note**

The sixth layer must cover the full length of the exposed insulated conductors.

24. Apply two half-lapped layers of high-temperature tape over the full length of each conductor to seal and secure the high-modulus tape.
25. Loosen the cable clamps and carefully stretch both cables to their original orientation. Remove the tape that was securing the jacket while the conductors were spread apart. Clean the exposed jacket by wiping with a clean cloth moistened with CRC (contact cleaner) or a naphtha-based solvent. Allow to dry completely.
26. Cut three lengths of Impervoil filler strip. The strip should extend the full distance of the insulated conductors between the cut ends of the jacket. Clean each length of filler strip with a cloth moistened with CRC (contact cleaner) or a naphtha-based solvent. Place filler strips between the conductors and secure with high-modulus tape.
27. Apply two half-lapped layers of high-modulus tape over the conductors and filler strips. The first layer should extend 1.5 in over the jacket at each end of the splice. The second layer, wrapped in the opposite direction, should cover the full length of the splice up to, but not on, the armor.
28. Unwrap one loop of armor at each end of the splice. Apply two half-lapped layers of glass tape over the completed taped area extending under the unwrapped armor. Bend the unwrapped armor back to fit snugly around the conductor.
29. Cut a 46-in (1.2-m) length of the appropriate armor. Pull out the tucked end; trim and flatten it. Use a round wooden rod or a piece of electrical conduit with a slightly smaller OD than the ID of the armor as a wrapping lever. Starting 2 in (51 mm) from the splice, form the armor around the splice in a spiral using the stick for leverage to draw the armor tight. Adjust the profile to expand around the spliced section while maintaining the interlock. Continue wrapping 2 in (51 mm) beyond the existing armor.

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30. Cut and secure both ends by soldering the last three convolutions with acid-core solder. Wipe away excess flux with a damp cloth and remove any excess solder from the armor with a surform file.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Avoid heat damage to the cable under the soldered area. Do not use a torch. Cool the soldered joint immediately with a water-wetted cloth. Avoid contact with hot surfaces.

31. Test the spliced cable to check electrical integrity. Test procedures are provided in .

5.5.2.4

Splicing Procedure, Flat Cable to Flat Cable

1. Position the ends of the two power cables on the splice bench and secure the clamps onto the armor. Do not overtighten the clamps.
2. Prepare the cable ends. Figure 5-26 shows dimensions for cables without barriers. Figure 5-27 shows dimensions for cables with a taped or extruded barrier between the insulation and jacket. Carefully remove the armor to the dimension shown in the figures without cutting into the jacket. Remove any burrs or sharp edges on the cut and bend the armor to fit snugly around the jacket.

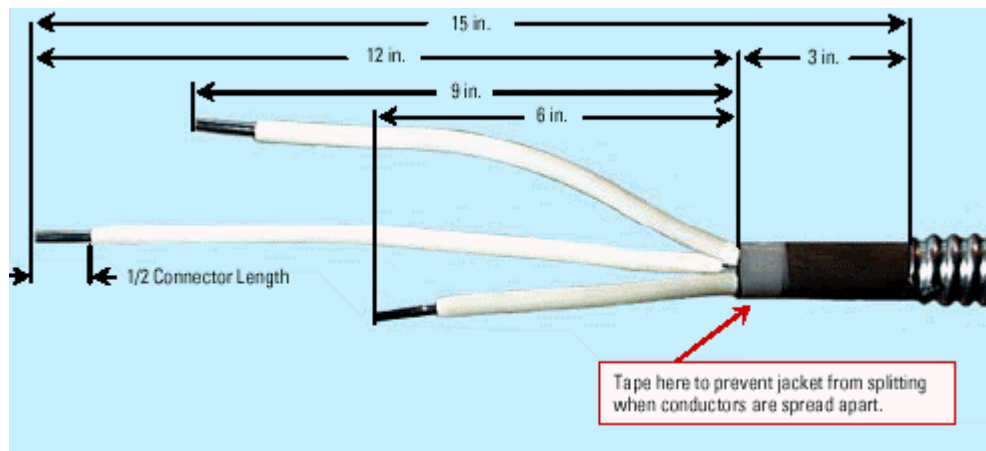


Figure 5-26: Conductor preparation for cables without barriers

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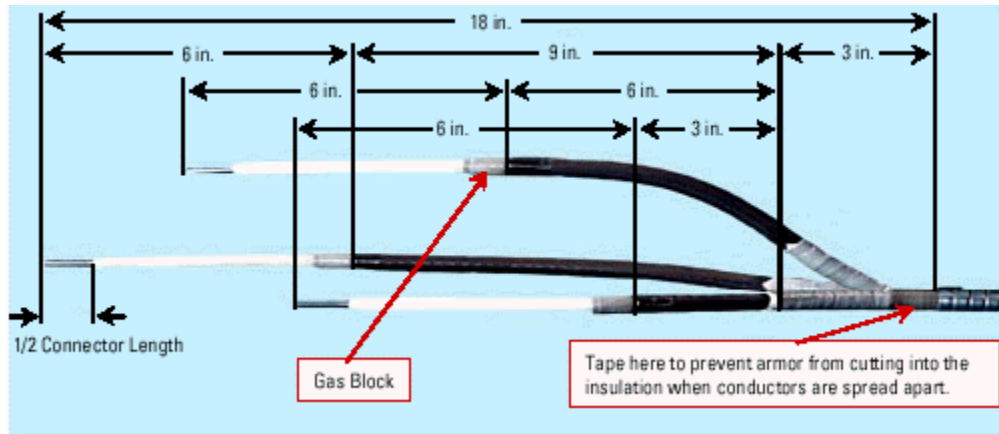


Figure 5-27: Conductor preparation for flat cables with barriers

3. Remove any braid with scissors to the dimension shown in the preceding figures. Clean the conductors using CRC or a naphtha-based solvent and a clean rag.
4. Remove the tape barrier back to the dimension shown in the preceding figures, Seal the barrier to the insulation with two wraps of high-temperature tape.
5. Remove the jacket back to the dimension shown in the preceding figures using extreme care to not damage the insulation. There must be no cuts in the insulation beyond the bare conductor area.



Caution

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Cuts in the conductor insulation are not permitted past the bare conductor area. If the insulation is damaged when removing the jacket, remove the damaged area and start a new splice.

6. Cut the conductors with a hacksaw back to lengths shown in the preceding figures.



Caution

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

When cutting with the hacksaw be careful not to damage the insulation on the adjacent conductors.

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7. Obtain the appropriate splicing sleeves. Descriptions and part numbers for the sleeves are shown in the *Splicing Sleeves* table in the *Parts, Tools and Materials* section. Remove the insulation to the dimension shown in the preceding figures.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Be careful not to nick or ring cut the copper when removing the insulation. This can create a hot spot which can result in a failure.

8. Ensure the sleeve fits all the way on and snugly against the insulation. If too much insulation was removed, use a file to trim the copper back to the proper dimension.
9. Remove filler from the outside diameter of any stranded conductors.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Do not remove the alloy coating covering the copper conductor.

10. Prepare the end of the cable to be spliced. Spread the conductors apart slightly and place the cable ends together such that one conductor on the second cable butts against the shortest conductor on the first cable. Mark the other two conductors and cut where indicated. (See the preceding figures) After cutting, check to ensure all three conductors butt up to each other when placed tightly together.
11. Remove the insulation from the ends of the second cable. Taper the insulation ends that butt up against the splicing sleeve using 100-grit emery cloth. Clean the exposed conductors with a clean cloth that has been moistened with CRC cleaner. Allow to completely dry.

**Note**

5000-volt cable with EPDM-type insulation does not require tapered ends.

12. Use a megger to test both cables to ensure electrical integrity before proceeding.

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13. Install the splicing sleeves over the conductors. Do not remove the abrasive material from the inside of the sleeves. Its purpose is to puncture the surface of the copper allowing a connection with very low resistance. Some sleeves also contain an antioxidant to prevent corrosion. Do not remove the antioxidant.
14. Use a crimping tool (nicopress tool) to place crimps in the sleeve. Refer to [Figure 5-28](#). Place a crimp at position **A**. Rotate the crimping tool 90 degrees and place a crimp at position **B**. Rotate the crimping tool 90 degrees and place a crimp at position **C**.
Place a crimp at position **D**. Rotate the crimping tool 90 degrees and place a crimp at position **E**. Rotate the crimping tool 90 degrees and place a crimp at position **F**.

**Note**

Crimp **A** and crimp **D** should be at the very end of the sleeve so there is no flare.

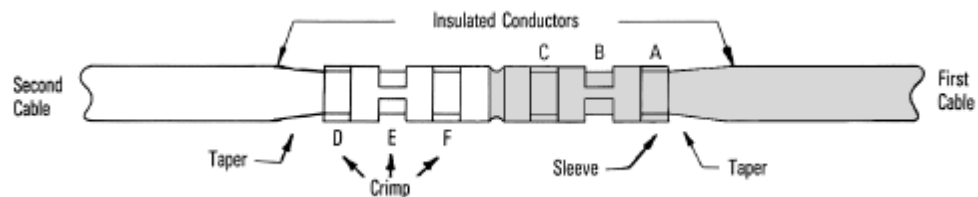


Figure 5-28: Splicing sleeve installation

15. Check the outside diameter of the pressed crimps with crimping gauge. The gauge must pass over the pressed crimp without binding.
16. Repeat the splicing sleeve installation procedure for the remaining conductors.

**Caution**

Potential Severity: Light
Potential Loss: Assets
Hazard Category: Electrical

Improper installation of splicing sleeves can result in splice failure.

17. Use a strip of 100-grit emery cloth on each sleeve to remove any burrs or sharp edges. Discard the emery cloth and clean the splice with CRC or naphtha-based solvent.
18. Use a new strip of 100-grit emery cloth to roughen all exposed conductor insulation.

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**Note**

Do not use emery cloth on the insulation after it has been used on the splicing sleeves or a lead jacket.

19. Clean exposed insulation and splicing sleeves by wiping with a clean cloth that has been moistened with CRC or a naphtha-based solvent. Allow to dry completely.
20. Apply several layers of high-modulus tape over the insulation adjacent to the jacket. This will create a gas block. Butt one edge of the tape against the cut end of the jacket (see the conductor preparation for flat cables with barriers figure). Apply enough full-lapped layers over the insulation until the last layer of tape is even with the outside diameter of the jacket. Use four layers of high-temperature tape to cover the joint between the cut end of the jacket and the tape covering the insulation.
21. Cover each splicing sleeve with five layers of high-temperature tape. Start the first layer of tape on the insulation 0.5 in (13 mm) from one end of the sleeve. Continue over the sleeve with half-lapped wraps until 0.5 in (13 mm) of the insulation is covered beyond the other end of the sleeve. Wrap the second layer in the opposite direction filling the void area in the taper flush with the nicopress sleeve. Extend each succeeding layer of tape 0.5 in (13 mm) beyond the ends of the previous layer. The tape must be wound continuously in a smooth spiral with no gaps or wrinkles.
22. For cables that have a separate jacket on all three conductors, cover the exposed area of each splice with six layers of high-modulus tape. Start the first layer of tape on the insulation 1 in (25 mm) beyond the ends of the high-temperature tape. The tape should be held taut when applying and snugly wrapped around the insulation in a smooth spiral with no gaps or wrinkles. Continue over the sleeve with half-lapped wraps until 1 in (25 mm) of insulation is covered beyond the other end of where the high-temperature tape finished. Cut the tape and begin a second layer wrapped in the opposite direction. Extend each successive layer 0.5 in (13 mm) out from the previous, cutting the tape at the end of each layer until the final layer covers the full length of the conductor and butts up against the end of the armor.

For cables that have a jacket integral to all three phases, apply the six layers of high-modulus tape such that the final layer covers the full length of the conductor and butts up against the end of the jacket.

**Note**

The sixth layer must cover the full length of the exposed insulated conductors.

23. Apply two half-lapped layers of high-temperature tape over the full length of each conductor to seal and secure the high-modulus tape.
24. Loosen the cable clamps and carefully stretch both cables to their original orientation.
25. Weave the three phases together using one-in wide Teflon tape to hold the conductors parallel.
26. For cables that have a jacket integral to all three phases, apply a small amount of Pliobond® in the ends of the jacket to help seal the jacket to the insulation.
27. Apply two half-lapped layers of high-modulus tape over the full length of the splice. Start the first layer of tape on the jacket 1.5 in (38 mm) from the armor and continue over the splice until 1.5 in (38 mm) from the other end of the armor. The second layer, wrapped in the opposite direction, should cover the full length of the splice up to, but not on, the armor.
28. Unwrap one loop of armor at each end of the splice. Apply two half-lapped layers of glass tape over the completed taped area extending under the unwrapped armor. Bend the unwrapped armor back to fit snugly around the conductor.
29. Cut a 46-in (1.2-m) length of the appropriate armor. Pull out the tucked end; trim and flatten it. Use a round wooden rod or a piece of electrical conduit with a slightly smaller OD than the ID of the armor as a wrapping lever. Starting 2 in (51 mm) from the splice, form the armor around the splice in a spiral using the stick for leverage to draw the armor tight. Adjust the profile to expand around the spliced section while maintaining the interlock. Continue wrapping 2 in (51 mm) beyond the existing armor.
30. Cut and secure both ends by soldering the last three convolutions with acid-core solder. Wipe away excess flux with a damp cloth and remove any excess solder from the armor with a surform file.



Caution

Potential Severity: Light
Potential Loss: Assets
Hazard Category: Electrical

Avoid heat damage to the cable under the soldered area. Do not use a torch. Cool the soldered joint immediately with a water-wetted cloth. Avoid contact with hot surfaces.

31. Test the spliced cable to check electrical integrity. Test procedures are provided in the *Cable Testing — Field Insulation Resistance Values* section.

5.5.2.5 Splicing Procedure, Flat Cable to Round Cable

Perform the appropriate conductor preparation procedures provided in the *Splicing Procedure, Round Cable to Round Cable* and *Splicing Procedure, Flat Cable to Flat Cable* sections. When forming the cable together, make the transition as uniform as possible. Use only half the required length of Impervoil filler (refer to step 26 in the *Splicing Procedure, Round Cable to Round Cable* section).

5.5.2.6 Splicing Procedure, Lead Jacket Power Cable and Motor Lead Extension



Warning

Potential Severity: Serious
 Potential Loss: Personnel
 Hazard Category: Toxic corrosive hazardous substances

Lead is a recognized hazard to occupational health. The WCP-QHSE-S003 Personnel Lead Protection Standard is available from InTouchSupport.com (content ID 3846334).

1. Position the ends of the two power cables on the splice bench and secure the clamps onto the armor. Do not overtighten the clamps.
2. Remove the armor to the dimension shown in [Figure 5-29](#). Do not cut into the lead jacket. Remove any burrs or sharp edges on the cut.

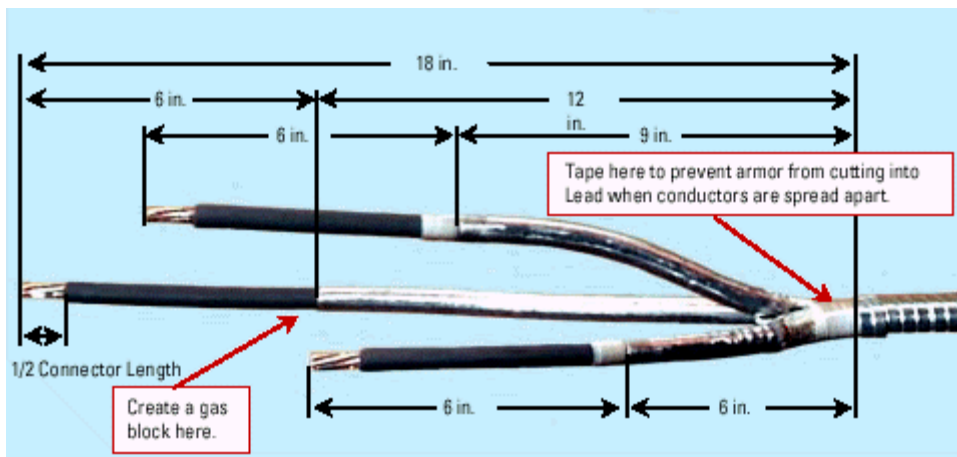


Figure 5-29: Conductor preparation for lead jacket cables and motor lead extensions

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3. Using high-modulus tape, tuck a single wrap underneath the armor followed by two or three wraps extending 1 in (25 mm) out around all three conductors as shown in [Figure 5-29](#). This tape will help protect the lead when the conductors are spread apart.
4. On cables incorporating a Tedlar® tape and braid such as ELTB and KELTB, use scissors to terminate the braid 1 in (25 mm) from the armor and the Tedlar tape 3 in (76 mm) from the armor. Seal the Tedlar to the jacket using two wraps of high-temperature tape as shown in [Figure 5-29](#).
5. Remove the lead jacket in accordance with the dimension shown in [Figure 5-29](#). Use extreme care not to damage the insulation.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Cuts in the conductor insulation are not permitted past the bare conductor area. If the insulation is damaged when removing the jacket, remove the damaged area and start a new splice.

6. Cut the conductors with a hacksaw back to lengths shown in [Figure 5-29](#).

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

When cutting with the hacksaw be careful not to damage the insulation on the adjacent conductors.

7. Obtain the appropriate splicing sleeves. Descriptions and part numbers for the sleeves are shown in the *Splicing Sleeves* table in the *Parts, Tools and Materials* section. Remove the insulation to the dimension shown in [Figure 5-29](#).

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Be careful not to nick or ring cut the copper when removing the insulation. This can create a hot spot which can result in a failure.

8. Ensure the sleeve fits all the way on and snugly against the insulation. If too much insulation was removed, use a file to trim the copper back to the proper dimension.

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**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Remove the Kapton® insulating tape from the conductor on motor lead extensions using a scraping tool and 150-grit emery cloth.

9. Remove filler from the outside diameter of any stranded conductors.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Do not remove the alloy coating covering the copper conductor.

10. Prepare the end of the cable to be spliced. Spread the conductors apart slightly and place the cable ends together such that one conductor on the second cable butts against the shortest conductor on the first cable. Mark the other two conductors and cut where indicated. (Refer to [Figure 5-29](#).) After cutting, check to ensure all three conductors butt up to each other when placed tightly together.
11. Remove the insulation from the ends of the second cable. Clean the exposed conductors with a clean cloth that has been moistened with CRC cleaner. Allow to completely dry.
12. Use a megger to test both cables to ensure electrical integrity before proceeding.
13. Install the splicing sleeves over the conductors. Do not remove the abrasive material from the inside of the sleeves. Its purpose is to puncture the surface of the copper allowing a connection with very low resistance. Some sleeves also contain an antioxidant to prevent corrosion. Do not remove the antioxidant.
14. Use a crimping tool (nicopress tool) to place crimps in the sleeve. Refer to [Figure 5-30](#). Place a crimp at position **A**. Rotate the crimping tool 90 degrees and place a crimp at position **B**. Rotate the crimping tool 90 degrees and place a crimp at position **C**.
 Place a crimp at position **D**. Rotate the crimping tool 90 degrees and place a crimp at position **E**. Rotate the crimping tool 90 degrees and place a crimp at position **F**.

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**Note**

Crimp **A** and crimp **D** should be at the very end of the sleeve so there is no flare.

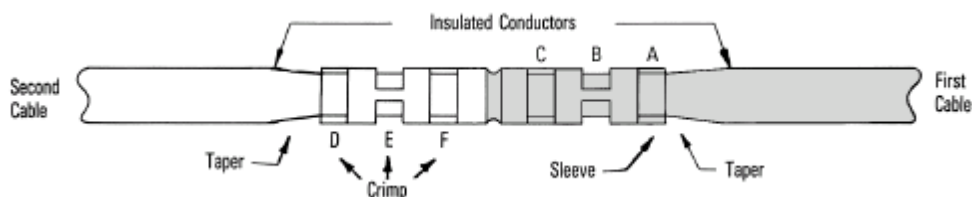


Figure 5-30: Splicing sleeve installation

15. Check the outside diameter of the pressed crimps with crimping gauge. The gauge must pass over the pressed crimp without binding.
16. Repeat the splicing sleeve installation procedure for the remaining conductors.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Improper installation of splicing sleeves can result in splice failure.

17. Use a strip of 100-grit emery cloth on each sleeve to remove any burrs or sharp edges. Discard the emery cloth and clean the splice with CRC or naphtha-based solvent.
18. Use a new strip of 100-grit emery cloth to roughen all exposed conductor insulation.

**Note**

Do not use emery cloth on the insulation after it has been used on the splicing sleeves or a lead jacket.

19. Use a new strip of 150-grit emery cloth to remove any oxidation on the lead jacket.
20. Clean exposed conductors by wiping with a clean cloth that has been moistened with CRC or a naphtha-based solvent. Allow to dry completely.
21. Apply several layers of high-modulus tape over the insulation adjacent to the lead jacket to create a gas block. One edge of the tape must be butted against the cut end of the lead sheath. Apply enough full-lapped layers over

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the insulation until the last layer of tape is even with the outside diameter of the jacket. Use four layers of high-temperature tape to cover the joint between the cut end of the jacket and the tape applied to the insulation.

**Note**

When applying the four layers of high-temperature tape, only tape up onto the lead jacket 1 in (25 mm). This will allow a bare location for solder when replacing the lead sheath.

22. Cover each splicing sleeve with five layers of high-temperature tape. Start the first layer of tape on the insulation 0.5 in (13 mm) from one end of the sleeve. Continue over the sleeve with half-lapped wraps until 0.5 in (13 mm) of the insulation is covered beyond the other end of the sleeve. Wrap the second layer in the opposite direction filling the void area in the taper flush with the nicopress sleeve. Extend each succeeding layer of tape 0.5 in (13 mm) beyond the ends of the previous layer. The tape must be wound continuously in a smooth spiral with no gaps or wrinkles.
23. Cover the exposed area of each splice with six layers of high-modulus tape with the exception of a 1- to 2-in area at the end of the lead jacket. Start the first layer of tape on the insulation 1 in (25 mm) beyond the ends of the high-temperature tape. Continue over the sleeve with half-lapped wraps until 1 in (25 mm) of insulation is covered beyond the other end of where the high-temperature tape finished. Cut the tape and begin a second layer wrapped in the opposite direction. Extend each successive layer 0.5 in (13 mm), cutting the tape at the end of each layer until the sixth layer covers 1 in (25 mm) of the lead jacket. Leave 1 in (25 mm) exposed for a soldering area. Start again with one half-lap layer of high-modulus tape and apply out to the end of the braid.
24. Tin the 1-in area of lead at each end of tape with liquid solder. Cut lead strip to fit from the middle of the 1-in tinned area to the other 1-in area on each phase. Make the lead wide enough to overlap 1/8 to 1/4 inch. Remove any oxidation along the edges of the lead strips. Shape the strips around the wire to fit as tight as possible leaving the seam at the top.
25. Solder around the ends beginning at the 6 o'clock position working the solder up to the 12 o'clock position. Inspect both ends to ensure no voids or pin holes exist. When soldering the top seam, work the solder across the splice and stop 1/8 inch from completely sealing the splice. Wait 5 minutes to allow the hot gases created during the soldering process to escape, then proceed to seal the remaining 1/8-inch seam. File any sharp points of solder then buff with 150-grit emery cloth.
26. Use high-temperature tape to fill in and smooth out the bottleneck at each end of the lead sleeve for an approximate length of 2 in (51 mm).

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27. Apply two half-lap layers of high-modulus tape across the lead sleeves and beyond the high-temperature tape.



Caution

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

No voids or pin holes should exist in the solder. This is critical.

28. Loosen the cable clamps and carefully stretch both cables to their original orientation.
29. Weave the three phases together using 1-in wide Teflon® tape to hold the conductors parallel.
30. Apply two half-lapped layers of high-modulus tape over the full length of the splice. Start the first layer of tape on the jacket 1.5 in (38 mm) from the armor and continue over the splice until 1.5 in (38 mm) from the other end of the armor. The second layer, wrapped in the opposite direction, should cover the full length of the splice up to, but not on, the armor.
31. Unwrap one loop of armor at each end of the splice. Apply two half-lapped layers of glass tape over the completed taped area extending under the unwrapped armor. Bend the unwrapped armor back to fit snugly around the conductor.
32. Cut a 46-in (1.2-m) length of the appropriate armor. Pull out the tucked end; trim and flatten it. Use a round wooden rod or a piece of electrical conduit with a slightly smaller OD than the ID of the armor as a wrapping lever. Starting 2 in (51 mm) from the splice, form the armor around the splice in a spiral using the stick for leverage to draw the armor tight. Adjust the profile to expand around the spliced section while maintaining the interlock. Continue wrapping 2 in (51 mm) beyond the existing armor.
33. Cut and secure both ends by soldering the last three convolutions with acid-core solder. Wipe away excess flux with a damp cloth and remove any excess solder from the armor with a surform file.



Caution

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Avoid heat damage to the cable under the soldered area. Do not use a torch. Cool the soldered joint immediately with a water-wetted cloth. Avoid contact with hot surfaces.

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34. Test the spliced cable to check electrical integrity. Test procedures are provided in the *Cable Testing — Field Insulation Resistance Values* section.

5.5.2.7

Splicing Procedure, Cable with Injection Tube

These procedures assume that the conductors and the injection tube require splicing. Certain situations may only require splicing of the injection tube or the cable conductors; the procedures can be modified for a particular situation.

Cable Preparation

1. Prepare the end on the first length of cable to be spliced. Remove 20.125 in (51 cm) of armor. (All of the conductors will extend 5 in (13 cm) more from the armor than on a typical cable splice.) The end of the tube should be the same length as the longest conductor. The conductor next to the tube should be the shortest of the three conductors. Trim the conductors to the lengths shown in [Figure 5-31](#).



Note

If the cable is a flat cable and has a layer of tape over a jacket or lead barrier, the tape should be trimmed 2 in (51 mm) from the end of the jacket or lead. If the cable is a round cable and the cable has a layer of tape, the jacket should be trimmed approximately 3 in (76 mm) from the armor and the tape should extend at least 2 in (51 mm) beyond the end of the jacket.

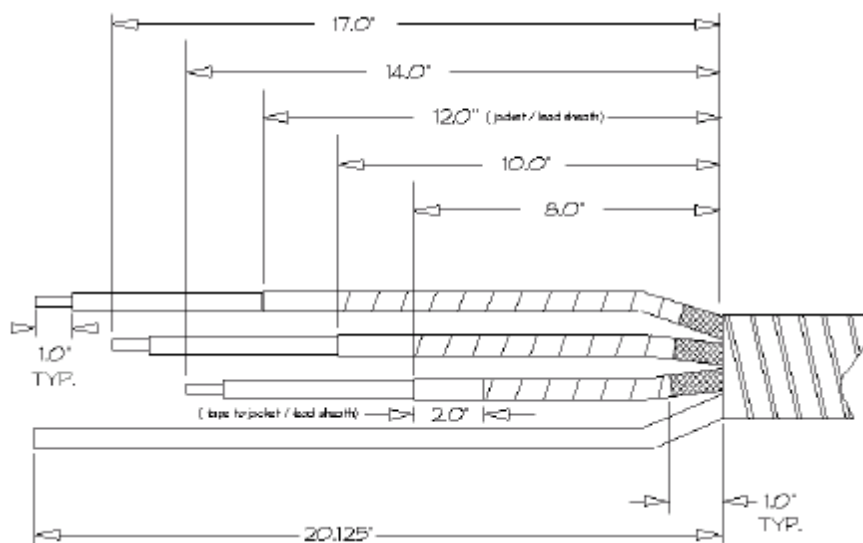


Figure 5-31: Cable preparation for cables with injection tubes

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2. Spread the conductors and the tubing slightly apart. Bending the conductor sharply will damage the cable.
3. Prepare one end on the other cable. The conductors on the second cable should be cut so they butt against the “mirrored” conductors. The injection tubing of the second cable should be on the same sides as the tubing of the first cable and overlap the tube on the first cable by approximately 6 in (15.3 cm). Remove the insulation, jacket and braid to the dimensions shown in [Figure 5-31](#).
4. Splice the three conductors up to the point of applying the armor. Splicing procedures are provided in the *Splicing Procedures, Round Cable to Round Cable*, *Splicing Procedures, Flat Cable to Flat Cable* and the *Splicing Procedures, Flat Cable to Round Cable* sections. Do not apply the armor at this time.
5. Pull the three spliced conductors together and apply two half-lapped layers of glass tape over the three conductors.
6. Straighten the injection tubes of both cables so that they are straight and lay against the three spliced conductors. Cut the tube on the end of the second cable with a gap between the ends of the tubes for proper dimensions. Do not remove too much tubing. The ends of both tubes must be square and smooth.
7. Remove 1 in (25 mm) of the lead jacket or EPDM (if present) from the end of each injection tube. Do not cut completely through the lead or EPDM. Do not cut or mark the injection tubes. Carefully remove the lead or EPDM without nicking the tube. A raised or recessed area on the tubing may prevent the connector from sealing.

**Warning**

Potential Severity: Serious
 Potential Loss: Personnel
 Hazard Category: Toxic corrosive hazardous substances

Lead is a recognized hazard to occupational health. The WCP-QHSE-S003 Personnel Lead Protection Standard is available from InTouchSupport.com (content ID 3846334).

8. Select the appropriate Swagelok® connector and slide the connector onto the end of one of the tubes. Insert the end of the second tube into the other end of the connector. The ends of both tubes should rest firmly against the shoulder inside the connector. Tighten the nuts finger-tight.
9. While holding the connector body with a wrench, tighten the nut 1¼ turns. Check to see that a “flat side” of the nut that will be in contact with the adjacent conductor. If necessary, continue tightening the nut until the next flat is exposed. Do not tighten the nut more than one additional ¼ turn.

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**Tip**

Before tightening the nut, scribe a line on the nut. Watch the line as the nut is turned. This way, there will be no doubt that the fitting has been properly installed.

10. Verify that the connector has been correctly installed by placing the appropriate Swagelok Gap Inspection Gage between the nut and the body hex. If the gage will not fit, the fitting is tightened sufficiently.
11. Tighten the other end of the connector by repeating step 9. Verify the installation.
12. Use a file to remove the sharp edges on the nut. This will prevent cutting of the tape and gouging of the conductors.
13. Wrap two half-lapped layers of high-modulus tape followed by two half-lapped layers of glass tape over the Swagelok connector and exposed tube. The first layer of tape should extend at least 1 in (25 mm) beyond the end of the lead on the tubes. Each subsequent layer should extend 0.5 in (13 mm) beyond the previous layer of tape.

Flat-to-Flat Cable

1. Pull the taped conductors and tube together. Apply two half-lapped layers of glass tape over the tube and the conductors.
2. Cut a 48-in section of galvanized or Monel round armor. (Use galvanized armor over cable splices on cables using galvanized armor. Use Monel armor over cable splices on cables using Monel armor.) Pull out the tucked end; trim and flatten it. Use a round wooden rod or a piece of electrical conduit with a slightly smaller OD than the ID of the armor as a wrapping lever. Starting 2 in (51 mm) from the splice, form the armor around the splice in a spiral using the stick for leverage to draw the armor tight. Adjust the profile to expand around the spliced section while maintaining the interlock. Continue wrapping 2 in (51 mm) beyond the existing armor.
3. Cut and secure both ends by soldering the last three convolutions with acid-core solder. Wipe away excess flux with a damp cloth and remove any excess solder from the armor with a surform file.

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**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Avoid heat damage to the cable under the soldered area. Do not use a torch. Cool the soldered joint immediately with a water-wetted cloth. Avoid contact with hot surfaces.

Round-to-Flat Cable

1. Cut four lengths of Impervoil filler strips. The strips should reach from the cut end of the jacket on the round cable to the end of the armor on the flat cable. Clean each length of the filler strip with a cloth that has been moistened with REDA #1 thinner (or equivalent). Roughen the round jacket with emery cloth so adhesive tape can fit snugly around the jacket.
2. Apply two half-lapped layers of tape over the conductors and filler strips. The first layer should extend over the first 1 in (25 mm) of jacket on the round cable and to the armor on the flat cable. The second layer should extend to the armor at each end of the splice, using the procedure provided in the *Splicing Procedure, Round Cable to Round Cable* section.
3. Clean the exposed jacket by sanding off the splines and wiping with a clean cloth moistened with REDA #1 thinner (or equivalent). Allow to completely dry.
4. Cut four lengths of Impervoil filler strip. The strips should extend the full distance of the insulated conductors between the cut ends of the jacket. Place filler strips between the conductors and tube as shown in [Figure 5-32](#).

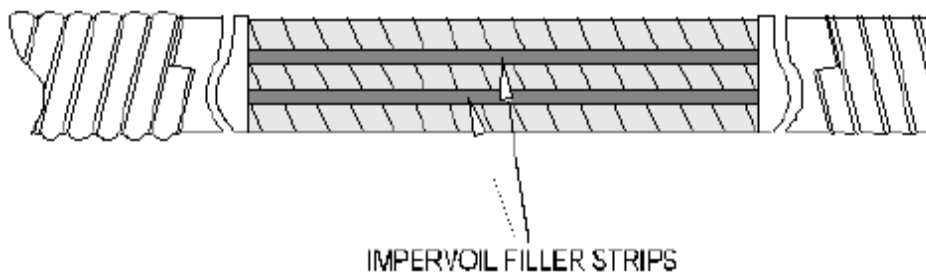


Figure 5-32: Placement of Impervoil filler strips

5. Use procedures provided in the *Splicing Procedure, Round Cable to Round Cable* section to complete the splice.

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Breakout of the Injection Tube

The most desirable place to break out the injection tube is at the motorlead splice since the cable armor is already broken. Swagelok connectors can be used to splice any additional tubing to obtain the desired length of tubing.

1. Splice the power cable and motorlead cable together using standard procedures. Allow the injection tube to extend a minimum of 6 in (15.3 cm) beyond the end of the motorlead armor.
2. Pull the taped conductors and the tube together. Apply two half-lapped layers of glass tape over the tube and the conductors.
3. Apply armor over the splice and the tube until the armor extends a minimum of 4 in (10.2 cm) beyond the end of the underlying motorlead armor. Solder the end on the armor.
4. Clamp any additional length of injection tubing to the equipment.



Note

Ground the tube during all tests.

5.5.3

HHS/HSM Wellhead Cable Pack-off Guidelines

Use the following guidelines for wellheads utilizing a split, rubber-type packing glands to seal off around the power cable. Illustrations of parts are available in Section 2.17 and 2.18.

- Armored round or flat cable requires armor removal through the gland area. For flat cable, remove several additional inches on either side of the gland so legs can be easily spread apart from each other. Apply high modulus tape on the conductors near the armor to prevent the armor from damaging the insulation when the conductors are spread apart.
- All round cables will be packed off against their outer jacket.
- Flat cables with tape and braid should have the tape and braid removed so the rubber gland material will pack off directly against the individual phase jacket.
- Lead jacketed cables should have the braid removed so that the gland material can smoothly seal directly against the lead jacket.
- Use caution when assembling the wellhead parts to avoid damage to lead/insulation and avoid sharp bends in the cable legs.

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After the wellhead assembly has been completed, the gland may be tightened. The HHS gland is tightened with eight flange bolts. The HSM gland is tightened with six gland cap screws. [2: Parts, Tools and Materials](#) chapter provides additional information.

- The gland should be tightened slowly and evenly. All bolts/cap screws should be hand tightened first. Then gently add tension to opposing bolts/cap screws until all are equally tight and the gap is the same all the way around the wellhead. This process should be conducted repeatedly until the appropriate tightness is achieved. Do not allow the gland to be overtightened as this can lead to premature failure.

**Note**

It is possible to deform and cut cable insulation if the wellhead packing is excessively tight. This is especially critical with flat cable. A snug gland is adequate.

- Extra care in preparing cable for pack off, proper gland assembly, and correct tightening will help to assure a quality installation.
- Adequately support the cable after the cable exits the wellhead. Avoid a short bending radius. Use a cable band or other material to secure the cable to the flowline for additional support.
- When the well has been started and fluid has reached surface, a check should be made to ensure that there are no leaks around each of the conductors.
- In an emergency situation, where the cable OD is slightly smaller than the packing gland ID, high-modulus tape can be used to build up the cable OD so pack off can be accomplished.
- If flat Redalene cable is being used and the gland hole ID is smaller than the jacketed phases, removal of the Nitrile jacket is acceptable to pack off directly on the Redalene.
- RTV silicone rubber sealer may be used to supplement sealing around worn or damaged packing glands. (New rubber packing elements are recommended.)

5.5.4 Testing Procedures

5.5.4.1 Overview

This section details standard cable testing procedures, using DC Hi Potential Tester Unit. Additional testing procedures and job aids are available on the [SOP Index](#). The Automated Cable Test System (ACTS) [InTouch ID 3455075](#) is the preferred testing system for ART Centers.

5.5.4.2 Cable Preparation

Use the following procedure to prepare cables for testing.

1. Remove 10 in (25.4 cm) of armor and jacket or lead sheath from the outside ends of the spliced cable.
2. Remove insulation to expose 0.5 in to 0.75 in (1.3 cm to 1.9 cm) of bare conductor.
3. Clean exposed conductors with a clean cloth that has been moistened with CRC (contact cleaner) or a naphtha-based solvent. Allow solvent to dry.
4. Spread the conductors so that the bare ends are at least 12 in (30.5 cm) apart and are not grounded.
5. Remove 12 in of lead from the other end of the cable (ends not being connected) and spread so they are at least 12 in apart and away from any ground (the reel, ground, metal building beams, etc.).

5.5.4.3 Cable Insulation Resistance Testing

After performing the [Cable Preparation](#) steps, use a Megger to measure the insulation resistance between each conductor and armor as ground. The insulation resistance should not be less than the values shown in .



Potential Severity:	Major
Potential Loss:	Personnel
Hazard Category:	Electrical

Never use a Megger on a rig floor and always discharge phases after testing.

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Table 5-9: Cable Insulation Resistance Values

New Cable	Installed Cable ¹
2000 Mega	20 Mega Ohms

¹ The installed cable value is the minimum value measured after the cable is installed. It includes the effects of well temperature on the insulation resistance.

5.6 Troubleshooting

5.6.1 Electrical Problems

- If high leakage or low megger readings are experienced on an MLE, remove moisture and clean the surface that has been touched with dirt (hands, oils, etc). Use a clean rag to wipe the insulation with CRC electrical contact cleaner or with a clean alcohol wipe. Allow cleaner to air dry before re-testing. Ensure the lead is cut back as per procedure in section for cable and MLE testing procedures and criteria.
- If the pothead has been exposed to a high humidity environment, use a hair/blow dryer to remove moisture.



Note

DO NOT use a heat gun as the temperature could get too hot and cause damage to the conductor.

- If a pothead is received with desiccant pack(s) that have turned pink (should be blue) this indicates excessive moisture. Use a hair/blow dryer to dry the surface of the conductors and front face of the pothead.
-



Note

DO NOT use a heat gun as the temperature could get too hot and cause damage to the conductor.

- Testing shows that high leakage readings for plug-in MLEs can be introduced by using a “bike pump” to perform the pressure seal check as the “bike pump” introduces moisture. The HiPot leakage can increase by 10 times if going from 45% to a 98% relative humidity. It is recommended to test with Nitrogen.
- If you experience high leakage or low megger readings on a cable, ensure the conductors are far enough apart, see section for cable and MLE testing procedures and criteria. For high humidity environments, put a clean bag over the 3 ends (not being tested).

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5.7 Parts, Spares, Tools, and Supplies

Refer to the [Part, Tools and Material](#) section for a listing of splicing tools and materials.

5.7.1 Nicopress Tools

5.7.1.1 Nicopress Crimp Gauge

Each type of nicopress tool is designed to quickly and easily compress the appropriate nicopress sleeve. To make a satisfactory splice it is important to keep the amount of pressure on the sleeve constant. Each new tool has an instruction sheet, a gauge and a wrench for adjusting the tool. Check splices with the gauge provided for this purpose.

When using the gauge, hold it so that it contacts the compressed portion of the sleeve at right angles to the fins. The compressed portion of the sleeve should enter the slot without forcing. If after long use the nicopress tool wears to such an extent that the compressed sleeve does not enter the gauge, it is a simple matter to readjust the tool. Be careful not to get the compressions too small. The sleeve should enter the gauge with very little clearance.

[Figure 5-33](#) shows a crimp gauge.

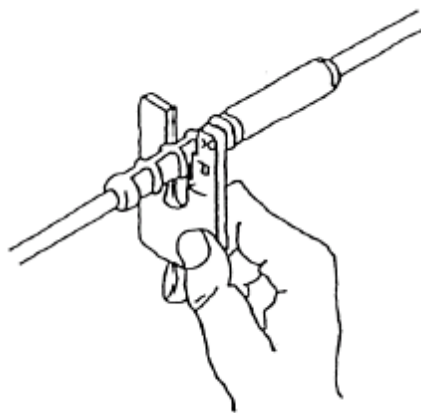


Figure 5-33: Crimp gauge used to check nicopress sleeves

5.7.1.2 Nicopress Tool Use, Care and Maintenance

When tools are used daily, they should receive a weekly inspection. Clean and lubricate the nicopress tool regularly.

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An empty tool should work freely with a slight spring at the final closing. If the tool binds, it can be eased by slightly loosening the particular hexagon head bolt which is causing the binding. Lock the bolt after each adjustment by bending up one of the lock washer ears.

All sleeves are marked with a tool groove letter. This letter must correspond with the letter stamped near the tool groove used for compressing the sleeve.

5.7.1.3 Nicopress Tool Adjustment

With the tool handles in the open position, use the wrench provided with the tool to first loosen the locking screw one or two turns. Then turn in (clockwise) the adjustment screw only a fraction of a turn.



Note

When holding the tool (shown in [Figure 5-34](#)), the bolt on the right-hand side of the tool has left-hand threads whereas the other two screws have right-hand threads.

Try the tool on a sleeve and check with gauge. Continue to turn the adjustment screw a small amount at a time until the pressed sleeve passes easily into the gauge. When the correct setting is obtained, turn up the locking screw so that the tool will hold its adjustment.

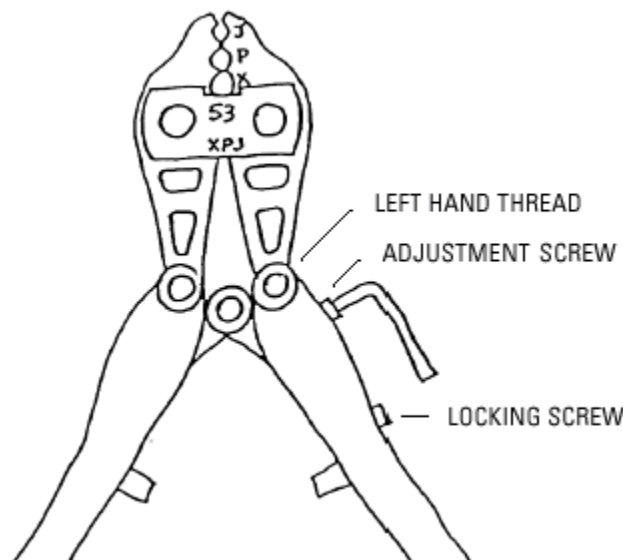


Figure 5-34: Nicopress tool adjustment

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5.8 References

Artificial Lift Cable Reference Page ([InTouch ID 3251798](#))

Artificial Lift Field Operations Reference Manual - Downhole (ALFORM-D)
([InTouch ID 3255850](#))

Artificial Lift Application Engineering Reference Manual ([InTouch ID 4227449](#))

5.8.1 Termination Procedures

[Power Cable Termination for Splicing \(100285583\)](#)

[Flat Cable Extensions Terminations for Splicing \(100285587\)](#)

[Splice Code And Nicopress Part Numbers \(100650432\)](#)

5.8.2 MLE Splicing Procedures

[MLE to Round Profile Power Cable Splice \(100716529\)](#)

[MLE to Flat Profile Power Cable Splice \(100357294\)](#)

[4/1 XTHLB \(or XHLTB\) MLE to 4/1 EHLTB Cable Splice \(100979795\)](#)

[Hotline II \(Hotline550\) MLE and Power Cable Splice \(100389530\)](#)

[Non-Lead Splice \(100795611\)](#)

[5 Conductor Flat Splice \(100357298\)](#)

5.8.3 MLE Electrical Testing

[Tape-in and Plug-in MLE Electrical Test \(100351056\)](#)

[Electrical test SAGD or H2S MLE \(100613136\)](#)

[Trident Pothead Electrical Test \(100845838\)](#)

[Electrical Test for Splice Cable \(100357305\)](#)

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5.8.4 **MLE Pressure Seal Testing**

[Standard MLE Pressure Seal Test \(100351045\)](#)

[Trident MLE/Pothead Pressure Seal Test \(100845088\)](#)

[Hotline SA3 MLE/Pothead Pressure Seal Test \(100757682\)](#)

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A Non-Standard Applications

This chapter contains non-standard applications.

Table A-1: Non-Standard Application References

InTouch Content ID	Description
3251799	Hotline I and Hotline 550 Equipment - Reference Page
5649843	HotlineSA3 Equipment - Reference Page
3844523	Poseidon Multiphase Pump - Product Bulletin
3861308	Advanced Protector - Product Bulletin

A.1 Hotline, Hotline550 and HotlineSA3

A.1.1 Systems Descriptions

The Hotline system is similar to a standard Schlumberger submersible lifting system but uses components that are rated for higher temperatures. The Hotline550 system is a fully integrated system specially designed for steam injection wells. The HotlineSA3 system improves reliability, increases the bottom-hole temperature rating, introduces extra functionality, and simplifies installation. Comparisons of various systems are shown in [Table A-2](#).

Table A-2: Comparisons of Various Lifting Systems

	Standard System	Hotline System	Hotline550 System	HotlineSA3 System
Maximum Bottomhole Temperature	300 degF (150 degC)	350 degF (177 degC)	425 degF (218 degC)	482 degF (250 degC)
Maximum Motor Winding Temperature	400 degF (204 degC)	550 degF (288 degC)	550 degF (288 degC)	572 degF (300 degC)
Motor Insulation	Kapton™/PEEK	proprietary insulation	proprietary insulation	proprietary insulation
Motor Bearing	SLK	SLK	SLK	SLK
Motor Elastomer Material	HSN/Aflas™	Chemraz™	Chemraz	Chemraz
Protector	modular protector	modular protector	advanced protector	integrated into motor
Protector Positive Seal Chamber Design	HSN/Aflas bag	Aflas bag	metal bellow	Metal bellows in compensator
Protector Radial Bearing	bronze	bronze	ceramic	none
MLE Configuration	KELB/KEOTB/ELB	KELB/ELB	KKPLB	XHLTB
Power Cable	various	ELB/ELBE	ELB/KKPLB	ELB/XTHLB
Motor Pothead	standard	special design (Hotline design)	special design (Hotline550 design)	special design (HotlineSA3 design)
Pump	standard	special design	special design	special design
Motor Oil	Reda No. 5	Reda No. 7	Reda No. 7	Purified Reda No. 7

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A.1.1.1 Hotline System

The Hotline system's critical motor and cable components are constructed of materials that have higher temperature ratings.

Features unique to the Hotline system are

- A combination of Aflas O-rings and high-temperature Chemraz O-rings are used throughout the system.
- special pothead design using a high-temperature Aflas seal
- Aflas protector bag
- Reda No. 7 oil for high-temperature applications
- proprietary insulation on the motor windings
- EHLB/EHLTB power cables and KELB/KEHLB MLEs
- special material selection and special design taking into consideration maximum expected different thermal expansions.

A.1.1.2 Hotline550 System

The Hotline550 system is a fully integrated system specially designed for steam injection wells.

Features unique to the Hotline550 system are

- motor tested to 550 degF (288 degC) internal motor temperature
- special high-temperature pothead and MLE designs
- high-temperature advanced protector that uses metal bellows
- Reda No. 7 oil for high-temperature applications
- high-temperature pump intake, gas separator
- AGH / Poseidon™ multiphase pump if required
- special Hotline550 centrifugal pumps
- high-temperature power cable and MLE
- accepts optional monitoring (Hotline550 gauge retrofit) system.

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A.1.1.3 HotlineSA3 System

Unlike the Hotline550 system, which uses traditional ESP architecture, the HotlineSA3 system has a radically new architecture. The HotlineSA3 integrated motor is made up of four sections a shaft seal module (SSM), motor section, gauge section and compensator (bellows) section. The functions of the protector have been split into a shaft seal module (SSM) above the motor and a compensator section below the motor. The MLE, pothead and pumps have been developed to cope with the high temperatures inherent to thermal applications and the abrasive conditions.

Features unique to the HotlineSA3 system are

- combines the protector, motor and gauge functions into a single assembly (integrated motor).
- motor tested to 572 degF (300 degC) internal motor temperature
- special high-temperature pothead (Chimera) and MLE designs
- high-temperature integrated motor that uses metal bellows
- Ultra purified Reda No. 7 oil for high-temperature applications
- high-temperature pump intake, gas separator
- special HotlineSA3 centrifugal pumps with ceramic bearings
- high-temperature power cable and MLE
- high temperature monitoring system (motor temperature, well pressure and temperature).

A.1.1.4 Hotline Gauge

The Hotline ESP monitoring system is an optional feature for the Hotline550 motor and is incorporated in all HotlineSA3 Integrated Motors. The gauge measures motor winding temperature, wellbore pressure, and wellbore temperature. Due to high downhole temperatures, the electronics are located at the surface and an independent downhole monitoring cable is required gauge communication and power.

There are two Hotline monitoring system configurations available. The standard configuration provides motor winding temperature and intake pressure/temperature measurements. The RTD-only configuration only provides motor winding temperature measurements.

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The Hotline ESP monitoring system is supported by the Hotline Extreme interface card. The card is installed in the UniConn which is normally incorporated into the switchboard or VSD. The monitoring cable transfers the power from the card to the downhole sensors and provides sense lines from the pressure gauge and the resistance temperature detectors (RTDs) for taking the measurements to surface. The monitoring cable is terminated in a junction box at surface, after it transitions through the wellhead. A flexible cable is then run through the cable trays to the Extreme card.

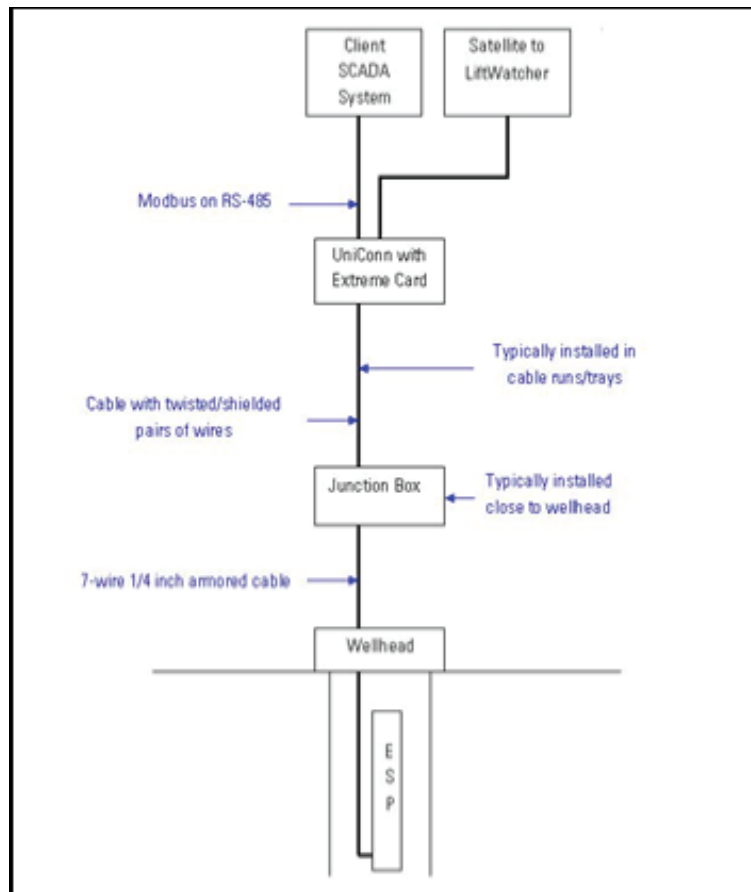


Figure A-1: High Level Diagram of Hotline ESP Monitoring System

The Hotline gauge portals provide the mechanical and electrical interface between the gauges and the monitoring cable. The RTD is secured to the end ESP motor coils to measure motor temperature. The leads from each device are terminated at a common connector inside the motor. The pressure sensor and RTDs are integrated in the motor during the manufacturing process by the product center or a qualified ART center.

The gauge portal kit is installed below the motor housing at the ART center, for retrofitted Hotline550 motors.

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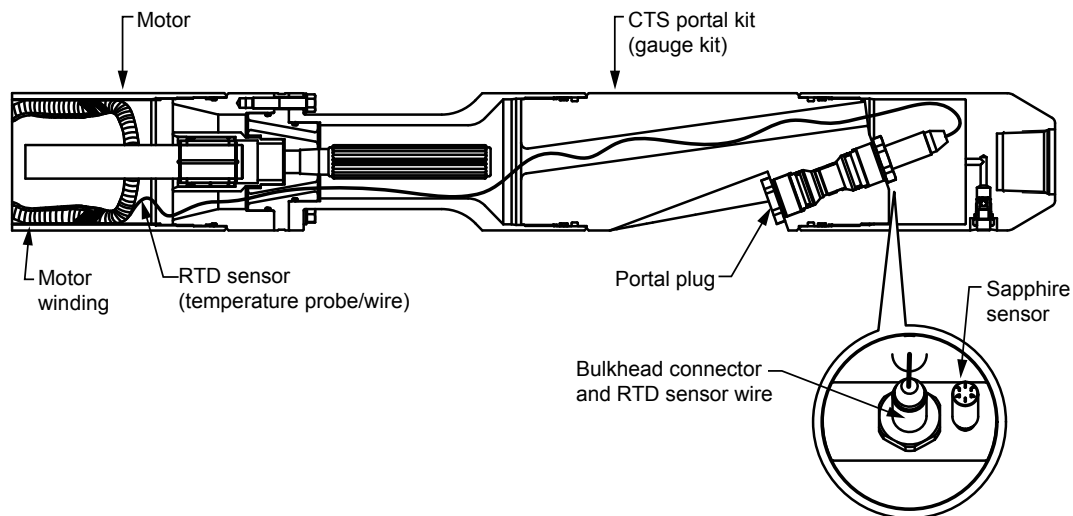


Figure A-2: Hotline550 Gauge Portal Kit

For the HotlineSA3 motors, the gauge portal is installed between the motor and compensator (bellows) section.

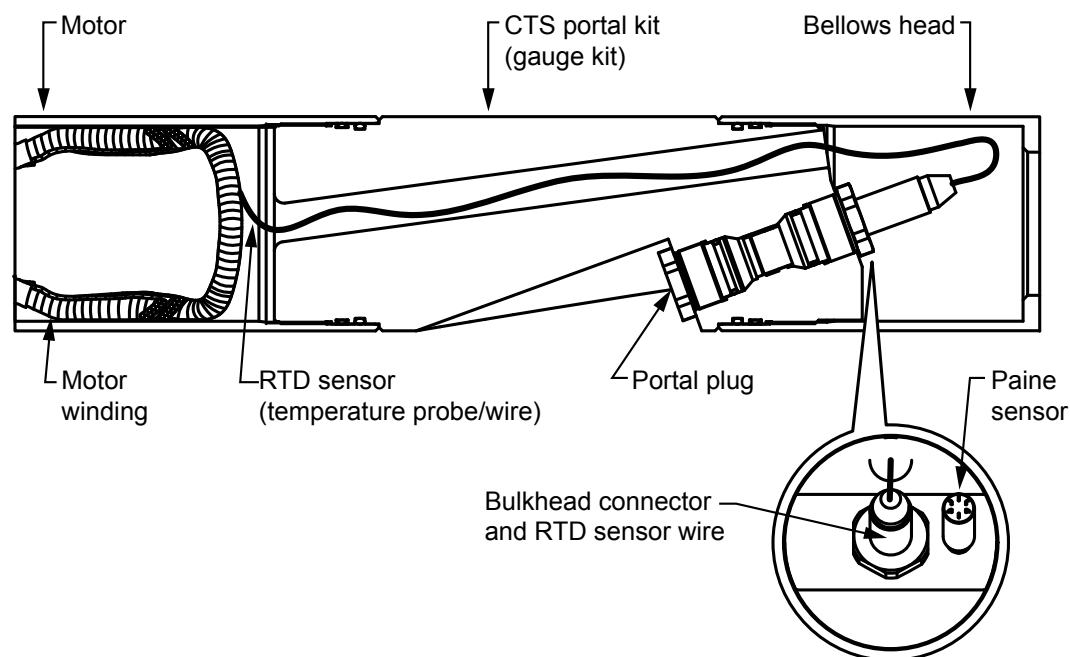


Figure A-3: HotlineSA3 Gauge Portal

Surface Acquisition System Compatibility

The Hotline ESP monitoring systems are supported by the UniConn only. The UniConn controller and Extreme card firmware required to support the Hotline Monitoring system is available via InTouchSupport.com.

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Configuration of the Extreme card is performed using StarView.

A.1.2 Systems Specifications

Specifications for the Hotline, Hotline550 and HotlineSA3 systems are shown in [Table A-3](#).

Table A-3: Hotline System Specifications

	Hotline System	Hotline550 System	HotlineSA3 System
Motor Series	562 series	562 Series	562 Series
Pump Series	400-, 540-, 538-series	400-, 540-, 538-series	400-, 540-, 538-series
Outside Diameter	5.62 in (14.27 cm)	5.62 in (14.27 cm)	5.62 in (14.27 cm)
Minimum Casing Size	7 in (178 mm)	7 in (178 mm)	7 in (178 mm)
Maximum Bottomhole temperature	350 degF (177 degC)	425 degF (218 degC)	482 degF (250 degC)
Maximum Horsepower	321 hp — single section (tandem motors available via Rapid Response)	321 hp — single section (tandem motors available via Rapid Response)	257 hp — single section

A.1.3 Transport and Handling

All Hotline, Hotline550 and HotlineSA3 equipment should follow normal equipment [transport and handling](#) guidelines. The equipment should be transported in steel shipping boxes and be carefully lifted to prevent damage to the components.

The Hotline and Hotline550 motors and protectors are filled with highly viscous oil (REDA #7 oil). The HotlineSA3 Integrated Motor uses special purified REDA #7 oil. The HotlineSA3 motor cannot be drained and refilled with regular REDA #7 oil and should not be 'topped off' with REDA #7.

It will be difficult to turn the shafts and fill the equipment after transportation to a wellsite in cold weather. Consideration should be given to methods of keeping the equipment above 32 degF (0 degC) when preparing for installation.

The equipment may be stored for a period of time before installation. Standard procedures for storage should be followed for this equipment. Most of the system components have unlimited shelf life. O-rings have a shelf life of five years from date of manufacture.

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A.1.3.1 Hotline Gauge

Gauge Specific QHSE Issues

The 7-wire armored cable is not a pressure barrier and if a leak occurs downhole, either by the armor being damaged or the connector leaking, wellbore high pressure and high temperature gases from the wellbore could be transported up the cable along the signal wires to the surface if a pressure barrier at the WHO is not present. Care must be taken when working on or around the cable end and when accessing the wires in the junction box.

Portal and Gauge

To prevent damage during handling of the motor/gauge portal, do not remove the dummy plug protecting the portal bulkhead connection. This plug prevents physical damage of the connector during transportation. It also prevents contaminants, including moisture, from entering the connection.

Monitoring Cable and Connector



Note

The cable connector is installed on the 7-wire monitoring cable in the shop. Refer to Monitoring Cable Male Connector Assembly, MPI-57151([GeMS document 100787432](#)) prior to shipping the cable to the wellsite. For recommended cable storage and handling procedure, refer to the MC Sensor Cable Storage and Handling Handbook ([GeMS document 100868612](#)).

The monitoring cable will be shipped on a reel. A portable spooler should be used to drive the reel and allow ease of cable feed into and out of the well. The reel will also protect the cable during transport and handling.

A.1.4 Installation Procedures

The Hotline and Hotline550 systems require some unique installation procedures. With the few exceptions noted below, the equipment installation shall follow procedures provided in the standard ESP [installation procedures](#).

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The HotlineSA3 system requires specific equipment for installation. Refer to the HotlineSA3 Technical Information Release (TIR) ([InTouch ID 5443035](#)) for installation procedures connection details and O-rings part numbers.

Field service technicians must have a sufficient supply of the special O-rings for installation. Exact O-ring sizing will vary depending on the pump series. Hotline O-rings and Hotline550 O-rings are shown in [Table A-4](#).

Table A-4: Hotline and Hotline550 O-rings

	O-ring Location	O-ring Type	Size	Part Number	Quantity Required
Motor	terminal cover	HSN 568	220	997577	1
	between housing and (motor head, support bearing)	Chemraz™ 605	9-250	100031429	3
	support bearing	Chemraz 526	9-032	100031421	2
Protector	between motor and protector	Aflas	2-237	1042605	2
	protector/motor head connection	Chemraz 605	2-153		1
	protector/motor head connection	Chemraz 605	2-237	2002250	2
Pothhead	pothead to motor head	MSE seal (obsolete)	double coil spring and Avalon 87 jacket, 1.519 in. by 1.644 in. (3.858 by 4.176 cm)	100014999	1
	dual O-ring design - near flange		2-029	100648275	1
	dual O-ring design - near motor		2-029	100648271	1

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A.1.4.1 **Motor**

Hotline motors and Hotline550 motors are filled with Reda No. 7 oil, which has a viscosity of 1223 cST at 70 degF (21 degC). Reda No. 7 oil is slightly less viscous than honey at this temperature. Motors are equipped with a valve to prevent oil loss from the bottom.

The pour point of Reda No. 7 oil is -32 degF (-36 degC). As the temperature approaches this value, the motor shaft will become much more difficult to turn. Be careful not to confuse this increased drag with that encountered by a damaged or bent motor. It is best to keep the motor at temperatures above 32 degF (0 degC) if possible.

A.1.4.2 **Protector**

The Advanced Protector is pre-filled with Reda No. 7 oil at the product center or service center. The wellsite installation procedures are very similar to those for standard protectors. For detailed installation information, refer to the standard [protector installation](#) procedures.

The pour point of Reda No. 7 oil is -32 degF (-36 degC). As the temperature approaches this value, the protector shaft will become much more difficult to turn. Be careful not to confuse this increased drag with that encountered by a damaged protector. It is best to keep the protector above 32 degF (0 degC) if possible when preparing for installation.

A.1.4.3 **Intake**

No special procedures are required for installing intakes for Hotline and Hotline550 applications.

A.1.4.4 **Pump**

Compression pumps are used in Hotline applications and Hotline550 applications. Standard installation procedures apply. Installation procedures and special shimming requirements are provided in the standard [pump shimming and installation](#) procedures.

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A.1.4.5 Pothead and Motor Lead Extension

The Hotline pothead design and the Hotline550 pothead design are different designs compared to the standard pothead design. The Hotline pothead uses an Aflas elastomer to seal the MLE to the motor. The Hotline550 pothead is a unique design developed for steam injection wells for operation at very high temperatures.

The maximum bottomhole temperature for the Hotline pothead is 350 degF (177 degC). The maximum conductor operating temperature is 450 degF (232 degC).

The maximum bottomhole temperature for the Hotline550 pothead is 425 degF (218 degC) and the maximum conductor operating temperature for the Hotline550 pothead is 500 degF (260 degC). The Hotline550 pothead installs in a 540/562 motorhead, designed with a smooth-bore pothole, sized to fit a “-029” O-ring. The Hotline550 pothead has a special configuration and will not fit a standard motor head; it can be used only on Hotline550 motors.

In addition to the pothead design, MLE material selection affects the temperature rating of the MLE and hence the application limitations. [Table A-5](#) shows the different options for power cable and MLE materials selection.

Table A-5: Power Cable and MLE Material Selection

Cable	Profile	Insulation 1	Insulation 2	Jacket	Barrier	MLE	Power Cable	Maximum Conductor Temperature
KELB	flat	Kapton	EPDM	—	lead	yes	—	450 degF (232 degC)
KEHLB	flat	Kapton	EPDM	—	heavy lead	yes	—	450 degF (232 degC)
ELB	flat	EPDM	—	—	lead	yes	yes	450 degF (232 degC)
KKPLB	flat	double Kapton	PEEK	—	lead	yes	on request	500 degF (260 degC)
ELBE	round	EPDM	—	EPDM	lead	—	yes	400 degF (204 degC)

The typical cable is described by:

Term	Definition
K	Kapton
KK	double Kapton
E	EPDM rubber insulation

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L	Lead
HL	Heavy Lead wall
B	Braid
P	PEEK

Standard cable armor options apply.

Hotline MLE

The Hotline MLE consists of a Hotline pothead and a length of AWG #4 KELB or ELB cable. Part numbers and descriptions for Hotline motor lead extensions are shown in [Table A-6](#).

Table A-6: Hotline MLE Part Numbers and Descriptions

Part Number	Description	Maximum Bottomhole Temperature	Maximum Conductor Operating Temperature	MLE Thickness (± 0.015 in. [mm])
100011602	MLE, 540 series Hotline, 200, 4 KELB .015 M5F	350 degF (177 degC)	450 degF (232 degC)	0.506 in. [12.85 mm]
100089348	MLE, 540 SERIES Hotline, 90, 4 KELB .015 M5F	350 degF (177 degC)	450 degF (232 degC)	0.506 in. [12.85 mm]
1258656	MLE: 540 SERIES, 100 FT, B TYPE, 4KV, MONEL ARMOR, SIZE 4 KEHLB	350 degF (177 degC)	450 degF (232 degC)	0.506 in. [12.85 mm]

The Hotline pothead can also be used on ELB power cable, with or without a splice to a larger power cable higher in the wellbore. ELB power cable can be used as a substitute for a KELB motor lead extension.

Tape-in procedures for the Hotline MLE are standard procedures and are provided in the [Cables section](#).

An ampacity plot for #4 AWG Redalead flat ELB motor lead and #4 AWG KELB motor lead is shown in [Figure A-4](#).

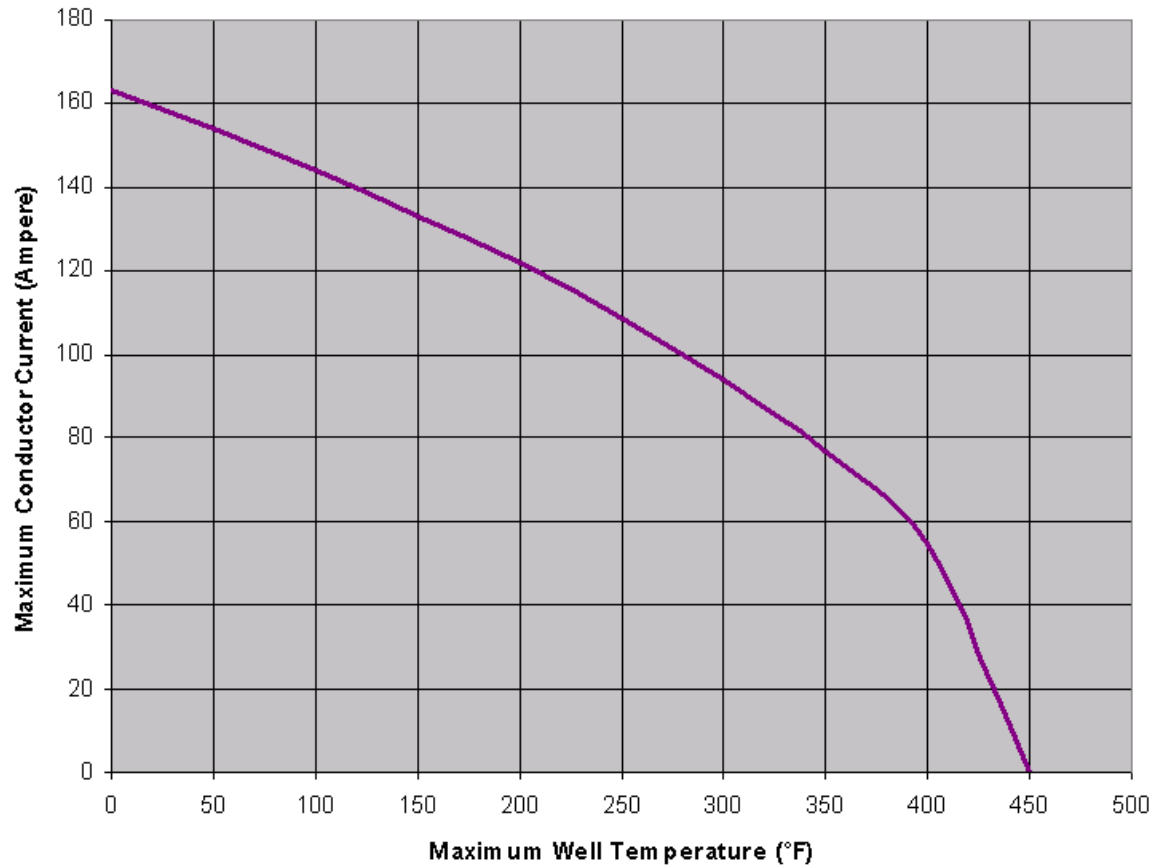


Figure A-4: Ampacity plot for #4 AWG Redalead flat ELB motor lead and #4 AWG KELB motor lead

Hotline550 MLE

The Hotline550 pothead design is significantly different from existing potheads and can only be used with Hotline550 motors. The part numbers and description for the Hotline550 (Hotline II) motor lead extension can be found in the OneCat. The MLE ampacity is available in the MLE ampacity spreadsheet ([InTouch ID 3016052](#)).

Hotline550 Pothead Installation Procedure

The Hotline550 pothead installs in a 540/562 motorhead, designed with a smooth-bore pothole, sized to fit a “-029” O-ring. The Hotline550 pothead has a special configuration and will not fit a standard motor head. The Hotline550 pothead can be used only on Hotline550 motors.

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There are two designs for the Hotline550 pothead. The original design, now obsolete, is a tape-in design with a MSE seal for the pothole. The second design has two O-rings, one for high temperature and one for low temperature. The high temperature O-ring is located nearest the electrical terminals and the low temperature O-ring is located nearest the flange of the pothead.

The (obsolete) Hotline550 pothead with MSE seal is installed using the following procedure.

1. Remove MLE pothead shipping cap and inspect for damage.
2. Confirm that pre-installation tests have been done previously (pressure and electrical test)
3. Remove the black rubber shipping cap O-ring.
4. Ensure motorhead area is clean and no fluid will drip onto the splice or motorhead prior to removing cover.
5. Install the white MSE seal with the open side towards the well. The visible spring inside the seal should be facing up, that is seen from the power cable side of the pothead.
6. When placing MSE seal on, slide the whole seal on carefully, ensuring even stretching of the spring.
7. After MSE seal set inside the groove, make sure it rotate freely.
8. Tape-in as per standard procedure. Use at least three layers of Teflon tape. Use four layers of Teflon tape if space permits.
9. Fill pothole with oil until no bubbles are seen.
10. Lubricate the pothole and MSE seal with Krytox Grease GPL – 206.
11. Ensure there no foreign materials in the hole and between the flanges.
12. Align the pothead with the threaded studs or bolt holes and insert the pothead carefully and evenly so the MSE seal installs without pinching or warping.

**Note**

Avoid shaking the cable as this may cause damage to the MSE seal.

13. Install lock washers and nuts. Tighten the pothead flange evenly from both sides few threads at a time.
14. Torque to specification.
15. Ensure the flanges match properly and there is no gap between them.
16. After complete servicing the protector, check pothead for any possible leak.

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The Hotline550 pothead with two O-ring grooves is installed using the following procedure.

1. Remove MLE pothead shipping cap and inspect for damage.
2. Confirm that pre-installation tests have been done previously (pressure and electrical test)
3. Remove shipping O-rings.
4. Install the high temperature O-ring. This goes in the first O-ring groove, nearest the electrical terminals, nearest to the motorhead.
5. Install the low temperature O-ring. This goes in the O-ring groove nearest the flange of the pothead. It can be installed by rolling it over the high temperature O-ring.
6. Tape-in as per standard procedure. Use at least three layers of Teflon tape. Use four layers of Teflon tape if space permits.
7. Fill pothead with oil until no bubbles are seen.
8. Lubricate the O-rings with Krytox Grease GPL – 206.
9. Ensure there are no foreign materials in the hole and between the flanges.
10. Align the pothead with the threaded studs or bolt holes and insert the pothead carefully and evenly to avoid damage of the two O-rings.
11. Install lock washers and nuts. Tighten the pothead flange evenly from both sides few threads at a time.
12. Torque to specification.
13. Ensure the flanges match properly and there is no gap between them.
14. After complete servicing the protector, check pothead for any possible leak.

HotlineSA3 MLE



Note

The HotlineSA3 pothead design is significantly different from existing potheads and can only be used with HotlineSA3 motors. The part numbers and description for the HotlineSA3 motor lead extension can be found in the OneCat.

The HotlineSA3 installation procedure is described in [InTouch Content ID 6058300](#) and the HotlineSA3 TIR [InTouch Content ID 5443035](#).

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A.1.4.6 ESP Power Cable

The cable splice is modified. It is similar to the standard Nicopress™ field splice, except a different type of tape is used (Hi-Mod with a silicone-based adhesive). The jackets must be fused together using a small jewelers torch with 95/5 solder. The solder joint is then checked with dye penetrant. This adds a level of complexity and required skill to successfully make these cable splices. The technique is suitable for a service center but not at the wellsite.

The high-temperature cable splicing procedure is provided in the following section.

Splicing Procedure, High-Temperature Lead Jacket Power Cables and MLE

Tools and materials required to perform high-temperature cable splices are shown in [Table A-7](#).

Table A-7: Tools and Materials Required to Perform High-Temperature Cable Splices

Description	Part Number
standard splicing tools and materials	—
dye penetrant test kit	—
jeweler's torch	—
95Pb–5Sn solder	—
acid-core solder	562678
lead sleeves	9990305
silicone adhesive high-modulus tape	R006164
acrylic adhesive high-modulus tape	748137
high-temperature tape	823435
Teflon™ tape	521427
glass tape	684811

Procedure

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1. Position the ends of the cables on the splice bench and secure the clamps onto the armor being careful not to over-tighten the clamps. (Overtightening the clamps will distort the armor.) Remove the armor to the dimensions shown in [Figure A-5](#) being extremely cautious to avoid cutting into the lead jacket. Remove any burrs or sharp edges on the cut.

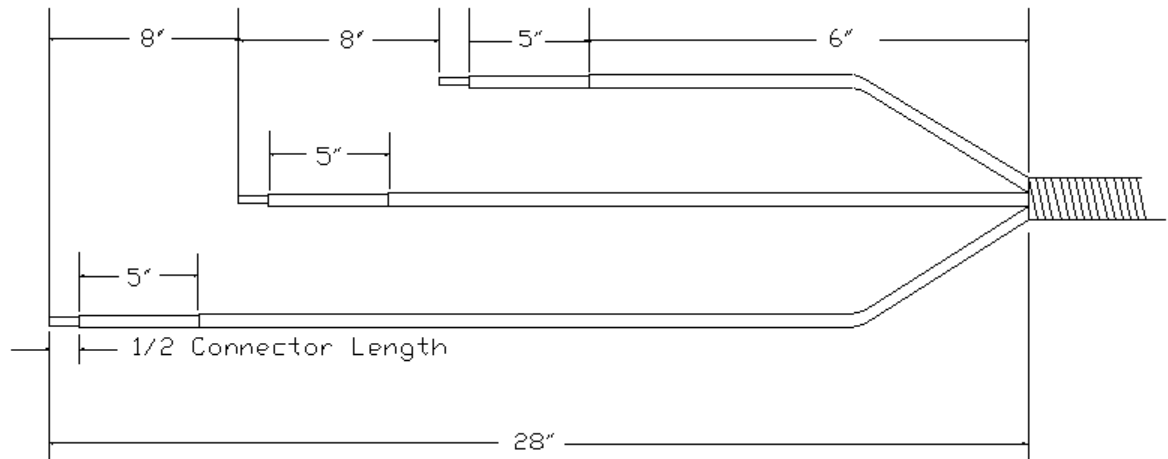


Figure A-5: Conductor preparation for high-temperature lead-jacket cables and MLE

2. Note the position of the part number on the pothead flange and orientate the part on the underside of the cable wrap as it is placed on the cable reel. The part number must be on the outboard side of pothead flange when it is installed on the motor.
3. Using acrylic adhesive high-modulus tape, tuck a single wrap underneath the armor followed by two or three wraps extending 1 in (2.54 cm) out around all three conductors. This tape will help to protect the lead when the conductors are spread apart.
4. Remove the lead jacket in accordance with the dimensions shown in [Figure A-5](#). Use extreme care not to damage the insulation.



Note

Cuts in the conductor insulation are not permitted anywhere beyond the bare conductor area. If the insulation is damaged when removing the jacket, cut off the damaged area and start a new splice.

5. Use a hacksaw or wire cutters and file to cut the conductors to the lengths shown in [Figure A-5](#).

**Note**

When cutting with the hacksaw, be careful not to damage the insulation or jacket on the adjacent conductors.

6. Obtain the proper sized splicing sleeves. Remove the insulation to the dimension shown in [Figure A-5](#).

**Note**

When removing the insulation, be careful not to ring-cut the copper. Doing so can create a hot spot leading to a failure.

7. Using the appropriate splicing sleeve, check to ensure the sleeve fits all the way on and snugly against the insulation. If too much insulation was removed, use a file to trim the copper back to the proper dimension.

**Note**

On motor lead extensions, remove the polyimide insulating tape from the conductor using a scraping tool and 150-grit emery cloth.

8. Remove filler from the outside diameter of any stranded conductors.
9. Prepare the end of the second cable to be spliced as instructed in Step 1, Step 2, Step 3 and Step 4.

Place the cable ends together such that the longest conductor on one cable butts against the shortest conductor on the other cable. Mark the other two conductors and cut where indicated. After cutting, check to ensure all three conductors butt up to each other when placed together tightly.

10. Strip the insulation from the ends of the second cable as instructed in Step 6. Clean the exposed conductors with a clean cloth that has been moistened with CRC and allow to dry completely.
11. Using a Megger™ tester, test both cables to ensure electrical integrity before proceeding further.
12. Place lead sleeves over the conductors and position the sleeves on the long end of the conductors to clear the splice area. The lead jacket must be exposed for at least 1 in (2.54 cm) at the splice area.

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13. Install the splicing sleeves over the conductors. Do not remove the grit from the inside of the sleeves. Its purpose is to puncture the surface of the copper allowing a lower resistance connection. Some sleeves also contain an antioxidant to prevent corrosion. This should remain in the sleeve as well.

**Note**

Splice failure can result from improper installation of splicing sleeves.

14. Use a crimping tool to press crimps in the sleeve. Place a crimp at position A. Rotate the crimping tool 90° and place a crimp at position B. Rotate the crimping tool 90° and place a crimp at position C. Place crimp at position D. Rotate the crimping tool 90° and place crimp at position E. Rotate the crimping tool 90° and place a crimp at position F.

**Note**

Crimp A and D are to be at the very end of the sleeve so there is no flare.

15. Check the outside diameter of the pressed crimps with a crimp gauge. The gauge must pass over the pressed crimp without binding. Adjust the crimping tool, if necessary, to permit the gauge to operate properly. Nicopress tool use, care and maintenance procedures are provided in the [Cables section](#).
16. Repeat the splicing sleeve installation procedure for the remaining two conductors.
17. Using a strip of 100-grit emery cloth, sand each sleeve to remove any burrs or sharp edges on the splicing sleeve. Discard the emery cloth and clean thoroughly.

**Note**

Do not use the emery cloth on the insulation after it has been used on the splicing sleeves or lead jacket.

18. Using a new strip of 100-grit emery cloth, roughen up all exposed insulation.
19. Using a new strip of 150-grit emery cloth, remove any oxidation on the lead jacket.
20. Clean exposed conductors well by wiping with a clean cloth which has been moistened with CRC or a naphtha-based solvent and allow to completely dry.

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21. To create a gas block, apply several layers of silicone adhesive high-modulus tape over the insulation adjacent to the lead jacket. One edge of the tape must be butted against the cut end of the lead sheath. Apply enough full-lapped layers over the insulation until the last layer of tape is even with the outside diameter of the jacket. Use four layers of high-temperature tape to cover the joint between the cut end of the jacket and the high-modulus tape.

**Note**

When applying the four layers of high-temperature tape, only tape onto the lead jacket 0.5 in (1.27 cm). This will allow a bare location for solder when replacing the lead sheath.

22. Cover each splicing sleeve with five layers of silicone adhesive high-modulus tape. Start the first layer of tape on the insulation 0.5 in (1.27 cm) from one end of the sleeve. Continue over the sleeve with half-lapped wraps until 0.5 in (1.27 cm) of the insulation is covered beyond the other end of the sleeve. Wrap the second layer in the opposite direction filling the void area in the taper flush with the Nicopress sleeves. Extend each succeeding layer of tape 0.5 in (1.27 cm) beyond the end of the previous layer. The tape must be wound continuously in a smooth spiral with no gaps or wrinkles. Tension should be applied to stretch the tape such that the tape is uniformly wrapped without creases or wrinkles. The tape can be stretched to decrease the width to about 85 % of its original width. Excessive stretching is not desirable.
23. Cover the exposed area of each splice with five layers of silicone adhesive high-modulus tape with the exception of a one in. to 2 in (2.54 cm to 5.08 cm) area at the end of the lead jacket. Start the first layer of tape on the insulation 1 in (2.54 cm) beyond the ends of the previous layers. Continue over the sleeve with half-lapped wraps until 1 in (2.54 cm) of insulation is covered beyond the other end of where the previous layer finished. Cut the tape and begin a second layer wrapped in the opposite direction. Extend each successive layer 0.5 in (1.27 cm), cutting the tape at the end of each layer until the fifth layer covers 1 in (2.54 cm) of the lead jacket. Leave 1 in (2.54 cm) exposed for a soldering area.
24. Apply a small amount of silicone lubricant to the taped splice. Position the lead sleeve so the ends of the sleeve are positioned over the lead barrier. Fit the sleeve into place, cut excess sleeve and file seams smooth. Remove oxidation from the lead sleeve.
25. Clean off any oxidation on the lead jacket. Apply solder flux. Tin the 1 in (2.54 cm) area of lead at each end of tape with 95Pb-5Sn solder. Round to a smooth contour with emery cloth or file as necessary.

26. Apply solder flux to the lead sleeves and apply 95Pb-5Sn solder around both ends of the sleeve. Apply solder flux to the longitudinal seams. Solder the longitudinal seams with 95Pb-5Sn solder. Stop 0.125 in (0.32 cm) from completely sealing the splice. Wait five minutes to allow the hot gases to escape and then seal the remaining 0.125 in (0.32 cm) seam. File any sharp points and buff with 150-grit emery cloth.
27. Inspect the soldered joints for pinholes. Perform a dye penetrant inspection test to check for pinholes.

**Note**

It is critical that no voids or pinholes exist in the solder.

28. Apply one layer of half-lapped acrylic adhesive high-modulus tape over the braid to the armor on each end of the conductors.
29. Use high-temperature tape to fill in and smooth out the bottleneck at each end of the lead sleeve for an approximate length of 2 in (5.08 cm).
30. Apply two half-lapped layers of acrylic adhesive high-modulus tape across the lead sleeves and beyond the high-temperature tape.
31. Loosen the cable clamps and carefully stretch the spliced conductors to a parallel orientation.
32. Using 1-in (2.54-cm) wide Teflon™ tape, weave the three phases together to hold the conductors parallel.
33. Apply two half-lapped layers of acrylic adhesive high-modulus tape over the full length of the entire splice. Start the first layer of tape on the jacket 1.5 in (3.8 cm) from the armor and continue over the splice until 1.5 in (3.8 cm) remain at the other end of the armor. The second layer, wrapped in the opposite direction, should cover the full length of the splice up to, but not on the armor.
34. Unwrap one loop of armor at each end of the splice. Apply two half-lapped layers of glass tape over the completed taped area extending under the unwrapped armor.
35. Bend the unwrapped armor back to fit snugly around the conductor.

36. Cut a 64-in (1.6-m) length of the armor. Pull out the tucked end, trim and flatten it. Use a round wooden stick or a piece of electrical conduit with a slightly smaller outside diameter than the inside diameter of the armor as a wrapping lever. Starting 2 in (5.08 cm) from the splice, form the armor around the splice in a spiral using the stick or conduit for leverage to draw the armor tight. Adjust the profile to expand around the spliced section while maintaining the interlock. Continue wrapping until 2 in (5.08 cm) overlap the existing armor.
37. Cut and secure both ends by soldering the last three convolutions with acid-core solder. Wipe away excess flux with a damp cloth and remove any excess solder from the armor.

**Note**

Use care to avoid heat damage to the cable under the soldered area. Cool the soldered joint immediately with a water-wetted cloth. Be careful to avoid contact with hot surfaces.

A.1.4.7

Hotline Gauge**Part Numbers**

Listed here are the part numbers of Hotline monitoring equipment and spare parts.

Table A-8: Required Equipment Installation

Part Number	Description	Comments
100746999	KIT: Gauge Portal, MTR 562, Sapphire, Hotline 550	
100833037	KIT: Gauge Portal, MTR 562, RTD Only, Hotline 550	(Motor temp only)
100867720 (Rockbestos DC07022-000)	MC SENSOR CABLE - 250degC RATED - 7/c 22 AWG 19/s - NICKEL PLATED COPPER - 316L .028" ARMOR	7 conductor cable used for Motor Temperature Only systems too.
100408570	Extreme Card	Suggest having 1 spare during installation.
100241377	DHT Connector 2x4 Pin 3.5mm (connectors on back of Extreme Card)	Suggest having 2 spares during installation.
100871384 ¹	7 PIN Male Metal To Metal Seal Connector Mates With Bulkhead 100660109	Suggest having 1 spare during installation.

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Part Number	Description	Comments
	7 conductor Penetrator	

¹ Includes the male connector and consumables (100660108 is just the hardware). Seven-pin male metal-to-metal seal connector (100871384) mates with the female bulkhead (100871383) in the motor portal. The male connector should be assembled to monitoring cable following procedure detailed in GeMS document 100787432.

Table A-9: Required Wellsite Installation Tools

Part Number	Description
100266133	Fluke Multimeter
100923123	Tester: Insulation, Fluke 1503, 500V
1156926	Torque Wrench: ½ Inch Square Drive, 10-80 ft/lb Setting
100785772	Tool: Torque Adapter, 1-3/4 Inch Open End
B047177	Tube Cutter
90006-00G-00001	Tubing Cutter Gauge
100499488	400 Grit Emery Cloth
100499484	Fine Triangular File 6 Inch
100880310	Krytox Grease (ANTIGALL)
100782309	Krytox grease for o-rings
NA	Wire strippers
NA	Vise grip, #4 size
NA	Wellhead fittings, SwageLok
NA	Uniconn test box with Extreme Card

Pre-Installation Checks

Perform the following checks before running the ESP string into the well to verify that the gauge is operational.

Connector and Cable Check

1. Remove the protective cap from the cable connector and visually inspect the connector and pins for damage.
2. Measure the resistance on each monitoring cable conductor and verify continuity is good. Compare the measured resistance readings and verify with the reference cable length according to the cable electric properties (17 ohms per 1000 ft of cable at 20 degC).
3. Megger test the cable conductor to conductor and conductor to ground (cable armor) to verify that the insulation is intact. All measurements need to greater than 10 MOhms.

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**Caution**

Potential Severity: Light
Potential Loss: Assets
Hazard Category: Electrical

Only a 500-V megger should be used for testing the cable insulation.

Gauge Function Test

1. Remove the protective plug from the portal bulkhead and visually inspect the portal for damage. Connect the test cable connector to the portal, securing the connector with the securing nut. The gauge and monitoring cable are now ready to be function tested.

**Note**

Care should be taken not to scratch the conical sealing surface inside the portal when inserting the cable connector.

2. Connect surface equipment (UniConn with Extreme card) to the other end of the cable according to the following figures.

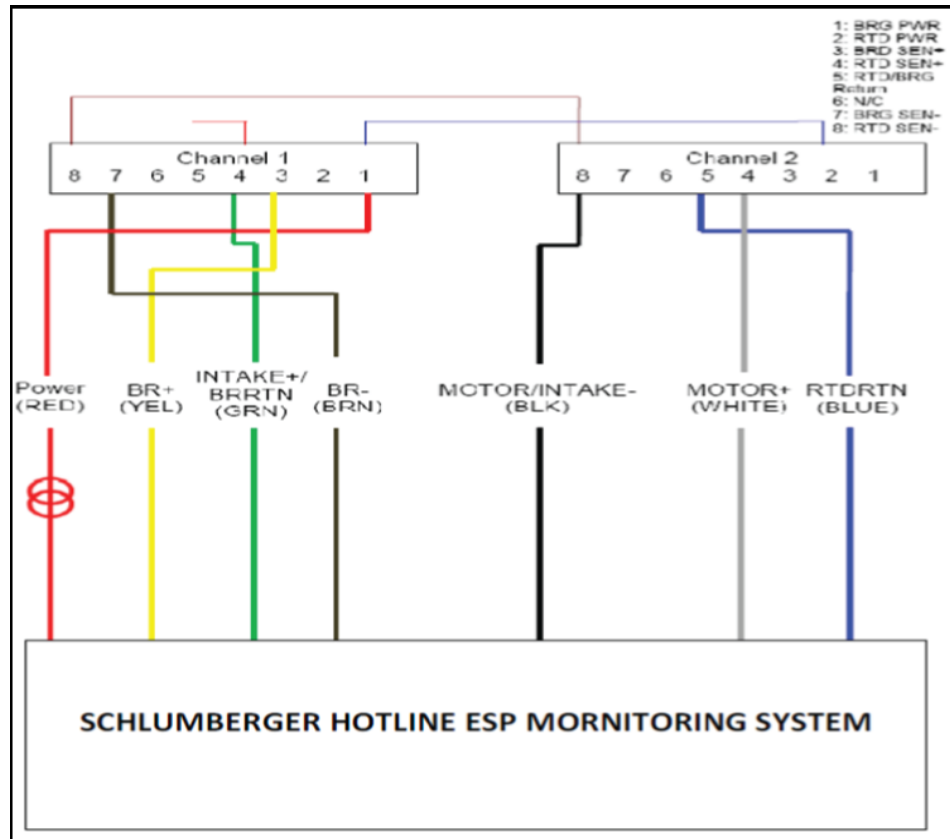


Figure A-6: Gauge Configuration

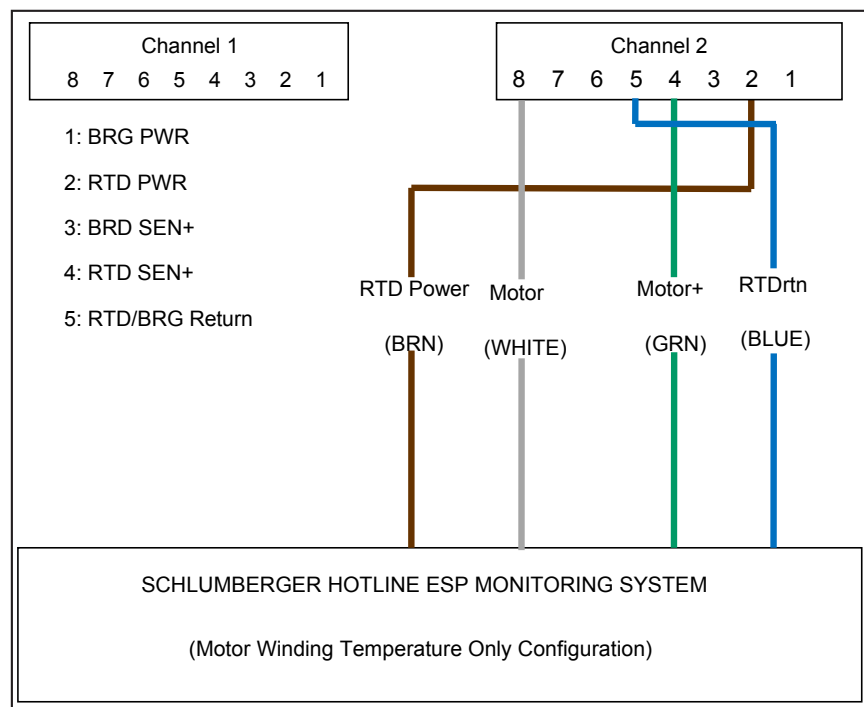


Figure A-7: Winding Temperature Configuration (RTD only)

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3. Verify that the correct coefficients have been downloaded on the Extreme card for the pressure sensor and RTD. Refer to the Hotline Extreme Card User Manual ([InTouch ID 5447042](#)) for details on how to download and verify coefficients.
4. Begin logging the data from either or both the Uniconn and ExtremeView using StarView.
5. Use a hydraulic pump or simple hand pump to apply pressure to the gauge through the gauge's pressure communication port. The port requires a Swagelok interface.
6. Confirm that the applied pressure corresponds with the values displayed on the UniConn and ExtremeView. (Non-Applicable for RTD only).

**Note**

The Hotline550 pressure sensors are only calibrated from 35 degC to 225 degC. Outside this temperature range, accurate pressure readings cannot be guaranteed. If a sensor will be installed in a cold environment, the gauge must first be checked at the shop and again at the wellsite.

Due to internal bellow in the Hotline550 pressure sensor (Sapphire), pressure readings at atmosphere might not be accurate (15 psi $\pm$ 10 psi). Apply pressure of about 200 psi and check for accuracy.

7. Power off all the equipment and record all function tests.
8. Carefully disconnect the test cable from the portal bulkhead and replace the protective plug to protect the portal connection.

Running in Hole

Equipment Assembly

1. The monitoring cable sheave positioning and suspension should be performed as outlined in the standard [transport and handling](#) procedures.

**Note**

Ensure that the cable connector is adequately protected during these operations. Should anything untoward occur during the makeup or hanging of the cable sheave a Megger test of the monitoring cable is recommended.

2. Power cable lifting should be performed as outlined in the standard [transport and handling](#) procedures.

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3. Motor installation performed using the alternate lifting method outlined in the standard [motor installation](#) procedures.
4. Connect the cable connector and portal bulkhead connection.
 - a. Remove the plug from the portal bulkhead and visually inspect the bulkhead and female connector inside.
 - b. Remove the shipping cap from Kemlon connector. Place one O-ring and two back-up rings into the groove on the connector body. Lubricate the O-ring with the Krytox Grease.

**Note**

For O-ring and back up rings installation, ensure that taper surface is always facing the O-ring. If a dot is found on the back up ring, it will be the flat surface. Rotate the scarf cut of back up rings 180° apart from each.

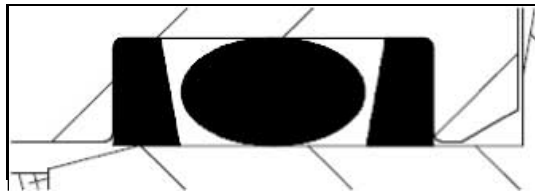


Figure A-8: Back-up rings and O-ring Orientation

- c. Insert the Kemlon connector in the gauge portal, aligning the connector keyway with the keyway inside the bulkhead. Be very careful while handling the cable and connector end to avoid placing any bending stresses on either.

**Note**

Once the key is aligned with the keyway, the Kemlon connector should slide in by hand, don't force the connection.

- d. Apply the Anti-gall threading compound to the secure nut. Thread the nut into the bulkhead with a 1-3/4 in wrench. The torque setting is 45 lbf-ft.
 - e. Connect the surface equipment to the end of the monitoring cable.
 - f. Power up the surface equipment and confirm gauge functions properly, recording all function test results.

Testing While RIH

1. After the ESP string is assembled, the gauge should be checked immediately to make sure nothing was damaged or disturbed during the assembly process.



Figure A-9: Surface Test Gauge

2. Connect the surface equipment to the end of the monitoring cable.
3. Power up the surface equipment and confirm gauge functions properly, recording all function test results.
4. Repeat function test every 20 joints or every hour (whichever comes first) to ensure that the gauge is functioning properly while running into the well.

Power Cable Banding

Power cable banding should be performed as outlined in the standard [power cable banding](#) procedures. The rig must be centered over the hole to provide adequate clearance for the power cable, monitoring cable, and cable bands/clamps. The monitoring cable should be run alongside of the power cable. This alignment should be checked periodically when running the equipment in the hole.

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**Note**

Always align the monitoring cable besides of the power cable, and secure them by using bands or clamps.

Landing Tubing Hanger

1. Make up the tubing hanger assembly to the tubing. Make sure the hanger seal is in place first.
 2. Determine the length of monitoring cable that will be required between the wellhead and the junction box.
 3. Spool off enough cable to cover the span plus an additional 5-ft safety margin.
-

**Note**

Do not spool off much more than is required. This will make handling the cable difficult.

4. Cut the cable using tubing cutters.
-

**Note**

Cut straight through the cable so that the end is flat. This will make it easier to feed the cable through the tubing hanger.

5. Feed the cut end of the monitoring cable through the hanger until all of the cable is through and the cable is straight and taut.
 6. Verify that the Swagelok fitting is in good condition and intact before you start to push it over the monitoring cable.
 7. Carefully pull the Swagelok fitting over the monitoring cable. This may be easier to do with the compression nuts opened.
 8. Make up the compression fitting to the top of the tubing hanger.
 9. Test the gauge again once the hanger landed.
-

Remove BOPs and Assemble the Wellhead

1. When the rig crew removes the BOP assembly, make sure the crane lifts the BOPs directly vertical to a height of approximately 2 m.
2. Carefully pull the excess monitoring cable through the BOPs. Try to keep the bending stress on the tubing hanger fitting to a minimum.

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3. Verify that the fitting (Swagelok or Conax) is in good condition and intact.
4. Install the fitting in the corresponding port on the top of the wellhead bonnet. Leave the nut of the fitting opened, this will be easier to feed the cable through.
5. Feed the cable through corresponding port in the bonnet using the same procedure as the tubing hanger.
6. Lower the bonnet onto wellhead flange carefully.
7. After the bonnet has been secured, make up the compression fittings on the top of wellhead bonnet.
8. Perform a gauge function test after the wellhead is connected.
9. Seal of the end of the monitoring cable with tape to avoid moisture entry.
10. Roll up and secure the monitoring cable in a safe place.

A.1.5

Equipment Retrieval Procedures

Work with the client to ensure that safe working procedures are in place and understood before initiating the retrieval process. Ensure a systematic and controlled method is used to cool the wellhead, production tubing and pumping system.



Danger

Potential Severity: Major
 Potential Loss: Personnel
 Hazard Category: Fire flammable

Be aware of the temperature and pressure of the wellhead and piping before loosening any connection. Liquids can easily flash to vapors when the pressure is relieved and cause severe burns.

The difference between the downhole operating temperature and the temperature at the surface is substantial. Procedures must be in place to ensure that the equipment is not “shocked” by a rapid temperature decrease.



Danger

Potential Severity: Major
 Potential Loss: Personnel
 Hazard Category: Fire flammable

Before loosening any connection on the pumping unit, make certain the temperature is cool to the touch. If there are pockets of contained liquids under pressure, these could flash to vapors and cause a severe injury.

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A.1.5.1 Motor, Protector, Intake and Pump

The motor, protector, intake and pump are unbolted and handled using the same procedures as a conventional units. Procedural information is provided in the standard [pulling equipment from the well](#) procedures.

A.1.5.2 Pothead

The pothead is unbolted from the motor using the same procedures as a conventional unit. A shipping cap should be placed on the pothead before returning it to the service center. These additional checks should be made:

- Position of conductor tips relative to the pothead bolting face.
- Ensure the gland and the hole is clean and free of scratches.

A.1.5.3 Cable

The cable should be handled using the same procedures as a conventional cable. Procedural information is provided in the [Cables section](#).

A.1.6 Servicing and Testing

All Hotline equipment and Hotline550 equipment can be repaired at a Schlumberger Service Center. Special instructions are provided in the following sections to determine whether the equipment can be reused.

A.1.6.1 Motor

The motor should be tested electrically in the same manner as a conventional motor. The motor can be tested using the Motor Idle Slant Test (MIST) procedure with the following changes.



Warning

Potential Severity:	Serious
Potential Loss:	Assets
Hazard Category:	Machinery equipment hand tools

Do not perform a MIST test on a motor filled with Reda No. 7 oil as serious personal injury or equipment damage may occur.

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- The motor shall be drained, then filled and tested using Reda No. 5 oil. Reda No. 5 oil is synthetic oil and will blend best with Reda No. 7 oil. There will be a viscosity reduction.
- The motor must pass the standard MIST test with the Reda No. 5 oil.
- Drain the Reda No. 5 oil immediately after the completion of the test while the motor is still warm. It is preferable to elevate the head and/or hang the motor vertically while draining the oil. It is also preferable to allow the motor to drain for at least 12 hours before refilling.
- Fill the motor with Reda No. 7 oil before shipping.

A.1.6.2 **Protector**

There will be more drag during a protector spin test due to the increased viscosity of the fluid. The protector must pass a helium leak test before reuse.

A.1.6.3 **Intake, Pump and Pothead**

The intake, pump and pothead are serviced and tested using standard service center procedures.

A.1.6.4 **Cable**

The cable is serviced and tested using standard service center procedures. Note that the braid may be carbonized to the point of disintegration for cables that have been installed in high-temperature environments. This should have no impact on the performance of the cable as the braid is used only to aid in the armoring process during fabrication.

A.2 **D8–42 Series Poseidon Multiphase Flow Pump**

A.2.1 **Product Description**

The presence of free gas in a submersible (centrifugal) pump causes degradation of head and flow performance, up to a level where gas locking occurs. Before now, conventional submersible pumps have not been considered for applications when more than 15% to 25% free gas is present at the pump intake. The D8–42

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Poseidon™ multiphase pump is a mechanical pre-conditioner/charge pump, bolted to the base of a submersible pump, which allows the submersible pump to pump fluids with free-gas percentages up to and above 75%.

In high-percentage free-gas applications, significant degradation of head and flow performance occurs in the lower stages of a submersible pump when used with the Poseidon multiphase pump, but gas locking should not occur. The energy transferred to the fluid in these lower stages of the centrifugal pump compresses the gas (head development). Once in the upper stages of the pump, the fluid (with gas compressed) behaves much like a single-phase fluid and pump performance also behaves as though handling single-phase fluids. To attain this level of gas handling requires a submersible pump design that includes bottom stages used to compress the gas that do not add appreciably to net lift.

A.2.2 Installation

The D8-42 Series Poseidon multiphase pump should always be installed using the standard [pump installation](#) procedures. It should always be installed between the intake/gas separator and submersible pump.

The D8-42 Series Poseidon multiphase pump CT version was designed to bolt directly to a D20-60 VGSA, which has a recessed shaft. This required that the end of the D8-42 shaft extend below its base flange. The recessed shaft allows the shaft coupling to be placed below the restricted inside diameter found in the D20-60 VGSA head.

The LT version was designed to be used in applications where gas cannot be vented out the casing.

Shimming

The D8-42 Series Poseidon utilizes the CR-style (compression ring) thrust design and requires shimming. It should be shimmed so that the bottom impeller thrust washer does not wear into the diffuser down-thrust pad, but instead just contacts the diffuser down-thrust pad. No seal is required between the down-thrust washer and down-thrust pad, as is typical with compression-type submersible pumps.

The average shaft end play of the D8-42 Series Poseidon multiphase pump is only 1/32 inch. It is essential that free shaft rotation is confirmed once the Poseidon multiphase pump is bolted to the intake/gas separator.

Installation Tools

Poseidon multiphase pumps use standard [pump installation tools](#).

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A.2.3 Connections

Connection details and part numbers are available in the ALFORM-D connection tables ([InTouch ID: 3255850](#)). The D8–42 Series Poseidon Multiphase Flow Pump can be used on 400–series (with DN-3000 stages) and larger pumps. For the 400–series and larger pumps the screw used to bolt it to those pumps must be a hex *socket* cap screw. For smaller pumps, the hex head cap screw can be used.

A.2.4 Poseidon Configurations (CT and LT versions)

The Poseidon is available in two configurations: center tandem (CT) and lower tandem (LT). Both versions have the top end of their shafts recessed into the head, like the D20-60 CT AGH. The Poseidon multiphase pump CT version has the bottom end of its shaft extended from its base flange like the 400–series (DN-4300 stages) CT pump. The LT version has an integral intake, and the bottom end of its shaft is flush with the base flange. The recessed shaft in the head of both versions requires long couplings for installation with pumps smaller than the 400–series (DN-4300 stages). See for coupling part numbers.

A.3 S50-90 Series Poseidon Multiphase Pump

A.3.1 Product Description

The presence of free gas in a centrifugal pump causes degradation of the head and flow performance. Conventional submersible pumps have not been considered for applications when more than 15% free gas is present at the intake. The S50–90 Series Poseidon multiphase pump is a mechanical pre-conditioner that increases the fluid mixture pressure in order to compress the gas to a point where the fluid can be pumped with a submersible pump. The free-gas bubbles are compressed until their volume decreases to a point where the vapor-to-liquid ratio becomes much smaller. Additional compression above bubblepoint causes the gas to go into solution, further decreasing the vapor volume while increasing the liquid volume. The fluid then begins to behave more like a single-phase fluid.

This technical information release provides information required to install the S50–90 Series Poseidon multiphase pump. The S50–90 Series Poseidon multiphase pump is specifically designed to be used with the higher volume 540– and 538–series submersible pumps.

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A.3.2 Installation

The S50–90 Series Poseidon multiphase pump should be installed using the pump installation procedures provided in the standard [pump shimming and installation](#) procedures. It should always be placed between the intake/gas separator and pumps.

The S50–90 Series Poseidon multiphase pump utilizes the CR-style thrust design and requires shimming. Shimming procedures are provided in the standard [pump shimming and installation](#) procedures.

Installation Tools

Poseidon multiphase pumps use standard pump [installation tools](#).

A.3.3 Connections

Connection details and part numbers are available in the ALFORM-D connection tables ([InTouch ID: 3255850](#)).

A.4 M90-470 Series Poseidon Multiphase Pump

A.4.1 Product Description

The presence of free gas in a submersible (centrifugal) pump causes degradation of head and flow performance, up to a level where gas locking occurs. Before now, conventional submersible pumps have not been considered for applications when more than 15% to 25% free gas is present at the pump intake. The M90-470 Poseidon™ multiphase pump is a mechanical pre-conditioner/charge pump, bolted to the base of a submersible pump, which allows the submersible pump to pump fluids with free-gas percentages up to and above 75%.

In high-percentage free-gas applications, significant degradation of head and flow performance occurs in the lower stages of a submersible pump when used with the Poseidon multiphase pump, but gas locking should not occur. The energy transferred to the fluid in these lower stages of the centrifugal pump compresses the gas (head development). Once in the upper stages of the pump, the fluid (with gas compressed) behaves much like a single-phase fluid and pump performance also behaves as though handling single-phase fluids. To attain this level of gas handling requires a submersible pump design that includes bottom stages used to compress the gas that do not add appreciably to net lift.

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A.4.2 Installation

The M90-470 Series Poseidon multiphase pump should always be installed using the pump installation procedures provided in the [pump shimming and installation](#) . The M90-470 Poseidon is designed to bolt directly to a 738 series protector. The Poseidon is only available in lower tandem (LT) configuration with integral intake.

A.4.2.1 Shimming

The M90-470 Series Poseidon utilizes the C-style (compression) design non-factory shimming and requires shimming. It should be shimmed so that the bottom impeller thrust washer does not wear into the diffuser down-thrust pad, but instead just contacts the diffuser down-thrust pad. No seal is required between the down-thrust washer and down-thrust pad, as is typical with compression-type submersible pumps.

The average shaft end play of the M90-470 Series Poseidon multiphase pump is only 1/32 in. It is essential that free shaft rotation is confirmed once the Poseidon multiphase pump is bolted to the protector.

Installation Tools

Poseidon multiphase pumps use standard pump [installation tools](#).

A.4.2.2 Connection Tables

Connection details and part numbers are available in the ALFORM-D connection tables ([InTouch ID: 3255850](#)).

A.5 Extended Runlife Protector

A.5.1 Product Description

The Extended Runlife Protector (also known as Advanced Protector) is a component of the Extended Runlife project. It is a new protector system that will substantially increase runlife in hostile applications.

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The function of the protector is to handle the down-thrust load from the pump section and to act as the compensation section for the pressure-balanced, oil-filled motor. Protector failure or even a small leakage past the seals allows well fluid into the high-voltage motor section which usually results in catastrophic system failure.

The current protectors, when only a single tandem is used, can handle benign environment wells but are in many cases the weak link in the submersible pump system. The use of tandem protectors has eliminated the majority of the system failures that are protector related, however, wells with high bottomhole temperature, wells with a high percentage of H₂S and wells with high abrasive content are still problematic even for tandem protectors. As such, Extended Runlife Protectors are targeted at the following type wells

- **Wells with high bottomhole temperatures** — The operating temperature limit of existing protectors is 400 degF (204 degC) and is limited by the elastomers used in the protectors. To exceed this limit, the High-Temperature Extended Runlife Protector uses metal bellows to replace elastomer bags and Chemraz™ seals to replace Aflas and HSN seals.
- **Wells with high H₂S** — For applications with a high percentage (up to 30%) of H₂S at low-to-moderate temperatures (100 to 300 degF [38 to 150 degC]) the elastomer bag currently being used is ineffective as an H₂S barrier. H₂S can easily migrate through the bag to attack the copper wires and other copper base components inside the protector and motor. The Corrosion-Resistant Extended Runlife Protector uses a metal bellows (Inconel 625 material) to block the migration of the H₂S.
- **Wells with high abrasive content** — Abrasives in well fluids tend to settle on the top end of the protector where it is connected to the pump intake. Accumulations of abrasives in this area will either directly damage the mechanical shaft seal located there or will damage the adjacent journal bearings that are used to support the shaft seal. The Abrasion-Resistant Extended Runlife Protector uses a new seal body design in conjunction with a shroud to reduce the accumulation of abrasives around the mechanical shaft seal. The Extended Runlife Protector also uses dual-head shaft bearings (shaft sleeves and bushings) made of abrasion-resistant tungsten carbide to reduce bearing wear and improve shaft stability.

Five configurations of the Extended Runlife Protector have been developed and are available to address runlife issues in the hostile well conditions noted above. The various configurations are shown in [Table A-10](#). Table nomenclature is defined in the last table row. Technical features are shown in [Table A-11](#). Table nomenclature is defined in the last table row.

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Table A-10: Extended Runlife Protector configurations.

	Abrasion Resistant (AR)	Corrosion Resistant (CR)		High Temperature (HT)	
		2–Chamber	3–Chamber	3–Chamber	4–Chamber
Configuration	AR-LSBPB-HL	CR-BSM-HL	CR-BSMPM-HL	HT-LSMPM	HT-LSMPMPM
Chamber Type					
Term	Definition				
B	Bag – indicates the chamber utilizes a positive seal or elastomer bag-type design				
L	Labyrinth – indicates the chamber utilizes a labyrinth-type design				
M	Metal Bellows – indicates the chamber utilizes a metal bellows-type design				
Connection Type					
Term	Definition				
P	Parallel – indicates the chambers are connected in a parallel configuration (bag or metal bellows chambers only)				
S	Series – indicates the chambers are connected in a series configuration				
Bearing Type					
Term	Definition				
HL	High Load – indicates the protector utilizes a high-load, high-temperature thrust bearing				
Protector Type					
Term	Definition				
AR	Abrasion Resistant				
CR	Corrosion Resistant				
HT	High-Temperature Resistant				

Table A-11: Extended Runlife Protector technical feature summary.

Feature	Abrasion Resistant (AR)	Corrosion Resistant (CR)	High Temperature (HT)
replacement of elastomer bag with metal bellows	no	yes	yes
use of metal bellows shaft seal	no	yes	yes
shaft material	Monel K500	Inconel 625	Inconel 625
housing material	1026 carbon steel	1026 carbon steel	1026 carbon steel
head / base / adapter material	8620 carbon steel	8620 carbon steel	8620 carbon steel

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Feature	Abrasion Resistant (AR)	Corrosion Resistant (CR)	High Temperature (HT)
up-thrust bearing type	nonstandard	nonstandard	nonstandard
down-thrust bearing type	high load	high load	KMC
dual head bearings for shaft seal support	yes	yes	yes
keyless tungsten carbide journal bearings for abrasion resistance	standard	corrosion resistant	standard
shroud over head bearings for abrasion resistance	yes	yes	yes
use of high-pressure relief valve to ensure valve closure	yes	yes	yes
factory shimmed shaft to eliminate field shimming	yes	yes	yes
oil type	Reda No. 5	Reda No. 5	Reda No. 7
o-ring type	Aflas	Aflas	Chemraz
bag type	Aflas	not applicable	not applicable

A.5.2 Health, Safety and Environment

This section reviews the manufacturing and field handling procedures and the potential hazards shop and field service personnel may experience while assembling, testing, installing, pulling and dismantling an Extended Runlife Protector.

A.5.2.1 HSE summary

The Extended Runlife Protector has undergone independent HSE reviews by SRC and BPC engineering, manufacturing and QHSE personnel. The findings of the review are

- No hazardous materials such as radioactive or explosive materials are used in any of the Extended Runlife Protectors.
- No electrical components are used in any of the Extended Runlife Protectors.
- No high-pressure components are used in any of the Extended Runlife Protectors.
- Only normal wellbore temperatures are experienced in the Extended Runlife Protectors.
- All of the Extended Runlife Protectors contain hydraulic oil and require normal hygiene precautions.

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- The Extended Runlife Protectors can be heavy. Manual lifting is not recommended. Standard lifting methods are provided in the [Wellsite Procedures](#) section.

A.5.2.2

Applicable standards

Standards applicable to the assembly, testing, installing, pulling and dismantling of the Extended Runlife Protector are shown in [Table 1-1](#).

Table A-12: Applicable Extended Runlife Protector assembly, testing, installing, pulling and dismantling standards.

Standard	Description
ED-105	O-ring installation in oil well equipment
ED-164	Chain torque wrench tighten of threaded joints
ED-174	Joint weld specification
GED-001	Equipment serial numbering system
MPI-53112	Protector assembly, modular
MPI-59102	Install shipping cap
MPI-59105	Paint unit, install nameplate, patent sticker and install screen
MPI-59110	Spot weld joints
MPI-2015017	MPI, ASSEMBLY, 3 CHAMBER ADVANCED PROTECTOR ASSEMBLY, 540
MPI-2015020	MPI, OIL-FILLING, 3 CHAMBER ADVANCED PROTECTOR, 540
MPI-2015022	MPI, DISMANTLE, 3 CHAMBER ADVANCED PROTECTOR, 540
QAD-3054	Material handling
QAI and TI-2027	SPS vibration analysis for vertically and horizontally tested equipment
QAI and TI-2028	Helium leak test
QAI and TI-2013	Modular protector testing, 400 and 540 series
QAI and TI-3019	Assembly quality plan
100010378	MPI, Assembly, Advanced Protectors, Four-Chambers, 540
100012094	MPI, Oil Filling, Four-Chamber Advanced Protector, 540
100012248	MPI, Dismantle, Four-Chamber Advanced Protector, 540

A.5.2.3

Assembly

Quality

All material handling shall be performed in accordance with QAD 3054. General assembly shall follow QAI and TI 3019 Assembly Quality Plans.

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- **Oil** — All current motor fluid used in protectors is considered nonhazardous.
- **Thread lube** — Current lubrication used on housings and head, base and body threads is food grade and is considered nonhazardous.
- **Loctite®** — Thread locking chemicals are considered nonhazardous.
- **Teflon® tape** — Nonhazardous.
- **Bellows** — The bellows are constructed of Inconel 625. This material is already common within the system and not considered hazardous.

Safety

- **Handling and lifting** — During the course of assembly, several awkward and/or heavy components will require lifting. The most notable of these is the housing and the assembled unit. In each case, lifting should be done with a lifting sling and a hoist in accordance with MPI 57101.

Other heavy items such as the head, base and shaft should be lifted and transported using proper stepping, handling and lifting procedures.

When possible all shop transport of long heavy objects should be done using a housing dolly.

- **Making up joints** — Protector joints are routinely tightened to only 200 ft-lbf. Appropriate documentation and tooling is available and in place.

A.5.2.4

Final test

Quality

Standard protector test parameters and procedures will apply except for the following:

- **Valve pressure setting** — An increase from 10 psi to 20 psi requires a change in the testing procedure for both the individual valve and the top-level assembly after the valve is installed. (See QAI 2013.)
- **Metal bellows** — A pressure and vacuum test needs to be performed on the bellows.

Safety

- **Lifting for test** — Lifting should be done with a lifting sling and a hoist in accordance with MPI 57101.

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- **Pinch points** — When making flange to flange connections, keep fingers clear while mating the units.

A.5.2.5 Shipping

Quality

All material handling should be performed in accordance with QAD 3054. Boxing and/or loading of equipment should be done in accordance with MAT-B019.

Safety

The protector should not be lifted by hand. Lifting of the unit either for loading in a shipping box or loading onto a truck shall be done with a hoist and sling in accordance with MPI 57101 or with a forklift in accordance with QAD 3054.

A.5.2.6 Equipment installation

Quality

The unit installation and wellsite procedures will be in accordance with current practices. Unit flange bolts will be tightened in accordance with the torque requirements in ED-119.

Safety

- **Lifting** — This unit should not be lifted by hand. All lifting clamps shall be used in accordance with FSB 119/rd and lifting should be performed in accordance with procedures provided in the [Wellsite Procedures](#) section.



Warning

Potential Severity:	Serious
Potential Loss:	Assets
Hazard Category:	Machinery equipment hand tools

Use caution not to damage the shaft spline or flange connections during lifting.

- **Pinch points** — When making flange-to-flange connections, keep fingers clear while lowering unit.

A.5.2.7 Equipment pull

Quality

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Equipment pulls should be performed in accordance with procedures provided in the [Wellsite Procedures](#) section.

A.5.2.8 Dismantle

Quality

The Extended Runlife Protector shall be dismantled per appropriate manufacturing process instruction (MPI).

A.5.3 Engineering Specifications

The engineering specifications and requirements for the Extended Runlife Protectors are described in detail in revision-controlled document 100024320 and are shown in [Table A-13](#), [Table A-14](#) and [Table A-15](#).

Table A-13: Extended Runlife Protector environmental and operational specifications.

Specification	Value
wellbore deviation	vertical to horizontal during run in hole
	vertical to horizontal at setting depth
pump intake pressure	up to 5000 psi (345 bar) without H ₂ S
	up to 2000 psi (138 bar) with H ₂ S
operating temperature	See Table A-14
well bore temperature	See Table A-14
surface temperature	-40 degF to 120 degF (-40 degC to 49 degC)
shaft requirements	shaft diameter is 1.1875 in (3.016 cm)
	end splines per reference drawing 1068378
	material — Inconel 625 per MS-03-168 for high-temperature and corrosion-resistant protectors
	material — Monel K500 per MS-03-109 for abrasion-resistant protectors
oil density	9 through 50 API gravity oils at setting depth conditions.
oil viscosity	dead oil viscosities at 100 degF (38 degC) from 1 to 10,000 cp.
water	0 % to 100 % water cut wells, with total dissolved solids content in the water ranging from fresh water classification to 250,000 ug/g.

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Specification	Value
scale	must function in the presence of standard downhole scales: CaCO_3, CaSO_4, FeCO_3, BaSO_4, etc. i Note specification not quantifiable but all current knowledge being utilized to provide improvement over existing systems.
paraffin/asphaltine	must function in the presence of crystallized oil paraffin i Note specification not quantifiable but all current knowledge being utilized to provide improvement over existing systems.
sand and solid content	specification not quantifiable but all current knowledge being utilized to provide improvement over existing protectors.
free gas	0 % to 100 % free gas at the pump intake
oil volume compensation	dependent on system configuration. See oil volume calculator application software for details.
motor temperature cycle	150 degF (66 degC) with basic unit
	450 degF (232 degC) by adding extra sections (for HT version only)

Table A-14: Extended Runlife Protector temperature specifications.

	Abrasion Resistant (AR)	Corrosion Resistant (CR)	High Temperature Resistant (HT)
operating temperature (internal to protector)	up to 400 degF (204 degC) with 0% H_2S	up to 400 degF (204 degC) with up to 0% H_2S	up to 475 degF (246 degC) with up to 0% H_2S
		up to 300 degF (149 degC) with up to 5% H_2S (100 psi [9 bar] at 2000 psi [138 bar] BHP)	up to 300 degF (149 degC) with up to 5% H_2S (100 psi [9 bar] at 2000 psi [138 bar] BHP)
		up to 250 degF (121 degC) with up to 30% H_2S (600 psi [41 bar] at 2000 psi [138 bar] BHP)	up to 250 degF (121 degC) with up to 30% H_2S (600 psi [41 bar] at 2000 psi [138 bar] BHP)

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	Abrasion Resistant (AR)	Corrosion Resistant (CR)	High Temperature Resistant (HT)
wellbore temperature (bottomhole temperature)	up to 350 degF (177 degC) with 0% H ₂ S	up to 350 degF (177 degC) with up to 0% H ₂ S	up to 425 degF (218 degC) with up to 0% H ₂ S
		up to 250 degF (121 degC) with up to 5% H ₂ S (100 psi [9 bar] at 2000 psi [138 bar] BHP)	up to 250 degF (121 degC) with up to 5% H ₂ S (100 psi [9 bar] at 2000 psi [138 bar] BHP)
		up to 200 degF (93 degC) with up to 30% H ₂ S (600 psi [41 bar] at 2000 psi [138 bar] BHP)	up to 200 degF (93 degC) with up to 30% H ₂ S (600 psi [41 bar] at 2000 psi [138 bar] BHP)

Table A-15: Extended Runlife Protector physical properties.

	Abrasion Resistant (AR)	Corrosion Resistant (CR)		High Temperature (HT)	
		2–Chamber	3–Chamber	3–Chamber	4–Chamber
configuration ^A	AR-LSBPB-HL	CR-BSM-HL	CR-BSMPM-HL	HT-LSMPM	HT-LSMPMPM
length	118 in (300 cm)	89 in (226 cm)	118 in (300 cm)	118 in (300 cm)	147 in (373 cm)
weight	390 lbm (177 kg)	303 lbm (137 kg)	395 lbm (179 kg)	386 lbm (175 kg)	478 lbm (217 kg)
^A Configuration nomenclature is provided in Table A-10 in the <i>Product Description</i> section.					

A.5.4 Parts, Tools and Materials

A.5.4.1 Installation tools

Part numbers and descriptions of the tools required to install Extended Runlife Protectors are shown in the [Parts, Tools and Materials](#) section.

A.5.4.2 Installation materials

[Table A-16](#) provides descriptions and part numbers for the Reda No. 7 oil used in High-Temperature Extended Runlife Protectors.

Table A-16: Descriptions and part numbers for Reda No. 7 oil.

Description	Part Number
Oil: Reda No. 7 (5 gal container)	100021260
Oil: Reda No. 7 (15 gal container)	100021263
Oil: Reda No. 7 (55 gal container)	100021264

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The [Technical and Engineering Information](#) section provides descriptions and part numbers for the Reda No. 5 oil used in Abrasion-Resistant Extended Runlife Protectors and Corrosion-Resistant Extended Runlife Protectors.

A.5.5 Technical and Engineering Information

A.5.5.1 Bolt flange and housing strengths

Bolt flange and housing strengths information is provided in the [Technical and Engineering Information](#) section.

A.5.5.2 Calculation of protector thrust loads

The [Technical and Engineering Information](#) section provides procedures for calculating protector thrust loads.

A.5.5.3 Equipment dimensions

Equipment dimensions are shown in the [Engineering Specifications](#) section in this document.

A.5.5.4 Oil selection

Reda No. 7 oil should be used in the High-Temperature Extended Runlife Protector. The No. 7 grade has higher viscosity and better lubrication properties at higher temperature compared to other Reda oils.

The oil selection criteria for the Abrasion-Resistant Extended Runlife Protectors and Corrosion-Resistant Extended Runlife Protectors should follow the guidelines provided in the [Technical and Engineering Information](#) section.

Ordering

The protector oil is application dependent and the type of oil must be specified when placing an order. If the type of oil is not specified when the order is placed, the Extended Runlife Protector will be shipped with the oil shown in [Table A-17](#).

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Table A-17: Standard oil types.

Protectors	Oil Type
540-series Abrasion-Resistant Extended Runlife Protector and Corrosion-Resistant Extended Runlife Protector	Reda No. 5
540 -series High-Temperature Extended Runlife Protector	Reda No. 7

Oil compatibility

The [Technical and Engineering Information](#) section provides oil compatibility information for protectors.

Other considerations

The [Technical and Engineering Information](#) section provides guidelines for other considerations for protectors.

A.5.5.5 Equipment serial numbering system

The equipment serial numbering system applicable to Extended Runlife Protectors is described in the [Technical and Engineering Information](#) section.

A.5.5.6 Connection tables

Connection Tables applicable to the Abrasion-Resistant Extended Runlife Protectors and Corrosion-Resistant Extended Runlife Protectors are provided in Artificial Lift Field Operation Manual - Connection Tables ([InTouch ID 3255850](#)).

Protector-to-protector connections

The Extended Runlife Protector configurations are designed to be run individually or in tandem with other Extended Runlife Protectors and standard protectors. High-Temperature Extended Runlife Protector-to-High-Temperature Extended Runlife Protector connection information is shown in [Table A-18](#). The Chemraz 605 o-ring listed in [Table A-18](#) is required for the High-Temperature Extended Runlife Protector connections.

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The O-ring sizes and part numbers for Abrasion-Resistant Extended Runlife Protector and Corrosion-Resistant Extended Runlife Protector connections and connections of these protectors to standard protectors are provided in the [Technical and Engineering Information](#) section.

Table A-18: High-Temperature Extended Runlife Protector-to-Protector Connections.

Connection	Shaft Diameter	Shaft Diameter	Size	O-Ring Part Number
				Chemraz 605
HT protector-to-HT protector	540-series 1.18 in (2.997 cm)	540-series 1.18 in (2.997 cm)	2-237	2002250

Protector-to-motor connections

All Extended Runlife Protector configurations are designed for use with any 562– or 540–series motor. High-Temperature Extended Runlife Protector-to-motor connection information is shown in [Table A-19](#). The Chemraz 605 o-rings listed in [Table A-19](#) are required for High-Temperature Extended Runlife Protector connections to motors.

O-ring sizes and part numbers to connect Abrasion-Resistant Extended Runlife Protectors and Corrosion-Resistant Extended Runlife Protectors to motors are provided in the [Technical and Engineering Information](#) section.

Table A-19: High-Temperature Extended Runlife Protector-to-Motor Connections.

Connection	Shaft Diameter	Shaft Diameter	Size	O-Ring Part Number
				Chemraz 605
HT protector-to-motor	540-series 1.18 in (2.997 cm)	375-series 0.87 in (2.21 cm)	2-237	2002250
		adapter-to-motor	2-231	2009952
HT protector-to-motor	540-series 1.18 in (2.997 cm)	456-series 1.12 in (2.845 cm)	2-237	2002250
		adapter-to-motor	2-231	2009952
HT HT protector-to-motor	540-series 1.18 in (2.997 cm)	540-series 1.18 in (2.997 cm)	2-237	2002250
HT protector-to-motor	540-series 1.18 in (2.997 cm)	540-series 1.18 in (2.997 cm)	2-237	2002250

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Connection	Shaft Diameter	Shaft Diameter	Size	O-Ring Part Number
				Chemraz 605
HT protector-to-motor	540-series 1.18 in (2.997 cm)	540-series 1.37 in (3.48 cm)	2-237	2002250
HT HT protector-to-motor	540-series 1.18 in (2.997 cm)	738-series 1.50 in (3.81 cm)	2-237	2002250
HT protector-to-motor	540-series 1.18 in (2.997 cm)	540-series 1.37 in (3.48 cm)	2-237	2002250
		adapter-to-motor	2-349	100035694
HT protector-to-motor	540-series 1.18 in (2.997 cm)	562-series 1.37 in (3.48 cm)	2-237	2002250

A.5.6 Installation Procedures

A.5.6.1 Transport and handling

Transport and handling procedures for the Extended Runlife Protector are provided in the [Wellsite Procedures](#) section.

A.5.6.2 Equipment preparation

Equipment preparation procedures for the Extended Runlife Protector are provided in the [Wellsite Procedures](#) section.

A.5.6.3 Lifting clamp installation

Lifting clamp installation procedures for the Extended Runlife Protector are provided in the [Wellsite Procedures](#) section.

A.5.6.4 Protector installation

Protector assembly procedures for the Extended Runlife Protector are provided in the [Wellsite Procedures](#) section.

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A.5.6.5 **Shimming procedures**

Extended Runlife Protectors are equipped with a factory-shimmed shaft; all Extended Runlife Protector shafts are shimmed at the manufacturing facility. Shaft-to-flange dimensions of the protector and the connecting equipment shall be rechecked at the wellsite. Apply shims when necessary. Shimming procedures (including dimension measurement) for the Extended Runlife Protector are provided in the [Wellsite Procedures](#) section.

A.5.6.6 **Oil filling procedures**

Protectors are filled with oil prior to shipment from the manufacturing facility. Typically, some oil is lost during field installation and refilling the protector is required. Oil refilling procedures at the wellsite are provided in the [Wellsite Procedures](#) section.

A.5.6.7 **Running equipment in the wellbore**

Run procedures for the Extended Runlife Protector are provided in the [Wellsite Procedures](#) section.

A.5.6.8 **Pulling equipment from the wellbore**

Pull procedures for the Extended Runlife Protector are provided in the [Wellsite Procedures](#) section.

A.5.7 **Disassembly and Reassembly**

A.5.7.1 **Disassembly**

Disassembly procedures for the Extended Runlife Protectors are provided in the following Manufacturing Process Instruction (MPI) documents located in the Global Engineering Management in Schlumberger (GeMS) database.

- GeMS No. MPI-2015022 Protector Dismantle, Advanced Protector, 3-Chamber
- GeMS No. 100012248 Protector Dismantle, Advanced Protector, 4-Chamber
- GeMS No. 100035127 Protector Dismantle, Advanced Protector, 2-Chamber

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A.5.7.2 Reassembly

Reassembly procedures for the Extended Runlife Protectors are provided in the following MPI documents:

- GeMS No. MPI-2015017, Protector Assembly, Advanced Protector, 3-Chamber
- GeMS No. 100010378 Protector Assembly, Advanced Protector, 4-Chamber
- GeMS No. 100035123 Protector Assembly, Advanced Protector, 2-Chamber

A.5.8 Troubleshooting

Troubleshooting procedures for Artificial Lift equipment are provided in the *Troubleshooting* section of the ALFORM-S . There are no specific troubleshooting procedures for the Extended Runlife Protector.

A.5.9 Field Inspection

Perform a visual inspection of the protector for physical damage and missing parts before installation.

A.5.10 Rerun Procedures

Extended Runlife Protectors should not be rerun in a well after being pulled. All pulled Extended Runlife Protectors should be sent to an Artificial Lift Service Center for disassembly, inspection and reassembly before being returned to service.

A.5.11 References

A.5.11.1 Artificial Lift Field Operations and Reference Manual - Downhole (ALFORM-D)

The Artificial Lift Field Operations and Reference Manual - Downhole, Connection Table and Surface are all available online ([InTouch ID 3255850](#)).

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A.5.11.2 Manufacturing Process Instructions

Manufacturing process instructions numbers and descriptions are shown in [Table A-20](#) (p. A-52).

Table A-20: Manufacturing process instructions (MPI) numbers and descriptions.

MPI Number	Description
2015017	Protector Assembly, Advanced Protector, 3-Chamber
2015020	Protector Oil-filling, Advanced Protector, 3-Chamber
2015022	Protector Dismantle, Advanced Protector, 3-Chamber
100010378	Protector Assembly, Advanced Protector, 4-Chamber
100012094	Protector Oil-filling, Advanced Protector, 4-Chamber
100012248	Protector Dismantle, Advanced Protector, 4-Chamber
100035123	Protector Assembly, Advanced Protector, 2-Chamber
100035125	Protector Oil-filling, Advanced Protector, 2-Chamber
100035127	Protector Dismantle, Advanced Protector, 2-Chamber

A.5.11.3 Assembly drawings

Extended Runlife Protector assembly drawings are available from the Global Engineering in Schlumberger (GeMS) database. Part numbers for the Extended Runlife Protector assembly drawings are shown in [Table A-21](#).

Table A-21: Extended Runlife Protector assembly drawing numbers.

GeMS Number	Description
2012101	Advanced Protector, High Temperature, 3-Chamber
100008798	Advanced Protector, High Temperature, 4-Chamber
2012624	Advanced Protector, Corrosion Resistant, 3-Chamber
100030878	Advanced Protector, Corrosion Resistant, 2-Chamber
2012510	Advanced Protector, Abrasion Resistant, 3-Chamber

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A.6 VGSA Product Update for Abrasive Applications

A.6.1 VGSA Product Update for Abrasive Applications

A.6.1.1 Summary

The VGSA design has changed to improve performance in abrasive applications.

A.6.1.2 Details

The stainless steel liners and compression tubes are being upgraded to a white iron material for all VGSA products. GeMs records are updated to reflect the changes. Manufacturing has qualified one vendor for the white iron components and is in the process of qualifying a second vendor.

The list of affected liners and compression tubes is shown in [Table A-22](#).

Table A-22: VGSA Affected Liners and Compression Tubes

Current Stainless Steel Part Number	Replacement White Iron Part Number	Description	Quantity
2003123	100109890	Liner, D20-60	2
2003128	100109887	Comp Tube, 400, D20-60	1
1307784	100109896	Liner S20-90	1
1297472	100109899	Liner S70-150	2
1307792	100109892	Comp Tube, 538, S20-90 and S70-150	1

A.6.1.3 Reason

Testing was performed on a D20-60 VGSA in response to a SQI submitted by WCP-AL operations. In the SQI, seven VGSA units showed excessive erosion of the stainless steel liner that extended through to the structural housing. In two of the seven units, the structural housings were severed which resulted in the lower parts of the units falling into the wellbore. Sand production rates varied from 300 to 750 ppm in these applications.

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Engineering testing confirmed the white iron liner is superior to the existing stainless steel liner in abrasive conditions.

A.6.1.4

Action

Based on test results, the engineering recommendation is to replace existing stainless steel liners with white iron liners in all VGSA units for new product sales. The bearing support material also changes from Ni-Resist to abrasion/corrosion resistant 5530 to increase the erosion resistance and improve the reliability of the VGSA in long-run abrasive applications. These changes are effective immediately for products ordered from the Bartlesville or Singapore product centers.

It is the responsibility of the field to determine if existing VGSA units in the field need to be upgraded to the latest design. Section 1.5 shows part numbers and descriptions for components required to upgrade existing VGSA units for abrasive service. The current VGSA units with stainless steel liners will work in applications where the sand production rate is less than 250 ppm.

For VGSA units that are installed in highly abrasive applications (greater than 250 ppm sand content), the use of silicon carbide bearings (ES-SS) is recommended to reduce the radial wear and further improved system reliability. [Table A-23](#) can be used as a guideline for selecting a VGSA for abrasive applications.

Table A-23: Guidelines for Selecting a VGSA for Abrasive Applications

Part Number	Sand Content	Series	Description	Shaft Size	Bearing	Type
2009915	< 250 ppm	387/387	D20-60	0.87 in	ES	CT
100103328	> 250 ppm	387/387	D20-60	0.87 in	ES-SS	CT
2003133	< 250 ppm	400/400	D20-60	0.87 in	ES	CT
100103337	> 250 ppm	400/400	D20-60	0.87 in	ES-SS	CT
2007915	< 250 ppm	400/400	D20-60	0.87 in	ES-ZS	CT
100103339	> 250 ppm	400/540	S20-90	0.87 in x 1.18 in	ES-SS	CT
2003573	< 250 ppm	400/540	S20-90	0.87 x 1.18	ES-ZS	CT
1270560	< 250 ppm	538/540	S20-90	1.18 in	ES	CT
1346550	> 250 ppm	538/540	S20-90	1.18 in	ES-SS	CT
2000037	> 250 ppm	538/540	S20-90	1.18 in	ES-SS	UT
1335389	< 250 ppm	538/540	S20-90	1.18 in	ES-ZS	CT
1313477	< 250 ppm	540/540	S20-90	1.18 in	ES	CT
100103341	> 250 ppm	540/540	S20-90	1.18 in	ES-SS	CT
2003677	< 250 ppm	540/540	S20-90	1.18 in	ES-ZS	CT

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Part Number	Sand Content	Series	Description	Shaft Size	Bearing	Type
100096919	> 250 ppm	538/540	S20-90R	1.18 in	ES-SS	CT
1293885	< 250 ppm	538/540	S70-150	1.18 in	ES	CT
2003763	> 250 ppm	538/540	S70-150	1.18 in	ES-SS	CT
2006260	< 250 ppm	538/540	S70-150	1.18 in	ES-ZS	CT
2011416	> 250 ppm	540/540	S70-150	1.18 in	ES-SS	CT
2003845	> 250 ppm	538/540	S70-150R	1.18 in	ES-SS	CT
2011175	> 250 ppm	538/540	S70-150R	1.18 in	ES-TT	CT

A.6.1.5 Components Required to Upgrade Existing VGSA Units for Abrasive Service

Liners and Compression Tube

Table A-24: Liners and Compression Tube

Current Stainless Steel Part Number	Replacement White Iron Part Number	Description	Quantity
2003123	100109890	Liner, D20-60	2
2003128	100109887	Comp Tube, 400, D20-60	1
1307784	100109896	Liner S20-90	1
1297472	100109899	Liner S70-150	2
1307792	100109892	Comp Tube, 538, S20-90 and S70-150	1

Support Bearing

Table A-25: Support Bearing (with Silicon Carbide Bushing and 5530)

Current Part Number, Ni-Resist		Replacement Part Number, 5530, SiC		Quantity
2003118	D20-60, ARZ	100100403	D20-60, 5530, SiC	2
1271980	S20-90, ARZ	100096929	S20-90, 5530, SiC	1
1346576	S20-90, SiC	100096929	S20-90, 5530, SiC	1
1293877	S70-150, ARZ	2003859	S70-150, 5530, SiC	2
2003757	S70-150, SiC	2003859	S70-150, 5530, SiC	2

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Sleeve

Table A-26: Sleeve (Silicon Carbide, > 250 ppm)

Current Part Number, ARZ	Replacement Part Number, SiC	Description
1127877	1065507	SLEEVE, D20-60, 0.87
1127893	1065499	SLEEVE, S20-90 and S70-150, 1.18

Bearing Housing

Table A-27: Bearing Housing (Silicon Carbide Bushing, > 250 ppm)

Current Part Number		Replacement 5530 Part Number	
2003116	D20-60, Ni-R, ARZ	100100772	D20-60, Ni-R, SiC
1271873	S20-90,S70-150, Ni-R, ARZ	1346568	S20-90,70-150, Ni-R, SiC
no existing part number	no existing part number	2003855	S20-90,S70-150, 5530, SiC

Base

Table A-28: Base (Silicon Carbide Bushing, > 250 ppm)

ARZ	SiC	Series
2009913	100100774	387
2003120	100100773	400
1271832	1346543	538
no existing part number	2000034	538, UT
1313444	2011423	540

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B Logging Bypass Section

B.1 Introduction

B.1.1 Description and Operation

The Logging By-pass System is a solution to enable intervention or logging with wireline or coiled tubing below an electrical submersible pump (ESP) without the need for workover

The junction in the bypass system is created by a standard or an Auto Y-Tool, which aligns the axis of the bypass tubing with the axis of the production tubing above, thus creating an intervention conduit to below the ESP.

By-pass tubing is clamped to the ESP. The ESP discharge fluid enters the tubing string through the eccentric leg of the Y-Tool and is prevented from re-circulating down the bypass tubing by either a diverter in an 'Auto Y-Tool', or by a blanking plug in a 'Standard Y-Tool'.

Wireline and Coiled Tubing plugs are available to enable intervention work to be carried out below the ESP under either static or dynamic conditions.

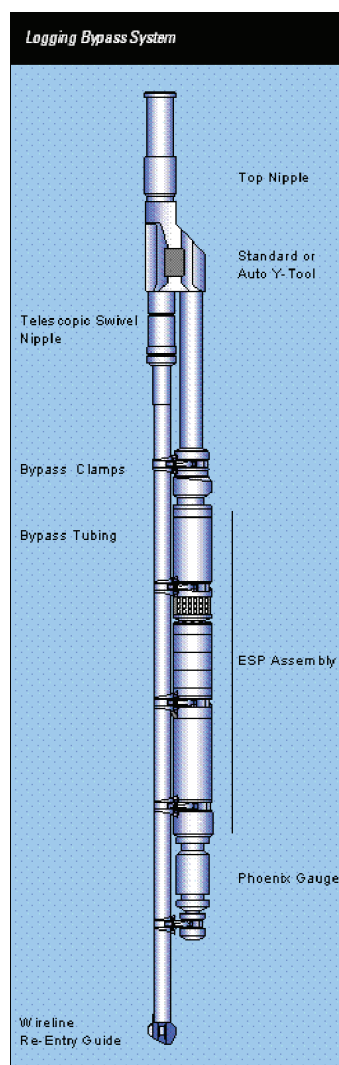


Figure B-1: Logging Bypass System

Applications

- Wireline logging below the pump
- Coiled tubing logging below the pump
- Retrieval of plugs below the pump
- Memory gauge deployment below the pump
- Back-up pump deployment on bypass tubing
- Bridge plug setting for water shutoff
- Tubing-conveyed perforating gun deployment and detonation below the pump
- Well stimulation through coiled tubing or by-pass tubing
- Wireline perforating below the pump
- Bottomhole samples deployment

B.2

Available Sizes

Bypass systems are available for an extensive range of casing sizes and weights. The size (OD) of the bypass tubing which can be accommodated for a given application is a function of the casing drift ID, and the ESP OD (Series). In standard logging bypass systems, the bypass tubing typically has a flush joint connection, so as to maximize the tubing OD, within the available casing space.

Table B-1 lists the possible flush joint bypass tubing sizes for some typical casing sizes.

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Table B-1: ESP OD Maximum. Bypass Tubing Sizes for Some Typical Casing Sizes

	3.750	4.000	4.562	5.130	5.400	5.440	5.625
Casing Size	Bypass Tubing OD (inch)						
6 5/8 - 20.0 lbf.ft	2.125	1.500					
6 5/8 - 24.0 lbf.ft	1.500	1.500					
6 5/8 - 28.0 lbf.ft	1.500	1.500					
6 5/8 - 32.0 lbf.ft	1.500	1.500					
7 - 23.0 lbf.ft	2.375	2.125					
7 - 26.0 lbf.ft	2.375	2.125					
7 - 29.0 lbf.ft	2.125	1.500					
7 - 32.0 lbf.ft	2.125	1.500					
7 - 35.0 lbf.ft	2.125	1.500					
7 5/8 - 26.4 lbf.ft	2.875	2.375	2.125	1.500			
7 5/8 - 29.7 lbf.ft	2.875	2.375	2.125	1.500			
7 5/8 - 33.7 lbf.ft	2.875	2.375	1.500	1.500			
8 5/8 - 28.0 lbf.ft	2.875	2.875	2.875	2.375	2.375	2.375	2.125
8 5/8 - 32.0 lbf.ft	2.875	2.875	2.875	2.375	2.375	2.125	2.125
8 5/8 - 36.0 lbf.ft	2.875	2.875	2.875	2.375	2.125	2.125	1.500
9 5/8 - 40.0 lbf.ft	2.875	2.875	2.875	2.875	2.875	2.875	2.875
9 5/8 - 43.5 lbf.ft	2.875	2.875	2.875	2.875	2.875	2.875	2.875
9 5/8 - 47.0 lbf.ft	2.875	2.875	2.875	2.875	2.875	2.875	2.875
9 5/8 - 53.5 lbf.ft	2.875	2.875	2.875	2.875	2.875	2.875	2.375
10 3/4 - 51.0 lbf.ft							3.500
10 3/4 - 55.5 lbf.ft		3500 OD, or larger, Bypass Tubing may be used.					3.500
10 3/4 - 60.7 lbf.ft							3.500

AL-ESP Field Service & Bypass Sys. \AL_Engineering\InTouch ID 3255850\3.2\Release Date:20-Jul-2011\EDMS UID: 274783215\Produced: 07-Mar-2013 21:09:14

Table B-2: Standard Bypass Tubing Data (inch)

3.500 OD x 2.992 ID - Drift 2.867
2.875 OD x 2.441 ID - Drift 2.347
2.750 OD x 2.362 ID - Drift 2.268
2.375 OD x 1.995 ID - Drift 1.901
2.125 OD x 1.869 ID - Drift 1.775
1.500 OD x 1.244 ID - Drift 1.150

B.3

Material Selection

Bypass system equipment is available in various materials to suit the operating environment of the application. Typically the bypass system metallurgy is categorized as low alloy steel for standard applications, or 13Cr compatible for corrosive environments. [Table B-3](#) shows the bypass system components and the material choices available.

**Note**

Note that other materials are available upon request.

Table B-3: Bypass Component Materials

Item	Standard Materials	13 Cr Compatible Materials
Y-Tool Sub Assembly		
Handling Sub	API 5CT L80 Type 1	API 5CT L80 Type 13Cr
Top Nipple	AISI 4140 18-22 RC	AISI 420 18-22RC
Y-Tool	Super 13 Cr 22 Rc	Super 13 Cr 22 Rc
Pump Sub	API 5CT L80 Type 1	API 5CT L80 Type 13Cr
Telescopic Swivel Nipple	AISI 4140 18-22RC	AISI 420 18-22RC
Blanking Plug (Standard Y-Tools only)	AISI 420 18-22RC	AISI 420 18-22RC
Bypass Tubing	API 5CT L80 Type 1	API 5CT L80 Type 13Cr
Wireline Re-Entry Guide	AISI 4140 18-22RC	AISI 420 18-22RC
Motor Base Plug	AISI 4140 18-22RC	AISI 420 18-22RC
Bypass Clamps and Cable Clips	Super 13 Cr 22 Rc	Super 13 Cr 22 Rc
Standing Valve	AISI 630 17-4 PH ST/ST	AISI 630 17-4 PH ST/ST
Isolation Tool	AISI 420 18-22RC	AISI 420 18-22RC
Wireline Logging Plug	AISI 630 17-4 PH ST/ST	AISI 630 17-4 PH ST/ST
Coiled Tubing Logging Plug	AISI 630 17-4 PH ST/ST	AISI 630 17-4 PH ST/ST

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Table B-4: Materials Specifications

Material	NACE Service	Conditon	Min Yield (Psi)	HRC	BHN
API 5CT L80 Type 1	MRO175	Quenched and Tempered	80000	23 max	241 max
AISI 4140 18-22RC	MRO175	Quenched and Tempered	80000	18/22	211/237
BS 3146 Part 1 CLAIC	MRO175	Doubled Annealed	n/a	22 max	237 max
AISI 410 18-22RC	MRO175	Quenched and Tempered	80000	22 max	237 max
API 5CT L80 Type 13Cr	MRO175	Quenched and Tempered	80000	23 max	241 max
AISI 420 18-22RC	MRO175	Quenched and Tempered	80000	18-22	211-237
AISI 630 17-4 PH ST/ST	MRO175	NACE MRO175 Proc 1	104000	28-33	261-301
BS 3146 Part 2 1975 ANC22	MRO175	Ref Standard	n/a	23-33	261-301

B.3.1 Elastomers

Where applicable the bypass system equipment is available with a choice of Viton or Aflas elastomers with Durometer 90 hardness.

B.4 Standard Y-Tool Bypass Systems

The Y-Tool is supplied as a part of an assembled, pressure tested sub assembly ready for use in the field; see Y-Tool Sub Assemblies.

The junction in the Bypass System is created by the Y-tool, which aligns the axis of the bypass tubing with the axis of the production tubing above, and thus creates an intervention conduit to below the ESP.

The standard Y-Tool requires a Blanking Plug to be set in the telescopic swivel nipple to prevent recirculation of the production fluids down the bypass tubing.

Where intervention work is required in a bypass system with a standard Y-Tool, the Blanking Plug must first be retrieved.

Y-Tools are available to suit a range of casing sizes (7 in, 7-5/8 in, 9-5/8 in, 10-3/4 in).

Table B-5: Specifications (9-5/8 in Standard Y-Tools)

Maximum OD	8.250 in
Minimum ID	2.992 in
Length	17.00 in
Weight	90 lbm
Maximum Differential Pressure	3000 psi
Threaded Connections	3-1/2 in 9.2 lbs/ft PMJ
Material	Super 13 Cr-22 Rc Max
MLE (FCE) Groove	2.70 in x 0.625 in

B.5 Auto Y-Tool Bypass Systems

The Auto Y-Tool is supplied as a part of an assembled, pressure tested sub assembly ready for use in the field; see Y-Tool Sub Assemblies.

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The junction in the Bypass System is created by the Auto Y-tool, which aligns the axis of the bypass tubing with the axis of the production tubing above, and thus creates an intervention conduit to below the ESP.

The Auto Y-Tool has an internal diverter, which seals over the bypass tubing when the ESP is operating to prevent recirculation, and automatically closes over the ESP discharge leg when the ESP is shut down, to allow intervention access through the bypass tubing. The function of the diverter replaces the Blanking Plug which is required in a Standard Y-Tool.

Where intervention work is required in a bypass system with a standard Y-Tool, the Blanking Plug must first be retrieved. In a bypass system with an Auto Y-Tool this is not necessary.

Y-Tools are available to suit a range of casing sizes (7 in, 7-5/8 in, 9-5/8 in, 10-3/4 in).

Table B-6: Specifications (9-5/8 in Auto Y-Tools)

Maximum OD	8.250 in
Minimum ID	2.992 in
Length	17.00 in
Weight	90 lbm
Maximum Differential Pressure	3000 psi
Flowrate to close diverter (min)	100 gal/min ¹
Pressure to keep diverter closed (min)	5 psi
Threaded Connections	3-1/2 in 9.2 lbs/ft PMJ
Material	Super 13 Cr Rc Max
MLE (FCE) Groove	2.70 in x 0.625 in

¹ The pump must be able to generate this flowrate at start up (no head). Check pump curve.

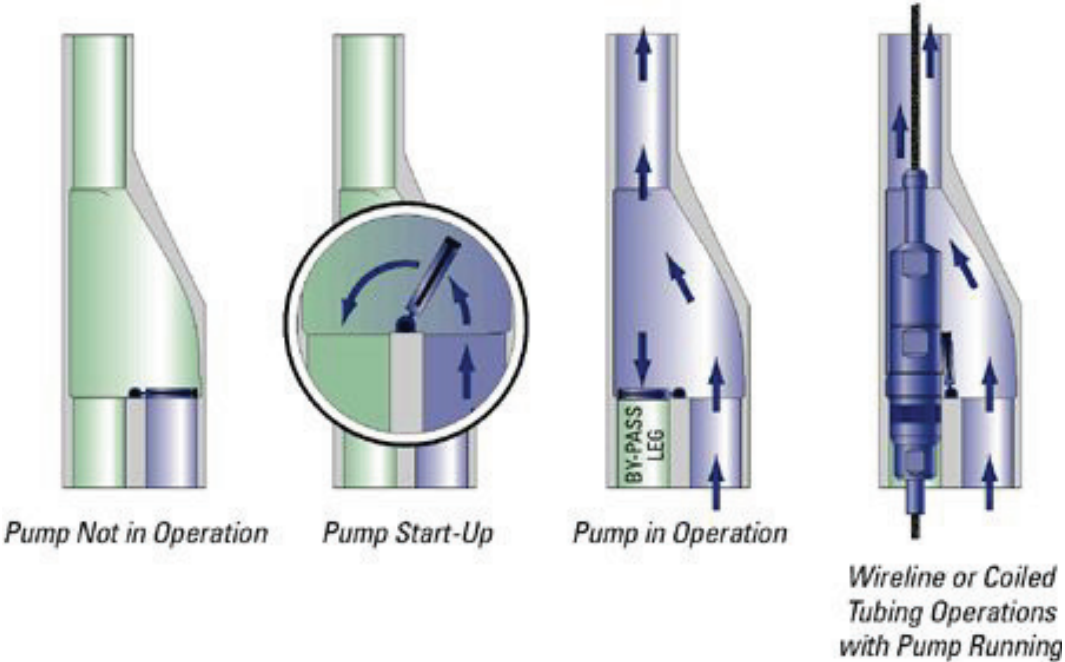


Figure B-2: Auto Y-Tool Operation

B.6 Logging Bypass System and Optional Accessories

B.6.1 Y-Tool Sub Assembly

The Y-Tool is supplied as a part of an assembled, torqued and pressure tested sub assembly ready for use in the field. The Y-Tool Sub Assemblies (standard or Auto) are supplied with the following:

Term	Definition
Handling Sub	This is available with a choice of connection for field make up. Additional crossovers can also be supplied to suit alternative production tubing threads.
Top Nipple Profile (Optional)	This accepts a Standing Valve for pressure testing tubing or is required in order to use an Isolation Tool
Y-Tool	Standard or Auto Y-Tool body. A cable clip is included to secure the MLE.
Pump Sub	For connection to ESP discharge head. Available with a choice of connection.
Telescopic Swivel Nipple	Available with a choice of seal bore sizes to accept bypass plugs, and a choice of bypass tubing connections.

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Blanking Plug (Standard Y-Tools only)

Blanking plugs are available with Lantern collets, or with a locking facility for use in free flowing wells.



Pump Discharge Connection

Figure B-3: Y-Tool Sub Assembly

B.6.1.1

Additional Features

- PMJ thread is used for inter connecting on 9-5/8 in systems.

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- Y-Tool body has AISI 410 22RC Metallurgy as standard
- Available with Low alloy steel or 13Cr for the remaining Y-tool sub assembly components.
- Available for 9-5/8 in, 7-5/8 in and 7 in casing.
- Selection of elastomers.
- 2-3/8 in or 2-7/8 in PTJ bypass tubing connections as standard for 9-5/8 in Systems.
- Y-Tool comes with an MLE Cable Clip fitted to one side as standard.
- An additional cable clip is required for the opposite side of the Y-Tool for applications with additional service lines, i.e. chemical injection lines. This is available separately (p/n 100238282).

B.6.1.2 Handling

The Y-Tool sub assembly is supplied as an assembled, torqued and pressure tested assembly ready to be used at the well site.

It is recommend to drift the Y-Tool sub assembly as part of pre-job checks. The drift size will depend on either the bypass tubing size, or the teleswivel seal bore diameter (whichever is smaller).

The Y-Tool sub assembly can be picked up on the rig using 3-1/2 in side door (square shoulder) elevators fitted to the 3-1/2 in handling sub.^{B-1} See [Figure B-4](#).

B-1. 9-5/8 in systems



Figure B-4: Handling

B.6.1.3 Identification and Selection

Each Y-Tool Sub Assembly configuration, metallurgy and elastomer selections can be identified by a single part number. A list of some of these is provided below, however refer to either the Y-Tool sub assembly datasheet available on [InTouch Content ID 4158997](#), or to OneCat for the latest information.

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Table B-7: Auto Y-Tool Type 1 Material (Low Alloy Steel). Artificial Lift Bypass Systems/A. Standard Bypass Sub Assemblies/9 5/8 in Auto Y-Tool Sub Assemblies

Part Number	Description	Casing Size (in)	Casing Weight Range (PPF)	Maximum OD (in)	Assembly Type	Thread - Connecting - Upper	Thread - Connecting - Lower	Thread Connecting - Discharge Leg	Material	Material/ Elastomers	Upper Nipple Size (in)	Lower Nipple Size (in)	Plug Type
100480378	9.625 AUTO Y, 3.5X2.88X3.5, 2.813X2.75 TYPE 1 AFLAS(A703)	9 5/8	40-53.5	8.25	AUTO	3.500, 9.3, EUE, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 Type 1 Rc 18-22	AFLAS	2.813	2.75	N/A
100316869	9.625 AUTO Y, 3.5X2.88X3.5, 2.813X2.75 TYPE 1 AFLAS (FL4S)(A703)	9 5/8	40-53.5	8.25	AUTO	3.500, 9.3, EUE, BOX	2.875, 6.5, FL4S PIN	3.500, 9.3, EUE, PIN	API 5CT L80 Type 1 Rc 18-22	AFLAS	2.813	2.75	N/A
100209903	9.625 AUTO Y, 3.5X2.88X3.5, N/AX2.75 TYPE 1 VIT (A703)	9 5/8	40-53.5	8.25	AUTO	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.2, NEW VAM, PIN	API 5CT L80 Type 1 Rc 18-22	VITON	N/A	2.75	N/A
100282146	9.625 AUTO Y, 3.5X2.88X3.5, 2.75X2.625 TYPE 1 VIT (A703)	9 5/8	40-53.5	8.25	AUTO	3.500, 9.2, NEW VAM, BOX	2.875, 6.5, STL, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 Type 1 Rc 18-22	VITON	2.75	2.625	N/A
100456531	9.625 AUTO Y, 3.5X2.88X3.5, N/AX2.625 TYPE 1 VITON (A703)	9 5/8	40-53.5	8.25	AUTO	3.500, 9.3, EUE, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 Type 1 Rc 18-22	VITON	N/A	2.625	N/A
100338323	9.625 AUTO Y, 3.5X2.88X3.5, 2.75X2.312 TYPE 1 VITON (A703)	9 5/8	40-53.5	8.25	AUTO	3.500, 9.3, EUE, BOX	2.375, 4.6, PTJ PIN	3.500, 9.3, EUE, PIN	API 5CT L80 Type 1 Rc 18-22	VITON	2.75	2.312	N/A
100282161	9.625 AUTO Y, 3.5X2.88X3.5, N/AX2.312 TYPE 1 VIT (A703)	9 5/8	40-53.5	8.25	AUTO	3.500, 9.2, NEW VAM, BOX	2.875, 6.5, STL, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 Type 1 Rc 18-22	VITON	N/A	2.312	N/A

Table B-8: Auto Y-Tools – 13Cr Material. Artificial Lift Bypass Systems / A. Standard Bypass Sub Assemblies / 9 5/8 in Auto Y-Tool Sub Assemblies

Part Number	Description	Casing Size (in)	Casing Weight Range (PPF)	Maximum OD (in)	Assembly Type	Thread - Connecting - Upper	Thread - Connecting - Lower	Thread Connecting - Discharge Leg	Material	Material/ Elastomers	Upper Nipple Size (in)	Lower Nipple Size (in)	Plug Type
100209908	9.625 AUTO Y, 3.5X2.88X3.5, 2.813X2.75 13CR AFL (A703)	9 5/8	40-53.5	8.25	AUTO	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.2, NEW VAM, PIN	API 5CT L80 13Cr Rc 18-22	AFLAS	2.813	2.75	N/A
100503800	9.625 AUTO Y, 3.5X2.88X3.5, 2.81X2.75 13CR AFL (A703)	9 5/8	40-53.5	8.25	AUTO	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 13Cr Rc 18-22	AFLAS	2.813	2.75	N/A
100507335	9.625 AUTO Y, 3.5X2.88X3.5, 2.81X2.75 13CR VIT (A703)	9 5/8	40-53.5	8.25	AUTO	3.500, 9.3, EUE, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 13Cr Rc 18-22	VITON	2.813	2.75	N/A
100335415	9.625 AUTO Y, 3.5X2.88X3.5, 2.813X2.75 13CR NITRILE (A703)	9 5/8	40-53.5	8.25	AUTO	3.500, 9.2, NEW VAM, BOX	2.875, 6.5, STL, PIN	3.500, 9.2, NEW VAM, PIN	API 5CT L80 13Cr Rc 18-22	NITRILE	2.813	2.75	N/A
100494754	9.625 AUTO Y, 3.5X2.88X3.5, N/AX2.75 TYPE 13CR VITON(A703)	9 5/8	40-53.5	8.25	AUTO	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.2, NEW VAM, PIN	API 5CT L80 13Cr Rc 18-22	VITON	N/A	2.75	N/A
100441734	9.625 AUTO Y, 4.5X2.88X3.5, N/AX2.75 TYPE 13CR AFLAS (A703)	9 5/8	40-53.5	8.25	AUTO	4.500, 12.6, VAM TOP, BOX	2.875, 6.5, STL, PIN	3.500, 9.2, NEW VAM, PIN	API 5CT L80 13Cr Rc 18-22	AFLAS	N/A	2.75	N/A

Table B-9: Standard Y-Tool Type 1 Material (Low Alloy Steel). Artificial Lift Bypass Systems / A. Standard Bypass Sub Assemblies / 9 5/8 in Y-Tool Sub Assemblies c/w blanking plug

Part Number	Description	Casing Size (in)	Casing Weight Range (PPF)	Maximum OD (in)	Assembly Type	Thread - Connecting - Upper	Thread - Connecting - Lower	Thread Connecting - Discharge Leg	Material	Material/ Elastomers	Upper Nipple Size (in)	Lower Nipple Size (in)	Plug Type
100051410	9.625 STD Y, 3.5X2.88X3.5, 2.813X2.75 TYPE 1 VIT LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	2.813	2.75	LANTERN
100295949	9.625 STD Y, 3.5X2.88X3.5, 2.813X2.75 TYPE 1 AFL LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	AFLAS	2.813	2.75	LANTERN
100293699	9.625 STD Y, 3.5X2.88X3.5, 2.813X2.75 TYPE 1 VIT LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.3, EUE, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	2.813	2.75	LANTERN
100281476	9.625 STD Y, 3.5X2.88X3.5, 2.813X2.75 TYPE 1 VIT LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	2.813	2.75	LOCK
100354168	9.625 STD Y, 3.5X2.88X3.5, 2.813X2.75 TYPE 1 AFL LOCK (A703)(FL4S)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.3, EUE, BOX	2.875, 6.5, FL4S, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	AFLAS	2.813	2.75	LOCK
100077721	9.625 STD Y, 3.5X2.88X3.5, N/AX2.75 TYPE 1 VIT LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	N/A	2.75	LANTERN
100425437	9.625 STD Y, 3.5X2.88X3.5, N/AX2.75 TYPE 1 AFL LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	AFLAS	N/A	2.75	LANTERN
100408011	9.625 STD Y, 4.5X2.88X3.5, N/AX2.75 TYPE 1 VIT LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	4.500, 12.75, EUE, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	N/A	2.75	LANTERN
100281480	9.625 STD Y, 3.5X2.88X3.5, N/AX2.75 TYPE 1 VIT LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.2, NEW VAM, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	N/A	2.75	LOCK

Part Number	Description	Casing Size (in)	Casing Weight Range (PPF)	Maximum OD (in)	Assembly Type	Thread - Connecting - Upper	Thread - Connecting - Lower	Thread Connecting - Discharge Leg	Material	Material/ Elastomers	Upper Nipple Size (in)	Lower Nipple Size (in)	Plug Type
100281354	9.625 STD Y, 3.5X2.88X3.5, N/AX2.75 TYPE 1 VIT LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	N/A	2.75	LOCK
100444953	9.625 STD Y, 3.5X2.88X3.5, N/AX2.75 TYPE 1 VIT LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.3, EUE, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	N/A	2.75	LOCK
100281458	9.625 STD Y, 3.5X2.88X3.5, 2.75X2.625 TYPE 1 AFL LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.2, NEW VAM, PIN	API 5CT L80 TYPE 1 Rc 18-22	AFLAS	2.75	2.625	LOCK
100273994	9.625 STD Y, 3.5X2.88X3.5, N/AX2.625 TYPE 1 VIT LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.3, EUE, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	N/A	2.625	LANTERN
100281474	9.625 STD Y, 3.5X2.88X3.5, N/AX2.625 TYPE 1 VIT LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.3, EUE, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	N/A	2.625	LOCK
100174390	9.625 STD Y, 3.5X2.38X3.5, 2.813X2.312 TYPE 1 VIT LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.375, 4.6, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	2.813	2.312	LANTERN
100174411	9.625 STD Y, 3.5X2.38X3.5, 2.813X2.312 TYPE 1 VIT LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.3, EUE, BOX	2.375, 4.6, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	2.813	2.312	LANTERN
100281383	9.625 STD Y, 3.5X2.38X3.5, 2.813X2.312 TYPE 1 VIT LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.375, 4.6, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	2.813	2.312	LOCK
100077728	9.625 STD Y, 3.5X2.88X3.5, 2.75X2.312 TYPE 1 VIT LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	2.75	2.312	LANTERN
100281371	9.625 STD Y, 3.5X2.88X3.5, 2.75X2.312 TYPE 1 VIT LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	2.75	2.312	LOCK

Part Number	Description	Casing Size (in)	Casing Weight Range (PPF)	Maximum OD (in)	Assembly Type	Thread - Connecting - Upper	Thread - Connecting - Lower	Thread Connecting - Discharge Leg	Material	Material/ Elastomers	Upper Nipple Size (in)	Lower Nipple Size (in)	Plug Type
100437453	9.625 STD Y, 3.5X2.88X3.5, 2.75X2.312 TYPE 1 VITON LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.3, EUE, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	2.75	2.312	LOCK
100365142	9.625 STD Y, 3.5X2.38X3.5, 2.750X2.312 TYPE 1 VIT LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.3, EUE, BOX	2.375, 4.6, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	2.75	2.312	LANTERN
100077736	9.625 STD Y, 3.5X2.88X3.5, N/AX2.312 TYPE 1 VIT LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	N/A	2.312	LANTERN
100281375	9.625 STD Y, 3.5X2.88X3.5, N/AX2.312 TYPE 1 VIT LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	N/A	2.312	LOCK
100175159	9.625 STD Y, 3.5X2.88X3.5, 2.312X1.875 TYPE 1 NIT LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.3, EUE, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	NITRILE	2.312	1.875	LANTERN
100281450	9.625 STD Y, 3.5X2.88X3.5, 2.312X1.875 TYPE 1 NIT LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.3, EUE, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	NITRILE	2.312	1.875	LOCK
100174407	9.625 STD Y, 3.5X2.88X3.5, N/AX1.88 TYPE 1 VIT LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.3, EUE, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	N/A	1.875	LANTERN
100281392	9.625 STD Y, 3.5X2.88X3.5, N/AX1.88 TYPE 1 VIT LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.3, EUE, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	N/A	1.875	LOCK
100281438	9.625 STD Y, 3.5X2.38X3.5, 2.813X2.312 TYPE 1 VIT LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.3, EUE, BOX	2.375, 4.6, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	2.813	2.312	LOCK
100240130	9.625 STD Y, 3.5X2.88X3.5, 2.75X2.625 TYPE 1 AFL LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.2, NEW VAM, PIN	API 5CT L80 TYPE 1 Rc 18-22	AFLAS	2.75	2.625	LANTERN

Part Number	Description	Casing Size (in)	Casing Weight Range (PPF)	Maximum OD (in)	Assembly Type	Thread - Connecting - Upper	Thread - Connecting - Lower	Thread Connecting - Discharge Leg	Material	Material/ Elastomers	Upper Nipple Size (in)	Lower Nipple Size (in)	Plug Type
100257821	9.625 STD Y, 3.5X2.88X3.5, N/AX2.75 TYPE 1 VIT LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.2, NEW VAM, PIN	API 5CT L80 TYPE 1 Rc 18-22	VITON	N/A	2.75	LANTERN

Table B-10: Standard Y-Tool 13Cr. Artificial Lift Bypass Systems / A. Standard Bypass Sub Assemblies / 9 5/8 in Y-Tool Sub Assemblies c/w blanking plug

Part Number	Description	Casing Size (in)	Casing Weight Range (PPF)	Maximum OD (in)	Assembly Type	Thread - Connecting - Upper	Thread - Connecting - Lower	Thread Connecting - Discharge Leg	Material	Material/ Elastomers	Upper Nipple Size (in)	Lower Nipple Size (in)	Plug Type
100295922	9.625 STD Y, 3.5X2.88X3.5, 2.813X2.75 TYPE 13CR AFL LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 13Cr Rc 18-22	AFLAS	2.813	2.75	LANTERN
100510691	9.625 STD Y, 3.5X2.88X3.5, 2.813X2.75 13CR VIT LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.3, EUE, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 13Cr Rc 18-22	VITON	2.813	2.75	LOCK
100408486	9.625 STD Y, 3.5X2.88X3.5, 2.813X2.75 TYPE 13CR AFLAS LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.3, EUE, BOX	2.875, 6.5, FL4S, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 13Cr Rc 18-22	AFLAS	2.813	2.75	LOCK
100397702	9.625 STD Y, 3.5X2.88X3.5, 2.813X2.75 TYPE 13CR VITON LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.5, FL4S, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 13Cr Rc 18-22	VITON	2.813	2.75	LOCK
100077744	9.625 STD Y, 3.5X2.88X3.5, N/AX2.75 TYPE 13CR VIT LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.2, NEW VAM, PIN	API 5CT L80 13Cr Rc 18-22	VITON	N/A	2.75	LANTERN
100077740	9.625 STD Y, 3.5X2.88X3.5, N/AX2.75 TYPE 13CR VIT LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 13Cr Rc 18-22	VITON	N/A	2.75	LANTERN
100281367	9.625 STD Y, 3.5X2.88X3.5, N/AX2.75 TYPE 13CR VIT LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.2, NEW VAM, PIN	API 5CT L80 13Cr Rc 18-22	VITON	N/A	2.75	LOCK
100281361	9.625 STD Y, 3.5X2.88X3.5, N/AX2.75 TYPE 13CR VIT LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 13Cr Rc 18-22	VITON	N/A	2.75	LOCK
100281453	9.625 STD Y, 3.5X2.88X3.5, N/AX2.75 13CR VIT LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.3, EUE, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 13Cr Rc 18-22	VITON	N/A	2.75	LOCK

Part Number	Description	Casing Size (in)	Casing Weight Range (PPF)	Maximum OD (in)	Assembly Type	Thread - Connecting - Upper	Thread - Connecting - Lower	Thread Connecting - Discharge Leg	Material	Material/ Elastomers	Upper Nipple Size (in)	Lower Nipple Size (in)	Plug Type
100334402	9.625 STD Y, 4.5X2.88X3.5, N/AX2.75 TYPE 13CR VIT LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	4.500, 12.6, VAM TOP, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.2, NEW VAM, PIN	API 5CT L80 13Cr Rc 18-22	VITON	N/A	2.75	LOCK
100286735	9.625 STD Y, 3.5X2.38X3.5, 2.75X2.312 TYPE 13CR AFLAS LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.375, 4.6, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 13Cr Rc 18-22	AFLAS	2.75	2.312	LANTERN
100174821	9.625 STD Y, 3.5X2.38X3.5, 2.75X2.312 TYPE 13CR AFL LANTERN FL4S(A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.375, 4.6, FL4S, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 13Cr Rc 18-22	AFLAS	2.75	2.312	LANTERN
100504019	9.625 STD Y, 3.5X2.88X3.5, N/AX2.31 13CR AFL LOCK (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.2, NEW VAM, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 13Cr Rc 18-22	AFLAS	N/A	2.312	LOCK
100377685	9.625 STD Y, 3.5X2.88X3.5, N/AX2.31 13CR VIT LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.3, EUE, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 13Cr Rc 18-22	VITON	N/A	2.312	LANTERN
100365145	9.625 STD Y, 3.5X2.38X3.5, 2.75X2.312 TYPE 13CR VITON LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.3, EUE, BOX	2.375, 4.6, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 13Cr Rc 18-22	VITON	2.75	2.312	LANTERN
100175168	9.625 STD Y, 3.5X2.88X3.5, N/AX2.75 13CR VIT LANTERN (A703)	9 5/8	40-53.5	8.25	STANDARD	3.500, 9.3, EUE, BOX	2.875, 6.4, PTJ, PIN	3.500, 9.3, EUE, PIN	API 5CT L80 13Cr Rc 18-22	VITON	N/A	2.75	LANTERN

B.6.2 Bypass Tubing

The Phoenix Taper Joint (PTJ) thread is the standard connection manufactured on the bypass tubing. The thread can be used on standard bypass systems or on more specialized bypass completions that require a structural and pressure tight bypass tubing thread^{B-2}.

For Standard Logging Bypass Systems the bypass tubing is supplied in 15 ft lengths, and can be made up to its optimum torque without the need to use power tongs. Some special applications may require joints of specific lengths, and the use of power tongs to perform make ups.

Swivel lifting nubbins should be used to pick up and handle the bypass tubing joints. These are available to suit the bypass tubing size and thread.

B-2. Although PTJ can be used for some specialist applications, alternative threads may still require to be used in some cases. This will be dependant on individual application requirements.

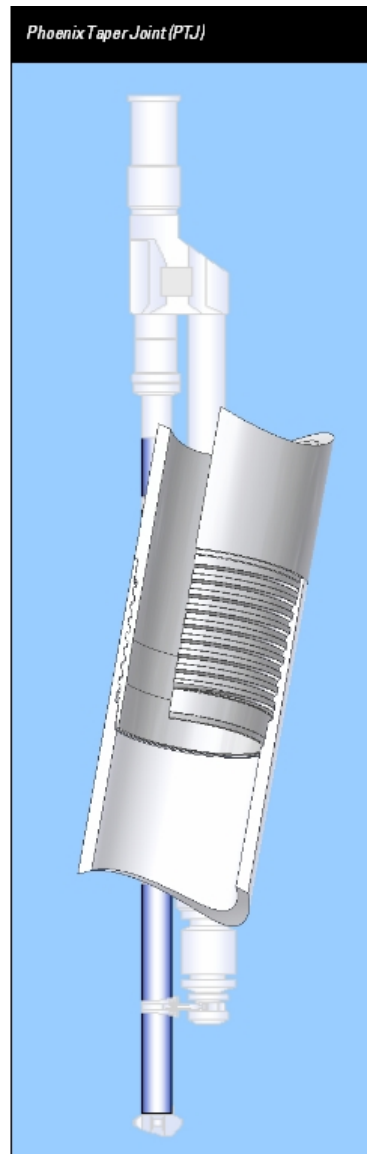


Figure B-5: PTJ Bypass Tubing

B.6.2.1

Features and Benefits

- Robust structural and pressure tight connection that can be made up to shoulder without the need to use power tongs
- 2.875 in 6.4 lb/ft and 2.375 in 4.6 lb/ft tubing sizes (Standard 9-5/8 in sizes)
- L80 Type 1 or L80 Type 13Cr Metallurgy
- Flush joint connection for maximum clearance
- Hooked thread design to provide maximum joint strength
- Primary Metal to metal nose seal

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- Secondary metal to metal external seal
- External torque shoulder to prevent back off and over torque of the connection.

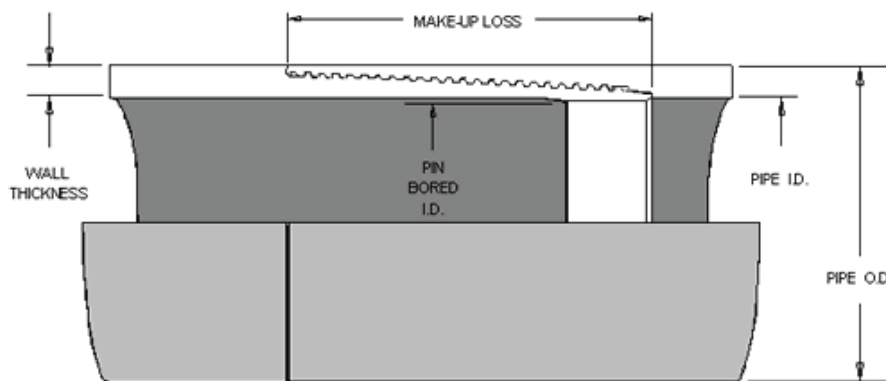


Figure B-6: 2-3/8 in and 2-7/8 in PTJ Technical Data (L80). Refer to [Table B-11](#), [Table B-12](#) and [Table B-13](#).

Table B-11: Dimensional Data

Pipe OD (in)	Nominal Weight (lbm/ft)	Wall Thickness (in)	Pipe ID (in)	Pin bored ID (in)	Make up Loss (in)
2.375	4.6	0.190	1.994	1.950	2.37
2.875	6.4	0.217	2.441	2.390	2.80

Table B-12: Make-up data

Pipe OD (in)	Nominal Weight (lbm/ft)	Shoulder Window		Make up torque	
		Minimum (ft-lb)	Maximum (ft-lb)	Minimum (ft-lb)	Maximum (ft-lb)
2.375	4.6	80	150	250	350
2.875	6.4	100	200	400	600

Table B-13: Physical properties

Pipe OD (in)	Nominal Weight (lbm/ft)	Pressure		Tension	
		Internal Yield (psi)	Collapse (psi)	Tensile Joint Strength (lb)	Minimum Parting Load (lb)
2.375	4.6	11,210	11,790	45,000	60,000
2.875	6.4	10,560	11,160	66,000	90,000

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B.6.2.2 Identification and Selection

A list of some of the bypass tubing joint part numbers is provided in [Table B-14](#); however, refer to either the PTJ (bypass tubing) data sheet which is available on [InTouch Content ID 4160429](#), or to OneCat for the latest information.

Table B-14: List of Bypass Tubing Joint Part Numbers

Part Number	Tubing OD	Drift	Thread	Overall Length	Make Up Length	Material
100273070	2.375 in	1.901	PTJ	15 ft	177.63 in	API 5CT Type 1 L80
100279531	2.375 in	1.901	PTJ	15 ft	177.63 in	API 5CT Type 13Cr L80
100257453	2.875 in	2.347	PTJ	15 ft	177.20 in	API 5CT Type 1 L80
100274975	2.875 in	2.347	PTJ	15 ft	177.20 in	API 5CT Type 13Cr L80
A036254-231 ¹	2.125 in	1.795	PFJ	15 ft	178 in	BS 6323 PART 4 1982 CFS 3 BK (Low alloy steel)
100007130 ²	2.125 in	1.795	PFJ	15 ft	178 in	316L stainless steel

¹ These items use the PFJ thread that does not have a seal. These are typically used as a conduit for 7 in and 7 5/8 in bypass systems.

² These items use the PFJ thread that does not have a seal. These are typically used as a conduit for 7 in and 7 5/8 in bypass systems.

B.6.3 Universal Bypass Clamp

The Schlumberger bypass clamps are fitted to each Electrical Submersible Pump (ESP) neck in logging bypass completions.^{B-3}

The clamp also serves to secure and protect cables and lines passing the ESP.

B-3. Bypass clamps should not be fitted directly below an intake. Where a bolt on intake will be used, the clamp can be fitted above. Where an integral intake is used, the clamp should be omitted.

For the most common 9-5/8 in casing applications, a universal bypass clamp is used. The universal clamp has adjustment to enable it to be fitted to a range of ESP necks. Two types of universal clamps are available to suit either 2-3/8 in or 2-7/8 in bypass tubing. Additional inserts are available for the 2-7/8 in clamp to convert it to smaller bypass tubing sizes.

The Universal Bypass clamp includes slots to secure 2 x 1/4 in control lines, 2 x 3/8 in control lines, and 2 x universal cable clips (one each side). Each universal clips can secure an MLE or flatpack with dimensions ranging from 0.400 in x 1.750 in to 0.610 in x 1.750 in (width x depth).

Individual ESP neck bypass clamps are available for those applications for which the universal clamp cannot be used.

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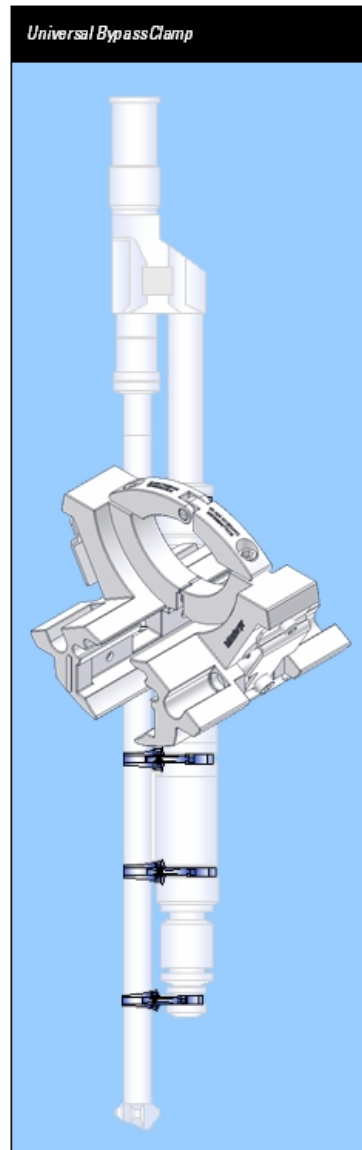


Figure B-7: Universal Bypass Clamp

B.6.3.1

Features and Benefits

- Suitable for 9-5/8 in casing up to 53.5 lb/ft ^{B-4}
- One clamp suits all 400 to 562 Series Reda Necks ^{B-5}
- Available for 2-3/8 in or 2-7/8 in bypass tubing (other sizes available upon request)

B-4. Refer to [Appendix B.2: Available Sizes](#).

B-5. 3. Universal clamps can also be fitted to all 400 – 544 Series Centrilift necks. For applications with larger series Centrilift ESP units, see OneCat or contact AL-CoE.

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- Super 13Cr (CA6NM Class B metallurgy)

Table B-15: Universal Bypass Clamp Identification

Part Number (2.375 in Bypass Tubing)	100394199
Part Number (2.875 in Bypass Tubing)	100831261, 100831262, 100831263, 100831264

Table B-16: Universal Bypass Clamp Technical Information

Maximum OD (in)	8.375
ESP series neck size range	Reda 400 series to 562 series
Body Material	Super 13Cr (CA6NM Class B) Investment casting
Bolting Material	AISI 420 22Rc Max, Tuftride QPQ finish
Control Line Slots	2 x ¼ in and 2 x 3/8 in
MLE/Flatpack Cable Clip Sizes	0.400 in x 1.750 in to 0.610 in x 1.750 in (2 off) (width x depth)

Table B-17: Universal Bypass Clamp Application

ESP Series	2 7/8 in Bypass Tubing	2 3/8 in Bypass Tubing
562	Weight range up to 47 lb/ft (8.525 in drift)	Weight range up to 53.5 lb/ft (8.379 in Drift)
540	Weight range up to 53.5 lb/ft (8.379 in Drift)	Weight range up to 53.5 lb/ft (8.379 in Drift)
456	Weight range up to 53.5 lb/ft (8.379 in Drift)	Weight range up to 53.5 lb/ft (8.379 in Drift)
400	Weight range up to 53.5 lb/ft (8.379 in Drift)	Weight range up to 53.5 lb/ft (8.379 in Drift)

**Note**

Refer to GeMs SOP 100476874 for fitting procedure.

B.6.4**Wireline Re-Entry Guide**

The Wireline Re-Entry Guide (WEG) is made up to the end of the bypass tubing. The purpose of the WEG is to ensure the logging string passes freely back into the bypass tubing upon retrieval.

WEGs are available to suit the bypass tubing thread and size. Two types of WEG are available, 'Plain', and 'Finned'.

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The finned WEG should be used when the bypass tubing tailpipe is to enter a lower liner or when stabbing past a sharp profile. Finned WEGs can be supplied in various ODs to suit the liner in these applications.

Both Plain and Finned WEGs are available in AISI 4140 Rc 18-22 (low alloy carbon steel), or AISI 420 Rc 18-22 (13Cr Stainless Steel) Metallurgy.

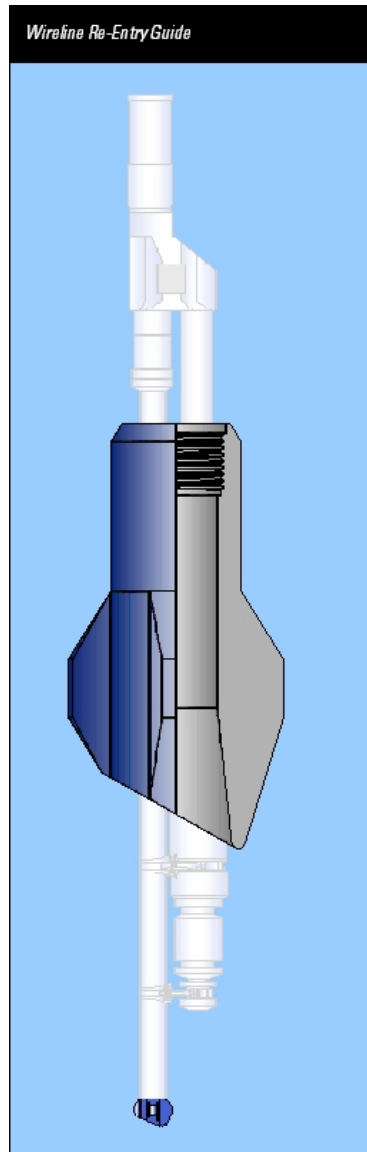


Figure B-8: Wireline Re-Entry Guide

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B.6.4.1 Identification and Selection

A list of some of WEG part numbers is provided [Table B-18](#); however, refer to either the WEG datasheet which is available on [InTouch Content ID 4161528](#), or to OneCat for the latest information.

Table B-18: List of WEG Part Numbers

Part Number	Type	Maximum OD (in)	Minimum ID (in)	Length (in)	Thread (Size, Wt., Type, Config.)	Material
100281008	Plain	2.875	1.995	10.00	2.375, 4.6, PTJ, Box	AISI 4140 18-22 Rc
100274268	Plain	2.875	1.995	10.00	2.375, 4.6, PTJ, Box	AISI 420 18-22 Rc
100257826	Finned	4.500	2.441	10.00	2.875, 6.4, PTJ, Box	AISI 4140 18-22 Rc
100274992	Plain	3.500	2.441	10.00	2.875, 6.4, PTJ, Box	AISI 4140 18-22 Rc
100274991	Plain	3.500	2.441	10.00	2.875, 6.4, PTJ, Box	AISI 420 18-22 Rc
A051021-231	Plain	2.450	1.864	10.00	2.125, 2.4, PFJ, Box	AISI 4140 18-22 Rc
A051021-431	Plain	2.450	1.864	10.00	2.125, 2.4, PFJ, Box	AISI 420 18-22 Rc
A051010-231	Finned	3.500	1.864	12.00	2.125, 2.4, PFJ, Box	AISI 4140 18-22 Rc
A051010-431	Finned	3.500	1.864	12.00	2.125, 2.4, PFJ, Box	AISI 420 18-22 Rc

1. AISI 4140 18-22Rc is low alloy carbon steel, and AISI 420 18-22Rc is 13Cr stainless steel.
2. Thread must match bypass tubing thread.

B.6.5 Motor Base Plug

The Motor Base Plug can be made up to the base of the ESP assembly (or Gauge) to provide an additional location for an MLE or bypass clamp. The motor base plug should be used in situations where there is no lower neck profile on the base of the lower motor. Single section or lower tandem motors typically do not have a lower neck. The motor base plug should also be used where a 'long' ESP gauge is fitted to base of the motor.

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The Motor Base Plug has a 2-3/8 in EUE pin connection for make up to a REDA motor base or gauge assembly. The plugs are available with different ODs to match the ESP motor series.

Universal bypass damp will fit to any REDA 456–562 Motor Base plug and any Centrilift 456–544 Motor Base plug.

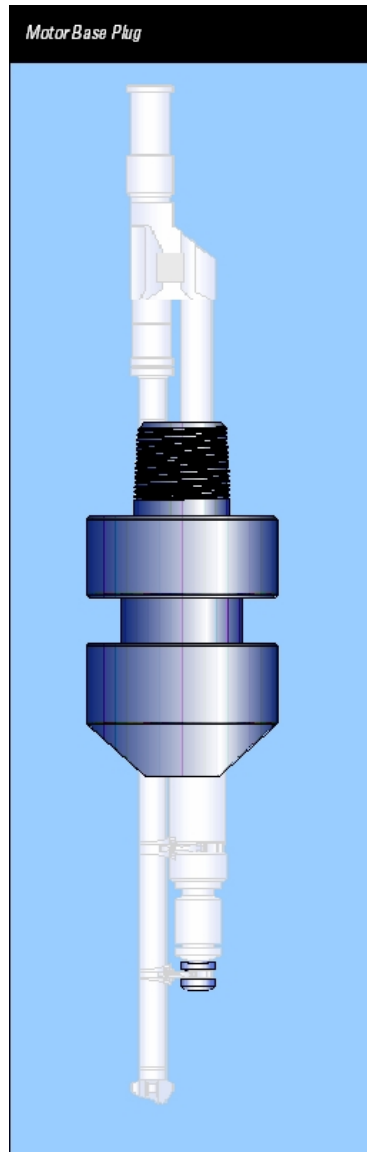


Figure B-9: Motor Base Plug

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B.6.5.1 Identification and Selection

A list of some of the Motor Base Plug part numbers is provided in [Table B-19](#); however, refer to either the Motor Base Plug which is available on [InTouch Content ID 4193307](#), or to OneCat for the latest information.

Table B-19: List of the Motor Base Plug Part Numbers

A054 PART NO	Type	Groove Diameter (IN)	Material/ Body Parts	Maximum OD (IN)	Length (IN)	Thread (Size, WT, Type, Config.)
A054004-231	REDA 375	2.440	AISI 4140 Rc 18-22	3.750	12	2.375, 4.7, EUE, PIN
A054004-431	REDA 375	2.440	AISI 420 Rc 18-22	3.750	12	2.375, 4.7, EUE, PIN
A054007-231	REDA 540	3.536	AISI 4140 Rc 18-22	5.400	12	2.375, 4.7, EUE, PIN
A054007-431	REDA 540	3.536	AISI 420 Rc 18-22	5.400	12	2.375, 4.7, EUE, PIN
A054008-231	CENTRILIFT 540	3.250	AISI 4140 Rc 18-22	5.400	12	2.375, 4.7, EUE, PIN
A054008-431	CENTRILIFT 540	3.250	AISI 420 Rc 18-22	5.400	12	2.375, 4.7, EUE, PIN
A054011-231	REDA 456	2.880	AISI 4140 Rc 18-22	4.500	10.5	2.375, 4.7, EUE, PIN
A054011-431	REDA 456	2.880	AISI 420 Rc 18-22	4.500	10.5	2.375, 4.7, EUE, PIN
A054012-231	REDA 540	3.536	AISI 4140 Rc 18-22	5.400	12	2.375, 4.7, EUE, PIN X 2.875, 6.4, NV BOX
A054013-231	REDA 540	3.536	AISI 4140 Rc 18-22	5.400	12	2.375, 4.7, EUE, PIN X 3.500, 6.4, NV BOX
A054013-431	REDA 540	3.536	AISI 420 Rc 18-22	5.400	12	2.375, 4.7, EUE, PIN X 3.500, 6.4, NV BOX
A054018-231	CENTRILIFT 562	4.250	AISI 4140 Rc 18-22	5.620	10	2.375, 4.7, EUE, PIN
A054018-431	CENTRILIFT 562	4.250	AISI 420 Rc 18-22	5.620	10	2.375, 4.7, EUE, PIN

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A054 PART NO	Type	Groove Diameter (IN)	Material/ Body Parts	Maximum OD (IN)	Length (IN)	Thread (Size, WT, Type, Config.)
A054021-231	REDA 562	3.875	AISI 4140 Rc 18-22	5.620	11	2.375, 4.7, EUE, PIN
A054021-431	REDA 562	3.875	AISI 420 Rc 18-22	5.620	11	2.375, 4.7, EUE, PIN
A054023-431	CENTRILIFT 375	2.875	AISI 420 Rc 18-22	3.750	14	2.375, 4.7, EUE, PIN
A054024-231	CENTRILIFT 540	3.250	AISI 4140 Rc 18-22	5.400	10.5	2.375, 4.7, EUE, PIN X 2.375, 4.7, EUE, BOX
 Note For part numbers ending with '-43x', 'x' denotes seal elastomer type, e.g. Vito-432; Aflas-433.						

 Note
Material

- AISI 4140 Rc 18-22 – Low Alloy Carbon Steel
- AISI 420 Rc 18-22 – 13Cr Stainless Steel

B.6.6

Blanking Plug - Lantern

The Blanking Plug is seated in the telescopic swivel nipple seal bore profile immediately below the Y-Tool. The plug prevents re-circulation of the produced fluids down the bypass tubing when the ESP is operating.

The plug can either be run pre-installed in the nipple, or set by slickline after installation to prevent fluids passing through the pump. The blanking plug with lantern collets is deployed and retrieved by standard slickline tools.

Upon retrieval of the plug, ports are opened which allows and hydrostatic pressure above to be equalized.

The blanking plug with lantern collets is not suitable when deployment on coiled tubing is necessary, or where the well can naturally free flow.

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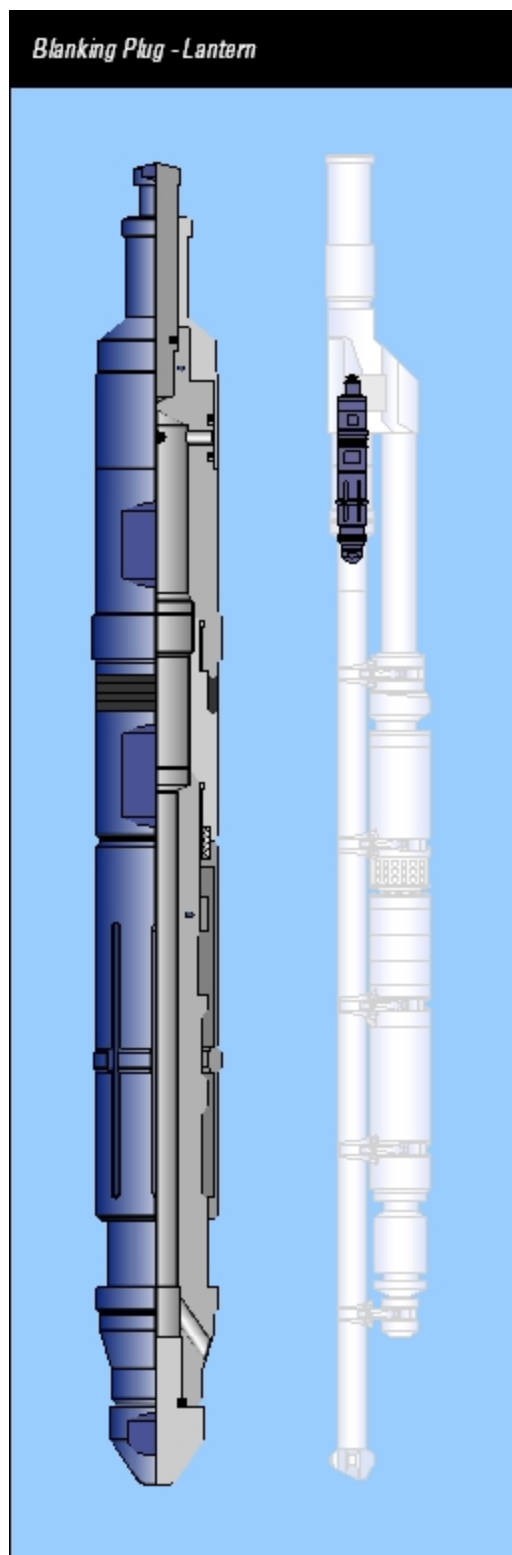


Figure B-10: Blanking Plug Lantern

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B.6.6.1 Features and Benefits

- Supplied as part of standard Y-Tool Sub-Assembly
- Prevents recirculation of production fluids down the bypass tubing during ESP operation.
- 3000 psi Maximum pressure from above
- Available in various sizes to suit the telescopic swivel nipple seal bore size.
- Top No-Go
- AISI 420 18–22 Rc/AISI 63017–4 PH Metallurgy
- Viton or Aflas Elastomers
- Redress kits available separately.

B.6.7 Blanking Plug - Locking

The Blanking Plug is seated in the telescopic swivel nipple seal bore profile immediately below the Y-Tool. The plug prevents recirculation of the produced fluids down the bypass tubing when the ESP is operating.

The plug can either be run pre-installed in the nipple, or set by slickline after installation to prevent fluids passing through the pump. The blanking plug with lock is deployed and retrieved by standard slickline tools, or can be deployed using coiled tubing.

Upon retrieval of the plug, ports are opened which allows and hydrostatic pressure above to be equalized.

The blanking plug with lock should be used when deployment on coiled tubing is necessary or where the well can naturally free flow.

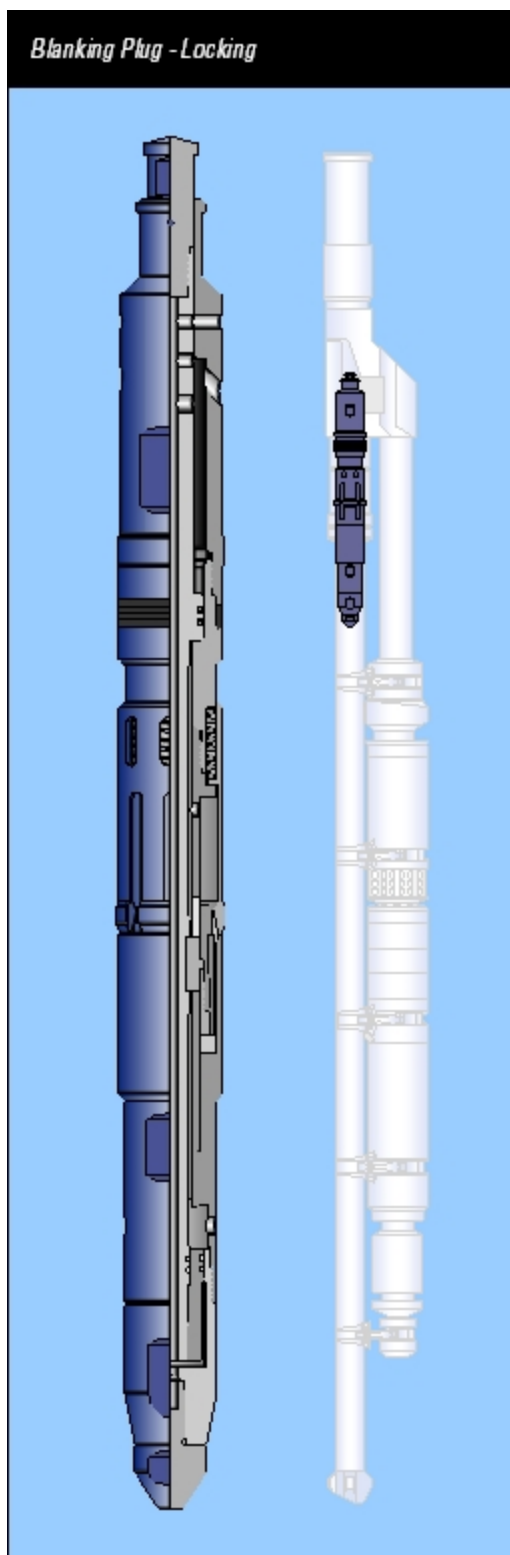


Figure B-11: Blanking Plug Locking

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B.6.7.1 Features and Benefits

- Supplied as part of standard Y-Tool Sub-Assembly
- Prevents recirculation of production fluids down the bypass tubing during ESP operation.
- Positive lock mechanism
- 5000 psi maximum pressure from above
- Available in various sizes to suit the telescopic swivel nipple seal bore size
- Top No-Go
- AISI 420 18-22 Metallurgy
- Viton or Aflas Elastomers
- Redress kits available separately

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Table B-20: Locking Blanking Plugs

Part Number	Seal Bore Diameter	Maximum OD	Minimum Recommended ID Restriction (Above Plug)	Running Neck	Wireline Pulling Tool	Fishing Neck	Wireline Pulling Tool	Maximum Test Pressure (Above Plug)
A072030-43x	2.750 in	2.802 in	2.812 in	1.750 in	2-1/2 in JDS	1.375 in	2 in JD (0.720 in Reach)	5000
A072055-43x	2.625 in	2.705 in	2.750 in	1.750 in	2-1/2 in JDS	1.375 in	2 in JD (0.720 in Reach)	5000
A072050-43x	2.312 in	2.346 in	2.441 in	1.750 in	2-1/2 in JDS	1.375 in	2 in JD (0.720 in Reach)	5000
A072032-43x	1.875 in	1.901 in	1.992 in	1.375 in	2 in JUS	1.187 in	1-1/2 in JUC	5000
<i>i</i> Note For part numbers ending with '-43x', 'x' denotes seal elastomer type, e.g. Viton-432; Aflas-433.								

B.6.8 PL Blanking Plug

- Base plug design that can be dressed for 2.312-in nipple sizes upwards by changing out Vee-rings, no-go, seal spacers (Dress Kit), and lock nut only (makes holding stock easier for field and manufacturing)
- Simple low cost locking plug design (6 main parts which don't change between seal sizes), which can be used for free flowing wells removing the option of choosing either lantern collet and locking plug designs (current options)
- Fast equalising
- Shear down to set, shear up to release using 2-in JDC running/pulling tool (CAMCO P/N 15156)
- No restriction when pulling out of hole in the unset position (collets fully close in both directions whether unset or set)
- Fits existing nipples
- Uses same dressing kit as PL Wireline Plug and PL Standing Valve
- Debris seal for cleaning / protecting nipple seal area
- Aflas seals as standard

B.6.8.1 Identification and Selection

A list of some PL Wireline Plug Assembly part numbers and PL Plug Dress kits for PL Blanking Plug Base Assembly part number 100562802 is provided below, however refer to OneCat for the latest information. Assembly and installation procedure is available in the InTouch Content ID 4761389 or in the latest release of the GeMS document 100564685.

PL Blanking plug part numbers:

- 2.875-in PL Blanking Plug Assembly (100571178)
- 2.750-in PL Blanking Plug Assembly (100571170)
- 2.625-in PL Blanking Plug Assembly (100571175)
- 2.312-in PL Blanking Plug Assembly (100571177)

PL Plug Dress kits for Blanking Plug Base Assembly (100562802):

- 2.875-in PL Plug Dress kit (100571179)
- 2.750-in PL Plug Dress kit (100564604)

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- 2.625-in PL Plug Dress kit (100564500)
- 2.312-in PL Plug Dress kit (100564497)
- 2.750-in to 2.875-in Locking Collar (100565531)
- 2.625-in and 2.312-in Locking Collar (100563729)

PL Blanking Plug redress kits:

- 2.875-in PL Blanking Plug redress kit (o-rings, vee rings and shear pins) (100571548)
- 2.750-in PL Blanking Plug redress kit (o-rings, vee rings and shear pins) (100571553)
- 2.625-in PL Blanking Plug redress kit (o-rings, vee rings and shear pins) (100571554)
- 2.312-in PL Blanking Plug redress kit (o-rings, vee rings and shear pins) (100571555)

B.6.9

Standing Valve

The Standing Valve is used in conjunction with the Top Nipple situated above the Y-Tool. The Standing Valve can be landed in the nipple against a full column of fluid and is used to pressure test the production tubing string to surface, or for setting of a hydraulic packer.

The standing valve can be run in hole installed in the Top Nipple, if required, saving a wireline run. The Standing valve has an internal ball valve which will allow any pressure below the valve to equalize.

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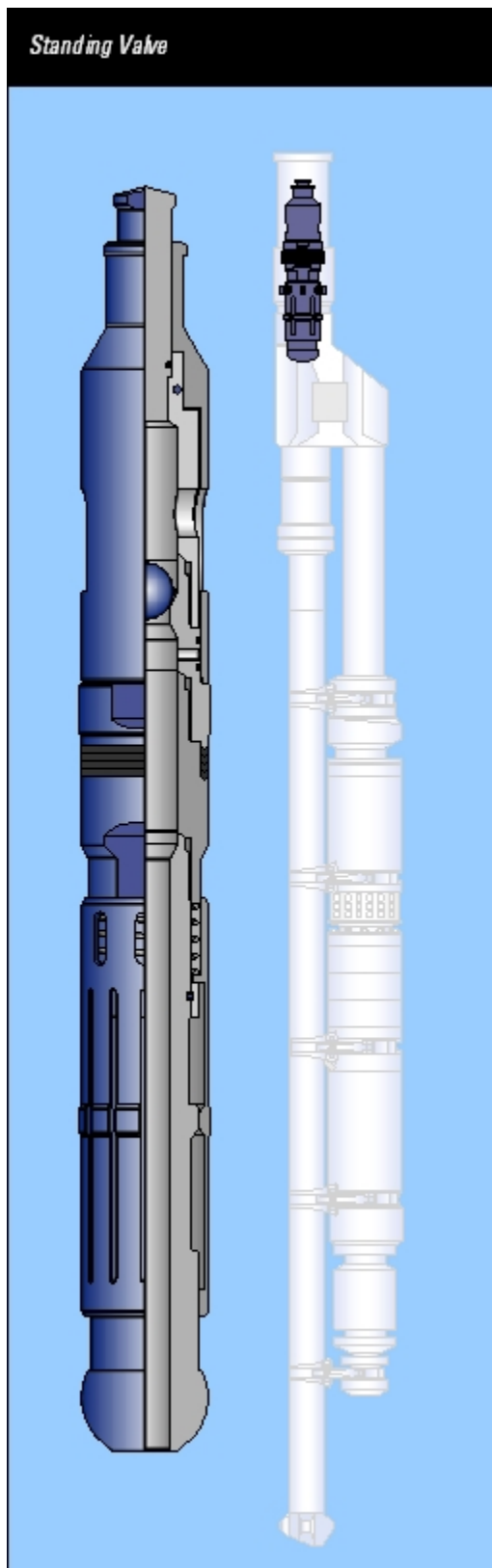


Figure B-12: Standing Valve

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B.6.9.1 Features and Benefits

- Enables pressure testing of production tubing string, or setting hydraulic packer
- 3000 psi Maximum Test Pressure
- Available in various sizes to suit the Top Nipple seal bore
- Top No-Go
- AISI 630 17-4 PH ST/ST Rc 28-33 Metallurgy
- Viton or Aflas Elastomers
- Redress kits available separately

B.6.9.2 Identification and Selection

A list of some of the Standing Valve part numbers is provided , however refer to either the Standing Valve datasheet which is available on [InTouch Content ID 4283109](#), or to OneCat for the latest information.

Table B-21: Standing Valves

Part Number	Seal Bore Diameter	Maximum OD	Minimum Recommended ID Restriction (Above Plug)	Running and Fishing Neck	Wireline Running and Pulling Tool	Maximum Test Pressure (Above Plug)	Elastomer
A071028-432	2.813 in	2.855 in	2.867 in	2.312 in	3 in JDS	3000 psi	Viton
A071028-433	2.813 in	2.855 in	2.867 in	2.312 in	3 in JDS	3000 psi	Aflas
A071027-432	2.750 in	2.802 in	2.813 in	2.312 in	3 in JDS	3000 psi	Viton
100016686	2.750 in	2.802 in	2.813 in	2.312 in	3 in JDS	3000 psi	Aflas

B.6.10 PL Standing Valve

- Base unit design
- Same dressing kits as the PL Wireline Plug and Blanking Plug
- Fits existing nipples
- Fast equalising design
- Dress Kits available for 2.312-in, 2.625-in, 2.687-in, 2.750-in, 2.813-in, 2.875-in, 3.125-in, 3.250-in, 3.500-in, and 3.687-in seal diameters

B.6.10.1 Identification and Selection

A list of some PL Plug Dress kits for PL Standing Valve Base Assembly (100591806) is provided below, however refer to OneCat for the latest information. Assembly and installation procedures are available in the InTouch content ID 4761381 or in the latest release of GeMS document 100594346.

- 2.875-in PL Plug Dress kit part number 100571179
- 2.750-in PL Plug Dress kit part number 100564604
- 2.625-in PL Plug Dress kit part number 100564500
- 2.312-in PL Plug Dress kit part number 100564497
- 3.688-in PL Plug Dress kit (can be used for PL blanking plug and wireline plug) 100592913
- 3.500-in PL Plug Dress kit (can be used for PL blanking plug and wireline plug) 100593273
- 3.250-in PL Plug Dress kit (can be used for PL blanking plug and wireline plug) 100593370
- 3.125-in PL Plug Dress kit (can be used for PL blanking plug and wireline plug) 100593389
- 2.812-in PL Plug Dress kit (can be used for PL blanking plug and wireline plug) 100593429
- 2.687-in PL Plug Dress kit (can be used for PL blanking plug and wireline plug) 100593428

PL Standing Valve redress kits:

- 2.875-in PL Standing Valve redress kit (O-rings, Vee rings) 100594326
- 2.750-in PL Standing Valve redress kit (O-rings, Vee rings) 100594323

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- 2.625-in PL Standing Valve redress kit (O-rings, Vee rings) 100594325
- 2.312-in PL Standing Valve redress kit (O-rings, Vee rings) 100594324
- 3.688-in PL Standing Valve redress kit (O-rings, Vee rings) 100594333
- 3.500-in PL Standing Valve redress kit (O-rings, Vee rings) 100594332
- 3.250-in PL Standing Valve redress kit (O-rings, Vee rings) 100594331
- 3.125-in PL Standing Valve redress kit (O-rings, Vee rings) 100594330
- 2.812-in PL Standing Valve redress kit (O-rings, Vee rings) 100594328
- 2.687-in PL Standing Valve redress kit (O-rings, Vee rings) 100594327

B.6.11

Isolation Tool

The Isolation Tool can be deployed in all standard bypass systems which are equipped with a Top Nipple.

The plug has two sets of seals and straddles the seal bores of the Top Nipple and Teleswivel Nipple. The plug has a through bore to allow bullheading or injection fluids to be injected through the bypass tubing, while fully isolating the ESP.

The plug can also be used in dual ESP bypass systems, where the flow rate may be too low for the Auto Y-Tool. The plug creates a flow path past the back up ESP, preventing recirculation. Locking plugs are available for this application.

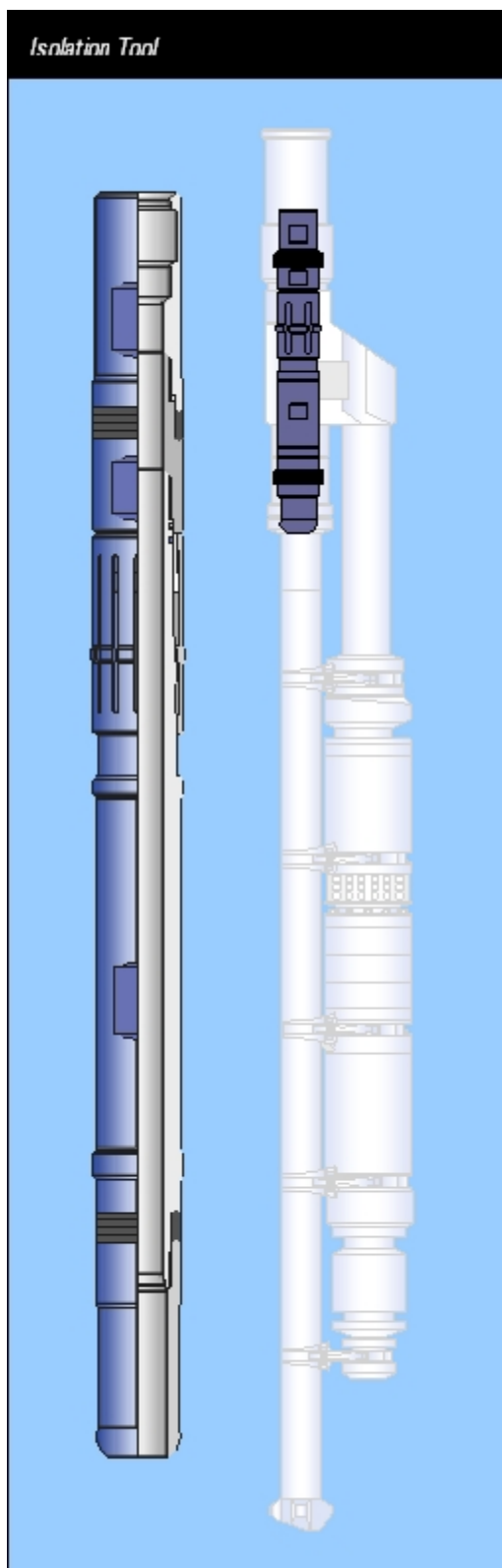


Figure B-13: Isolation Tool

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B.6.11.1 Features and Benefits

- Fully isolates the ESP during bull-heading/injection of fluids through production tubing
- Prevents unnecessary pump wear (wind-milling)
- Available in various sizes to suit the Top Nipple and Telescopic Swivel Nipple seal bores.
- Available with Locking Option where flow may be seen from below to above.
- Can be deployed and retrieved with slickline using standard Otis GS tools. Locking type can also be run on Coiled Tubing.
- AISI 420 ST/ST Rc 18-22 Metallurgy.
- Viton or Aflas Elastomers
- Redress kits available separately

B.6.11.2 Identification and Selection

A list of some of the Isolation Tool part numbers is provided below, however refer to either the Isolation Tool datasheet which is available on [InTouch Content ID 4283114](#), or to OneCat for the latest information.

Table B-22: Isolation Tools

PART NUMBER	TOP SEAL	BOTTOM SEAL	NO-GO	ID	TYPE	SEALS
100296877	2.687	2.312	2.425	1.625	LANTERN	VITON
100296893	2.687	2.312	2.425	1.625	LANTERN	AFLAS
100294380	2.813	2.750	2.800	1.875	LANTERN	VITON
100296849	2.813	2.750	2.800	1.875	LANTERN	AFLAS
100300608	2.063	1.875	2.030	1.125	LANTERN	VITON
100300613	2.063	1.875	2.030	1.125	LANTERN	AFLAS
100301466	2.750	2.312	2.380	1.780	LOCK	VITON
100301471	2.750	2.312	2.380	1.780	LOCK	AFLAS
100302070	2.750	2.625	2.700	1.812	LANTERN	VITON
100302076	2.750	2.625	2.700	1.812	LANTERN	AFLAS

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PART NUMBER	TOP SEAL	BOTTOM SEAL	NO-GO	ID	TYPE	SEALS
100304013	2.750	2.625	2.705	1.687	LOCK	VITON
100304015	2.750	2.625	2.705	1.687	LOCK	AFLAS
100304019	2.813	2.750	2.850	1.830	LOCK	VITON
100304022	2.813	2.750	2.850	1.830	LOCK	AFLAS

**Note**

The above plugs are suitable for use with the Y-Tool Sub Assemblies available in [OneCat](#) only. If the plug is required for an installed bypass system, please contact SPPC with details of the bypass assembly to ensure a compatible plug is provided.

B.6.12**Wireline Logging Plug**

The Wireline Logging Plug can be deployed in all standard bypass systems. The plug allows wireline strings to be suspended beneath the ESP through the bypass tubing.

The plug seals between the telescopic swivel nipple bore and the wireline outside diameter enabling wireline work to be carried out under both dynamic and static conditions.

Wireline grapples are required for seating wireline logging plugs. The grapple is made up to the wire at the same distance above the tool string, as the distance between the zone to be logged and the Y-Tool. When the logging tool string is on depth, the grapple lands on top of the W/L plug. The grapple ensures the W/L plug remains seated as the pump is switched on. When the pump is running the logging string can be moved up and down to log as required.

Inserts for the cable grapple and flow tubes for the logging plugs are ordered separately to suit the wireline/slickline diameter.

Grapples are not required for the Self-seating logging plugs as wireline tractors deploy these.

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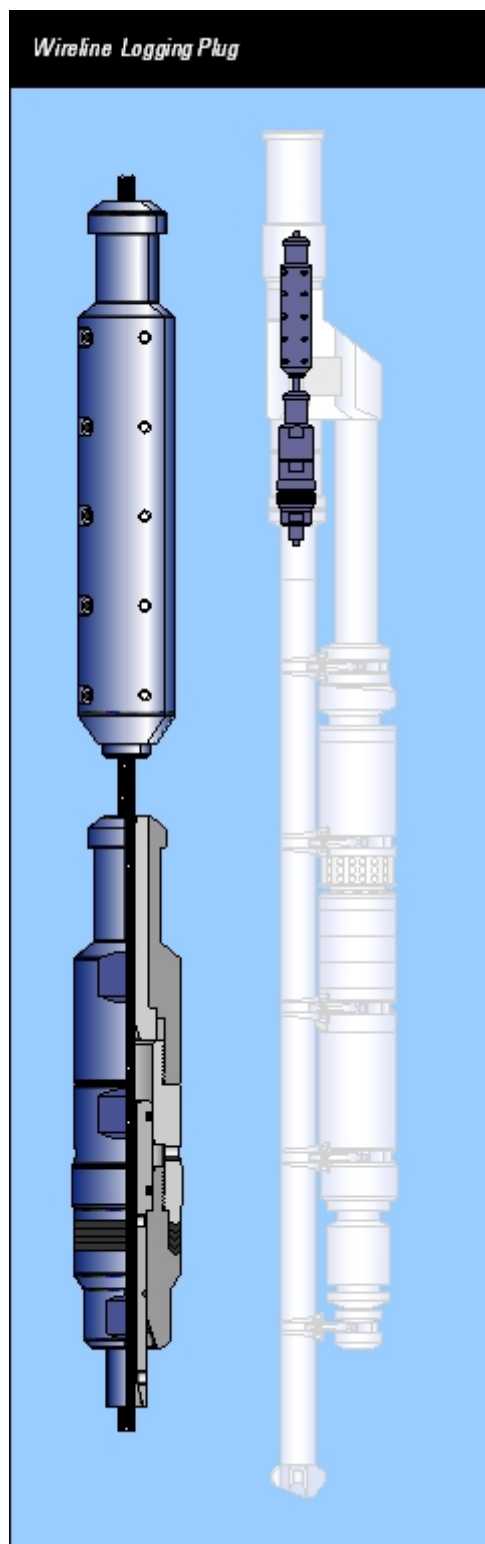


Figure B-14: Wireline Logging Plug

Private

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B.6.12.1 Features and Benefits

- Enables dynamic logging using wireline with an ESP
- No need to pull the completion to perform wireline operations
- Available in various sizes to suit the Telescopic Swivel Nipple seal bore
- Available with Tool String Catcher Facility
- Interchangeable flow tubes to suit various wire sizes
- Available in self seating option for deviated wells with the use of a wireline tractor.
- AISI 630 17-4 PH Stainless Steel Metallurgy
- Viton or Aflas Elastomers
- Redress kits available separately.

B.6.12.2 Identification and Selection

A list of some of the Wireline Logging Plug part numbers is provided below, however refer to either the Wireline Logging Plug datasheet which is available on [InTouch Content ID 4210085](#), or to OneCat for the latest information.

Table B-23: A073 Wireline Logging Plug Assembly with Catcher

Part Number	No-Go Diameter (in)	Seal Diameter (in)	Fishing Neck Size (in)	Fishing Neck Size (On Latch Body - Catcher Size) (in)	Working Pressure (Above) (psi)	Material/ Body Parts	Material/ Elastomers
A073021-431	1.990	1.875	1.187	1.187 Schlumberger	3000	AISI 630	NITRILE
A073036-431	2.346	2.312	1.000	1.000 Halliburton	3000	AISI 630	NITRILE
A073037-431	2.346	2.312	1.187	1.187 Schlumberger	3000	AISI 630	NITRILE
A073022-431	2.705	2.625	1.375	1.187 Schlumberger	3000	AISI 630	NITRILE
A073022-432	2.705	2.625	1.375	1.187 Schlumberger	3000	AISI 630	VITON
A073032-431	2.802	2.750	1.375	1.375 Western Atlas	3000	AISI 630	NITRILE
A073032-432	2.802	2.750	1.375	1.375 Western Atlas	3000	AISI 630	VITON
A073039-431	2.802	2.750	1.375	1.187 Schlumberger	3000	AISI 630	NITRILE
A073039-432	2.802	2.750	1.375	1.187 Schlumberger	3000	AISI 630	VITON
100241009	2.802	2.750	1.375	1.750 Schlumberger/ Halliburton	3000	AISI 4140	VITON
100324636	2.802	2.750	1.375	1.750 Schlumberger/ Halliburton	3000	AISI 4140	AFLAS

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Table B-24: A073 Wireline Logging Plug Assembly (without Catcher)

Part Number	No-Go Diameter (in)	Seal Diameter (in)	Fishing Neck Size (in)	Working Pressure (Above) (psi)	Material/ Body Parts	Material/ Elastomers
A073031-431	1.901	1.875	1.375	3000	AISI 630	NITRILE
A073031-432	1.901	1.875	1.375	3000	AISI 630	VITON
A073008-431	2.346	2.312	1.375	3000	AISI 630	NITRILE
A073008-432	2.346	2.312	1.375	3000	AISI 630	VITON
A073041-431	2.705	2.625	1.375	3000	AISI 630	NITRILE
A073041-432	2.705	2.625	1.375	3000	AISI 630	VITON
A073041-433	2.705	2.625	1.375	3000	AISI 630	AFLAS
A073014-431	2.802	2.750	1.375	3000	AISI 630	NITRILE
A073014-432	2.802	2.750	1.375	3000	AISI 630	VITON
A073014-433	2.802	2.750	1.375	3000	AISI 630	AFLAS

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Table B-25: D073 Flow Tube

Part Number	Wire Size (in)	Material/ Body Parts
D073-0005-601	0.092	BRASS
D073-0006-601	0.108	BRASS
D073-0015-601	0.125	BRASS
D073-0007-601	0.1875	BRASS
D073-0008-601	0.218	BRASS
D073-0009-601	0.266	BRASS
D073-0010-601	0.235	BRASS
100324993	0.250	BRASS
D073-0014-601	0.226	BRASS
D073-0011-601	0.320	BRASS
D073-0012-601	0.460	BRASS
100325037	0.469	BRASS

Table B-26: A074 Self Seating Logging Plug Assembly

Part Number	No-Go Diameter (in)	Seal Diameter (in)	Fishing Neck Size (in)	Fishing Neck Size (On Latch Body - Catcher Size) (in)	Working Pressure (Above) (psi)	Material/ Body Parts	Material/ Elastomers
A074036-431	2.346	2.312	1.187	1.187 Schlumberger	3000	AISI 630	NITRILE
A074042-431	2.705	2.625	1.187	1.750 Schlumberger/ Halliburton	3000	AISI 630	NITRILE
A074042-432	2.705	2.625	1.187	1.750 Schlumberger/ Halliburton	3000	AISI 630	VITON
A074042-433	2.705	2.625	1.187	1.750 Schlumberger/ Halliburton	3000	AISI 630	AFLAS
A074037-431	2.802	2.750	1.187	1.187 Schlumberger	3000	AISI 630	NITRILE
A074039-432	2.802	2.750	1.375	1.375 Western Atlas	3000	AISI 630	VITON
A074040-431	2.802	2.750	1.750	1.750 Schlumberger/ Halliburton	3000	AISI 630	NITRILE
A074040-432	2.802	2.750	1.750	1.750 Schlumberger/ Halliburton	3000	AISI 630	VITON
A074041-431	2.802	2.750	1.375	1.375 Western Atlas	3000	AISI 630	NITRILE
A074041-432	2.802	2.750	1.375	1.375 Western Atlas	3000	AISI 630	VITON
A074041-433	2.802	2.750	1.375	1.375 Western Atlas	3000	AISI 630	AFLAS

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Table B-27: D074 Flow Tube

Part Number	Wire Size (in)	Material/ Body Parts
D074-0065-601	0.082	BRASS
D074-0066-601	0.092	BRASS
D074-0067-601	0.108	BRASS
D074-0075-601	0.125	BRASS
D074-0068-601	0.1875	BRASS
D074-0069-601	0.218	BRASS
D074-0070-601	0.226	BRASS
D074-0071-601	0.235	BRASS
D074-0074-601	0.240	BRASS
D074-0072-601	0.312	BRASS
D074-0073-601	0.320	BRASS
100279521	0.375	BRASS
100280255	0.474	BRASS

B.6.13

PL Wireline Plug

- Base plug design that can be dressed for 2.312-in nipple sizes upwards by changing out Vee-rings and NO-GO (Dress Kit) (makes holding stock easier for field and manufacturing)
- Uses same dressing kit as PL Blanking Plug and PL Standing Valve
- Optional Tool Catcher (1-in to 1.75-in fish neck sizes)
- Spring loaded Equalising sleeve activated by PLT fish neck rope socket
- Fits existing nipple sizes
- Fast equalising
- Debris seal
- Aflas seals as standard

B.6.13.1

Identification and Selection

A list of some PL Plug Dress kits for PL Wireline Plug Base Assembly (100558739) is provided below, however refer to OneCat for the latest information. Assembly and installation procedures are available in the InTouch content ID 4761361 or in the latest release of GeMS document 100564617.

- 2.875-in PL Plug Dress kit part number 100571179

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- 2.750-in PL Plug Dress kit part number 100564604
- 2.625-in PL Plug Dress kit part number 100564500
- 2.312-in PL Plug Dress kit part number 100564497
- Other Dress Kit Sizes available (Refer to PL Standing Valve section of this manual)
- Universal Tool Catcher Kit 100564633

PL Wireline Plug redress kits:

- 2.875-in PL Wireline Plug redress kit (O-rings, Vee rings) 100571563
- 2.750-in PL Wireline Plug redress kit (O-rings, Vee rings) 100565517
- 2.625-in PL Wireline Plug redress kit (O-rings, Vee rings) 100565516
- 2.312-in PL Wireline Plug redress kit (O-rings, Vee rings) 100565514

B.6.14

Coiled Tubing Logging Plug

The Coiled Tubing Logging Plug can be deployed in all standard by-pass systems. The plug allows Coiled Tubing strings to be suspended beneath the ESP through the bypass tubing.

The plug seals between the telescopic swivel nipple bore and the Coiled Tubing outside diameter, enabling Coiled Tubing work to be carried out under both dynamic and static conditions.

The plug can be used for various combinations of coiled tubing strings and telescopic swivel nipple size.

It has been utilized successfully in the past for clean out operations, production-logging runs and also as a means of running downhole samplers below the ESP's on horizontal reservoir sections.

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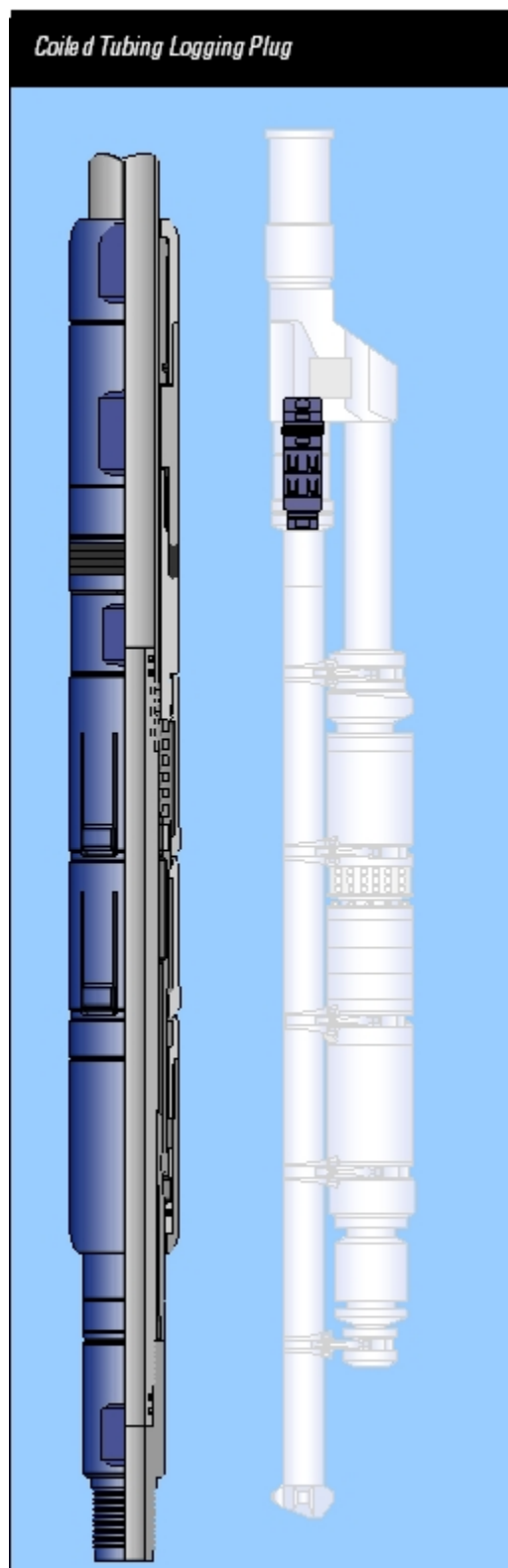


Figure B-15: Coiled Tubing Logging Plug

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B.6.14.1 Features and Benefits

- Enables dynamic logging using Coiled Tubing with an ESP
- No need to pull the completion to perform Coiled Tubing work
- Available in various sizes to suit the Telescopic Swivel Nipple seal bore.
- Available to suit different Coiled Tubing ODs and wall thickness
- AISI 630 17-4 PH ST/ST Rc 28-33 Metallurgy
- Viton or Aflas Elastomers
- Redress kits available separately

B.6.14.2 Coiled Tubing Sizes

The maximum coiled tubing size which can be used (for logging) is dependant on seal bore size of the coiled tubing plug and teleswivel, and the bypass tubing size. [Table B-28](#) and [Table B-29](#) show the maximum CT sizes for 2-3/8 in and 2-7/8 in bypass tubing.

Table B-28: 2-3/8 in Bypass Tubing

Teleswivel Seal Bore	Maximum CT Size
2.312 in	1.50 in
2.625 in	1.50 in
2.750 in	1.50 in

Table B-29: 2-7/8 in Bypass Tubing

Teleswivel Seal Bore	Maximum CT Size
2.312 in	1.50 in
2.625 in	1.75 in
2.750 in	2.00 in

B.6.14.3 Coiled Tubing Plug Connectors

Coiled tubing plug connectors are required to attach the coiled tubing plug to the coil, and a crossover is required to make up to the tool string. The required parts are:

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1. Roll-On Connector.

This is connected directly to the coil. The roll-on connector will be specified to match the CT size and wall thickness. The roll-on connector is formed onto the coiled tubing and will require to be replaced between operations.

2. Sleeve^{B-6}

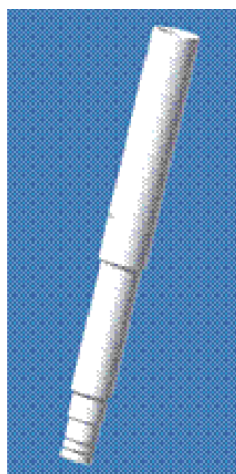
The sleeve makes up to the roll-on connector and has a profile to accept the coiled tubing plug retaining sleeve. This is specified by CT size

3. Crossover

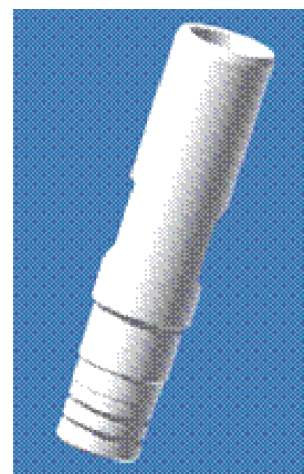
The crossover makes up to the sleeve. Crossovers are specified by CT size and by the different connections to make up to the coiled tubing tool string



Roll on connector



Sleeve



Crossover

B.6.14.4

Identification and Selection

A list of some of the Coiled Tubing Plug and Coiled Tubing Plug Connector part numbers is provided below, however refer to either the Coiled Tubing Logging Plug datasheet which is available on [InTouch Content ID 4199162](#), or to OneCat for the latest information.

B-6. In some cases the Sleeve and the Roll-On Connector are supplied as a single piece. Sleeve part number marked as NA in these cases.

Table B-30: Coiled Tubing Plugs

Part Number	Coiled Tubing Size (in)	Seal Diameter (in)	No-Go Diameter (in)	Material/ Body Parts	Material/ Elastomers	Notes
A075035-432	1.25	2.750	2.802	AISI 630	VITON	
A075048-432	1.25	2.625	2.705	AISI 630	VITON	
A075048-432	1.25	2.625	2.705	AISI 630	AFLAS	
A075046-432	1.25	2.312	2.346	AISI 630	VITON	
A075046-433	1.25	2.312	2.346	AISI 630	AFLAS	
A075037-432	1.25	1.875	1.901	AISI 630	VITON	
A075034-432	1.50	2.750	2.802	AISI 630	VITON	
A075034-433	1.50	2.750	2.802	AISI 630	AFLAS	
A075049-432	1.50	2.625	2.705	AISI 630	VITON	
A075049-433	1.50	2.625	2.705	AISI 630	AFLAS	
A075038-432	1.50	2.312	2.346	AISI 630	VITON	
A075038-433	1.50	2.312	2.346	AISI 630	AFLAS	
100279770	1.75	3.250	3.375	AISI 630	AFLAS	
A075044-432	1.75	2.750	2.802	AISI 630	VITON	
A075044-433	1.75	2.750	2.802	AISI 630	AFLAS	
A075042-432	1.75	2.625	2.705	AISI 630	VITON	
A075039-431	2.00	3.250	3.375	AISI 630	NITRILE	
A075043-432	2.00	2.750	2.802	AISI 630	VITON	
100106142	2.00	2.750	2.802	AISI 630	VITON	For low flow rates

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Table B-31: Connector Components for 1 ¼ in Coil Tubing

Coil Tubing size (in)	Wall Thickness (in)	Roll-on connector part number	Sleeve Part Number	Crossover Part Number	Thread (bottom of crossover)	Minimum ID (in)	Maximum OD (in)	Material (Elastomers)
1.25	0.156	A175035-232	A175033-432	A175013-232	1.500-10 Stub ACME PIN	0.562	1.900	Viton
1.25	0.156	A175035-232	A175033-433	A175013-233	1.500-10 Stub ACME PIN	0.562	1.900	Aflas

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Table B-32: Connector Components for 1 1/2 in Coil Tubing

Coil Tubing size (in)	Wall Thickness (in)	Roll-on connector part number	Sleeve Part Number	Crossover Part Number	Thread (bottom of crossover)	Minimum ID (in)	Maximum OD (in)	Material (Elastomers)
1.5	0.094	A175002-232	N/A	A175013-232	1.500-10 Stub ACME PIN	0.750	1.900	Viton
1.5	0.094	A175002-233	N/A	A175013-233	1.500-10 Stub ACME PIN	0.750	1.900	Aflas
1.5	0.102	A175041-232	A175033-232	A175013-232	1.500-10 Stub ACME PIN	0.750	1.900	Viton
1.5	0.102	A175041-233	A175033-233	A175013-233	1.500-10 Stub ACME PIN	0.750	1.900	Aflas
1.5	0.109	A175032-232	A175033-232	A175013-232	1.500-10 Stub ACME PIN	0.750	1.900	Viton
1.5	0.109	A175032-233	A175033-233	A175013-233	1.500-10 Stub ACME PIN	0.750	1.900	Aflas
1.5	0.125	A175036-232	A175033-232	A175013-232	1.500-10 Stub ACME PIN	0.750	1.900	Viton
1.5	0.125	A175036-233	A175033-233	A175013-233	1.500-10 Stub ACME PIN	0.750	1.900	Aflas
1.5	0.134	100990291	N/A	A175013-232	1.500-10 Stub ACME PIN	0.750	1.900	Viton
1.5	0.134	100990289	N/A	A175013-233	1.500-10 Stub ACME PIN	0.750	1.900	Aflas
1.5	0.156	A175031-232	N/A	A175013-232	1.500-10 Stub ACME PIN	0.750	1.900	Viton

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Coil Tubing size (in)	Wall Thickness (in)	Roll-on connector part number	Sleeve Part Number	Crossover Part Number	Thread (bottom of crossover)	Minimum ID (in)	Maximum OD (in)	Material (Elastomers)
1.5	0.156	A175031-233	N/A	A175013-233	1.500-10 Stub ACME PIN	0.750	1.900	Aflas
1.5	0.175	A175057-232	A175033-232	A175013-232	1.500-10 Stub ACME PIN	0.750	1.900	Viton

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Table B-33: Connector Components for 1-3/4 in Coil Tubing

Coil Tubing size (in)	Wall Thickness (in)	Roll-on connector part number	Sleeve Part Number	Crossover Part Number	Thread (bottom of crossover)	Minimum ID (in)	Maximum OD (in)	Material (Elastomers)
1.75	0.109	A175010-232	A175022-232	A175058-231	1.500-10 Stub ACME PIN	0.906	2.00	Viton
1.75	0.125	A175038-232	A175022-232	A175058-231	1.500-10 Stub ACME PIN	0.906	2.00	Viton
1.75	0.134	A175039-232	A175022-232	A175058-231	1.500-10 Stub ACME PIN	0.906	2.00	Viton
1.75	0.156	A175059-432	A175022-232	A175058-231	1.500-10 Stub ACME PIN	0.906	2.00	Viton

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Table B-34: Connector Components for 2 in Coil Tubing

Coil Tubing size (in)	Wall Thickness (in)	Roll-on connector part number	Sleeve Part Number	Crossover Part Number	Thread (bottom of crossover)	Minimum ID (in)	Maximum OD (in)	Material (Elastomers)
2.00	0.109	A175042-232 (Dimple Type)	A175043-432	A175045-131	1.500 AMMT PIN	0.906	2.37	Viton
2.00	0.109	A175042-233 (Dimple Type)	A175043-433	A175045-131	1.500 AMMT PIN	0.906	2.37	Aflas
2.00	0.109	A175025-232 (Roll-on Type)	N/A	A175040-432	1.8125-10 Stub ACME PIN	0.906	2.00	Viton
2.00	0.109	A175025-233 (Roll-on Type)	N/A	A175040-433	1.8125-10 Stub ACME PIN	0.906	2.00	Aflas
2.00	0.134	A175044-232 (Dimple Type)	A175043-432	A175045-131	1.500 AMMT PIN	0.906	2.37	Viton
2.00	0.134	A175044-233 (Dimple Type)	A175043-433	A175045-131	1.500 AMMT PIN	0.906	2.37	Aflas

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Table B-35: Connector Components for 2 in Coil Tubing

Coil Tubing size (in)	Wall Thickness (in)	Roll-on connector part number	Sleeve Part Number	Crossover Part Number	Thread (bottom of crossover)	Minimum ID (in)	Maximum OD (in)	Material (Elastomers)
2.00	0.134	A175026-232 (Roll-on Type)	N/A	A175040-432	1.8125-10 Stub ACME PIN	0.906	2.00	Viton
2.00	0.134	A175026-233 (Roll-on Type)	N/A	A175040-433	1.8125-10 Stub ACME PIN	0.906	2.00	Aflas
2.00	0.156	A175070-232 (Roll-on Type)	N/A	A175040-432	1.8125-10 Stub ACME PIN	0.906	2.00	Viton

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**Note**

Where Sleeve P/N shows N/A, the sleeve and roll on connector are one piece.

Table B-36: Miscellaneous Crossovers

Part Number	Description	Where used
A175016-231	1.25-10 STUB ACME box X 1 in 1.8# Hydril CS pin	Fits sleeve P/N A175033
A175017-231	1.50-10 STUB ACME box X 1.25 in 2.4# Hydril CS pin	Fits sleeve P/N A175022 and roll on connector P/Ns A175026, A175025
A175018-231	1.50-10 STUB ACME box X 1.25 AMMT pin	Fits sleeve P/N A175022 and roll on connector P/Ns A175026, A175025
A175024-231	1.50-10 STUB ACME box X 1.50 AMMT pin	Fits sleeve P/N A175022 and roll on connector P/Ns A175026, A175025

**Note**

Other crossovers are available by special request.

B.6.15**Handling Equipment**

Specialist handling equipment is available to assist safe installation of the logging bypass system. The handling equipment for a bypass system consists of:

1. Swivel Lift Nubbin
2. Safety Clamp
3. Stove Pipe (ESP/ Bypass worktable)

B.6.15.1**Swivel Lift Nubbin**

The swivel lift nubbin is used to handle and pick up the joints of bypass tubing. The swivel lift nubbin has a pin thread for making up to the tubing, and a swivel shackle for attaching to the tugger line/air winch. It is recommended that two swivel lift nubbins be available at the well site. This will allow one to be in use on the rig floor, and one made up to the next bypass joint required saving time.

Swivel lift nubbins are available in different threads and sizes to match the bypass tubing.

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For specialist applications where the weight of the bypass tubing string exceeds load rating of the standard swivel lift nubbin, an alternative design is used which fits into 3-1/2 in side door elevators to support the string weight.

The standard Swivel lift nubbin is rated to 8960 lb (2-3/8 in and 2-7/8 in versions).



Figure B-16: Swivel Lift Nubbin

Table B-37: Swivel Lift Nubbin Assemblies

Part Number	Type	Material / Body Parts	Thread (Size, Wt., Type, Config.)	Safe Working Load (lb)
A030004-231	Standard	AISI 4145 Rc 30-36	2-1/8 in 2.5 lbf.ft PFJ Pin	3360 (On Shackle)
100281015	Standard	AISI 4145 Rc 30-36	2-3/8 in 4.6 lbf.ft PTJ Pin	8960 (On Shackle)
100258779	Standard	AISI 4145 Rc 30-36	2-7/8 in 6.4 lbf.ft PTJ Pin	8960 (On Shackle)
100304586	Dual / Heavy Duty*	AISI 4145 Rc 30-36	2-7/8 in 6.4 lbf.ft PTJ Pin	8960 (On Shackle)

B.6.15.2

Safety Clamp

The safety clamp is designed to secure the bypass tubing, while alongside the ESP assembly. This enables subsequent bypass tubing joint make ups. The clamp rests on the stovepipe and supports the weight of the bypass tubing string. Clamps are available to suit the various bypass tubing ODs.

For specialist applications where the weight of the bypass tubing string exceeds load rating of the standard safety clamp, alternative designs are used which can support higher tensile loads.

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The standard Safety Clamp^{B-7}, uses 'die' type inserts to achieve a load rating of 44,800 lb (2-3/8 in and 2-7/8 in versions).



Figure B-17: Safety Clamp

Table B-38: Safety Clamp Assemblies

Part Number	Bypass Tubing OD	Material/Body Parts	Safe Working Load (lb)
100349630	2-3/8 in	MDS-106 (AISI 4140)	44,800
100366943	2-7/8 in	MDS-106 (AISI 4140)	44,800



Note

The safety clamp body is common to both 2-3/8 in and 2-7/8 in tubing sizes. The slip dies fitted to the safety clamp can be replaced should it be required to change to another size of bypass tubing. Replacement slip die insert part numbers shown below:

Part No. 100350543: Safety Clamp Slip Die Insert for 2 3/8 in tubing

Part No. 100350545: Safety Clamp Slip Die Insert for 2-7/8 in tubing

B-7. Always confirm the load rating for the exact safety clamp part number you will be using. Previous designs which may still be used in the field may have a lower load rating than the standard part currently supplied.

B.6.15.3 Stove Pipe (ESP/ Bypass worktable)

The stovepipe is a worktable designed for the installation of ESP bypass systems. The stovepipe is positioned over the rotary table and is used as a work platform to land the ESP lifting clamps and bypass tubing safety clamp between make ups.

The standard stovepipe has a 7 in profile to allow up to 675 series ESP equipment and is rated to 22,400 lb.

For specialist applications where the weight of the string exceeds load rating of the standard stovepipe, alternative designs are used which can support higher tensile loads.

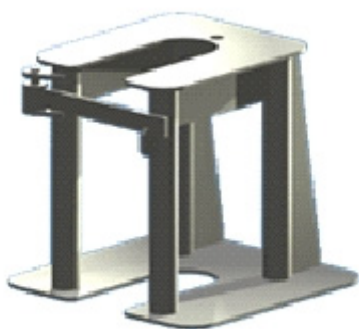


Figure B-18: Stove Pipe

Table B-39: Stove Pipe (Worktable) Assemblies

Part Number	Cut Out Size	Material / Body Parts	Safe Working Load (lb)
A030031-231	7 in	Structural Steel	56,000
A030059-231	9 in	Structural Steel	56,000



Note

Dual/Heavy Duty Swivel Lift Nubbin has a shoulder for 3-1/2 in side door elevators. When held in elevators (and not by Shackle) the load rating is same as bypass tubing connection tensile strength. Refer to PTJ datasheet for details.

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B.7 ESP Bypass System Installation Procedure

B.7.1 Overview

The following procedures apply to the installation of a Generic SPPC By-Pass System.

As these procedures are generic only, consideration should be made for the exact configuration of the individual ESP and bypass assembly. The procedures should be reviewed and modification may be necessary.

B.7.1.1 Responsibilities

It is the responsibility of the testing engineer to ensure that all relevant permits are in place and that concerned parties are notified of any work being undertaken.

It is the responsibility of the testing engineer to ensure that any problems encountered are documented and submitted in Quest. The product center should be notified of any problems related to bypass equipment quality.

B.7.1.2 Safety

Toolboxes, A-Frames with control lines etc. should be laid out on the rig floor so as not to pose a hazard to personnel.

Tools should be kept clean and in the Toolbox when not in use. Damaged or worn tools should not be used.

Hand tools should be tied off to avoid the potential of objects being lost downhole.

An on-site risk assessment should be carried out prior to commencing any work on the installation of any equipment. Any generic risk assessments used should be reviewed to reflect job specific hazards and risks.

A toolbox talk should be held prior to commencement of the installation to brief all involved parties on the operation of installing the bypass equipment.

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B.7.2

Special Handling Equipment

Table B-40: Special Handling Equipment

Description	Check
Safety Clamp of Correct Size for Bypass Tubing	
Swivel Lifting Nubbin of Correct Size and Thread for Bypass Tubing	
ESP Bypass Worktable (Stove Pipe)	
3-1/2 in Elevators, Slips and Dog Collar	
API Tubing Drift for Bypass Tubing	
0 – 250 ft.lbs Torque Wrench with 1-7/8 in Socket	
0-80 ft lbs Torque Wrench with 6 mm, 8 mm, 12 mm Hex Bits	
Coppaslip grease or equivalent	
2 x Pipe Wrenches (24 in and 36 in)	
Assorted Hand Tools (Spanners, Allen Keys, Speed Wrenches etc)	

**Note**

Equipment for bypass system only. ESP and MultiSensor will require additional equipment.

B.7.3

Tubular Torque Values

Connection	Minimum, ft. lbs	Optimum, ft.lbf	Maximum, ft.lbf
2-3/8 in, 4.6 lbs/ft PTJ	250	300	350
2-7/8 in, 6.4lbs/ft, PTJ	400	500	600

B.7.4

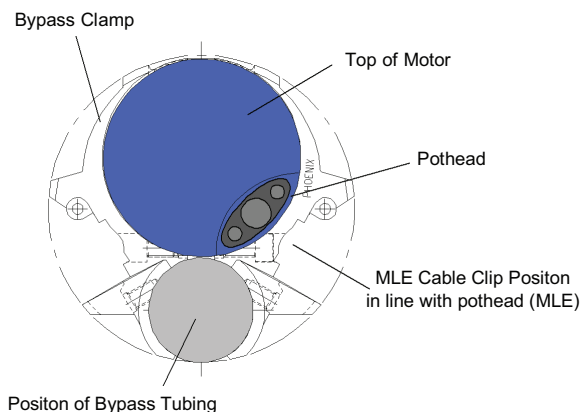
Preparation

1. Ensure that all completion sub-assemblies are torqued, measured, drifted, and pressure tested prior to running the completion.
2. If the Y-Tool sub assembly is a standard type, with Blanking Plug, then determine if it is required to run the completion with the Blanking Plug installed or removed. The plug can be removed or set in the Y-Tool assembly on deck using Stem Bar, Spang Jars, and the pulling and running tools specified in the correct Blanking Plug SOP.

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3. If the Y-Tool assembly includes a Top Nipple profile for a Standing Valve, then determine if it is required to run the completion with the Standing Valve installed or removed. The plug can be removed or set in the Y-Tool assembly on deck using Stem Bar, Spang Jars and the pulling and running tools specified in the correct Standing Valve SOP.
4. Stroke open the Teleswivel to mid-stroke position. Normally 7.5 in stroke open for mid position (15 in full stroke).
5. Measure (tally) the make up lengths of all the bypass tubing and ESP sections.
6. Refer to Appendix 1 to create a bypass tubing space out diagram.
7. Mark the pick up order of the bypass tubing as per the space out diagram.
8. Referring to the space out diagram, determine where the ESP connections will lie relative to the bypass tubing alongside. Mark the bypass joints showing where the ESP connections will lie when alongside.
9. Fit the WEG to the first joint of bypass tubing using wrenches.
10. Locate the Pothead on the upper motor. Use a bypass clamp to establish the position of the bypass tubing relative to the Pothead. Refer to .



11. Draw a vertical line down the full length of the motor to mark the position of the bypass tubing. If there is more than one motor section in the string, use the head and base bolt patterns to continue the line for the bypass tubing position down the length of each motor section.
12. If there are additional lines to be run past the ESP, for example a gauge pressure discharge line, and/or chemical injection lines, then use the bypass clamp cable clips to determine where these will be positioned. The position of MLE and control lines secured by the clamp will depend on the clamp type, and the application. Make any further markings required on the ESP units to aid positioning of these lines during the assembly.

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13. If applicable, fit the gauge to base of the lower motor. If the gauge has a pressure discharge, or Remote Sub line then ensure the gauge is correctly aligned on the motor, for the position of the line.
14. Fit the motor base plug (if included) to the bottom of the motor, or gauge (if included) using wrenches.

B.7.5 ESP Bypass Assembly

1. Fit the swivel lift nubbin to the first bypass joint. Pick up the assembly with the tugger / air winch. Position over the well and RIH.
2. Position the ESP worktable over the well. Fit the safety clamp to the bypass tubing and torque to 200 ft.lbf. Land on the ESP worktable.
3. Remove the swivel lift nubbin and fit to the next bypass tubing joint. Pick up the joint with the tugger / air winch and make up to the assembly using wrenches.
4. Pick up to remove the safety clamp. POOH until the mark on the bypass tubing for the bottom of the motor is at working height.
5. Fit the lifting clamp to the ESP motor and pick up with the elevators.
6. Manoeuvre motor over the well. If applicable make up the pressure discharge line or remote sub line to the gauge.



Note

If a pressure discharge line is fitted to the gauge, it should be pressure tested at this point.

7. Position the motor alongside the bypass tubing and orientate such that the vertical line drawn on the motor is in line with the bypass tubing. Adjust the height of the motor such that the base is in line with the marking made on the bypass tubing.
8. Referring to SOP 100476874, fit a bypass clamp to the base of the motor (or motor base plug if fitted). Securing control lines if applicable.
9. Slowly RIH with the assembly servicing the motor as required.
10. Land the motor on the ESP worktable / stovepipe.
11. Pick up and make up the next section of ESP. Pick up to remove lifting clamp. Referring to SOP 100476874, fit a bypass clamp to the base of the ESP unit, securing lines if applicable.

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12. RIH with the assembly to the top of the bypass tubing joint. Fit the safety clamp to the bypass tubing.
13. Remove the swivel lift nubbin. Pick up the next section of bypass tubing with the tugger / air winch, and make up with wrenches. Remove safety clamp.
14. Continue to RIH with the assembly making up ESP sections and bypass tubing joints as required. As each unit is added, ensure the bypass tubing has not slipped by confirming the ESP make up points match the markings made on the bypass tubing.
15. Once the Protectors have been fitted tie in the MLE and perform electrical tests. Refer to SOP 100476874 to fit a bypass clamp to each ESP neck, securing MLE cable and any additional lines.

**Note**

A bypass clamp should not be fitted directly below the pump intake. If a bolt on intake is used, a clamp can be fitted above the intake, however if the intake is integral, then the clamp should be omitted.

16. Once all the ESP units and bypass tubing joints have been made up, continue to RIH and land the pump lifting clamp on the ESP worktable / stovepipe.
17. If applicable make up the pressure discharge sub, or remote sub to the top of the pump observing correct alignment for the position of the line.
18. Pick up the Y-Tool sub assembly with the elevators. Position the discharge head over ESP assembly. Align the Y-Tool sub assembly so that the teleswivel is in alignment with the bypass tubing.
19. Slowly lower the Y-Tool assembly and make up the discharge head to the top of the ESP assembly.
20. Pick up the assembly to remove the ESP lifting clamp.
21. Slowly lower the assembly so that the teleswivel is at working height. Fit the safety clamp to the bypass tubing and torque to 200 ft.lbf. Remove the lifting nubbin and make up the teleswivel to the bypass tubing. Tighten the teleswivel sleeve and secure lock ring.
22. POOH to the discharge head. If applicable terminate pressure discharge or remote sub line.
23. Referring to SOP 100476874, fit a bypass clamp to the discharge head, securing lines if applicable.

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24. Slowly RIH with the assembly to the Y-Tool. Secure the MLE in the cable clip at one side of the Y- Tool. If applicable, secure control lines in cable clip at opposite side of Y-Tool.
25. Slowly RIH with the assembly and set the slips on the handling sub.

B.7.6

Constructing a Bypass Space Out Drawing

1. Use to sketch the ESP units of the string, including MultiSensor, Pressure Discharge Sub or Remote Sensor etc.
2. Mark on the lengths, of the pump sub, and all the ESP units. Assume the base of the Y-Tool to be depth 0.00 and working from top down, add up the lengths of the ESP string, noting each flange or make up point as a depth from 0.00. See .
3. Add the length of the Teleswivel at mid stroke to the sketch.
4. Working from the Teleswivel down, add lengths of bypass tubing to the sketch. As each length is added, calculate the depth relative to 0.00 at the make up point.
5. It is recommend that the bypass tubing extend at least 10 ft below the base of the ESP. This should be discussed with the client in case of special requirements.
6. Calculate where each ESP flange (make up point) will lie with respect to the bypass tubing joints alongside, and note on the drawing. See finished example in [Figure B-21](#).

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Notes:
 2-3/8" 4.6lbs/ft PTJ Make Up Loss: 2.370"
 2-7/8" 6.4lbs/ft PTJ Make Up Loss: 2.800"

Dimension Units:

Depth	Length	Length	Depth
-------	--------	--------	-------

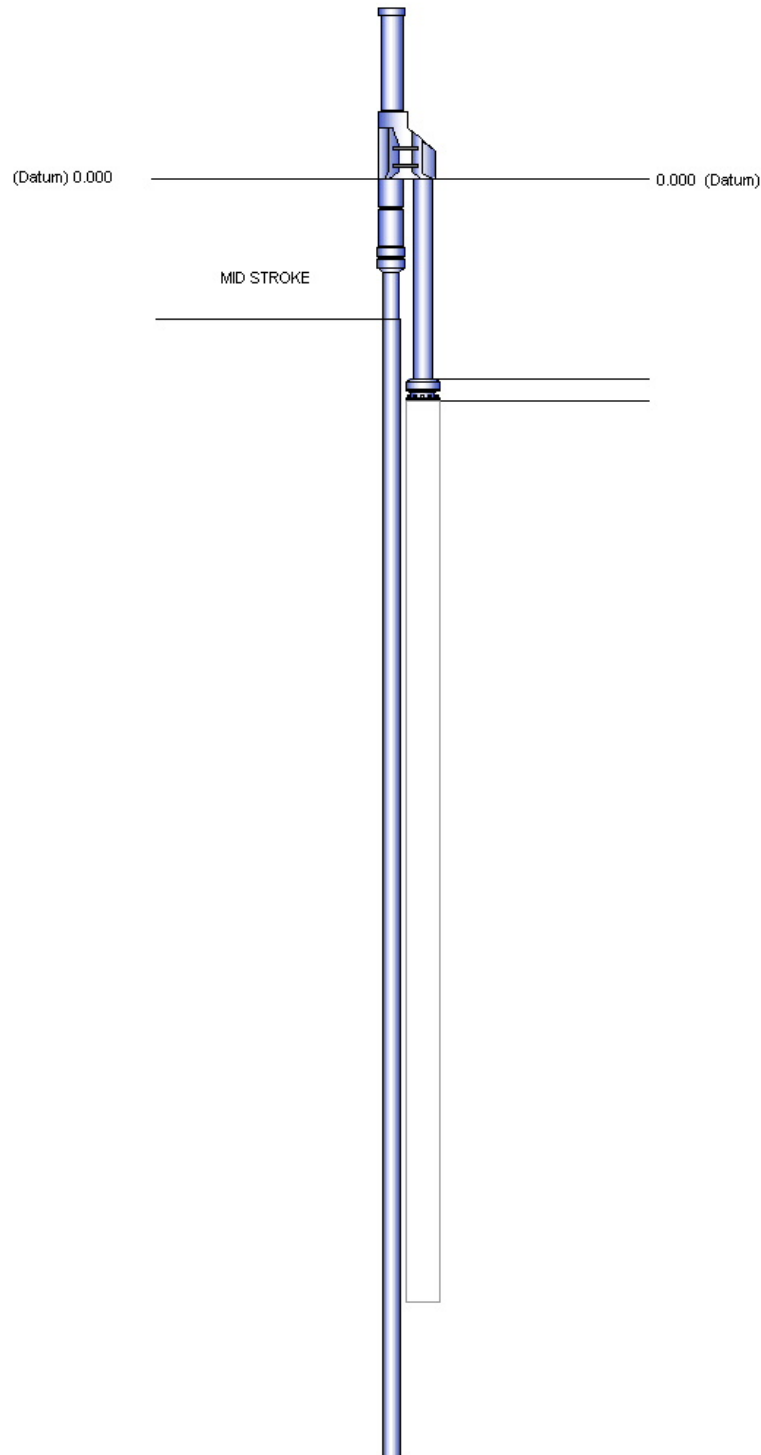


Figure B-19: Bypass Space Out Drawing Template

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Notes:
 2-3/8" 4.6lbs/ft PTJ Make Up Loss: 2.370"
 2-7/8" 6.4lbs/ft PTJ Make Up Loss: 2.800"

All dimensions in feet unless stated

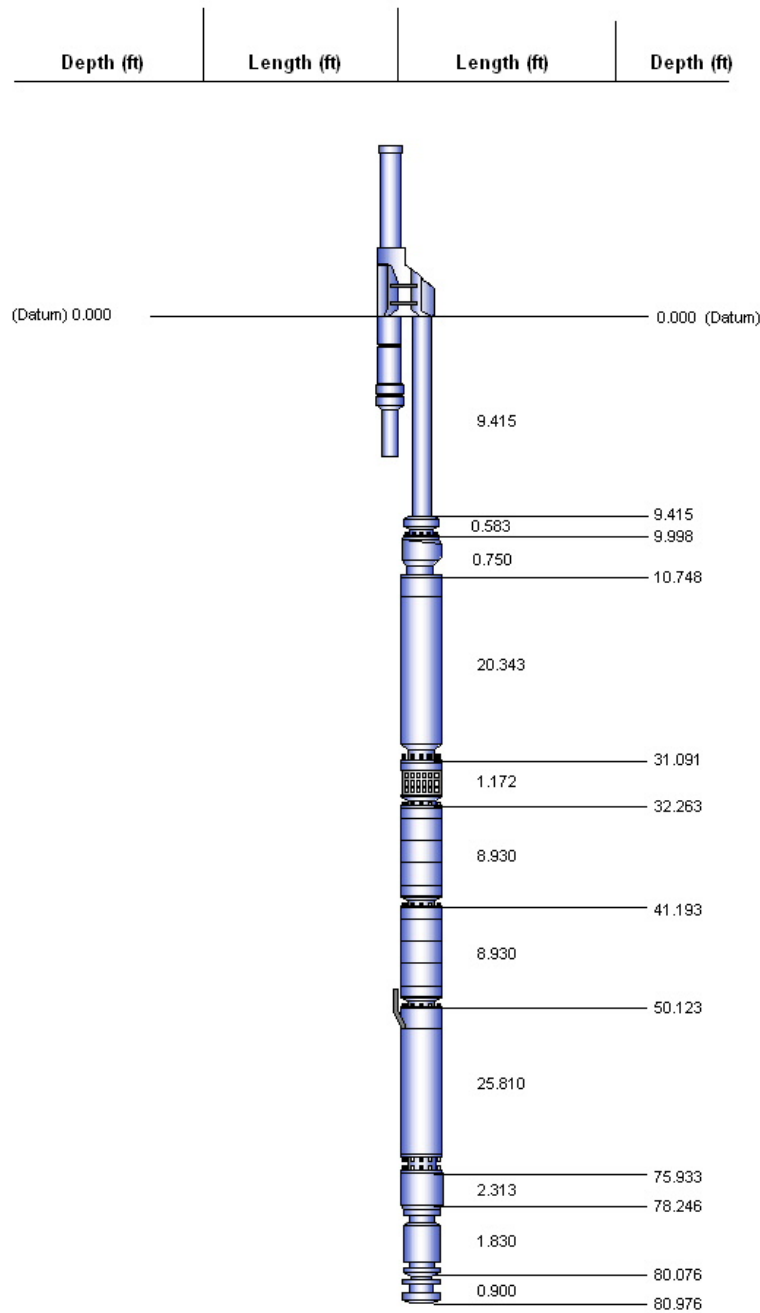


Figure B-20: Completed Space Out Template

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Notes:
 2-3/8" 4.6lbs/ft PTJ Make Up Loss: 2.370"
 2-7/8" 6.4lbs/ft PTJ Make Up Loss: 2.800"

All dimensions in feet unless stated

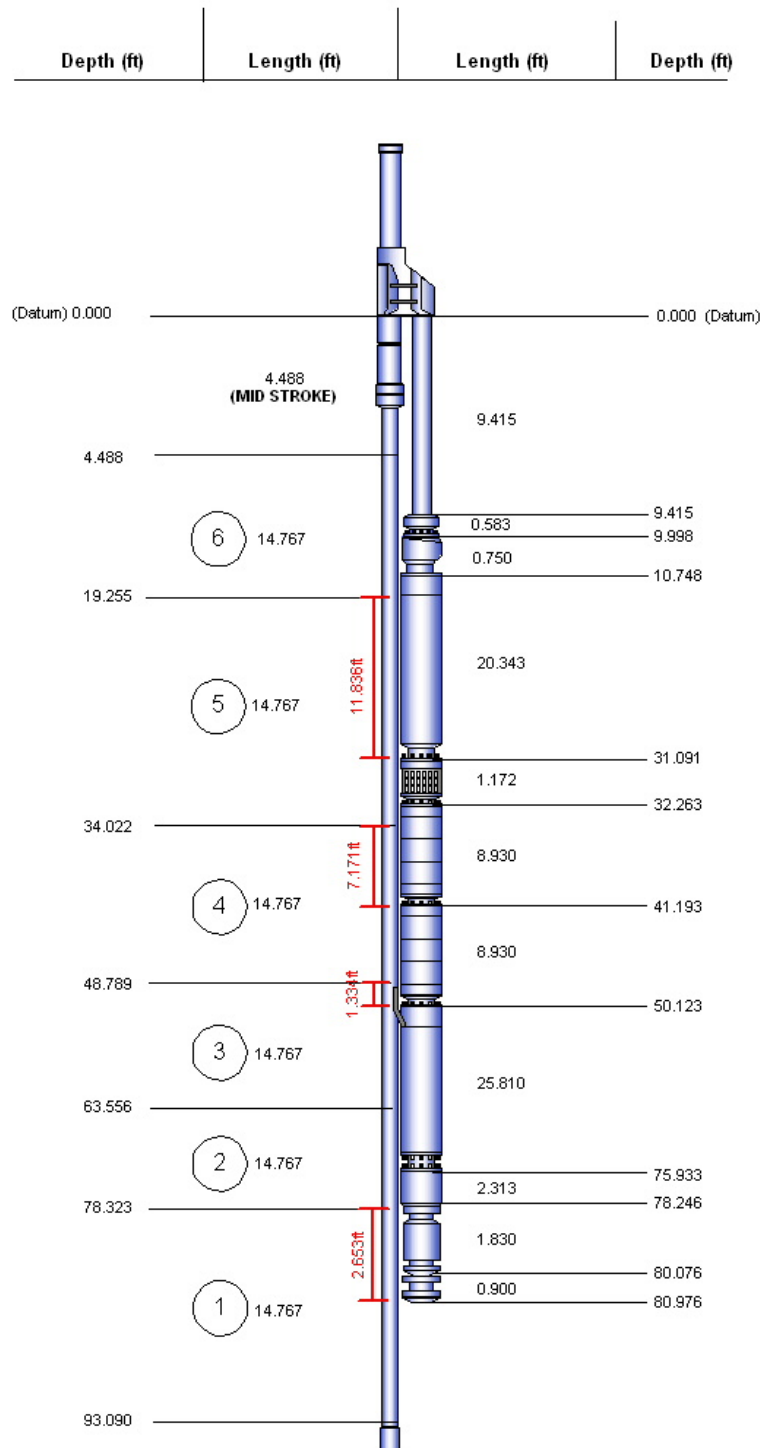


Figure B-21: Bypass Space Out Drawing Template

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B.7.7

**Universal Bypass Clamp Installation Procedure
(SOP 100476874)**

For installation procedure on how to install universal bypass clamp refer to [GeMS](#) SOP 100476874.

Maximus

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C Maximus

C.1 Introduction

The standard Maximus equipment installation is almost identical to other equipment. The exceptions are:

- The ProMotor is factory-filled at the manufacturing plant and does not require oil servicing at the wellsite.
- Maximus protectors are factory-filled and factory-shimmed, no oil filling or shimming measurements are required at the wellsite because they arrive to the wellsite ready to plug-and-play.
- The MaxLok ESP motor pothead design eliminates the need for taping pothead electrical connections.

The Trident motor head is available for high amperage and extreme reliability requirements such as subsea installations. The Trident motor head is compatible with the Trident MLE which has three individual conductors and potheads.

Detailed specifications are available in the Product Bulletins [REDA Maximus Motors \(InTouch Content ID 4206720\)](#) and [Protectors \(InTouch Content ID 4206743\)](#) for internal reference and in Product data sheets for external reference.

C.2 Installation

C.2.1 Connections information

Connection Information for Maximus equipment is listed in the ALFORM Connection Table ([InTouch ID: 3255850](#)).



Note

Installation of the 375/325/375 Maximus protector to the 375 AS motor requires an adapter, as shown in the following figure. The parts required are listed in the ALFORM connection table.

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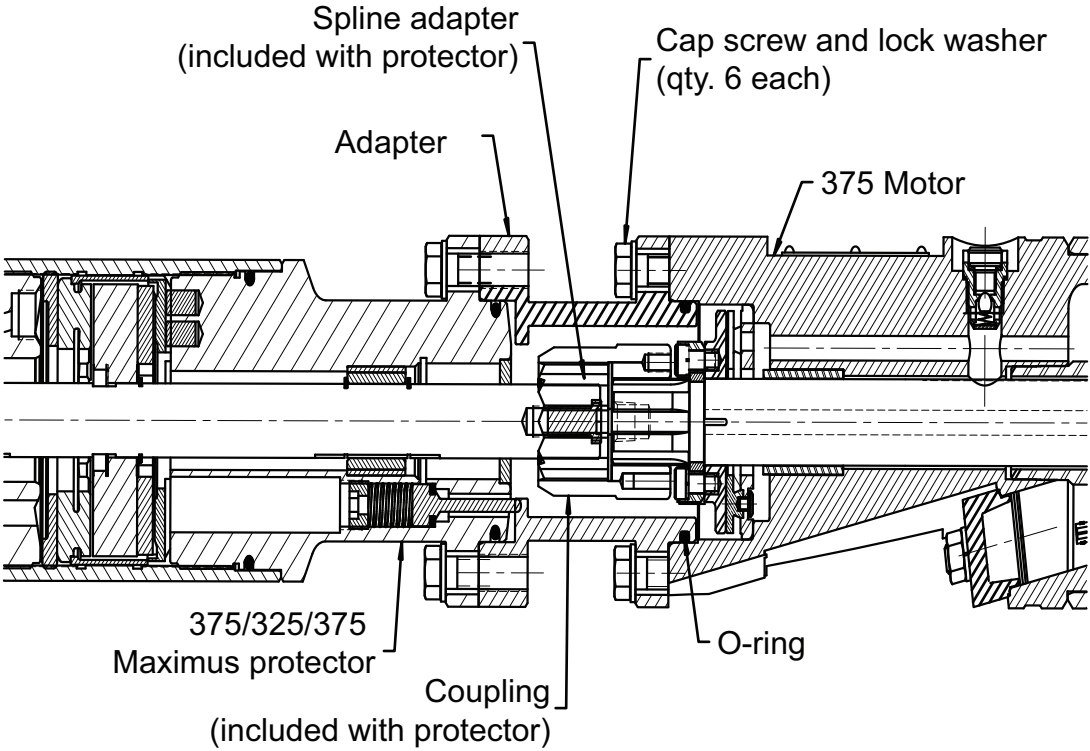


Figure C-1: 375/325/375 Protector to 375 AS Motor connection

C.2.2 Installation tools

When installing Maximus motors and protectors, there are some additional tools required. [Table C-1](#) and [Table C-2](#) show tools required in addition to tools used for the installation of standard motors and protectors.

Table C-1: Additional installation tools (MaxLok MLE)

Item	Description	Part Number	Series
1	Spline alignment tool for protector-to-motor	100326236	456
2	Spline alignment tool for protector-to-motor	100326237	562
3	Spline alignment tool for motor-to-motor	100298670	375 ¹
4	Spline alignment tool for motor-to-motor	100351834	456
5	Spline alignment tool for motor-to-motor	100305504	562
6	Spline Wrench .87-6A male and female for protector-to-motor	100642812	456
7	Spline Wrench 1.19-6B male and female for protector-to-motor	100644839	562

¹ Non-Maximus motor with compensating shipping cap

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Table C-2: Additional installation tools (Trident MLE)

Item	Description	Part Number	Series
1	TEST PROBE:738/562 MOTOR WITH TRIDENT POTHEAD, 3FT, PEEKLB M, 5KV, 1/1, P/I	101049527	738/562
2	WRENCH, SNAP-ON, 1/1 PEEKLB 5KV MLE WITH TRIDENT POTHEAD	100793643	738/562
3	TOOL: COUPLING & SPLINE WRENCH, 2.000-20 INV, T-HANDLE	100835923	738/562
4	TOOL: SPLINE ALIGNMENT, UT MTR TO LT PROT, 738 MAXIMUS, 2.000-20 INV	100831872	738/562
5	HYDRAULIC PUMP FOR PRESSURE TEST AFTER INSTALLING TRIDENT-300 POTHEAD	N/A	738

C.3 Wellsite Procedures

C.3.1 General Notes on MaxJoints and Shipping Caps

The following general notes are applicable to all joints between Maximus motors, protectors and ProMotors, which are known as MaxJoints.

1. The MaxJoint allows making up a flange without trapping air or loss of required oil. Therefore, Maximus motors, protectors and ProMotors do not normally require field refilling upon installation unless oil is lost.
2. A compensating shipping cap is used on the head of all Maximus motors, protectors and ProMotors, except for a few cases where it is not required. For example, 400 series UT protectors with a bag on top do not require a compensating shipping cap because the bag acts as a compensator and another protector cannot be mated to it. The compensating shipping cap compensates for the oil thermal expansion and contraction during the storage and transportation of the units.

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Figure C-2: Compensating shipping cap on UT motor

3. On receipt of the unit (motor, protector or ProMotor) at the wellsite inspect the unit and the shipping box, or container for signs of oil leakage. It is normal for residual oil to drain from between the housing and the bag of the compensating shipping. However, if a significant amount of oil has leaked from other locations on the unit, it needs to be refilled according to the oil-filling procedures. Refer to the standard [wellsite procedures](#) in the unless specified otherwise.
4. When lifting a unit equipped with a compensation shipping cap, a Spreader Lifting Chain (100367280) must be used to lift Maximus Equipment. In case of emergency, if the Spreader Lifting Chain is not available, two normal chains (0902437) can be connected together on the clamp with a 3/4-in link (0902262) to make a chain long enough to clear the compensating shipping cap. In cases where the 3/4-in link is not available, then three normal chains (0902437) can be connected together to make a chain long enough to clear the compensating shipping cap.



Danger

Potential Severity: Major
 Potential Loss: Assets, Personnel
 Hazard Category: Land transport

The compensating shipping cap must not be used as a lifting point for lifting the unit because it can break and let the equipment fall, endangering personnel and equipment.

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5. If the conventional shipping cap on the equipment base is difficult to remove because of a partial vacuum. The drain-and-fill valve on the shipping cap may be loosened to relieve the vacuum.
6. After the shipping cap has been removed from a base and the oil contained within the shipping cap has drained, the oil communication valve and the shaft seal in the base should prevent continued loss of oil. If oil continues to drain and more oil is lost than the compensating shipping cap on the head of the unit can make up, the unit may need to be refilled as discussed by the following.
7. When the compensating shipping cap is removed from the head, the head should be overflowing with oil. Watch the oil level for a few minutes and wait for any bubbles in the system to rise to the surface. The length of time required for this will vary according to equipment type, length and temperature. If the oil level falls more than 0.5 in. (13 mm) or bubbles larger than 0.5 in (13 mm) diameter continue to rise, the unit may not be properly filled with oil or may have lost oil. It should be refilled with the proper oil per the standard [wellsite procedures](#), in the same manner as Dominator motors or Modular protectors. If the oil level continues to fall, check for a leaking drain-and-fill valve or vent plug or reject the unit.
8. After removing a compensating shipping cap, it should be cared for and returned for salvage or reuse. Set the compensating shipping cap upside down to minimize oil loss and avoid contamination. Protect it by installing the shipping cap and gasket removed from the base of the unit above it.
9. If oil is displaced, lowering the oil level more than 0.12 in (3 mm) below the flange face, while preparing to make up a flange connection (checking shaft rotation, aligning splines, etc.), top it off by pouring more of the proper oil into the head. Refer to the oil specification for the equipment.
10. When a flange is made up, an actuator ring in the head of the lower unit lifts the oil communication valve in the base of the unit above it to establish communication. Make sure the actuator ring is present in the head of the lower unit. Also avoid any damage to the communication valve in the base of the upper unit.
11. When a flange is made up, excess oil will be displaced from the head. If the joint is reopened, the head must be refilled with oil before making it up again. Otherwise, air will be trapped in the joint.

Refer to [Table C-3](#) for Maximus compensating shipping caps and hardware part numbers. Refer to the ALFORM-D connection tables ([InTouch ID: 3255850](#)) for a complete list of O-ring part numbers, coupling and bolt torque information.

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Table C-3: Maximus Compensating Shipping Cap Connections

Equipment	Compensating Shipping Cap	O-ring			Carbon Steel		Monel	
		Size	Aflas	HSN	Screw	Lock-washer	Screw	Lock-washer
325, 387, 400 Series Maximus Protector ¹	100230733	2-227	1042555	996082	68965 (6)	306993 (6)	118182 (6)	309740 (6)
540 Series Maximus Protector	100264210	2-237	1042605	812776	68973 (11)	307223 (11)	118190 (11)	310086 (11)
456 Series Maximus Motor S and UT	100251368	2-231	1042589	831248	68965 (6)	306993 (6)	118182 (6)	309740 (6)
456 Series Maximus Motor LT	100112112	2-040	1135128	100078629	68965 (6)	306993 (6)	118182 (6)	309740 (6)
562 Series Maximus Motor S and UT	100264210	2-237	1042605	812776	68973 (11)	307223 (11)	118190 (11)	310086 (11)
562 Series Maximus Motor CT and LT	100118869	2-154	1105725	1018761	68973 (11)	307223 (11)	118190 (11)	310086 (11)

¹ ALL 325-, 387-, 400-series Maximus protectors use compensating shipping caps, except for the following. 387-series and 400-series Maximus protectors, with BPBSL-UT chambers configuration, DO NOT use compensating shipping caps.

AL-ESP Field Service & Bypass Sys. \AL_Engineering\InTouch ID 3255850\3.2\Release Date:20-Jul-2011\EDMS UID: 274783215\Produced: 07-Mar-2013 21:09:14

C.3.2 Clamp installation

Choose the proper motor, protector or ProMotor lifting clamp from [Table C-4](#). Install the lifting clamp so the shoulder of the clamp rests inside the recess of the motor or protector head. Clamping in the undercut helps prevent the clamp from slipping. Tighten the bolts evenly to ensure equal spacing between the clamp faces.



Note

When lifting a unit equipped with a compensating shipping cap, two chains (0902437) must be connected together on the clamp with a 3/4-in link (0902262) to make a chain long enough to clear the compensating shipping cap.



Danger

Potential Severity:	Major
Potential Loss:	Assets, Personnel
Hazard Category:	Land transport

The compensating shipping cap must not be used as a lifting point for lifting the unit because it can break and let the equipment fall, endangering personnel and equipment.

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Table C-4: Maximus lifting clamps

Maximus Installation Clamps	Part Number
Clamp: 325/375 protector, shoulder	0041236
Clamp: 325/375 protector, Shoulder	0609016
Clamp: Tail, 456 Series Motor, Friction	0301002
Clamp: Motor, 456/450 Series Head	607010
Clamp: Motor, 562 Series, UT	0004218 ¹
Clamp: Motor, 562 Series, CT/LT ProMotor	1072842 (Suitable for UT)
Clamp: Motor, 562 Series, CT/LT Maximus	1215060 (Suitable for UT)
Clamp: Protector, 540/562 Series Friction (Same clamp as 540 Pump)	27441
Clamp: Pump, 387 Series Shoulder (Also use on 387/456 ProMotor Protector)	1142298
Clamp: Pump, 387 Series Friction	46748
Clamp: Pump, 538 Series Friction	1148899
Chain: Spreader Lifting Chain, Certified, WLL , 13,900 lbm	100367280
Chain	0902437
Link	0902262

¹ Some UT Maximus motors may have a 5.62 in head which is unsuitable for the shoulder clamp (0004218). Verify head before using clamp. Refer to following note.



Note: Some UT Maximus motors may have a 5.62 in head which for motor clamp (0004218) is unsuitable.

1. Check the head OD at the flange for your UT Maximus motor.
2. If the head OD is 5.44 in the shoulder clamp (0004218) is suitable, if it is 5.62 in, friction clamp (1215060) must be used.
3. To install MLE,
 - Use a friction clamp at the housing and then set the motor on the work table to connect the pothead. Or,
 - Tandem the motor to a protector before installing the MLE. Then while holding the Motor and Protector using a Protector Head clamp, install the MLE.

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C.3.3 Installation of Phoenix Gauge

The Phoenix gauge can be installed onto the Gauge-Ready-Base (GRB) motors at the wellsite. Installation of the gauge can also be done in the shop. The gauge can be installed either vertically or horizontally. When the lower shipping cap is taken off for gauge installation, the check valve inside the base prevents the oil from leaking out off the motor. After the gauge is installed, only the small space between the base and the gauge needs to be filled. The motor itself does not need to be refilled.



Figure C-3: Gauge Ready Base (GRB) in Maximus S and LT motors

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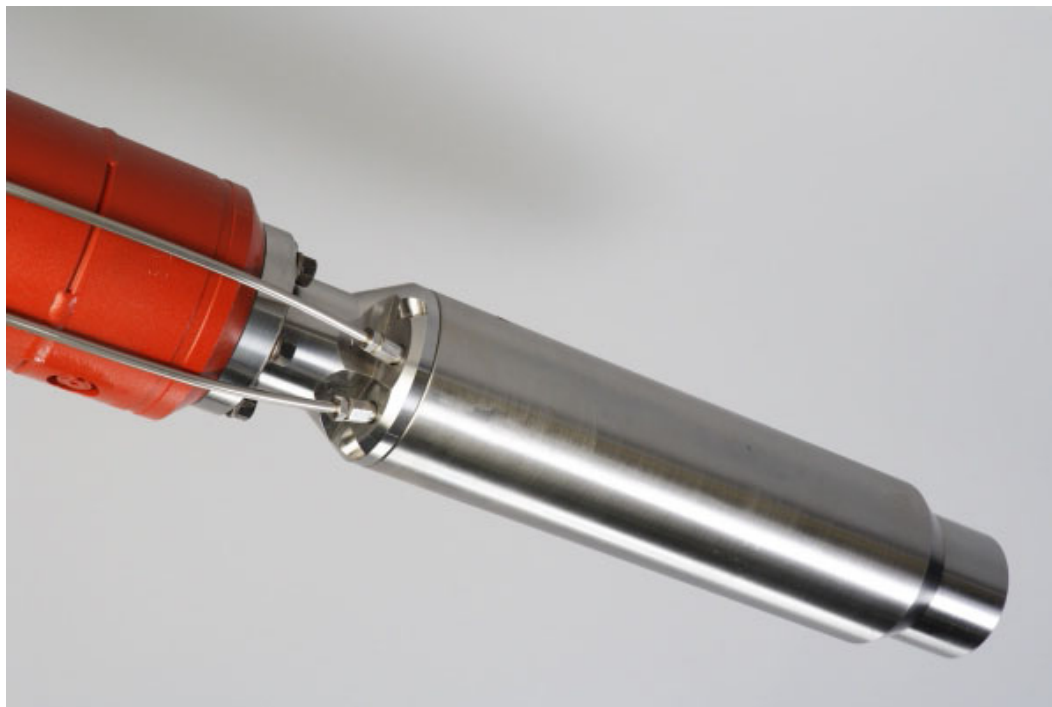


Figure C-4: Maximus LT - GRB Motor Connected to a Phoenix Select Gauge



Note

There are two drain-and-fill valves in the GRB, marked with “G” for communication to gauge, and “M” for communication to motor.

Section [Appendix C.3.3.1: Installing Gauge Vertically](#) and [Appendix C.3.3.2: Installing Gauge Horizontally](#) describe the gauge installation options.

C.3.3.1

Installing Gauge Vertically

1. Remove the vent plug from the GRB to relieve the vacuum inside the base, so that the cap can be easily removed.
2. Remove the shipping cap from the GRB.
3. Uncoil the I-wire and thermocouple wires from inside of the GRB.
4. Lift the gauge close to the base.
5. Connect the boots of the I-wire and the thermocouple wires to the corresponding terminals in the gauge head.
6. Install the gauge onto the base and fasten it to the base with the specified washers and bolts.

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7. Tighten the bolts with torque setting refer to the DME manual for that particular equipment.
8. Fill the space between the base and the gauge by pumping oil into the D and F valve marked with "G" until oil runs clear out off the vent hole.
9. Install plugs into the vent holes and drain-and-fill valves with new lead gaskets.

**Note**

Make sure the vent plugs are torqued to 15 ft-lbf and the drain-and-fill valves are torqued to 35 ft-lbf.

C.3.3.2 Installing Gauge Horizontally

1. Remove the vent plug from the GRB to relieve the vacuum inside the base, so that the cap can be easily removed.
2. Remove the shipping cap from the GRB.
3. Uncoil the I-wire and thermocouple wires from inside of the GRB.
4. Lift the gauge close to the base.
5. Connect the boots of the I-wire and the thermocouple wires to the corresponding terminals in the gauge head.
6. Install the gauge onto the base and fasten it to the base with the specified washers and bolts.
7. Tighten the bolts with torque setting refer to the DME manual for that particular equipment.
8. Roll the motor so that the vent hole in the base is pointing up.
9. Fill the space between the base and the gauge by pumping oil into the D and F valve marked with "G" until oil runs out off the vent hole. Alternatively, oil may be poured into the vent hole until it overflows.
10. Install plugs into the vent holes and drain-and-fill valves with new lead gaskets.

**Note**

Make sure the vent plugs are torqued to 15 ft-lbf and the drain-and-fill valves are torqued to 35 ft-lbf.

C.3.4 Motor Installation

C.3.4.1 Installation of Single Section Motor and Single Section ProMotor

**Caution**

Potential Severity: Light
Potential Loss: Information
Hazard Category: Machinery equipment hand tools

Before proceeding, read and follow section [Appendix C.3.1: General Notes on MaxJoints and Shipping Caps](#).

To install a single section motor or single section ProMotor, perform the following steps:

1. Raise the S (single section) motor (or S ProMotor throughout this section) over the wellhead and remove the tail clamp before lowering it into the wellhead.
2. Lower the S motor into the wellhead until the clamp rests on the work table. While lowering it, the plugs and drain-and-fill valves in the base and the head should be inspected for signs of leakage.
 - a. Tighten the valve body to 35 lbf-ft.
 - b. Tighten the vent plug in the valve to 15 lbf-ft.

**Note**

If the clamp prevents access to the head drain-and-fill valve, check it later after the clamp has been removed.

3. Remove the compensating shipping cap from the motor head. Check for free shaft rotation.
4. It is now ready to receive the next piece of the string, (e.g., protector, intake, gas device or pump). Place a clean spare conventional shipping cap on the head with two screws to prevent contamination while preparing the next piece.

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C.3.4.2 Installation of Lower Tandem motor

**Caution**

Potential Severity:	Light
Potential Loss:	Information
Hazard Category:	Machinery equipment hand tools

Before proceeding, read and follow the general notes above on MaxJoints and compensating shipping caps.

To install a lower tandem Maximus motor, perform the following steps:

1. Raise the LT motor over the wellhead and remove the tail clamp before lowering it into the wellhead.
2. While lowering the LT motor, the plugs and drain-and-fill valves in the base and the head should be inspected for signs of leakage.
 - a. Tighten the valve body to 35 lbf-ft.
 - b. Tighten the vent plug in the valve to 15 lbf-ft.

**Note**

If the clamp prevents access to the head drain-and-fill valve, check it later after the clamp has been removed.

3. Lower the LT motor into the wellhead until the head clamp is approximately 18 in from the top of the work table.

**Note**

For use in temperatures below 5 degF (-15 degC), the hydraulic motor jacks should be filled with a low temperature oil, such as Exxon Unavis HV113 (B044780). The oil normally supplied with the jacks may become too viscous for the jacks to operate.

4. Slide the motor jack assembly (44826) into position on the work table. Ensure that both hydraulic jacks are in their retracted position.
5. Lower the motor into the wellhead until the clamp rests in the cradle on top of the jacks.
6. Remove the compensating shipping cap from the motor head.

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7. Check for free shaft rotation.
8. It is now ready to receive the next piece of the string (e.g., CT motor, UT motor or UT ProMotor). Place a clean spare conventional shipping cap on the head with two screws to prevent contamination while preparing the next piece.

C.3.4.3

Installation of CT Motor, UT Motor or UT ProMotor**Caution**

Potential Severity: Light
 Potential Loss: Information
 Hazard Category: Machinery equipment hand tools

Before proceeding, read and follow the general notes above on MaxJoints and compensating shipping caps.

To install CT/UT Motors (CT motor, UT Motor or UT ProMotor) perform the following steps:

1. Raise the CT/UT Motor over the wellhead. Remove the tail clamp and the lower shipping cap.
2. Use the spline alignment tool listed in [Table C-4](#) to align the flange
 - a. Insert the male end of the alignment tool into the coupling on the LT shaft.
 - b. Turn the shaft until the LT alignment stud enters the hole in the tool.
 - c. Remove the tool without rotating the shaft.
 - d. Place the female end of the tool over UT shaft.
 - e. Turn the shaft until the tool pin enters the correct bolt hole in the UT flange for the LT alignment stud, adjacent to the alignment groove on the flange OD.
 - f. Remove the tool without rotating the shaft.

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3. Select the correct O-ring(s) from [Table C-3](#) (456 Series uses 996058, 562 Series uses 1018761). Refer to the ALFORM-D connection tables ([InTouch ID: 3255850](#)) for a complete list of torque specifications. Lubricate the O-ring with Dow Corning Lubricant (0553420). Remove old O-ring and carefully install the new one into the correct position.

**Caution**

Potential Severity: Light
 Potential Loss: Assets, Process
 Hazard Category: Machinery equipment hand tools

It is important when connecting tandem motors that the rig is centered over the wellbore and that the traveling blocks are not swaying. In strong wind conditions, it may be necessary to have someone at the monkey board holding the blocks stationary, centered over the wellhead. Alternatively, this can be done from the floor with a rope or cable around the unit and tied back to the rig.

4. Inspect the terminal pins and insulators to ensure they are not bent or broken.
 - a. Have the rig operator lower the CT/UT Motor approximately 4- to 6-in above the LT Motor.
 - b. Pump both jacks simultaneously to slowly raise the LT Motor and engage the coupling.

Make sure the alignment pin slides into the respective hole and that the motor terminals are properly engaged.

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Electrical

Use care not to damage the terminal insulators or O-rings.

5. Install the bolts and lock washers. The torque specifications for 456 and 562 Series is 26 lbf-ft. Refer to the ALFORM-D connection tables ([InTouch ID: 3255850](#)) for a complete list of torque specifications.
6. Lower the CT/UT Motor into the wellhead until the clamps rest on the work table. While lowering the motor, each drain-and-fill valve and protector vent plug should be inspected for signs of leakage.
 - a. Tighten the valve body to 35 lbf-ft.
 - b. Tighten the vent plug in the valve to 15 lbf-ft.
7. Remove the shipping cap from the motor (or protector of a ProMotor) head and check rotation.

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8. It is now ready to receive the next piece of string (e.g., protector, intake, gas device, pump). Place a clean spare conventional shipping cap on the head with two screws to prevent contamination while preparing the next piece.

C.3.4.4

Installation of UT Motor with Trident Motor Head

Currently, two types of Trident UT motors are available the 738 Maximus motor (Trident-300) and 562 Maximus motor (Trident-400).

1. Lower UT motor and set the lifting clamp on to the hydraulic jack cradles and plate which are set in position on the top of the work table, remove spreader bar shackles from UT clamps.



Note

Ensure the hole covers are installed around the shroud to eliminate any debris/tooling falling into the shroud and capsule.

2. Fit the appropriate clamp onto the upper part of the compensating shipping cap (below shoulder on motor head).



Note

The standard 3/8-in plug (on the side of the head) at the top of the compensation shipping cap must be removed before attempting to disassemble the shipping cap from the ESP section. A vacuum may be present and it's possible that the ESP section can be picked up if the unit is not allowed to equalize.

3. Secure the shackles of the spreader bar onto the friction clamp.
4. Remove compensating shipping cap bolts from the flange and prepare to remove the compensating shipping cap.



Note

Make sure there is a spare 5 gallon plastic oil-bucket to catch the excess oil from the compensating shipping cap when remove it from the UT motor.

5. Slowly pick up compensating shipping cap and fit lower flange cap, lay equipment off to the side of the work area.

C.3.5 Protector Installation

By virtue of MaxJoint technology, Maximus protectors are ready for installation upon arrival at the wellsite. With no oil servicing required, the only procedure remaining at wellsite is to plug-and-play the motor and protector.

The screws at the lower and upper ends of Maximus protector shafts are shimmed by manufacturing to a very tight tolerance, ± 0.003 in. Therefore, it is not necessary to take any shimming measurements at the wellsite.

C.3.5.1 Tandem Maximus Protectors

In tandem Maximus protector applications, additional shims are required between the lower and upper protector due to thermal expansion. The additional shims should be calculated and prepared in the ART Center prior to the installation. If this is not possible, the shims calculation must be done in the field.

The additional shims for each tandem protector application are calculated by considering the well fluid temperature at pump setting depth (BHT), lower protector make-up length and material. The formulas for calculating the additional shims are listed in section [Appendix C.3.5.2: Tandem Maximus Protector Shimming Calculations](#).



Note: Communicate shimming requirements

The Application Engineer must be aware of any additional shimming requirements, and should communicate the requirements to the field crew. The Field Engineer/Specialist must communicate with the Application Engineer to obtain the correct amount of shims required for each tandem protector application.



Figure C-5: Maximus protector base with MaxJoint connection technology

C.3.5.2

Tandem Maximus Protector Shimming Calculations

The amount of shims required between the upper and lower protector is calculated for each tandem protector application according to the well fluid temperature at pump setting depth (BHT), the lower protector make-up length and the material. [Table C-5](#) provides the Maximus protector make-up length obtained from OneCAT, according to the series and number of chambers in the protectors. Verify this information in OneCat prior to installation.

Table C-5: Maximus Protectors Make-up Length

Series	2 Chambers (in)	3 Chambers (in)
375/325/375	68	96
387	70	96
400	70	96
540	78	107
562	-	108

The following are the formulas for calculating the amount of shims required between tandem Maximus protectors, according to the lower protector housing and shaft materials.

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Equation C-1: Lower Protector with Carbon Steel housing and Monel shaft material

$$Shims = 0.095 \text{ in} - \frac{L * (BHT - AT)}{1,000,000}$$

Equation C-2: Lower Protector with Carbon Steel housing and Inconel shaft material

$$Shims = 0.095 \text{ in} - \frac{2 * L * (BHT - AT)}{10,000,000}$$

Equation C-3: Lower Protector with Redalloy housing and Monel shaft material

$$Shims = 0.095 \text{ in} - \frac{1.9 * L * (BHT - AT)}{1,000,000}$$

Equation C-4: Lower Protector with Redalloy housing and Inconel shaft material

$$Shims = 0.095 \text{ in} - \frac{1.1 * L * (BHT - AT)}{1,000,000}$$

Term	Definition
Shims	Total thickness of shims (in)
L	Make-up length of lower protector (in)
BHT	Well fluid temperature at pump setting depth (degF)
AT	Ambient temperature (degF) at the manufacturing center. Use 75 degF.

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**Example**

Conditions: Two 540-series protector are going to be installed in tandem. The lower protector has a two chamber configuration and is built using a Redalloy housing and Inconel shaft material. From Application Engineer information, the BHT of the well is 300 degF.

Question: What shim(s) should be added between the upper and lower protectors shaft?

Answer: From [Table C-5: Maximus Protectors Make-up Length](#), the make-up length of a two chamber 540-series Protector is 78 in. By using [Equation C-4: Lower Protector with Redalloy housing and Inconel shaft material](#), the shims required can be calculated as follows:

$$\text{Shims} = 0.095 \text{ in} - \frac{1.1 * 78 * (300 - 75)}{1,000,000} = 0.0757 \text{ in}$$

Shims = One 1/16-in shim plate + Three 1/200-in shim plates ~ 0.0775 in

**Note**

Convert the result of the shim calculation into fraction value (i.e., 1/16-in, 1/32-in, 1/64-in or 1/200-in) and use the shim plates provide in [Table 4-23: Shim Material](#). Always round up the result from the calculation to maximum of 1/200 in (0.005 in).

C.3.5.3

Installation Single or Lower Tandem Protector**Caution**

Potential Severity:	Light
Potential Loss:	Information
Hazard Category:	Machinery equipment hand tools

Before proceeding, read and follow the general notes in section [Appendix C.3.1: General Notes on MaxJoints and Shipping Caps](#).

1. Lift the protector to a position 2 to 3 ft above the motor.
2. Remove the shipping cap and gasket from the base of the protector.

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3. Remove the O-ring and carefully install a new O-ring. The ALFORM-D connection tables ([InTouch ID: 3255850](#)) provide O-ring part numbers, coupling and bolt torque information.

**Note**

A second backup O-ring is NOT required above the groove O-ring on 540- and 562-series protectors.

4. Use spline wrenches to ensure the shafts of the motor and protector turn freely and smoothly.
5. Ensure the coupling is in place on the motor shaft.
6. Have the rig operator slowly lower the protector until the shafts are engaged.
 - a. Turn the protector so that the flat cable slots on the protector flange and motor flange line up with each other.
 - b. Lower the protector until the flanges meet, making sure not to damage the O-ring.
7. Insert the bolts and washers and torque them to specification.
8. Pick up the assembly by the protector clamp.
9. Remove the motor clamp
10. Lower the assembly until the protector clamp rests on the work table. While lowering the protector, inspect each drain-and-fill valve and vent plug for signs of leakage.
 - a. Tighten the valve body to 35 lbf-ft.
 - b. Tighten the vent plug in the valve to 15 lbf-ft.
11. Remove the compensating shipping cap from the protector head.
12. Using a spline wrench, check rotation of the assembly.
13. The protector is now ready for the installation of the next component of the string (e.g., intake, gas device, UT protector, or LT pump). Place a clean spare standard shipping cap on the head of the protector and tighten with two bolts to protect it from contamination while preparing the next string component.

C.3.5.4 Installation of Upper Tandem Protector

**Note**

375/325/375-series, 540-series and 562-series Maximus protectors do not have a UT or LT configuration. If these types of protectors are installed in tandem, the top (uppermost) protector will be called the UT and the bottom (lowermost) protector will be called the LT.

387-series UT and 400/400-series UT Maximus protectors have a 400-series flange at the base that can be connected directly to the head of any 400/456-series S/LT protector. 387-series and 400/400-series UT Maximus protectors come equipped with a tandem protector-to-protector coupling (P/N 0304444). No other couplings or adapters are required.

1. Lift the UT protector to a position 2 to 3 ft above the LT protector (or ProMotor throughout this section).
2. Remove the O-ring and carefully install a new O-ring. The ALFORM-D connection tables ([InTouch ID: 3255850](#)) provide O-ring part numbers, coupling and bolt torque information.

**Note**

A second backup O-ring is NOT required above the groove O-ring on 540- and 562-series protectors.

3. Use spline wrenches to ensure the shafts of the UT and LT protector turn freely and smoothly.
4. Ensure the coupling is in place on the S/LT protector shaft and place the required shims in the coupling of the S/LT protector. Refer to the following note.

**Note: Required shims**

The required shims for Maximus tandem protectors are explained in section [Appendix C.3.5: Protector Installation](#), while the formulas for calculating the required shims are available in section [Appendix C.3.5.2: Tandem Maximus Protector Shimming Calculations](#).

5. Have the rig operator slowly lower the UT protector until the shafts are engaged.
 - a. If a cable slot is present on the UT protector head or base, rotate the UT protector so that the cable slots line up with the LT protector or motor cable slot.
 - b. Lower the protector until the flanges meet, making sure not to damage the O-ring.
6. Insert the bolts and washers and torque them to specification.
7. Pick up the assembly by the UT protector clamp.
8. Remove the LT protector clamp and lower the assembly until the UT protector clamp rests on the work table. While lowering the UT protector, inspect each drain-and-fill valve and vent plug for signs of leakage.
 - a. Tighten the valve body to 35 lbf-ft.
 - b. Tighten the vent plug in the valve to 15 lbf-ft.
9. Remove the compensating shipping cap from the UT protector head. Using a spline wrench, check rotation of the assembly.
10. The protector is now ready for the installation of the next component of the string (e.g., intake, gas device, or pump). Place a clean spare standard shipping cap on the head of the protector and tighten with two bolts to protect it from contamination while preparing the next string component.

C.3.6 **Shimming Procedures for Protectors-to-Pumps**

Maximus protectors include the factory shimming feature and should be installed accordingly. Refer to the [pump shimming and installation procedures](#) in the [Wellsite Procedures](#) section.

C.3.7 **MaxLok Plug-in Pothead Installation**

The MaxLok ESP motor pothead design eliminates the need for taping pothead electrical connections considerably reducing the time required to make the connection while maintaining the same performance and reliability.

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Figure C-6: Maximus Motor Lead Extension (MLE) with MaxLok motor pothead



Caution

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

Do not install a non-Maximus plug-in MLE/pothead into a Maximus motor. This combination will not seal properly and a failure will, likely, result. All Maximus MLEs/potheads must be checked prior to installation to confirm that they are compatible with Maximus motor.

The Maximus pothead seals with a O-ring contained on a groove, as shown in the figure [Figure C-6](#). Non-Maximus potheads utilize a “trap” seal and do not have an O-ring groove. The two are NOT interchangeable and cannot be mixed

Equipment tests and assembly procedures for the plug-in design are similar to tests and procedures for the standard plug-in design.

Raise the string until the pothole on the UT or S motor (or UT or S ProMotor) is at a comfortable working height.

1. While holding the pothole shipping cover down, remove the nuts.
2. Remove the shipping cover.
3. Remove the shipping cap from the flat cable extension.
4. Remove the O-ring.
5. Apply a coating of Dow #4 or Dow #111 grease to a new O-ring. Install the O-ring in the pothead O-ring gland.

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6. Inspect the inside of the pothead flange to verify that a black rubber gasket is installed. The light gray color of the pothead front terminal block will be observed if a gasket is not installed.

**Note**

If a gasket is found in the shipping cap rather than installed in the pothead:

- a. Ensure the gasket is clean and undamaged.
 - b. Apply a coating of Dow #4 or Dow#111 grease to the gasket.
 - c. Push the gasket into the pothead cavity. It will be pushed into the proper location when the pothead is plugged to the motor terminal block.
-

**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

Do not install two gaskets. The installation of two gaskets may crack the terminal block. This will result in insufficient terminal engagement, a short to ground and disassembly problems when the pothead is unplugged from the motor.

**Caution**

Potential Severity: Light
 Potential Loss: Assets, Reputation
 Hazard Category: Electrical

Ensure the PEEK shroud external surface is free from moisture, condensation and contamination. This surface is sensitive to arc tracking and must be protected by motor oil or silicon grease.

7. Fill the remaining pothead cavity with Dow #4 grease or Reda #5 motor oil.

- **Motor oil:** Flood the pothole and the pothead front cavity with Reda #5 motor oil or apply a coating of Dow #4 or Dow #111 to the front cavity of the pothead.



Caution

Potential Severity: Light

Potential Loss: Assets

Hazard Category: Machinery equipment hand tools, Temperature

Do not use grease in temperatures below 5 degF (-15 degC). Instead, flush the pothead and pothole with the proper motor oil immediately before plugging-in. In cold weather, the grease can be so stiff that it interferes with electrical contact.

- **Grease:** Fill the front cavity of the pothead evenly with Dow #4 or Dow #11 grease. The small gaps between the PEEK shroud/flange and between the three PEEK shrouds must be filled with the grease. Displace as much air as possible from the pothead cavity with the grease.

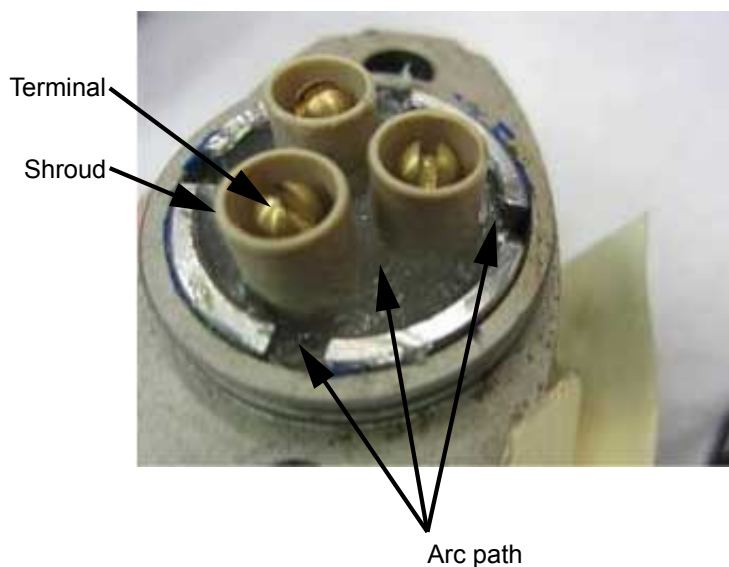


Figure C-7: Fill Pothead Cavity. Fill the cavity with Reda #5 motor oil or coat with Dow #4 or Dow #111 grease.

8. Ensure that no moisture or contamination has been introduced to the pothead.
9. Align the terminals, then quickly and carefully push the pothead assembly into the motor pothole by hand.
10. Hold the pothead in position and install the retaining nuts.
11. Tighten the retaining nuts down evenly and torque to 16 lbf-ft.

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**Caution**

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools

The pothole is equipped with a valve that prevents oil loss while the pothole is open, but allows pressure equalization when the pothead is attached. While pothead is half engaged, oil can seep out, so the pothead should be plugged in quickly to minimize oil loss. Hold the pothead fully engaged while installing or removing the nuts but do not leave the pothead half engaged. Loss of a few spoonfuls of oil would not be a problem, but if a cupful was lost, the protector should be refilled per the standard [oil filling procedures for protectors](#) like a Modular protector.

The string is now ready for electrical testing through the cable, tubing installation and cable banding per the standard [cable testing procedures](#) and [wellsite procedures](#).

C.3.8

Trident Plug-in Pothead Installation

Currently, there are two types of Trident MLE are available: Trident-300 and Trident-400. The Trident-300 is used with the 738 Maximus motor for off-shore applications and has pothead pressure test ports on the motor head. The Trident-400 is used with the 562 High Ampacity Maximus motor and does not have pressure test ports.

The Trident-300 pothead and Trident-400 pothead have identical geometry and dimensions. The only differences between the two potheads are the O-ring materials and configurations. The Trident-300 pothead uses two size 2-022, HNBR, 70 Durometer O-rings. The Trident-400 pothead uses one size 2-022, Low Temperature 75 Durometer O-ring and one PEEK gland filling ring. This is because 562 Maximus motor head does not have pressure test ports.

1. Install UT motor with Trident motohead. Refer to [Appendix C.3.4.4: Installation of UT Motor with Trident Motor Head](#).
2. Perform UT motor electrical checks.
 - a. Remove the three pothole caps from the UT motor, visually inspect internal electrical connections.

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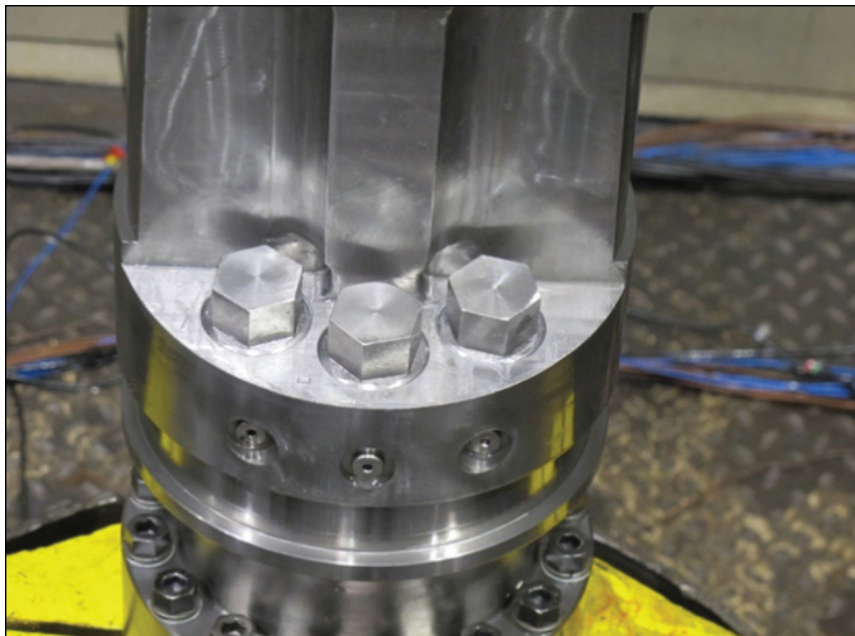


Figure C-8: Maximus Trident motor head

- b.** Install special UT motor test probes (P/N 101049527) into the motor head potholes.

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Figure C-9: Test Motor using Trident Test Probes

i Note

The Trident test probes allow the testing of the UT motor and reduce the risk of contamination during testing. Record all electrical data on ESP system integrity sheet.

- c. Perform ground resistance tests.
 - d. Perform phase resistance tests.
 - e. Perform functional tests on downhole gauge.
3. Remove electrical test probes and replace the pothole caps.

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4. Perform UT motor mechanical checks.
 - a. Verify that the flange and seal face are not damaged.
 - b. Ensure the coupling part number is checked against the power string number and installed correctly on the motor shaft.
 - c. Check motor shaft rotation using special tools (100835923).
 - d. Align UT motor shaft for matching procedure using special tool (100831872).
 - e. Service the UT motor head using fresh dielectric oil.
 - f. Cover and protect UT motor head in preparation to install the LT protector.
5. Install the LT protector.

**Note**

The Trident MLE must be installed after the installation of LT protector and before the installation of the tandem protector. This prevents the hydrostatic force of the motor oil in the protector from forcing the motor oil out of the trident potholes when the pothole caps are removed. Also, the MLE will cause problems if the LT protector is installed to the motor after the MLE.

6. Prepare the Trident MLE for installation.
 - a. Pull the MLE from the reel and temporarily secured the MLE above motor head.
 - b. Identify the phase sequence A-B-C on pothead.

**Note**

Phase sequence should be A-B-C from left to right.

- c. Separate the MLE, allow enough separation and flexibility while preparing the pothead.
- d. Check the hard stamp A-B-C mark, from left to right, on the potholes of the motor head.
- e. Remove pothead shipping caps and visually check for any damage on sealing surfaces and terminals.

**Caution**

Potential Severity: Light
Potential Loss: Assets
Hazard Category: Machinery equipment hand tools

Always keep the shipping cap closed unless the pothead is being tested or installed. The terminal insulator (shroud) of the pothead is very sensitive to moisture and contamination. NEVER contact the shroud with dirty hands.

7. Remove and discard original O-rings installed on the pothead for shipping purpose.
8. Install the supplied new O-rings into the O-ring grooves.
 - a. **Trident-300 pothead:** Install two new size 2-022 HNBR O-rings (100781769) into the two O-ring grooves on each pothead.

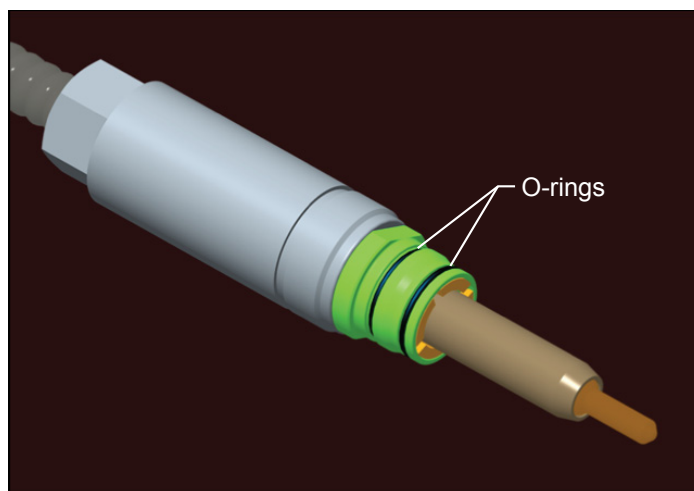


Figure C-10: Trident-300 pothead

- b. **Trident-400 pothead:** Install one new size 2-022 Viton O-ring (100925883) into the small O-ring groove on the lower end of each pothead. Verify a PEEK gland filling ring (with scuff cut) has been installed in the larger O-ring groove on the upper end of each pothead. It is factory installed to prevent the installation of an O-ring in this groove by mistake. Do NOT install an O-ring in this groove.

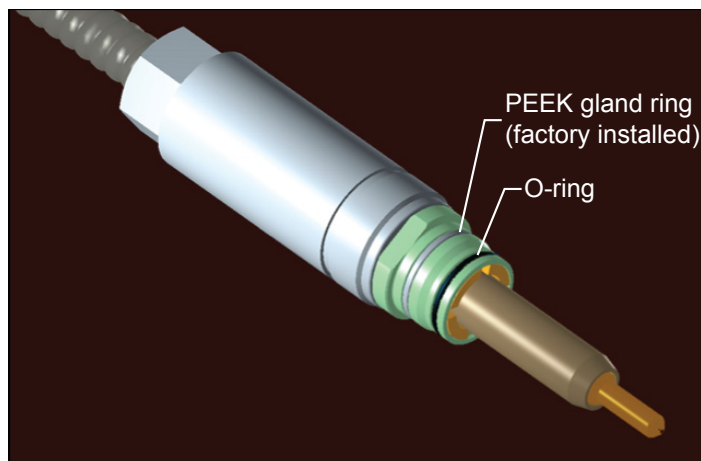


Figure C-11: Trident-400 pothead



Caution

Potential Severity: Light
 Potential Loss: Assets
 Hazard Category: Machinery equipment hand tools, Pressure

The Trident-300 potheads only use size 2-022 O-rings, HNBR 70 Durometer (100781769). The Trident-400 potheads only use size 2-022 O-rings, 75 Durometer, GT 761 (100925883). Do not use any O-rings made from other materials, or by any other suppliers.

9. Apply a small amount of Dow Corning 111 grease onto the O-ring surfaces.
10. Verify that the pothead terminals and surrounding area are clean for the electrical connection.
11. Insert each Trident pothead into the motor head and hand-tighten the pothead retaining nut ([Figure C-12: Trident Pothead Connections](#)).

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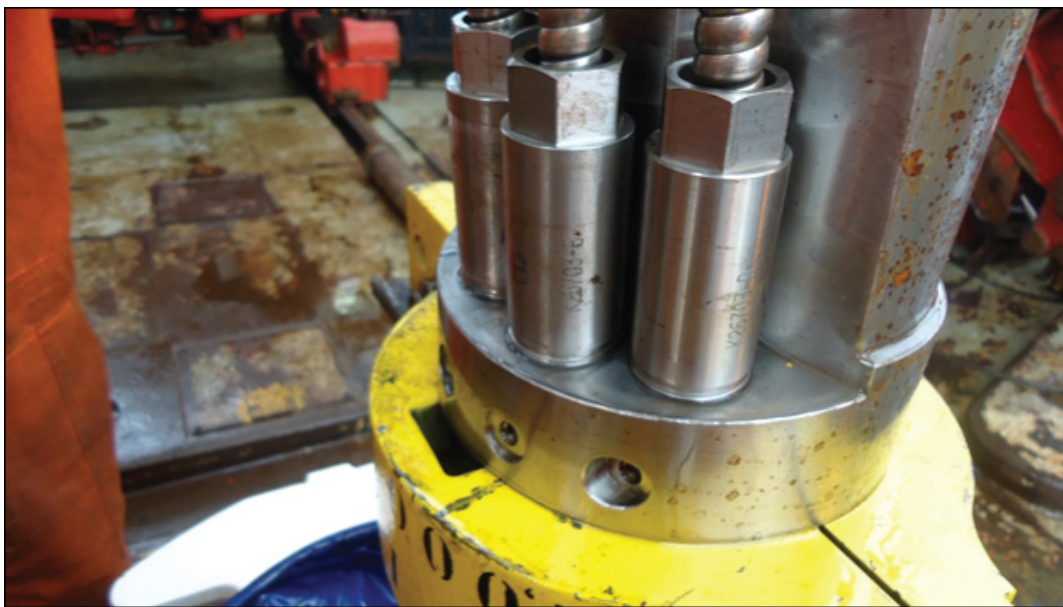


Figure C-12: Trident Pothead Connections



Note

Make sure O-rings are fully engaged. Be careful not damage the O-rings while installing the potheads in the potholes.

12. Tighten each 1-3/16-in pothead retaining nut to specified torque 35 ft-lbf with the pre-set torque wrench (100793643).



Figure C-13: Torque Trident Connection

13. Verify that all three pothead connections are installed correctly.
14. Pothead installation is complete.

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**Note: Pressure test Trident-300 pothead**

Each Trident-300 pothead connection must be pressure tested before continuing to the next step. Refer to [Appendix C.3.8.1: Trident-300 Pressure Test](#) for pressure test procedure.

The Trident-400 pothead connections are not pressure tested. The 562 Maximus motor head does not have pressure test ports.

15. The string is now ready for electrical testing through the cable, tubing installation, and cable banding per the standard [wellsite procedures](#).

C.3.8.1 Trident-300 Pressure Test

Each Trident-300 pothead connection must be pressure tested using the pressure test ports on the motor head.

1. Install the pressure test manifold to the SAE port in the UT motor head.
2. Connect hydraulic pump to the pressure test manifold. Pump and bleed the hose from pump to manifold before connecting.



Figure C-14: Pressure Test Trident Pothead Connection

3. Pressure up to 100 psi; stabilize pressure and hold for 5 minutes.

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**Note**

Do not over pressurize the gauge. Ensure you have a good view of the gauge as it is very easy to over pressurize the gauge and damage it.

4. Record the pressure after 5 minutes.
5. Lower pressure to 10 psi; stabilize pressure and hold for another 5 minutes.
6. Record the pressure after 5 minutes.
7. If the Trident pothead successfully passed the pressure test, continue to next step.
If the pothead failed the pressure test,
 - a. Remove the pothead from the motor head.
 - b. Thoroughly check and analyze the causes of the failure, and solve problems.
 - c. Reinstall pothead and pressure test the pothead connection again.
Repeat steps 8 through 13 in the section [Appendix C.3.8: Trident Plug-in Pothead Installation](#).
8. After a successful pressure test, remove the manifold and equipment and replace the SAE plugs.

**Note**

Do NOT install O-rings on the SAE plugs. Verify that the plug (100813229) has a groove on the internal face that allows a communication path. This will ensure that the pressure will not build between the O-rings during the operation.

9. The string is now ready for electrical testing through the cable, tubing installation, and cable banding per the standard [wellsite procedures](#).

Abbreviated Terms

Term	Definition
BOH	Bolt-On Head
ESP	Electric Submersible Pump
FCE	Flat Cable Extension
ID	Inside Diameter
MLE	Multilead Extension
OD	Outside Diameter
PIP	Pump Intake Pressure
PIT	Pump Intake Temperature
PSI	Pressure Sensing Instrument
SPS	Submersible Pumping System
TDH	Total Dynamic Head
THP	Tubing Head Pressure
WCP	Well Completions and Productivity

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